

means that the observed number of double crossovers was only 50 percent of the number we would expect to observe if crossing-over in the two regions were independent. The value of the interference depends on the distance between the genetic markers and on the species. In some species, the interference increases as the distance between the two outside markers becomes smaller, until a point is reached at which double crossing-over is eliminated; that is, no double crossovers are found, and the coefficient of coincidence equals 0 (or, to say the same thing, the interference equals 1). In *Drosophila*, this distance is about 10 map units. In yeast, by contrast, interference is incomplete even over short distances. For markers separated by 3 map units, the interference is in the range 0.3 to 0.6; for those separated by 7 map units, it is in the range 0.1 to 0.3. In most organisms, when the total distance between the genetic markers is greater than about 30 map units, interference essentially disappears and the coefficient of coincidence approaches 1.

Genetic Mapping Functions

The effect of interference on the relationship between genetic map distance and the frequency of recombination is illustrated in Figure 4.15. Each curve in Figure 4.15 is an example of a **mapping function**, which is the mathematical relation between the genetic distance across an interval in map units (centimorgans) and the observed frequency of recombination across the interval. In other words, a mapping function tells you how to convert a *map distance* between genetic markers into a *recombination frequency* between the markers. As we have seen, when the map distance between the markers is small, the recombination frequency equals the map distance. This principle is reflected in the curves in Figure 4.15 in the region in which the map distance is smaller than about 10 cM. At less than this distance, all of the curves are nearly straight lines, which means that map distance and recombination frequency are equal; 1 map unit equals 1 percent recombination, and 10 map units equal 10 percent recombination.

For distances greater than 10 map units, the recombination frequency becomes smaller than the map distance. How much smaller it is, for any given map distance, depends on the pattern of interference along the chromosome. Each pattern of interference yields a different mapping function. In Figure 4.15, three types of mapping functions are shown. The upper curve is based on the assumption of com-

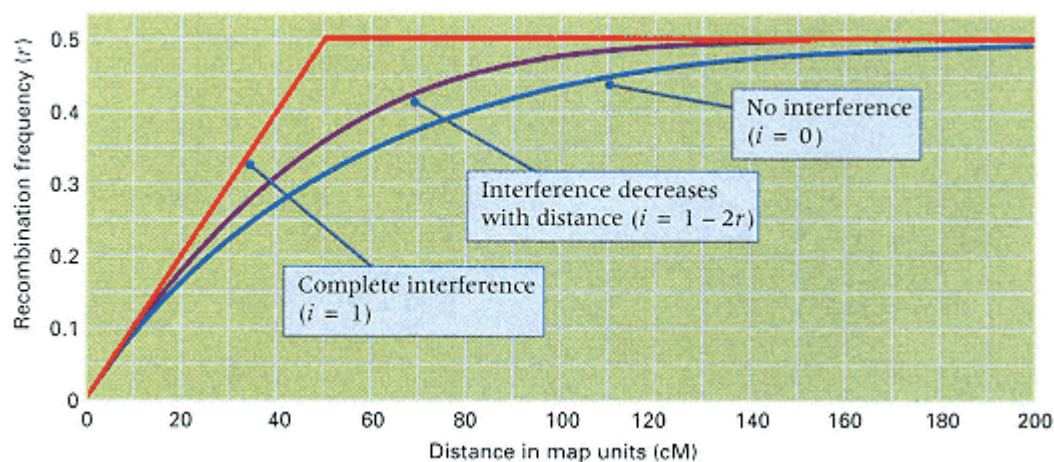


Figure 4.15

A mapping function is the relation between genetic map distance across an interval and the observed frequency of recombination across the interval. Map distance is defined as one-half the average number of crossovers converted into a percentage. The three mapping functions correspond to different assumptions about interference, i . In the top curve, $i = 1$ (complete interference); in the bottom curve, $i = 0$ (no interference). The mapping function in the middle is based on the assumption that i decreases as a linear function of distance.

plete interference i , so that $i = 1$. With this mapping function, the linear relation holds all the way to a map distance of 50 cM, for which the recombination frequency is 50 percent; for map distances larger than 50 cM, the recombination frequency remains constant at 50 percent.

The bottom curve in Figure 4.15 is usually called **Haldane's mapping function** after its inventor. It assumes no interference ($i = 0$), and the mathematical form of the function is $r = (1/2)(1 - e^{-d/50})$, where d is the map distance in centimorgans. Any mapping function for which i is between 0 and 1 must lie in the interval between the top and bottom curves. The example shown is **Kosambi's mapping function**, in which the interference is assumed to decrease as a linear function of distance according to $i = 1 - 2r$. Although simple in its underlying assumptions, the formula for Kosambi's function is not simple. (The formula is in one of the problems at the end of the chapter.)

Haldane developed his mapping function in 1919, Kosambi his in 1943. Between these years and long afterward, geneticists had little interest in different mapping functions other than as curiosities, because there were few sets of data large enough to distinguish one reasonable function from the next. In recent years, with the explosion in the number of genetic markers available in virtually all organisms, and with the resurgence of interest in genetic mapping because of its role in identifying the position of mutations as precisely as possible prior to cloning (isolating the DNA), mapping functions have again become moderately fashionable. Checked against large data sets, none of the simple mapping functions in Figure 4.15 fits perfectly, but alternatives that fit better are much more complex even than Kosambi's mapping function.

Most mapping functions are almost linear near the origin, as are those in Figure 4.15. This near linearity implies that for map distances smaller than about 10 cM, whatever the pattern of chromosome interference, there are so few double recombinants that the recombination frequency in percent essentially equals the map distance. Hence the map distance between two widely separated genetic markers can be estimated with some confidence by summing the map distances across smaller segments between the markers, provided that each of the smaller segments is less than about 10 map units in length.

Genetic Distance and Physical Distance

Generally speaking, the greater the physical separation between genes along a chromosome, the greater the map distance between them. Physical distance and genetic map distance are usually correlated because a greater distance between genetic markers affords a greater chance for a crossover to take place; crossing-over is a physical exchange between the chromatids of paired homologous chromosomes.

On the other hand, the general correlation between physical distance and genetic map distance is by no means absolute. We have already noted that the frequency of recombination between genes may differ in males and females. An unequal frequency of recombination means that the sexes can have different map distances in their genetic maps, although the physical chromosomes of the two sexes are the same and the genes must have the same linear order. An extreme example of a sex difference in recombination is in *Drosophila*, in which there is no recombination in males (as we noted earlier). Hence, in *Drosophila* males, the map distance between any pair of genes located in the same chromosome is 0. (Genes on different chromosomes do undergo independent assortment in males.)

The general correlation between physical distance and genetic map distance can even break down in a single chromosome. For example, crossing-over is much less frequent in certain regions of the chromosome than in other regions. The term **heterochromatin** refers to certain regions of the chromosome that have a dense, compact structure in interphase; these regions take up many of the standard dyes used to make chromosomes visible. The rest of the chromatin, which becomes visible only after chromosome condensation in mitosis or meiosis, is called **euchromatin**. In most organisms, the major heterochromatic regions are adjacent to the centromere; smaller blocks are present at the ends of the chromosome arms (the telomeres) and interspersed with the euchromatin. In

general, crossing-over is much less frequent in regions of heterochromatin than in regions of euchromatin.

Because there is less crossing-over in heterochromatin, a given length of heterochromatin will appear much shorter in the genetic map than an equal length of euchromatin. In heterochromatic regions, therefore, the genetic map gives a distorted picture of the physical map. An example of such distortion appears in Figure 4.16, which compares the physical map and the genetic map of chromosome 2 in *Drosophila*. The physical map is depicted as the chromosome appears in metaphase of mitosis. Two genes near the tips and two near the euchromatin-heterochromatin junction are indicated in the genetic map. The map distances across the euchromatic arms are 54.5 and 49.5 map units, respectively, for a total euchromatic map distance of 104.0 map units. However, the heterochromatin, which constitutes approximately 25 percent of the entire chromosome, has a

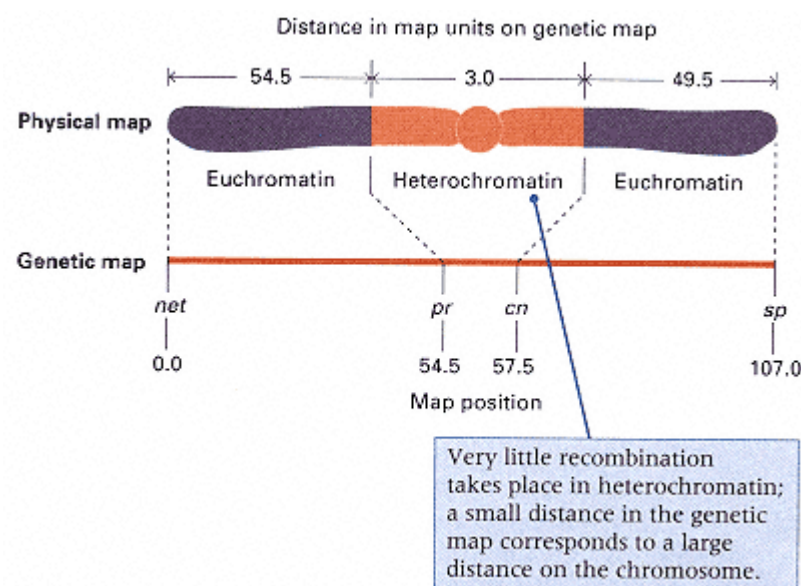


Figure 4.16

Chromosome 2 in *Drosophila* as it appears in metaphase of mitosis (physical map, top) and in the genetic map (bottom). Heterochromatin and euchromatin are in contrasting colors. The genes indicated on the map are *net* (net wing veins), *pr* (purple eye color), *cn* (cinnabar eye color), and *sp* (speck of wing pigment). The genes *pr* and *cn* are actually in euchromatin but are located near the junction with heterochromatin.

The total map length is $54.5 + 49.5 + 3.0 = 107.0$ map units. The heterochromatin accounts for $3.0 / 107.0 = 2.8$ percent of the total map length but constitutes approximately 25 percent of the physical length of the metaphase chromosome.

genetic length in map units of only 3.0 percent. The distorted length of the heterochromatin in the genetic map results from the reduced frequency of crossing-over in the heterochromatin. In spite of the distortion of the genetic map across the heterochromatin, in the regions of euchromatin there is a good correlation between the physical distance between genes and their distance in map units in the genetic map.

4.4—

Genetic Mapping in Human Pedigrees

Before the advent of recombinant DNA, mapping genes in human beings was very tedious and slow. There were numerous practical obstacles to genetic mapping in human pedigrees:

1. Most genes that cause genetic diseases are rare, so they are observed in only a small number of families.
2. Many genes of interest in human genetics are recessive, so they are not detected in heterozygous genotypes.
3. The number of offspring per human family is relatively small, so segregation cannot usually be detected in single sibships.
4. The human geneticist cannot perform testcrosses or backcrosses, because human matings are not manipulated by an experimenter.

In recent years, because recombinant-DNA techniques allow direct access to the DNA, genetic mapping in human pedigrees has been carried out primarily by using genetic markers present in the DNA itself, rather than through the phenotypes produced by mutant genes. There are many minor differences in DNA sequence from one person to the next. On the average, the DNA sequences at corresponding positions in any two chromosomes, taken from any two people, differ at approximately one in every thousand base pairs. Most of the differences in DNA sequence are not associated with any inherited disease or disability. Indeed, many of the differences

are found in DNA sequences that do not code for proteins. Nevertheless, all of these differences can serve as convenient genetic markers, and differences that are genetically linked to genes causing hereditary diseases are particularly important. Some differences in nucleotide sequence are detected by means of a type of enzyme called a **restriction endonuclease**, which cleaves double-stranded DNA molecules wherever a particular, short sequence of bases is present. For example, the restriction enzyme *EcoRI* cleaves DNA wherever the sequence GAATTC appears in either strand, as illustrated in Figure 4.17. Restriction enzymes will be considered in detail in Chapter 5. For now, we simply note that their significance is related to the fact that a difference in DNA sequence that eliminates a cleavage site can be detected because the region lacking the cleavage site will be cleaved into one larger fragment instead of two smaller ones (Figure 4.18). More rarely, a mutation in DNA sequence will create a new site rather than destroy one already present. Organisms, including human beings, frequently have minor differences in DNA sequence that are present in homologous, and otherwise completely identical, regions of DNA; any difference that alters a cleavage site will also change the length of the DNA fragments produced by cleavage with the corresponding restriction enzyme. The different DNA fragments can be separated by size by an electric field in a supporting gel and detected by various means. Differences in DNA fragment length produced by the presence or absence of the cleavage sites in DNA molecules are known as **restriction fragment length polymorphisms (RFLPs)**.

RFLPs are typically formed in one of two ways. A mutation that changes a base sequence may result in loss or gain of a cleavage site that is recognized by the restriction endonuclease in use. Figure 4.19A gives an example. On the left is shown the relevant region in the homologous DNA molecules in a person who is heterozygous for such a sequence polymorphism. The homologous chromosomes in the person are distinguished by the letters *a* and *b*. In the region of interest, chromosome *a* contains two cleavage sites and chromosome *b* contains three. On the right is shown the position of the DNA fragments produced by cleavage after separation in an electric field. Each fragment appears as a horizontal band in the gel. The fragment from chromosome *a* migrates more slowly than those from chromosome *b* because it is larger, and larger fragments move more slowly through the gel. In this example, DNA from a person heterozygous for the *a* and *b* types of chromosomes would yield three bands in a gel. Similarly,

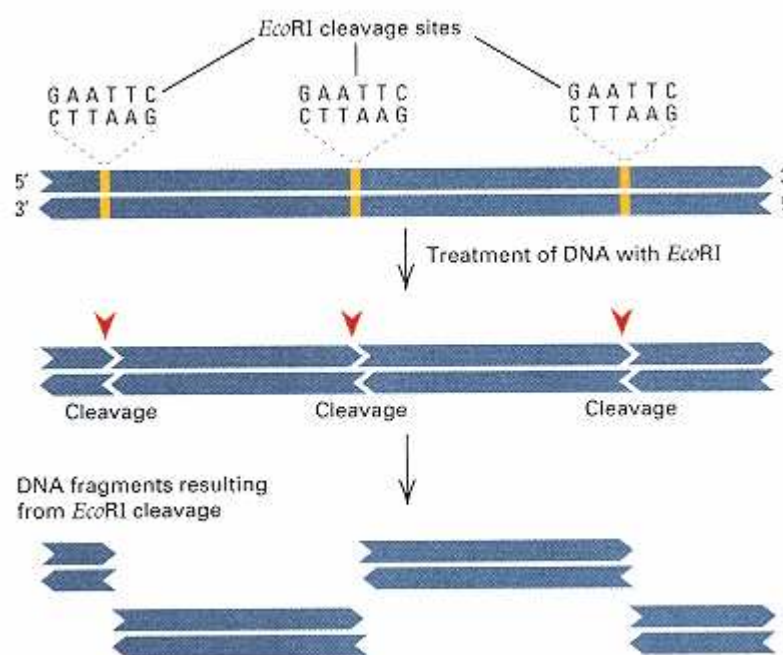


Figure 4.17

The restriction enzyme *EcoRI* cleaves double-stranded DNA wherever the sequence 5'-GAATTC-3' is present. In the example shown here, the DNA molecule contains three *EcoRI* cleavage sites, and it is cleaved at each site, producing a number of fragments.

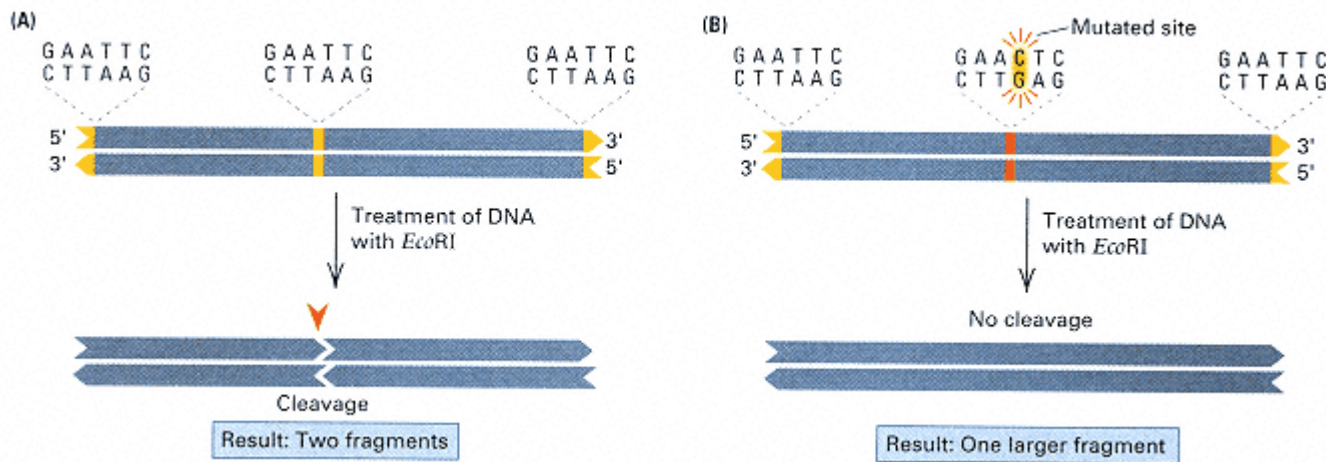


Figure 4.18

A minor difference in the DNA sequence of two molecules can be detected if the difference eliminates a restriction site. (A) This molecule contains three restriction sites for *EcoRI*, including one at each end. It is cleaved into two fragments by the enzyme. (B) This molecule has a mutated base sequence in the *EcoRI* site in the middle. It changes 5'-GAATTC-3' into 5'-GAACTC-3', which is no longer cleaved by *EcoRI*. Treatment of this molecule with *EcoRI* results in one larger fragment.

DNA from homozygous *aa* would yield one band, and that from homozygous *bb* would yield two bands.

A second type of RFLP results from differences in the number of copies of a short DNA sequence that may be repeated many times in tandem at a particular site in a chromosome (Figure 4.19B). In a particular chromosome, the tandem repeats may contain any number of copies, typically ranging from ten to a few hundred. When a DNA molecule is cleaved with a restriction endonuclease that cleaves at sites flanking the tandem repeat, the size of the DNA fragment produced is determined by the number of repeats present in the molecule. Figure 4.19B illustrates homologous DNA sequences in a heterozygous person containing one chromosome *a* with two copies of the repeat and another chromosome *b* with five copies of the repeat. When cleaved and separated in a gel, chromosome *a* yields a shorter fragment than that from chromosome *b*, because *a* contains fewer copies of the repeat. An RFLP resulting from a **variable number of tandem repeats** is called a **VNTR**. The utility of VNTRs in human genetic mapping derives from the very large number of alleles that may be present in the human population. The large number of alleles also implies that most people will be heterozygous, so their DNA will yield two bands upon cleavage with the appropriate restriction endonuclease. Because of their high degree of variation among people, VNTRs are also widely used in DNA typing in criminal investigations (Chapter 15).

In genetic mapping, the phenotype of a person with respect to an RFLP is a pattern of bands in a gel. As with any other type of gene, the genotype of a person with respect to RFLP alleles is inferred, insofar as it is possible, from the phenotype. Linkage between different RFLP loci is detected through lack of independent assortment of the alleles in pedigrees, and recombination and genetic mapping are carried out using the same principles that apply in other organisms except that, in human beings, because of the small family size, different pedigrees are pooled together for analysis. Primarily through the use of RFLP and VNTR polymorphisms, genetic mapping in humans has progressed rapidly.

A three-generation pedigree of a family segregating for several alleles at a VNTR locus is illustrated in Figure 4.20. In this example, each of the parents is heterozygous, as are all of the children. Yet every person can be assigned his or her genotype because the VNTR alleles are codominant. At present, DNA polymorphisms are the principal types of genetic markers used in

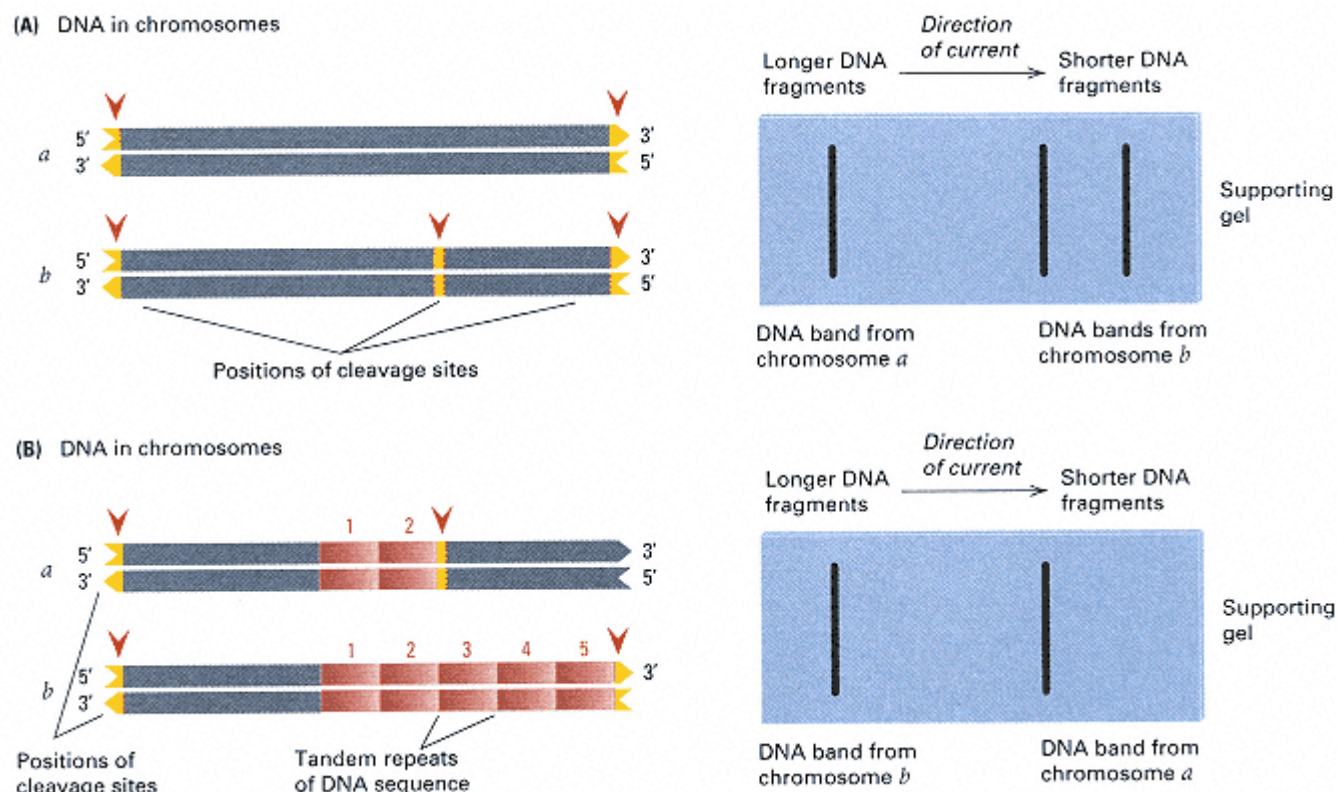


Figure 4.19

Two types of genetic variation that are widespread in most natural populations of animals and plants. (A) RFLP (restriction fragment length polymorphism), in which alleles differ in the presence or absence of a cleavage site in the DNA. The different alleles yield different fragment lengths (shown in the gel pattern at the right) when the molecules are cleaved with a restriction enzyme. (B) VNTR (variable number of tandem repeats), in which alleles differ in the number of repeating units present between two cleavage sites

genetic mapping in human pedigrees. Such polymorphisms are prevalent, are located in virtually all regions of the chromosome set, and have multiple alleles and so yield a high proportion of heterozygous genotypes. Furthermore, only a small amount of biological material is needed to perform the necessary tests. Many of the polymorphisms that are most useful in genetic mapping result from variation in the number of

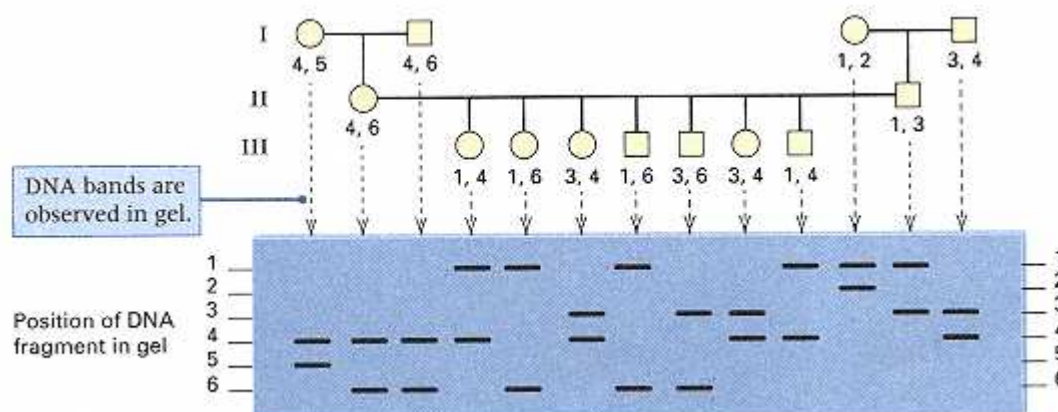


Figure 4.20

Human pedigree showing segregation of VNTR alleles. Six alleles (1–6) are present in the pedigree, but any one person can have only one allele (if homozygous) or two alleles (if heterozygous).

tandem copies of a simple repeating sequence present in the DNA, such as

5' - . . . TGTGTGTGTG . . . -3'

A simple-sequence polymorphism of this type is called a **simple tandem repeat polymorphism**, or **STRP**. The utility of STRPs in genetic mapping derives from the large number of alleles present in the population and the high proportion of genotypes that are heterozygous for two different alleles.

The present human genetic map is based on more than 5000 genetic markers, primarily STRPs, each heterozygous in an average of 70 percent of people tested. Because there is more recombination in females than in males, the female and male genetic maps differ in length. The female map is about 4400 cM, the male map about 2700 cM. Averaged over both sexes, the length of the human genetic map for all 23 pairs of chromosomes is about 3500 cM. Because the total DNA content per haploid set of chromosomes is 3154 million base pairs, there is, very roughly, 1 cM per million base pairs in the human genome.

4.5—

Mapping by Tetrad Analysis

In some species of fungi, each meiotic tetrad is contained in a sac-like structure called an **ascus** and can be recovered as an intact group. Each product of meiosis is included in a reproductive cell called an **ascospore**, and all of the ascospores formed from one meiotic cell remain together in the ascus (Figure 4.21). The advantage of using these organisms to study recombination is the potential for analyzing all of the products from each meiotic division. Two other features of the organisms are especially useful for genetic analysis: (1) They are haploid, so dominance is not a complicating factor because the genotype is expressed directly in the phenotype; and (2) they produce very large numbers of progeny, making it possible to detect rare events and to estimate their frequencies accurately.

The life cycles of these organisms tend to be short. The only diploid stage is the zygote, which undergoes meiosis soon after it is formed; the resulting haploid meiotic products (which form the ascospores) germinate to regenerate the vegetative stage (Figure 4.22). In some species, each of the four products of meiosis subsequently undergoes a mitotic division, with the result that each member of the tetrad yields a *pair* of genetically identical ascospores. In most of the organisms, the meiotic products, or their derivatives, are not arranged in any particular order in the ascus. However, bread molds of the genus *Neurospora* and related organisms have the useful characteristic that the meiotic products are arranged in a definite order directly related to the planes of the meiotic divisions. We will examine the ordered system after first looking at unordered tetrads.

The Analysis of Unordered Tetrads

In the tetrads, when two pairs of alleles are segregating, three patterns of segregation are possible. For example, in the cross $AB \times ab$, the three types of tetrads are

$AB\ AB\ ab\ ab$ referred to as **parental ditype**, or PD.
Only two genotypes are represented,
and their alleles have the same
combinations found in the parents.

$Ab\ Ab\ aB\ aB$ referred to as **nonparental ditype**, or
NPD. Only two genotypes are
represented, but their alleles have
nonparental combinations.

$AB\ Ab\ aB\ ab$ referred to as **tetratype**, or TT. All four
of the possible genotypes are present.

It is because of the following principle that tetrad analysis is an effective way to determine whether two genes are linked.

When genes are *unlinked*, the parental ditype tetrads and nonparental ditype tetrads are expected in equal frequencies (PD = NPD).

The reason for the equality PD = NPD for unlinked genes is shown in Figure 4.23A for two pairs of alleles. *A*, *a* and *B*, *b* located

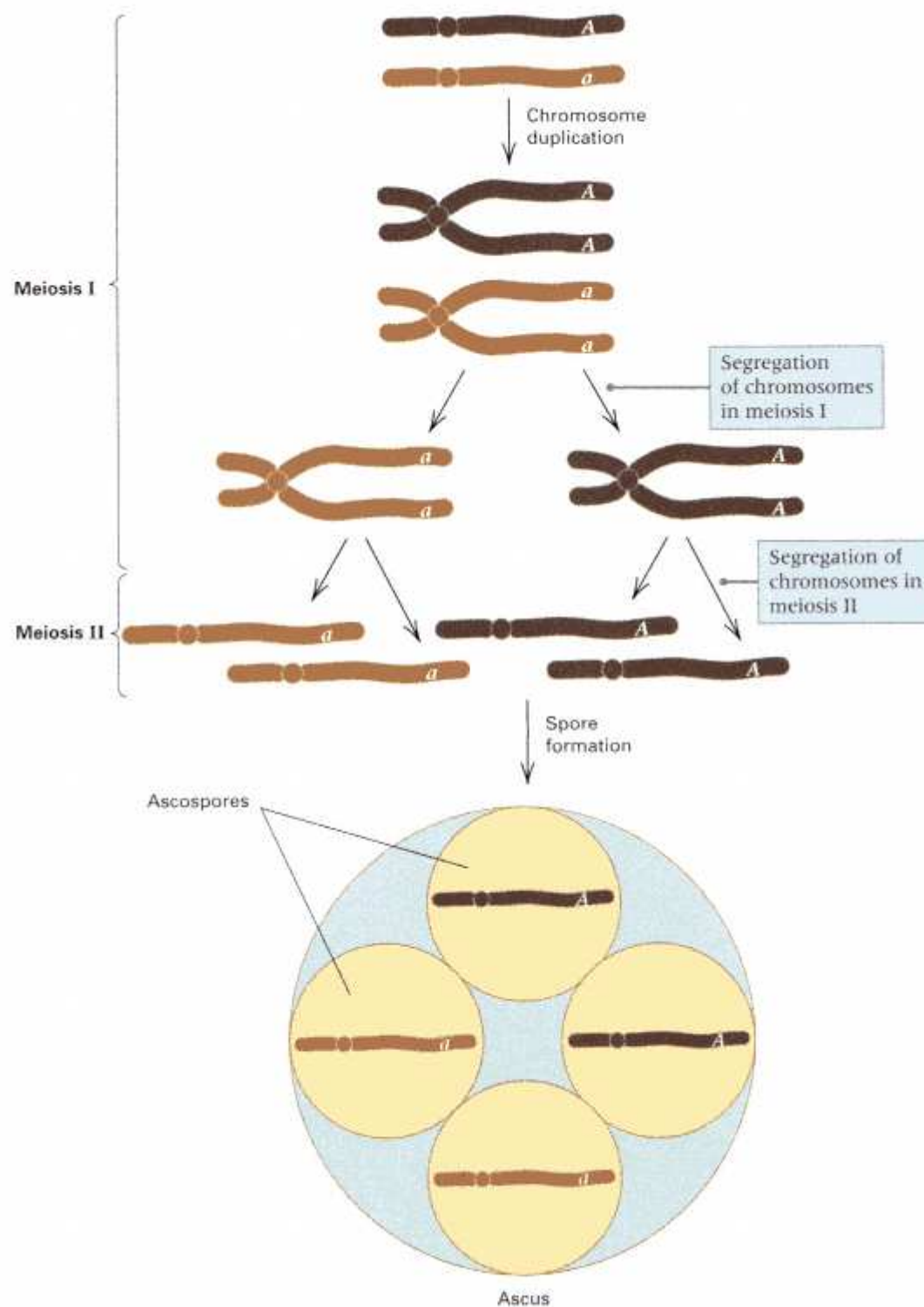


Figure 4.21
Formation of an ascus containing all of the four products of a single meiosis. Each product of meiosis forms a reproductive cell called an ascospore; these cells are held together in the ascus. Segregation of one chromosome pair is shown.

in different chromosomes. In the absence of crossing-over between either gene and its centromere, the two chromosomal configurations are equally likely at metaphase I, so PD = NPD. When there is crossing-over between either gene and its centromere (Figure 4.23B), a tetratype tetrad results, but this does not change the fact that PD = NPD.

In contrast, when genes are linked, parental ditypes are far more frequent than nonparental ditypes. To see why, assume that the genes are linked and consider the events required for the production of the

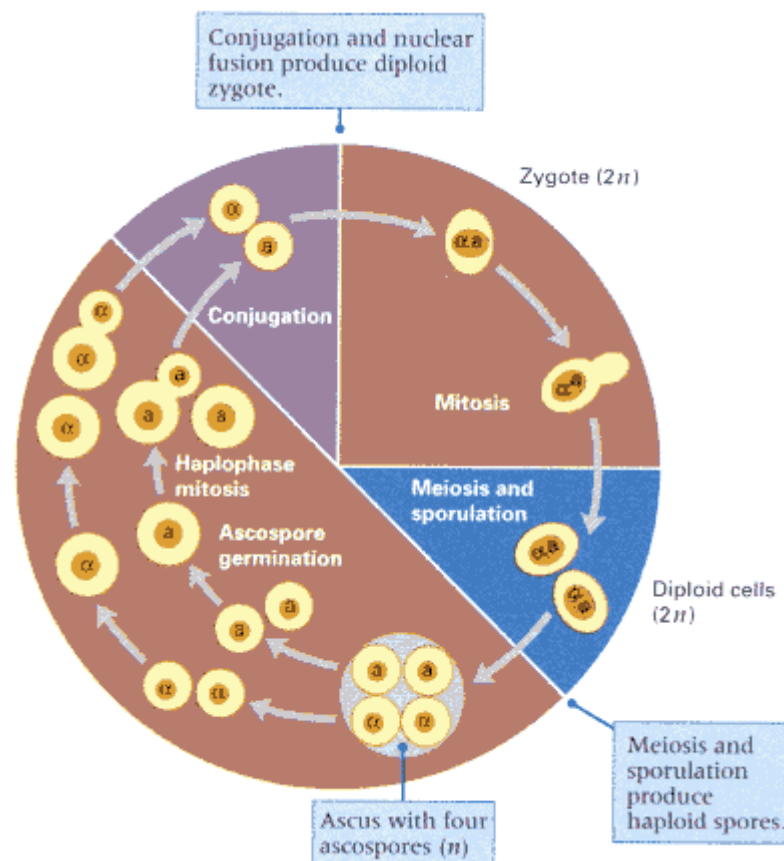


Figure 4.22

Life cycle of the yeast *Saccharomyces cerevisiae*. Mating type is determined by the alleles **a** and **α**. Both haploid and diploid cells normally multiply by mitosis (budding). Depletion of nutrients in the growth medium induces meiosis and sporulation of cells in the diploid state. Diploid nuclei are red; haploid nuclei are yellow.

three types of tetrads. Figure 4.24 shows that when no crossing-over takes place between the genes, a PD tetrad is formed. Single crossing-over between the genes results in a TT tetrad. The formation of a two-strand, three-strand, or four-strand double crossover results in a PD, TT, or NPD tetrad, respectively. With linked genes, meiotic cells with no crossovers will always outnumber those with four-strand double crossovers. Therefore,

Linkage is indicated when nonparental ditype tetrads appear with a much lower frequency than parental ditype tetrads ($\text{NPD} \ll \text{PD}$).

The relative frequencies of the different types of tetrads can be used to determine the map distance between two linked genes, assuming that the genes are sufficiently close that triple crossovers and higher levels of crossing-over can be neglected. To determine the map distance, the number of NPD tetrads is first used to estimate the frequency of double crossovers. The four types of double crossovers in Figure 4.24 are expected in equal frequency. Thus the number of tetrads resulting from double crossovers (parts C, D, E, and F of Figure 4.24) should equal four times the number of NPD tetrads (part F), or $4 \times [\text{NPD}]$, in which the square brackets denote the observed number of NPD tetrads. At the same time, the number of TT tetrads resulting from three-strand double crossovers (parts D and E) should equal twice the number of NPD tetrads, or $2 \times [\text{NPD}]$ (part F). Subtracting $2 \times [\text{NPD}]$ from the total number of tetratypes yields

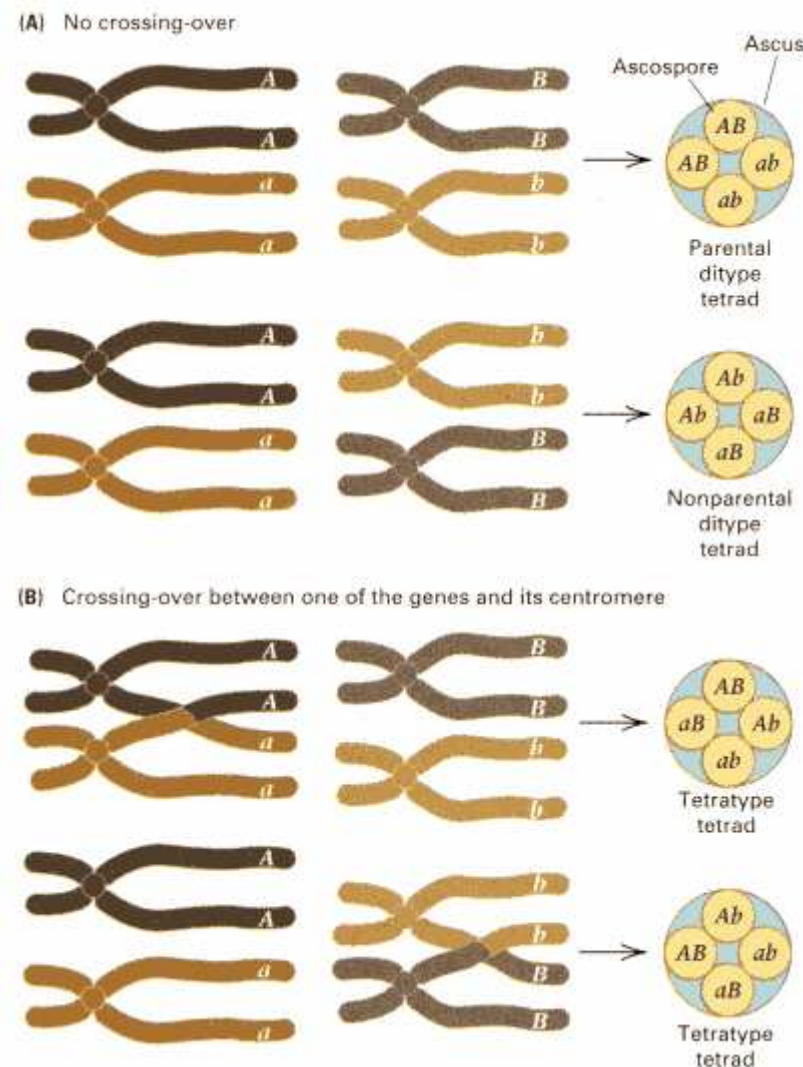


Figure 4.23

Types of unordered asci produced with two genes in different chromosomes. (A) In the absence of crossing-over, random arrangement of chromosome pairs at metaphase I yields two different combinations of chromatids, one yielding PD tetrads and the other NPD tetrads. (B) When crossing-over takes place between one gene and its centromere, the two chromosome arrangements yield TT tetrads. If both genes are closely linked to their centromeres (so that crossing-over is rare), then few TT tetrads are produced.

the number of tetratype tetrads that originate from single crossovers, or $[TT] - 2[NPD]$. By definition, the map distance between two genes equals one-half the frequency of single-crossover tetrads ($[TT] - 2[NPD]$) plus the frequency of double-crossover tetrads ($4[NPD]$). Thus

Map distance

$$\begin{aligned}
 &= \frac{(1/2)([TT] - 2[NPD]) + 4[NPD]}{\text{Total number of tetrads}} \times 100 \\
 &= \frac{(1/2)[TT] + 3[NPD]}{\text{Total number of tetrads}} \times 100 \\
 &= \frac{(1/2)([TT] + 6[NPD])}{\text{Total number of tetrads}} \times 100 \quad (1)
 \end{aligned}$$

The 1/2 in this equation corrects for the fact that only half of the meiotic products in a TT tetrad produced by single crossing-over are recombinant. In the special case where the linked genes are close enough that $[NPD] = 0$ (no double crossovers), the map distance equals the percentage of TT tetrads divided by 2. (The division by 2 is necessary because only half the ascospores in TT tetrads are recombinant for the alleles.)

A systematic format for analyzing linkage relationships for allelic pairs in unordered tetrads is presented in the

form of a tree in Figure 4.25. All of the equations are based on the assumption that there is no chromatid interference. The right-hand side summarizes the equations for linked genes that we derived on the basis of Figure 4.25. The left-hand side pertains to unlinked genes, and it shows how the frequency of tetratypes can be used to make

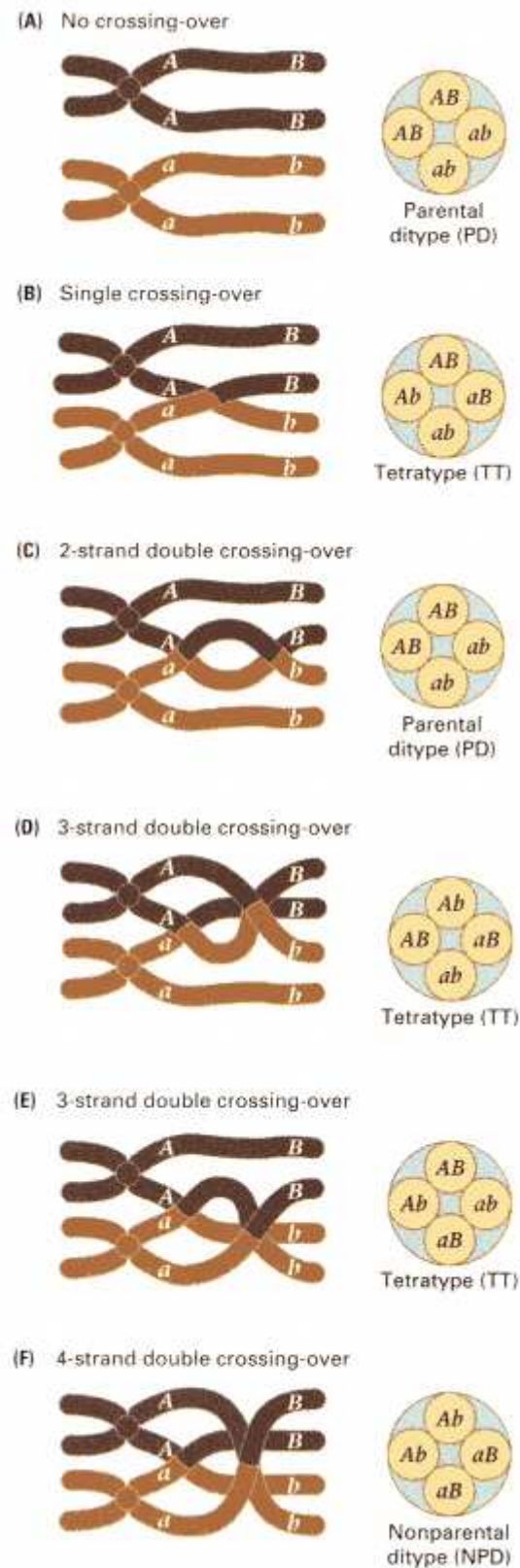


Figure 4.24
Types of tetrads produced with two linked genes. In the absence of crossing-over (A), a PD tetrad is produced. With a single crossover between the genes (B), a TT tetrad is produced. Among the four possible types of double crossovers between the genes (C, D, E, and F), only the four-strand double crossover in part F yields an NPD tetrad.

inferences about the distance of each gene from its own centromere.

As an example of the calculations for linked genes, consider a two-factor cross that yields 112 PD, 4 NPD, and 24

TT tetrads. The fact that NPD << PD (that is, 4 << 112) indicates that the two genes are linked and leads us down the right branch of the tree. Because both TT and NPD tetrads are recovered, we take the first left fork, and because there are NPD tetrads, we also take the next left fork. The appropriate equation is therefore Equation (1). Substitution of the values yields a map distance of

Map distance

$$\begin{aligned} &= \frac{[(1/2) \times 24] + (3 \times 4)}{112 + 4 + 24} \times 100 \\ &= 17.1 \text{ cM} \end{aligned}$$

Note that this mapping procedure differs from that presented earlier in the chapter in that recombination frequencies are not calculated directly from the number of recombinant and nonrecombinant chromatids, but rather from the inferred types of crossovers.

As Figure 4.25 indicates, tetrad analysis yields a great deal of information about the linkage relationship between genetic markers and the distance of unlinked markers to their respective centromeres. However, it is not necessary to carry out a full tetrad analysis for estimating linkage. The alternative is to examine spores chosen at random after allowing the tetrads to break open and disseminate their spores. This procedure is called **random-spore analysis**, and the linkage relationships are determined exactly as described earlier for *Drosophila* and *Zea mays*. In particular, the frequency of recombination equals the number of spores that are recombinant for the genetic markers divided by the total number of spores.

The Analysis of Ordered Tetrads

In *Neurospora crassa*, a species used extensively in genetic investigations, the products of meiosis are contained in an *ordered* array of ascospores (Figure 4.26). A zygote nucleus contained in a sac-like ascus undergoes meiosis almost immediately

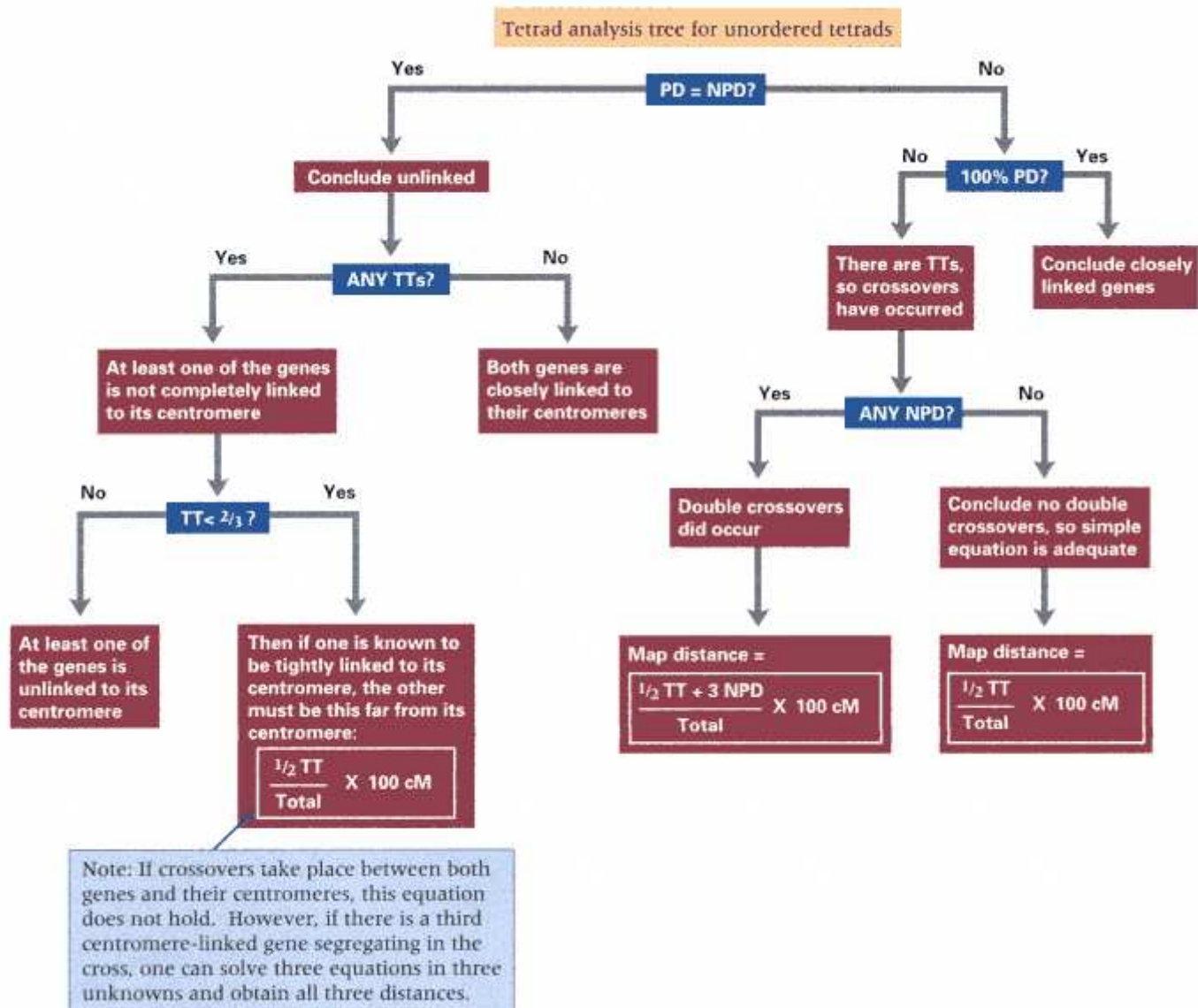


Figure 4.25
Tree diagram for analyzing linkage relations from unordered tetrads.

after it is formed. The four nuclei produced by meiosis are in a linear, ordered sequence in the ascus, and each of them undergoes a mitotic division to form two genetically identical and adjacent ascospores. Each mature ascus contains eight ascospores arranged in four pairs, each pair derived from one of the products of meiosis. The ascospores can be removed one by one from an ascus and each germinated in a culture tube to determine its genotypes.

Ordered asci also can be classified as PD, NPD, or TT with respect to two pairs of alleles; hence the tree diagram in Figure 4.25 can be used to analyze the linkage data. In addition, the ordered arrangement of meiotic products makes it possible to determine the recombination frequency between any particular gene and its centromere. The logic of the mapping technique is based on the feature of meiosis shown in Figure 4.27:

Homologous centromeres of parental chromosomes separate at the first meiotic division; the centromeres of sister chromatids separate at the second meiotic division.

Thus, in the absence of crossing-over between a gene and its centromere, the alleles of the gene (for example, *A* and *a*) must separate in the first meiotic division; this separation is called **first-division segregation**. If, instead, a crossover is formed between the gene and its centromere, the *A*

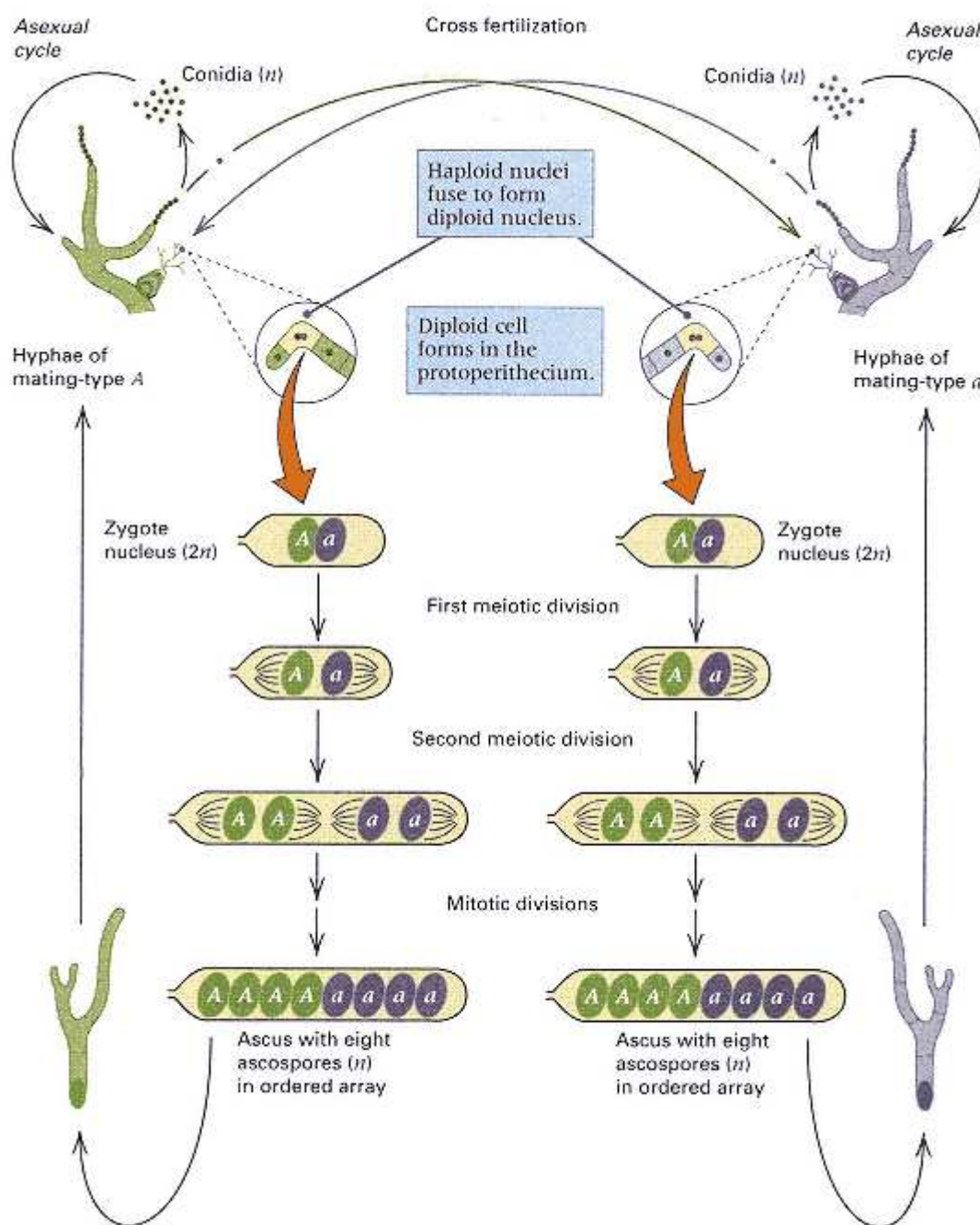


Figure 4.26

The life cycle of *Neurospora crassa*. The vegetative body consists of partly segmented filaments called hyphae. Conidia are asexual spores that function in the fertilization of organisms of the opposite mating type. A protoperithecium develops into a structure in which numerous cells undergo meiosis.

and a alleles do not become separated until the second meiotic division; this separation is called **second-division segregation**. The distinction between first-division and second-division segregation is shown in Figure 4.27. As shown in Figure 4.27A, only two possible arrangements of the products of meiosis can yield first-division segregation: $AAaa$ and $aaAA$. However, four patterns of second-division segregation are possible because of the random arrangement of homologous chromosomes at metaphase I and of the chromatids at metaphase II. These four arrangements, shown in Figure 4.27B, are $AaAa$, $aAaA$, $AaAa$, and $aAaA$.

The percentage of asci with second-division segregation patterns for a gene can be used to map the gene with respect to its centromere. For example, let us assume that 30 percent of a sample of asci from a cross have a second-division segregation pattern for the A and a alleles. This means that 30 percent of the cells undergoing

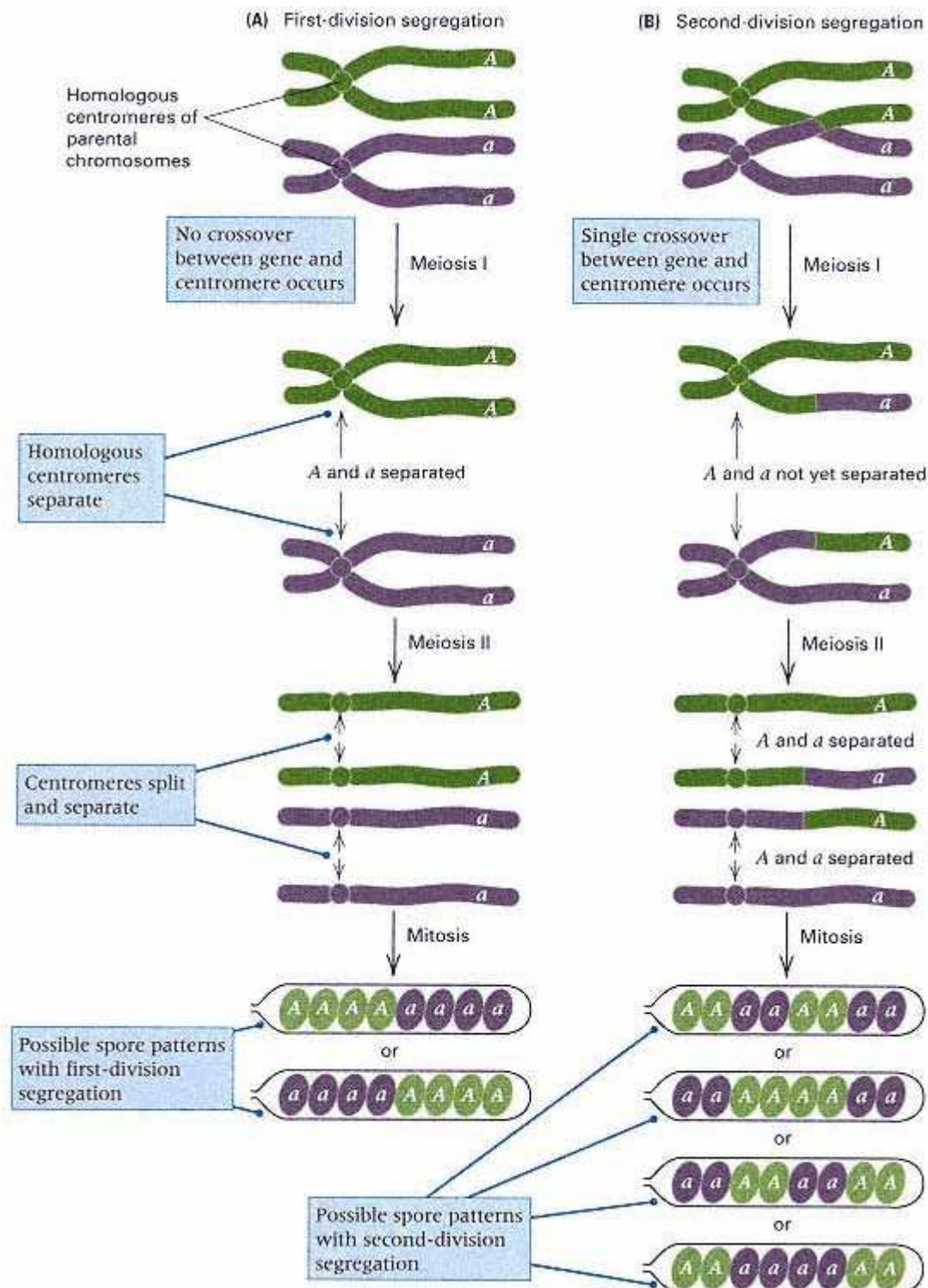


Figure 4.27

First- and second-division segregation in *Neurospora*. (A) First-division segregation patterns are found in the ascus when crossing-over between the gene and centromere does not take place. The alleles separate (segregate) in meiosis I. Two spore patterns are possible, depending on the orientation of the pair of chromosomes on the first-division spindle. (B) Second-division segregation patterns are found in the ascus when crossing-over between the gene and the centromere delays separation of A from a until meiosis II. Four patterns of spores are possible, depending on the orientation of the pair of chromosomes on the first-division spindle and that of the chromatids of each chromosome on the second-division spindle.

meiosis had a crossover between the A gene and its centromere. Furthermore, in each cell in which crossing-over takes place, two of the chromatids are recombinant and two are nonrecombinant. In other words, a frequency of crossing-over of 30 percent corresponds to a recombination frequency of 15 percent. By convention, map distance refers to the frequency of recombinant meiotic products rather than to the frequency of cells with crossovers. Therefore, the map distance

between a gene and its centromere is given by the equation

$$\frac{(1/2)(\text{Asci with second-division segregation patterns})}{\text{Total number of asci}} \times 100 \quad (2)$$

This equation is valid as long as the gene is close enough to the centromere that multiple crossing-over can be neglected. Reliable linkage values are best determined for genes that are near the centromere. The location of more distant genes is then accomplished by the mapping of these genes relative to genes nearer the centromere.

If a gene is far from its centromere, crossing-over between the gene and its centromere will be so frequent that the *A* and *a* alleles become randomized with respect to the four chromatids. The result is that the six possible spore arrangements shown in Figure 4.27 are all equally frequent. This equal likelihood reflects the patterns that can result from choosing randomly among 2 *A* and 2 *a* spore pairs in the ascus, as shown by the branching diagram in Figure 4.28. Therefore, in the absence of chromatid interference,

The maximum frequency of second-division segregation asci is 2/3.

4.6—

Mitotic Recombination

Genetic exchange can also take place in mitosis, although at a frequency about 1000-fold lower than in meiosis and probably by a somewhat different mechanism than meiotic recombination. The first evidence for mitotic recombination was obtained by Curt Stern in 1936 in experiments with *Drosophila*, but the phenomenon has been studied most carefully in fungi such as yeast and *Aspergillus*, in which the frequencies are higher than in most other organisms. Genetic maps can be constructed from mitotic recombination frequencies. In some organisms in which a sexual cycle is unknown, mitotic recombination is the only method of obtaining linkage data. In organisms in which both meiotic recombination and mitotic recom-

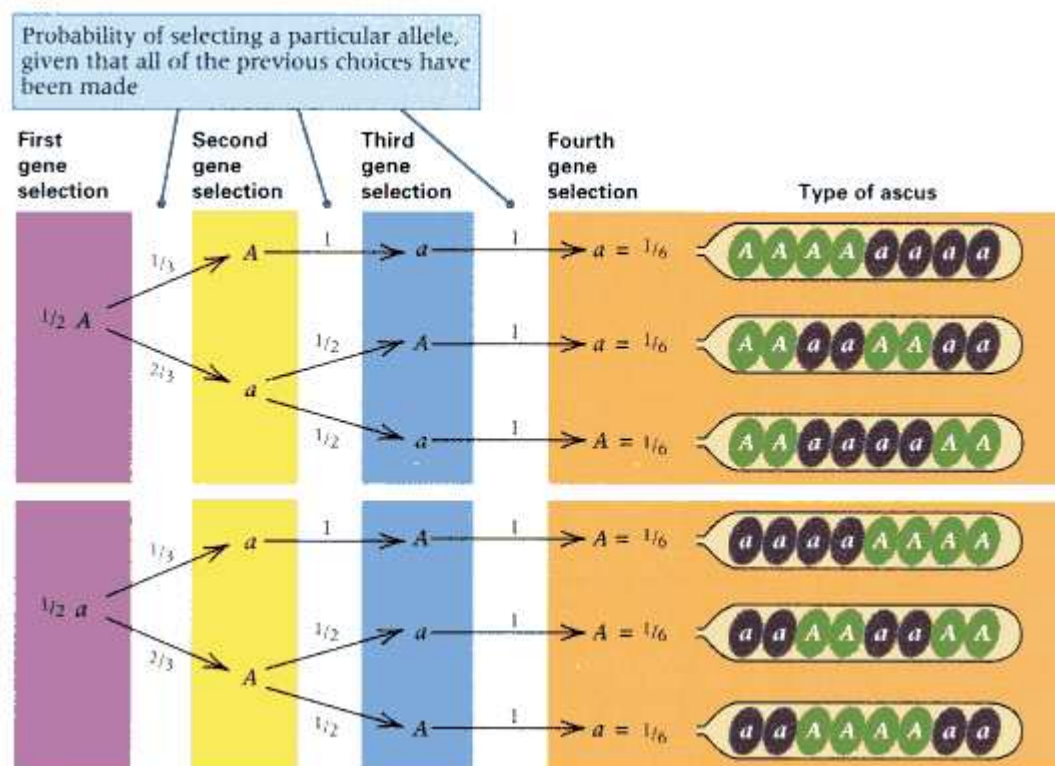


Figure 4.28

Diagram showing the result of free recombination between an allelic pair, *A* and *a*, and the centromere. The frequency of second-division segregation equals 2. This is the maximum frequency of second-division segregation, provided that there is no chromatid interference.

bination are found, mitotic recombination is always at a much lower frequency. In mitotic recombination, the relative map distances between particular genes sometimes correspond to those based on meiotic recombination frequencies, but for unknown reasons, the distances are often markedly different. The discrepancies may reflect different mechanisms of exchange and perhaps a nonrandom distribution of potential sites of exchange.

The genetic implications of mitotic recombination are illustrated in Figure 4.29. Each of the homologous chromosomes has two genetic markers distal to the site of the breakage and reunion. At the following anaphase, each centromere splits and the daughter cells receive one centromere of each color along with the chromatid attached to it. The separation can happen in two ways, as shown on the right. If the two nonexchanged and the two exchanged chromatids go together (top), the result of the exchange is not detectable genetically. However, if each nonexchanged chromatid goes with one of the exchanged chromatids, then the result is that the region of the chromosome distal to the site of the exchange becomes homozygous. With an appropriate configuration of genetic markers, the exchange is detectable as a **twin spot** in which cells of one homozygous genotype (in this example, aB/aB) are adjacent to cells of a different homozygous genotype (in this example, Ab/Ab). In the original experiments with *Drosophila*, Stern used the X-linked markers $A = y^+$ and $a = y$ (y is yellow body color) along with $B = sn^+$ and $b = sn$ (sn is singed bristles). In females, the result of mitotic recombination was observed as a twin spot in which a patch of cuticle with a yellow color and normal bristles was adjacent to a patch of cuticle with normal color and singed bristles.

The rate of mitotic recombination can be increased substantially through x-ray treatment and the use of certain mutations.

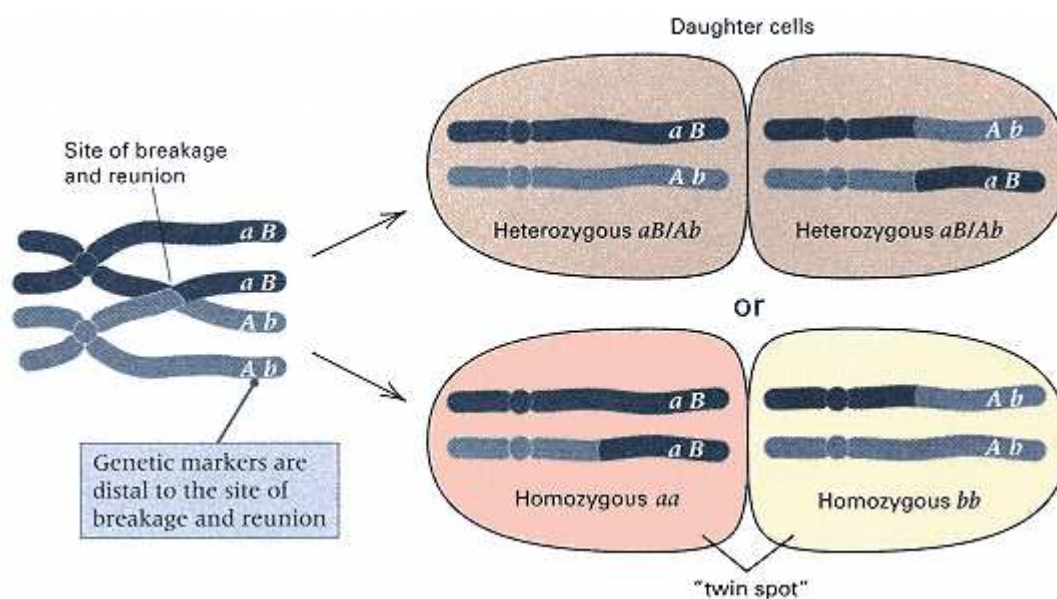


Figure 4.29

Mitotic recombination. Homologous chromosomes in prophase of mitosis are shown in light and dark blue. At anaphase, each centromere splits, and each daughter cell receives one centromere of each color and its attached chromatid. When a rare reciprocal exchange takes place between non-sister chromatids of homologous chromosomes, the daughter cells can be either of two types. If the nonexchanged chromatids and the exchanged chromatids go together (top right), both cells are genetically like the parent; in this example, $A b/a B$. If the nonexchanged and exchanged chromatids go together (bottom right), the result is that alleles distal to the point of exchange become homozygous. In this example, one daughter cell is $a B/a B$ and the other $A b/A b$. Adjacent cell lineages are therefore homozygous for either aa or bb and can be detected phenotypically as a "twin spot" of aa somatic tissue adjacent to bb somatic tissue.

Mitotic recombination is a useful tool in genetics because it results in the production of **somatic mosaics**, organisms that contain two or more genetically different types of tissue. For example, the mosaic *Drosophila* females discussed in the previous paragraph consist of predominantly wildtype tissue but include some patches of $y^+ sn/y^+ sn$ and some patches of $y sn^+/y sn^+$. Each twin spot of mutant tissue derives from repeated division of a single cell that became homozygous through a mitotic exchange. Hence the mutant tissue resulting from mitotic recombination can be used to trace the movement and differentiation of particular cell lineages in development.

4.7—

Recombination Within Genes

Genetic analyses and microscopic observations made before 1940 led to the view that a chromosome was a linear array of particulate units (the genes) joined in some way resembling a string of beads. The gene was believed to be the smallest unit of genetic material capable of alteration by mutation and the smallest unit of inheritance. Recombination had been seen in all regions of the chromosome but not within genes, and the idea of the indivisibility of the gene developed. This classical concept of a gene began to change when geneticists realized that mutant alleles of a gene might be the consequence of alterations at different mutable sites within the gene. The evidence that led to this idea was the finding of rare recombination events within several genes in *Drosophila*. The initial observation came in 1940 from investigations by C. P. Oliver of a gene in the X chromosome of *D. melanogaster* known as *lozenge* (*lz*). Mutant alleles of *lozenge* are recessive, and their phenotypic effects include a disturbed arrangement of the facets of the compound eye and a reduction in the eye pigments. Numerous *lz* alleles with distinguishable phenotypes are known.

Females heterozygous for two different *lz* alleles, one in each homologous chromosome, have lozenge eyes. When such heterozygous females were crossed with males that had one of the *lz* mutant alleles in the X chromosome, and large numbers of progeny were examined, flies with normal eyes were occasionally found. For example, one cross carried out was

$$\begin{array}{c} \text{X} \quad + \quad lz^{BS} \quad + \quad + \\ \text{X} \quad ct \quad + \quad lz^g \quad v \end{array} \times \begin{array}{c} + \quad + \quad lz^g \quad v \\ \text{X} \end{array}$$

in which lz^{BS} and lz^g are mutant alleles of *lozenge*, and *ct* (cut wing) and *v* (vermillion eye color) are genetic markers that map 7.7 units to the left and 5.3 units to the right, respectively, of the *lozenge* locus. From this cross, 134 males and females with wildtype eyes were found among more than 16,000 progeny, for a frequency of 8×10^{-3} . These exceptional progeny might have resulted from a reverse mutation of one or of the other of the mutant *lozenge* alleles to lz^+ , but the observed frequency, though small, was much greater than the known frequencies of reverse mutations. The genetic constitution of the maternally derived X chromosomes in the normal-eyed offspring was also inconsistent with such an explanation: The male offspring had cut wings and the females were *ct*/+ heterozygotes. That is, all of the rare nonlozenge progeny had an X chromosome with the constitution

$$\underline{ct \quad ++ \quad +}$$

which could be accounted for by recombination between the two *lozenge* alleles. Proof of this conclusion came from the detection of the reciprocal recombinant chromosome

$$\underline{+ \quad lz^{BS} lz^g \quad v}$$

in five male progeny that had a lozenge phenotype distinctly different from the phenotype resulting from the presence of either a lz^{BS} or lz^g allele alone. The exceptional males also had vermillion eyes, as expected from reciprocal recombination. The observation of intragenic recombination indicated that genes have a fine structure and that the multiplicity of allelic forms of some genes might be due to mutations at different sites in the gene. As we have seen in Chapter 1, these sites are the nucleotide constituents that make up DNA.

4.8—

A Closer Look at Complementation

The discovery of intragenic recombination emphasized the question "What is a gene?" A gene cannot be defined solely on the basis of a mutant phenotype. On the one hand, mutations in different genes can yield similar phenotypes. On the other hand, multiple alleles of the same gene can have quite different phenotypes, depending on the residual function of the mutation and the time- and tissue-specificity of its expression.

An alternative definition of a gene is based on recombination: Two mutations with similar phenotypes are alleles of the same gene if they fail to recombine. The discovery of recombination between alleles of *lozenge* and between alleles of other genes in *Drosophila* showed that this definition of a gene is unacceptable. Allelic mutations can recombine. The theoretical problem posed by recombination between alleles is indicated by the term *pseudoalleles* used to designate alleles that recombined. However, we now know that there is nothing "pseudo" about pseudoalleles. They are perfectly legitimate alleles that are far enough apart along the DNA to allow the detection of recombination.

The experimental resolution of the problem of the definition of a gene was the complementation test discussed in Chapter 2. The test is the same sort of test one would use to find out whether holes in each member of a pair of socks were in the same place. Put both socks on the same foot. If the holes are in the same place, bare skin shows through where the holes overlap. If the holes are in different places, each hole is covered by the fabric present in the other sock. In the complementation test, two recessive alleles are regarded as alleles of the same gene if, when they are present together in the same organism, they fail to complement each other and so yield a mutant phenotype, the "bare skin."

When a set of recessive mutations is tested in all pairwise combinations, the complementation test enables them to be separated into groups called complementation groups (Chapter 2). For example, in a genetic analysis of flower color in peas, a geneticist may isolate many independent recessive mutations, each of which causes the flowers to be white instead of purple. To group the mutations into complementation groups, strains that are homozygous for different mutations are crossed in all possible pairs. If the progeny of a cross has white flowers, then the mutations in the parents are said to fail to complement. These mutations are classified as belonging to the same complementation group, which means that they are alleles of the same gene. On the other hand, if the progeny of a cross between two mutant strains has purple flowers, then the mutations in the parents are said to complement. The mutations are assigned to different complementation groups, which means that they are alleles of different genes.

The molecular basis of complementation is illustrated in Figure 4.30. Figure 4.30A depicts the situation when two mutations, m_1 and m_2 , each resulting in white flowers, are different mutations of the same gene. (The site of each mutation is indicated by a cross.) In a genotype in which m_1 and m_2 are in the *trans*, or repulsion, configuration, m_1 codes for a protein with one type of defect and m_2 codes for a protein with a different type of defect, but both types of protein are nonfunctional. Hence alleles in the same gene yield a mutant phenotype (white flowers), because neither mutation is able to support the production of a wildtype form of the protein.

When the mutations are alleles of different genes, the situation is as depicted in Figure 4.30B. Because the mutations are in different genes, the homozygous m_1 genotype is also homozygous for the wildtype allele of the gene mutated in m_2 ; likewise, the homozygous m_2 genotype is also homozygous for the wildtype allele of the gene mutated in m_1 . Hence the same cross that yields the genotype m_1/m_2 in the case of allelic mutations (Figure 4.30A), in the case of different genes yields the genotype $m_1/+ m_2/+$, where the plus signs represent the wildtype alleles of the genes not mutated in m_1 and m_2 . In this case, the mutations do complement each other and yield an organism with a wildtype phenotype (purple flowers). With respect to the protein rendered defective by m_1 , there is a functional form encoded by the wildtype allele brought in from the m_2m_2 parent.

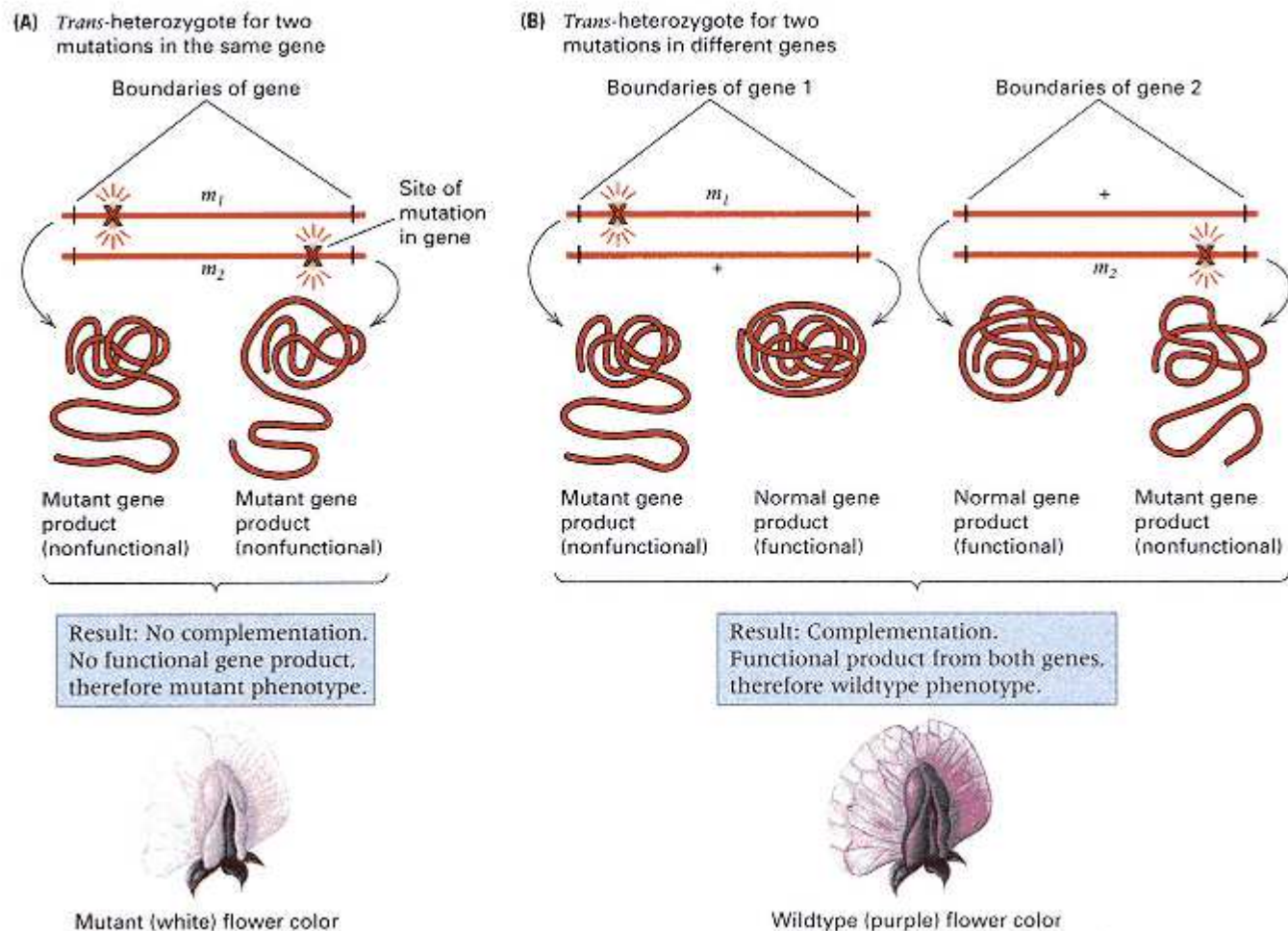


Figure 4.30

The basis for interpretation of a complementation test used to determine whether two mutations are alleles of the same gene (A) or alleles of different genes (B).

With respect to the protein rendered defective by m_2 , there is again a functional form encoded by the wildtype allele brought in from the $m_1 m_1$ parent. Because a functional form of both proteins is produced, the result is a normal phenotype, or complementation.

Complementation and recombination must not be confused. Complementation is inferred from the phenotype of the *trans*-heterozygote having a mutation in each of two different genes (Figure 4.30B). Recombination is inferred from genetic analysis of the progeny of heterozygous genotypes.

Chapter Summary

Nonallelic genes located in the same chromosome tend to remain together in meiosis rather than to undergo independent assortment. This phenomenon is called linkage. The indication of linkage is a significant deviation from the 1 : 1 : 1 : 1 ratio of phenotypes in the progeny of a cross of the form $Aa Bb \times aa bb$. When alleles of two linked genes segregate, more than 50 percent of the gametes produced have parental combinations of the segregating alleles, and fewer than 50 percent have nonparental (recombinant) combinations of the alleles. The recombination of linked genes results from crossing-over, a process in which nonsister chromatids of the homologous chromosomes exchange corresponding segments in the first meiotic prophase. At the molecular level, the Holliday model explains recombination as a result of a single-strand break in each of two homologous DNA molecules, interchange of pairing partners between the broken strands, and repair of the gaps, followed by strand migration and a second breakage and reunion to connect the originally unbroken DNA strands. This model, although not correct in all its details, is nevertheless the basis on which more accurate and complex models of recombination have been developed.

The frequency of recombination between different genes can be used to determine the relative order and locations of the genes in chromosomes. This type of analysis is called genetic mapping. Distance between adjacent genes in such a

map (a genetic or linkage map) is defined to be proportional to the frequency of recombination between them; the unit of map distance (the map unit or centimorgan) is defined as 1 percent recombination. One map unit corresponds to a physical length of the chromosome in which a crossover event takes place, on the average, once in every 50 meiotic divisions. For short distances, map units are additive. (For example, for three genes with order $a b c$, if the map distances a to b and b to c are 2 and 3 map units, respectively, then the map distance a to c is $2 + 3 = 5$ map units.) The recombination frequency underestimates actual genetic distance if the region between the genes being considered is too great. This discrepancy results from multiple crossover events, which yield either no recombinants or the same number produced by a single event. For example, two crossovers in the region between two genes may yield no recombinants, and three crossover events may yield recombinants of the same type as that from a single crossover.

When many genes are mapped in a particular species, they form linkage groups equal in number to the haploid chromosome number of the species. The maximum frequency of recombination between any two genes in a cross is 50 percent; this happens when the genes are in nonhomologous chromosomes and assort independently or when the genes are sufficiently far apart in the same chromosome that at least one crossover is formed between them in every meiosis. The map distance between two genes may be considerably greater than 50 centimorgans because the map distance is equal to half of the average number of crossovers per chromosome times 100. A mapping function is the mathematical relation between the genetic map distance across an interval and the observed percent recombination in the interval.

In many organisms, including model experimental organisms, agricultural animals and plants, and human beings, the genetic map includes hundreds or thousands of genetic markers distributed more or less uniformly throughout the euchromatin. Some of the most useful genetic markers are changes in base sequence present in wildtype organisms that are not associated with any phenotypic abnormalities. Prominent among these are nucleotide substitutions that create or destroy a particular cleavage site recognized by a restriction endonuclease. Such mutations can be detected because different chromosomes yield restriction fragments that differ in size according to the positions of the cleavage sites. Genetic variation of this type is called restriction fragment length polymorphism (RFLP). Most species also have considerable genetic variation in which one allele differs from the next according to the number of copies of a tandemly repeated DNA sequence it contains (variable number of tandem repeats, or VNTR). Many of the genetic markers in the genetic map of human beings consist of RFLPs and VNTRs. Because of their high degree of variability among people, VNTRs are also employed in DNA typing.

The four haploid products of each of a series of meiotic divisions can be used to analyze linkage and recombination in some species of fungi and unicellular algae. The method is called tetrad analysis. In *Neurospora* and related fungi, the meiotic tetrads are contained in a tubular sac, or ascus, in a linear order, which makes it possible to determine whether a pair of alleles segregated in the first or the second meiotic division. With such asci, it is possible to use the centromere as a genetic marker; in fact, the centromere serves as a reference point to which all genes in the same chromosome can be mapped. Linkage analysis in unordered tetrads is based on the frequencies of parental ditype (PD), nonparental ditype (NPD), and tetratype (TT) tetrads. The observation of $NPD \ll PD$ is a sensitive indicator of linkage.

Recombination between alleles of the same gene demonstrated that genes are composed of a linear array of subunits, now known to be the nucleotides in the DNA. The functional definition of a gene is therefore not based on mutation or on recombination but on complementation (by means of the complementation test). If two recessive mutations, a and b , are present in a cell or organism in the *trans* configuration and the phenotype is wildtype (complementation), the mutations are in different genes. If the phenotype is mutant (noncomplementation), the mutations are in the same gene. At the molecular level, lack of complementation implies that allelic mutations impair the function of the same protein molecule.

Key Terms

ascospore	Haldane's mapping function	recombination
ascus	heterochromatin	repulsion
attached-X chromosome	Kosambi's mapping function	restriction endonuclease

centimorgan	linkage	restriction fragment length polymor-
chromosome inteference	linkage group	phism (RFLP)
chromosome map	linkage map	second-division segregation
chromatid interference	locus	simple tanderm repeat polymorphism
coefficient of coincidence	map unit	(STRP)
compound-X chromosome	mapping function	sometic mosaics
condensation	negative chromatid interference	tetratype (TT)
coupling	nonparental ditype (NPD)	three-point cross
euchromain	parental combinations	twin spot
first-division segregation	parental ditype (PD)	variable number of tanderm repeats
frequency of recombination	positive chromatid interference	(VNTR)
genetic map	random-spore analysis	
genetic market	recombinat	

Chapter 4 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

<http://www.jbpub.com/genetics>

Select the link to **Genetics: Principles and Analysis** and then choose the link to **GeNETics on the web**. You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor

1. The absence of crossing-over in males of *Drosophila* is a great convenience for detecting **linkage**. You can see this for yourself at the keyword site. Mate females having a black body and dumpy wings with wildtype males. Then cross the F₁ male progeny with the black, dumpy females. What are the phenotypes of the progeny and in what numbers do they occur? Does this result tell you anything about the linkage of black and dumpy? If assigned to do so, mate wildtype females with black, dumpy males. Make a diagram of the matings, indicating all genotypes and phenotypes and the numbers observed in each generation of progeny. Calculate the frequency of recombination between black and dumpy.

2. The keyword **Saccharomyces** will connect you with the genome database of the yeast *Saccharomyces cerevisiae*. Select **Maps** for graphical views of yeast chromosomes and **Genetic Map** to access the linkage map of each chromosome individually. If assigned to do so, use the site to find the relation between map length (in centimorgans) and physical length (in base pairs) of each of the yeast chromosomes. Make a bar graph of this information.

3. Detecting genetic linkage in **human pedigrees** requires special methods because human sibships are typically quite small. For each

(text box continued on next page)

Review the Basics

- What does it mean to say that two genes have a recombination frequency of 12 percent? How is the recombination frequency estimated?
- For a region between two genes in which no more than one crossing over can take place in each cell undergoing meiosis, the frequency of recombination is equal to one-half the frequency of meiotic cells in which a chiasma occurs in the region. Explain why the factor of one-half occurs when we compare recombination frequency to chiasma frequency.
- What is a double crossover? How many different kinds of double crossovers are possible? Draw a diagram of each kind of double crossover in a bivalent in a region between two genes, each heterozygous.
- If each cell undergoing meiosis had two crossovers between the genes, and the types of double crossovers were equally frequent, what would be the frequency of recombination between the genes?
- What is interference? In a region of high interference, would you observe more or fewer double crossovers?

- Explain why the recovery of ordered tetrads allows any gene to be mapped relative to the centromere of the chromosome on which the gene is located.

Guide to Problem Solving

Problem 1: In *Drosophila*, the genes *cn* (cinnabar eyes) and *bw* (brown eyes) are located in the same chromosome but so far apart that they appear unlinked. Compare the possible offspring produced by the cross

$$\frac{cn\ bw}{++} \text{♀} \times \frac{cn\ bw}{cn\ bw} \text{♂}$$

and the cross

$$\frac{cn\ bw}{cn\ bw} \text{♀} \times \frac{cn\ bw}{++} \text{♂}$$

The + symbols denote the wildtype alleles. Both *cn* and *bw* are recessive. Flies with the genotype *cn cn* have cinnabar

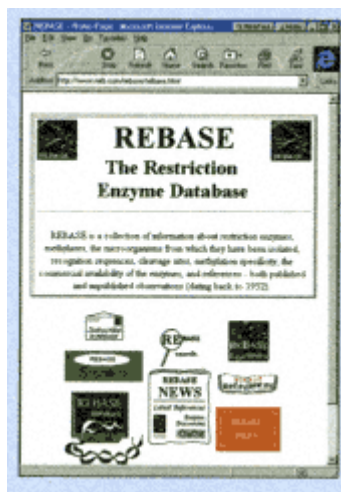
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genetic marker in a pedigree in which a trait, for example, a genetic disease, is present, the likelihood of linkage is assessed by a "lod score." The principal advantage of lod scores is that data from individual pedigrees can be summed. Use the keyword to learn more about this approach and examine the sample lod score calculation for a simple pedigree. If assigned to do so, write a 250-word summary of the method and the logic on which it is based.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 4, and you will be linked to the current exercise that relates to the material presented in the chapter.

PIC SITE



The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 4.

(bright red) eyes, *bw bw* flies have brown eyes, and the double mutant *cn bw/cn bw* has white eyes.

Answer: The key to this problem is to remember that crossing-over does not take place in *Drosophila* males. When the male is *cn bw/+ +*, his gametes are $1/2$ *cn bw* and $1/2$ *+ +*, so the progeny are *cn bw/cn bw* (white eyes) and *+ +/cn bw* (wildtype eyes) in equal proportions. Crossing-over does happen in female *Drosophila*, but *cn* and *bw* are so far apart that they undergo independent assortment. When the female is *cn bw/+ +*, her gametes are $1/4$ *cn bw*, $1/4$ *cn +*, $1/4$ *+ bw*, and $1/4$ *+ +*, so the progeny are *cn bw/cn bw* (white eyes), *cn +/cn bw* (bright red), *+ bw/cn bw* (brown), and *+ +/cn bw* (wildtype) in equal proportions.

Problem 2: One of the earliest reported cases of linkage was in peas in the year 1905 between the gene for purple versus red flowers and the gene for elongate versus round pollen. These genes are separated by about 12 map units. Plants of genotype *B-* have purple flowers, and plants of the genotype *bb* have red flowers. Plants of genotype *E-* have elongate pollen, and those of genotype *ee* have round pollen. Determine what genotypes and phenotypes among the progeny would be expected from crosses of F_1 hybrids $\times$ red, round when the F_1 hybrids are obtained as follows

(a) purple, elongate $\times$ red, round $\rightarrow$ purple, elongate F_1

(b) purple, round $\times$ red, elongate $\rightarrow$ purple, elongate F_1

Answer: The red, round parents have genotype $b\ e/b\ e$. You must deduce the genotype of the F_1 parent. Although both F_1 hybrids have the dominant purple, elongate phenotype, they are different kinds of double heterozygotes. **(a)** In this mating, both the b and e alleles come from the red, round parent, so the F_1 genotype is the *cis*, or coupling, heterozygote $B\ E/b\ e$. **(b)** In this mating, the b allele comes from one parent and the e allele from the other, so the F_1 genotype is the *trans*, or repulsion, heterozygote $B\ e/b\ E$. To calculate the types of progeny expected, use the fact that 1 map unit corresponds to 1 percent recombination: Because the genes are 12 map units apart, 12 percent of the gametes will be recombinant (6 percent of each of the reciprocal classes) and 88 percent will be nonrecombinant (44 percent of each of the reciprocal classes). Therefore, the two types of heterozygotes produce gametes with the following expected percentages:

<i>Gametes</i>	<i>BE/be</i>	<i>Be/bE</i>
<i>BE</i>	44	6
<i>Be</i>	6	44
<i>bE</i>	6	44
<i>be</i>	44	6

The answers to the specific questions are as follows.

(a) BE/be (purple, elongate) 44 percent; Be/be (purple, round) 6 percent; bE/be (red, elongate) 6 percent; be/be (red, round) 44 percent.

(b) BE/be (purple, elongate) 6 percent; Be/be (purple, round) 44 percent; bE/be (red, elongate) 44 percent; be/be (red, round) 6 percent.

Problem 3: In *Drosophila*, the genes ct (cut wing margin), y (yellow body), and v (vermillion eye color) are X-linked. Females heterozygous for all three markers were mated with wildtype males, and the following male progeny were obtained. As is conventional in *Drosophila* genetics, the wildtype allele of each gene is designated by a + sign in the appropriate column.

ct	y	v	4
ct	y	+	93
ct	+	v	54
ct	+	+	349
+	y	v	331
+	y	+	66
+	+	v	97
+	+	+	6
Total			1000

(a) What was the genotype of the female parents?

(b) What is the order of the genes?

(c) Calculate the recombination frequencies and the interference between the genes.

(d) Draw a genetic map of the genes.

Answer: This is a conventional kind of three-point linkage problem. The way to solve it is to start by deducing the genotype of the triply heterozygous parent. This is done by noting that *the most common classes of offspring are the nonrecombinants*, in this case $ct^+ +$ and $+y v$. Then deduce the order of the genes. This is done by noting that *the least common classes of offspring are the double recombinants*, in this case $ct y v$ and $+ + +$. Moreover, *each class of double recombinants will be identical with one of the nonrecombinants except for the marker that is in the middle*. In this case, comparing $ct y v$ with $+ y v$ and $+ + +$ with $ct + +$ shows that ct is in the middle. **(a)** Therefore, the answer to part (a) is that the mothers' genotype was $+ ct +/y + v$. **(b)** The answer to part (b) is that ct is in the middle. (It is not possible to determine from these data whether the gene order is $y ct v$ or $v ct y$.)

The next step is to rearrange the data with the genes in the correct order, with reciprocal chromosomes adjacent and with nonrecombinants at the top, single recombinants for region 1 and 2 in the middle, and nonrecombinants at the bottom.

+	a	+	349	Nonrecombinant
y	+	v	331	Nonrecombinant
y	a	+	93	Region 1 crossing-over

+	+	v	97	Region 1 crossing-over
+	a	v	54	Region 2 crossing-over
y	+	+	66	Region 2 crossing-over
y	a	v	4	Region 1 + Region 2 crossing-over
+	+	+	6	Region 1 + Region 2 crossing-over

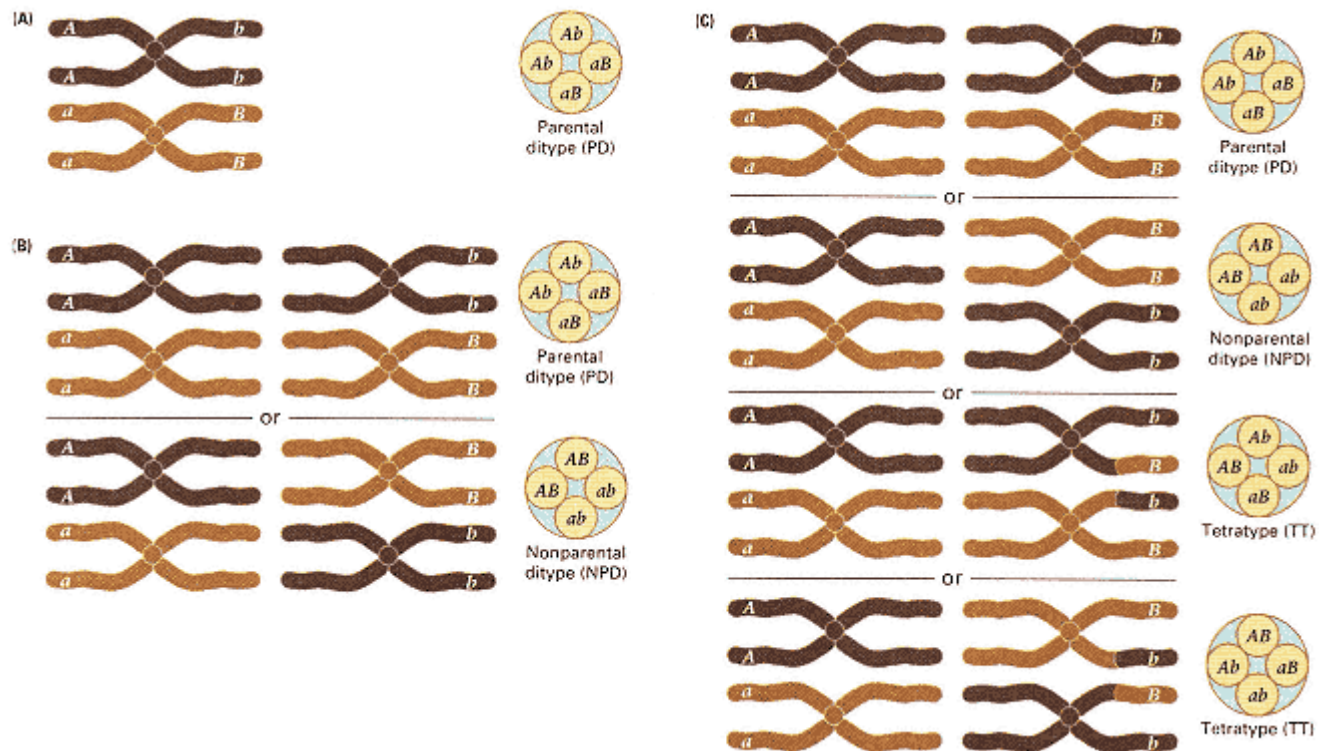
(c) Now, to answer part (c), calculate the recombination frequencies. The most common mistake is to ignore the double recombinants. In fact, the double recombinants are recombinant in *both* regions and so must be counted twice—once in calculating the recombination frequency in region 1 and again in calculating the recombination frequency in region 2. The frequency of recombination in region 1 (*y-ct*) is $(93 + 97 + 4 + 6)/1000 = 0.20$, or 20 percent. The frequency of recombination in region 2 (*ct-v*) is $(54 + 66 + 4 + 6)/1000 = 0.13$, or 13 percent. To calculate the interference, note that the expected number of double recombinants is $0.20 \times 0.13 \times 1000 = 26$, whereas the actual number is 10. The coincidence is the ratio $10/26 = 0.385$, and therefore the interference across the region is $1 - 0.385 = 0.615$. (d) For part (d), the genetic map based on these data is shown below.



Problem 4: In the yeast *Saccharomyces cerevisiae*, the gene *MET14* codes for an enzyme used in synthesis of the amino acid methionine and is very close to the centromere of chromosome XI. For simplicity, denote *MET14* with the symbol *a*, and suppose that *a* is used in mapping a new mutation—say, *b*—whose location in the genome is unknown. Diploids are made by mating *a B* × *A b*, in which the uppercase symbols denote the wildtype alleles, and asci are examined. In this mating, what kinds of asci correspond to parental ditype (PD), nonparental ditype (NPD) and tetratype (TT)? What relative proportions of PD, NPD and TT asci would be expected in each of the following cases:

- (a) *b* closely linked with *a*?
- (b) *b* on a different chromosome from the one *a* is on but close to the centromere?
- (c) *b* on a different chromosome from the one *a* is on and far from the centromere?

Answer: The genotype of the diploid is the *trans*, or repulsion, heterozygote *a B/A b*, so the parental ditype (PD) tetrads contain two *a B* spores and two *A b* spores; the nonparental ditype (NPD) tetrads contain two *a b* spores and two *A B* spores; and the tetratype (TT) tetrads contain one spore each of *a B*, *A b*, *a b*, and *A B*. In this problem, the specific questions illustrate the principle that *any marker tightly linked to its centromere in a fungus with unordered tetrads serves to identify first-division and second-division segregation of every other gene in the organism*. This principle makes possible the mapping of other genes with respect to their centromeres. Reasoning through problems of tetrad analysis is best approached by considering the possible chromosome configurations during the meiotic divisions. The various possibilities are illustrated in the adjoining figure, in which parts A through C correspond to questions (a) through (c).



(a) Part A shows the genes closely linked to each other (and therefore to the same centromere). They are shown in different arms, but they could just as well be in the same arm. With no crossing-over between either gene and the centromere, all tetrads are PD.

(b) In part B, the genes are closely linked to different centromeres. With no crossing-over between either gene and its centromere, half the tetrads are PD and half NPD. (c) In part C, the *b* gene is not centromere-linked. Again PD = NPD (indicating that A and B are in different chromosomes), but crossing-over between the gene and the centromere yields TT tetrads. *The frequency of TT tetrads increases with the recombination frequency between the gene and the centromere, to a maximum of 2/3 when the gene and centromere undergo free recombination.*

Analysis and Applications

4.1 Each of two recessive mutations, m_1 and m_2 , can produce albino guinea pigs. The genes are close together in one chromosome and may be allelic. The cross $m_1 m_1 \times m_2 m_2$ produces progeny with wildtype pigmentation. Are the mutations in the same gene?

4.2 What gametes are produced by an individual of genotype $A b/a B$ if the genes are

(a) in different chromosomes?

(b) in the same chromosome with no recombination between them?

4.3 What gametes are produced by an individual of genotype $A b/a B$ if recombination can occur between the genes? Which gametes are the most frequent?

4.4 In the absence of recombination, what genotypes of progeny are expected from the cross $A B/a b \times A b/a B$? If A and B are dominant, what ratio of phenotypes is expected?

4.5 Which of the following arrays represents the *cis* configuration of the alleles A and B: $a b/A B$ or $a B/A b$?

4.6 What gametes, and in what ratios, are produced by male and female *Drosophila* of genotype $A B/a b$ when the genes are present in the same chromosome and the frequency of recombination between them is 5 percent?

4.7 If a large number of genes were mapped, how many linkage groups would be found in

(a) a haploid organism with 17 chromosomes per somatic cell?

- (b) a bacterial cell with a single, circular DNA molecule?
- (c) the rat, with 42 chromosomes per diploid somatic cell?

4.8 If the recombination frequency between genes *A* and *B* is 6.2 percent, what is the distance between the genes in map units in the linkage map?

4.9 Consider an organism heterozygous for two genes located in the same chromosome, *A B/a b*. If a single crossing-over occurs between the genes in every cell undergoing meiosis, and no multiple crossing-overs occur between the genes, what is the recombination frequency between the genes? What effect would multiple crossing-over have on the recombination frequency?

4.10 Normal plant height in wheat requires the presence of either or both of two dominant alleles, *A* and *B*. Plants homozygous for both recessive alleles (*a b/a b*) are dwarfed but otherwise normal. The two genes are in the same chromosome and recombine with a frequency of 16 percent. From the cross *A b/a B × A B/a b*, what is the expected frequency of dwarfed plants among the progeny?

4.11 In the yellow-fever mosquito, *Aedes aegypti*, sex is determined by a single pair of alleles, *M* and *m*. Heterozygous *Mm* mosquitos are male, homozygous *mm* are female. The recessive gene *bz* for bronze body color (the dominant phenotype is black) is linked to the sex-determining gene and recombines with it at a frequency of 6 percent. In the cross of a heterozygous *bz⁺ M/bz m* male with a bronze female, what genotypes and phenotypes of progeny are expected?

4.12 Two genes in chromosome 7 of corn are identified by the recessive alleles *gl* (glossy), determining glossy leaves, and *ra* (ramosa), determining branching of ears. When a plant heterozygous for each of these alleles was crossed with a homozygous recessive plant, the progeny consisted of the following genotypes with the numbers of each indicated:

Gl ra/gl ra 88 *gl Ra/gl ra* 103

Gl Ra/gl ra 6 *gl ra/gl ra* 3

Calculate the frequency of recombination between these genes.

4.13 The recessive alleles *b* and *cn* of two genes in chromosome 2 of *Drosophila* determine black body color and cinnabar eye color, respectively. If the genes are 8 map units apart, what are the expected genotypes and phenotypes and their relative frequencies among the progeny of the cross *+ +/b cn × + +/b cn*? (Remember that crossing-over does not occur in *Drosophila* males.)

4.14 In a testcross of an individual heterozygous for each of three linked genes, the most frequent classes of progeny were *A B c/a b c* and *a b C/a b c*, and the least frequent classes were *A B C/a b c* and *a b c/a b c*. What was the genotype of the triple heterozygote parent, and what is the order of the genes?

4.15 Construct a map of a chromosome from the following recombination frequencies between individual pairs of genes: *r—c*, 10; *c—p*, 12; *p—r*, 3; *s—c*, 16; *s—r*, 8. You will discover that the distances are not strictly additive. Why aren't they?

4.16 Yellow versus gray body color in *D. melanogaster* is determined by the alleles *y* and *y⁺*, vermilion versus wildtype eyes by the alleles *v* and *v⁺*, and singed versus straight bristles by the alleles *sn* and *sn⁺*. When females heterozygous for each of these X-linked genes were testcrossed with yellow, vermilion, singed males, the following classes and numbers of progeny were obtained:

yellow, vermilion, singed 53

yellow, vermilion 108

yellow, singed	331
yellow	5
vermilion, singed	3
vermilion	342
singed	95
wildtype	63

(a) What is the order of the three genes? Construct a linkage map with the genes in their correct order, and indicate the map distances between the genes.

(b) How does the frequency of double crossovers observed in this experiment compare with the frequency expected if crossing-over occurs independently in the two chromosome regions? Determine the coefficient of coincidence and the interference.

4.17 In corn, the alleles *C* and *c* result in colored versus colorless seeds, *Wx* and *wx* in nonwaxy versus waxy endosperm, and *Sh* and *sh* in plump versus shrunken endosperm. When plants grown from seeds heterozygous for each of these pairs of alleles were testcrossed with plants from colorless, waxy, shrunken seeds, the progeny seeds were as follows:

Colorless, nonwaxy, shrunken	84
colorless, nonwaxy, plump	974
colorless, waxy, shrunken	20
colorless, waxy, plump	2349
colored, waxy, shrunken	951
colored, waxy, plump	99
colored, nonwaxy, shrunken	2216
colored, nonwaxy, plump	15
Total	6708

Determine the order of the three genes, and construct a linkage map showing the genetic distances between adjacent genes.

4.18 The accompanying diagram summarizes the recombination frequencies observed in a large experiment to study three linked genes.



What was the observed frequency of double crossing-over in this experiment? Calculate the interference.

4.19 The recessive mutations *b* (black body color), *st* (scarlet eye color), and *hk* (hooked bristles) identify three autosomal genes in *D. melanogaster*. The following progeny were obtained from a testcross of females heterozygous for all three genes.

black, scarlet	243
black	241
black, hooked	15
black, hooked, scarlet	10
hooked	235
hooked, scarlet	226
scarlet	12
wildtype	18

What conclusions are possible concerning the linkage relations of these three genes? Calculate any appropriate map distances.

4.20 In the nematode *Caenorhabditis elegans*, the mutations *dpy-21* (dumpy) and *unc-34* (uncoordinated) identify linked genes that affect body conformation and coordination of movement. The frequency of recombination between the genes is 24 percent. If the heterozygote *dpy-21* *+/+* *unc-34* undergoes self-fertilization (the normal mode of reproduction in this organism), what fraction of the progeny is expected to be both dumpy and uncoordinated?

4.21 Dark eye color in rats requires the presence of a dominant allele of each of two genes, *R* and *P*. Animals homozygous for either or both of the recessive alleles have light-colored eyes. In one experiment, homozygous dark-eyed rats were crossed with doubly recessive light-eyed rats, and the resulting F_1 animals were testcrossed with rats of the homozygous light-eyed strain. The progeny from the test-cross consisted of 628 dark-eyed and 889 light-eyed rats. In another experiment, *R p R p* animals were crossed with *r P r P* animals, and the F_1 progeny were crossed with animals from an *r p r p* strain. The progeny consisted of 86 dark-eyed and 771 light-eyed rats. Determine whether the genes are linked, and if they are, estimate the frequency of recombination between the genes from each of the experiments.

4.22 A *Drosophila* geneticist exposes flies to a mutagenic chemical and obtains nine mutations nine mutations in the X chromosome that are lethal when homozygous. The mutations are tested in pairs in complementation tests, with the results shown in the accompanying table. A + indicates complementation (that is, flies carrying both mutations survive), and a - indicates noncomplementation (flies carrying both mutations die). How many genes (complementation groups) are represented by the mutations, and which mutations belong to each complementation group?

	1	2	3	4	5	6	7	8	9
1	-	+	-	+	+	+	-	+	+
2		-	+	+	+	+	+	+	+
3			-	+	+	+	-	+	+
4				-	+	-	+	+	+
5					-	+	+	+	-
6						-	+	+	+
7							-	+	+
8								-	+
9									-

4.23 The following classes and frequencies of ordered tetrads were obtained from the cross $a^+ b^+ \times a b$ in *Neurospora*. (Only one member of each pair of spores is shown.)

Spore pair				Number of asci
1-2	3-4	5-6	7-8	
$a^+ b^+$	$a^+ b^+$	ab	ab	1766
$a^+ b^+$	ab	$a^+ b^+$	ab	220
$a^+ b^+$	ab^+	$a^+ b$	an	14

What is the order of the genes in relation to the centromere?

4.24 Three genes in chromosome 9 of corn determine shrunken *sh* versus plump *Sh* kernels, waxy *wx* versus nonwaxy *Wx* endosperm, and glossy *gl* versus nonglossy *Gl* leaves. The genetic map of this chromosomal region is $sh-30-wx-10-gl$. From a plant of genotype $Sh wx Gl/sh Wx gl$, what is the expected frequency of $sh wx gl$ gametes:

(a) in the absence of interference?

(b) assuming 60 percent interference?

4.25 The following spore arrangements were obtained in the indicated frequencies from ordered tetrads in a cross between a *Neurospora* strain *com val (c v)*, which exhibits a compact growth form and is unable to synthesize the amino acid

valine, and a wildtype strain, + +. (Only one member of each pair of spores is shown.)

Spore pair	Accus composition				
1-2	cv	c+	cv	+v	cv
3-4	cv	c+	c+	c+	+v
5-6	++	+v	+v	cv	c+
7-8	++	+v	++	++	++
Number:	34	36	20	1	9

What can you conclude about the linkage and location of the genes with respect to each other and to the centromere?

Challenge Problems

4.26 In *Neurospora* several genes are needed for production of black pigment in spores. Recessive alleles at any one of these can result in tan spores. Six possible arrangements of spore colors are possible in crosses segregating for one of these allelic pairs, as shown in the accompanying diagram. Three different allelic pairs were analyzed in crosses 1, 2, and 3, respectively. The results are tabulated in the illustration.

	Asci observed	Cross 1	Cross 2	Cross 3
①		6	0	20
②		6	0	20
③		48	60	20
④		6	0	20
⑤		6	0	20
⑥		48	60	20

For each cross, indicate how the different arrangements arise and what you can deduce about the location of the segregating gene. In crosses in which it is relevant, calculate a map distance.

4.27 In baker's yeast (*Saccharomyces cerevisiae*), which produces unordered asci, the following cross was made:

leu2 trp1 + × + + *met14*

The resulting diploid was induced to undergo meiosis, and the asci were dissected and spore clones analyzed. The following tetrad types were found, with the observed number of each shown below. For convenience, *leu2*, *trp1*, and *met14* are abbreviated *l*, *t*, and *m*, respectively.

<i>l t m</i>	<i>l t +</i>	<i>+ t +</i>	<i>+ t m</i>	<i>+ t m</i>	<i>+ t +</i>
<i>l t m</i>	<i>l t +</i>	<i>+ t +</i>	<i>+ t m</i>	<i>l t m</i>	<i>l t +</i>
<i>+ + +</i>	<i>+ + m</i>	<i>l + m</i>	<i>l + +</i>	<i>+ + +</i>	<i>+ + m</i>
<i>+ + +</i>	<i>+ + m</i>	<i>l + m</i>	<i>l + +</i>	<i>l + +</i>	<i>l + m</i>
230	235	215	220	54	46

Analyze these data as fully as you can with respect to the linkage of each gene with respect to every other gene and

with respect to its centromere. Where a genetic linkage is indicated, calculate the map distance.

4.28 In baker's yeast (*Saccharomyces cerevisiae*), which produces unordered asci, the following cross was made:

rad6 trp5 leu1 + × *+ + + met14*

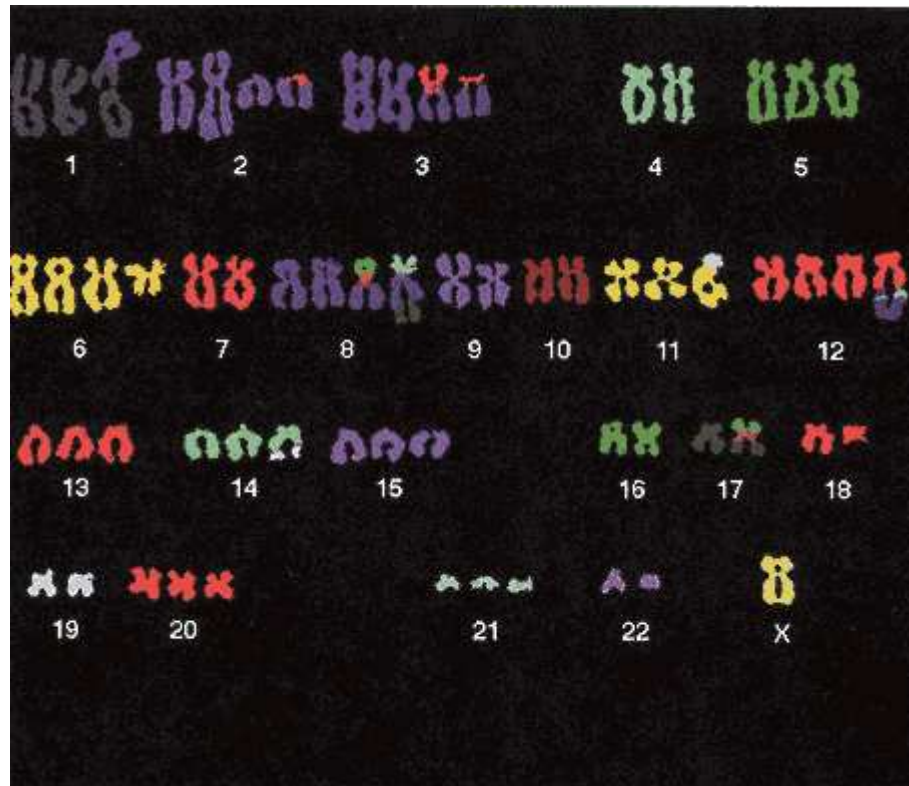
The resulting diploid was induced to undergo meiosis, the asci were dissected, and the spore clones analyzed. The following tetrad types were found in the numbers shown below each tetrad. For convenience, *rad6*, *trp5*, *leu1*, and *met14* are abbreviated as *r*, *t*, *l*, and *m*, respectively.

<i>r t l +</i>	<i>r t l m</i>	<i>r t l +</i>	<i>r t l m</i>	<i>r t l +</i>
<i>r t l +</i>	<i>r t l m</i>	<i>+ t l +</i>	<i>+ t l m</i>	<i>+ + l +</i>
<i>+ + + m</i>	<i>+ + + +</i>	<i>r + + m</i>	<i>r + + +</i>	<i>r t + m</i>
<i>+ + + m</i>	<i>+ + + +</i>	<i>+ + + m</i>	<i>+ + + +</i>	<i>+ + + m</i>
188	206	105	92	140
<i>r t l m</i>	<i>r t l +</i>	<i>r t l m</i>	<i>r + l +</i>	<i>+ t l m</i>
<i>+ + l m</i>	<i>+ + + +</i>	<i>+ + l m</i>	<i>+ t l +</i>	<i>+ + l m</i>
<i>r t + +</i>	<i>r t l m</i>	<i>+ t + +</i>	<i>r t + m</i>	<i>r + + +</i>
<i>+ + + +</i>	<i>+ + + m</i>	<i>r + + +</i>	<i>+ + + m</i>	<i>r t + +</i>
154	109	3	2	1

Use what you learned about *met14* in Challenge Problem 4.27 to analyze these data as fully as you can for linkage of the genes to one another and of each to its respective centromere. When genetic linkage is indicated, calculate the map distance.

Further Reading

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Human chromosomes isolated from cells of an ovarian cancer. Each chromosome is labeled with a different color. Note the many abnormal chromosomes, containing two or more colors, that have been formed by breakage of other chromosomes and Reunion of their broken parts in abnormal combinations. Note also that this cell has 63 chromosomes (instead of the normal 46) and that one of the X chromosomes is missing.

[Courtesy of David C. Ward and Michael R. Speicher.]

Chapter 5— The Molecular Structure and Replication of the Genetic Material

CHAPTER OUTLINE

5-1 The Chemical Composition of DNA

5-2 The Physical Structure of the Double Helix

5-3 What a Genetic Material Needs That DNA Supplies

5-4 The Replication of DNA

The Basic Rule for the Replication of Nucleic Acids

The Geometry of DNA Replication

5-5 DNA Synthesis

5-6 Discontinuous Replication

Fragments in the Replication Fork

Initiation by an RNA Primer

The Joining of Precursor Fragments

Other Proteins Needed for DNA Replication

5-7 The Isolation and Characterization of Particular DNA Fragments

Denaturation and Renaturation

Nucleic Acid Hybridization

Restriction Enzymes and Site-Specific DNA Cleavage

Gel Electrophoresis

The Southern Blot

5-8 The Polymerase Chain Reaction

5-9 Determination of the Sequence of Bases in DNA

The Sequencing Procedure

Clinical Use of Dideoxynucleoside Analogs

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Challenge Problems

Further Reading

GeNETics on the web

PRINCIPLES

- A DNA strand is a polymer of A, T, G and C deoxyribonucleotides joined 3' to 5' by phosphodiester bonds.
- The two DNA strands in a duplex are held together by hydrogen bonding between the A–T and G–C base pairs.
- DNA replication is semiconservative and takes place only in the 5' to 3' direction; successive nucleotides are added only at the 3' end.
- Each type of restriction endonuclease enzyme cleaves double-stranded DNA at a particular sequence of bases usually four or six nucleotides in length.
- Separated strands of DNA or RNA that are complementary in nucleotide sequence can come together (hybridize) spontaneously to form duplexes.
- In the polymerase chain reaction, short oligonucleotide primers are used in successive cycles of DNA replication to amplify selectively a particular region of a DNA duplex.
- The DNA fragments produced by a restriction enzyme can be separated by electrophoresis, isolated, sequenced, and manipulated in other ways.

CONNECTIONS

CONNECTION: The Double Helix

James D. Watson and Francis H. C. Crick 1953
A structure for deoxyribose nucleic acid

CONNECTION: Replication by Halves

Matthew Meselson and Franklin W. Stahl 1958
The replication of DNA in Escherichia coli

Analysis of the patterns of inheritance and even the phenotypic expression of genes reveals nothing about gene structure at the molecular level, how genes are copied to yield exact replicas of themselves, or how they determine cellular characteristics. Understanding these basic features of heredity requires identification of the chemical nature of the genetic material and the processes through which it is replicated. In Chapter 1, we reviewed the experimental evidence demonstrating that the genetic material is DNA. The structure of DNA was described as a helix of two paired, complementary strands, each composed of an ordered string of nucleotides bearing A (adenine), T (thymine), G (guanine), or cytosine (C). Watson-Crick base pairing between A and T and between G and C in the complementary strands holds the strands together. The complementarity also holds the key to replication, because each strand can serve as a template for the synthesis of a new complementary strand. In this chapter, we take a closer look at DNA structure and its replication. We also consider how our knowledge of DNA structure and replication has been used in the development of laboratory techniques for isolating fragments that contain genes or parts of genes of particular interest and for determining the sequence of bases in DNA fragments.

5.1—

The Chemical Composition of DNA

DNA is a polymer—a large molecule that contains repeating units—composed of 2'-deoxyribose (a five-carbon sugar), phosphoric acid, and the four nitrogen-containing bases denoted A, T, G, and C. The chemical structures of the bases are shown in Figure 5.1. Note that two of the bases have a double-ring structure; these are called **purines**. The other two bases have a single-ring structure; these are called **pyrimidines**.

- The purine bases are adenine (A) and guanine (G).
- The pyrimidine bases are thymine (T) and cytosine (C).

In DNA, each base is chemically linked to one molecule of the sugar deoxyribose, forming a compound called a **nucleoside**. When a phosphate group is also attached to the sugar, the nucleoside becomes a **nucleotide** (Figure 5.2). Thus a nucleotide is a nucleoside plus a phosphate. In the conventional numbering of the carbon atoms in the sugar in Figure 5.2, the carbon atom to which the base is attached is the 1' carbon. (The atoms in the sugar are given

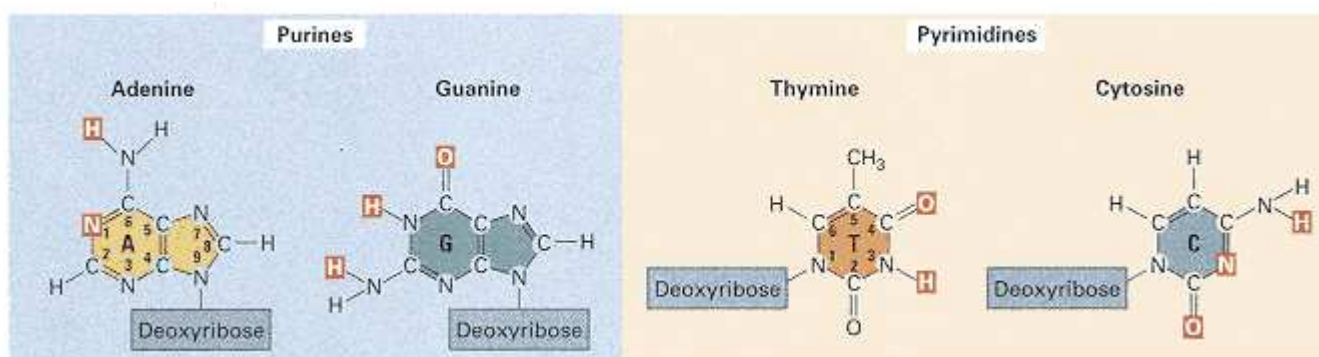


Figure 5.1

Chemical structures of adenine, thymine, guanine, and cytosine, the four nitrogen-containing bases in DNA.

In each base, the nitrogen atom linked to the deoxyribose sugar is indicated. The atoms shown in red participate in hydrogen bonding between the DNA base pairs, as explained in Section 5.2.

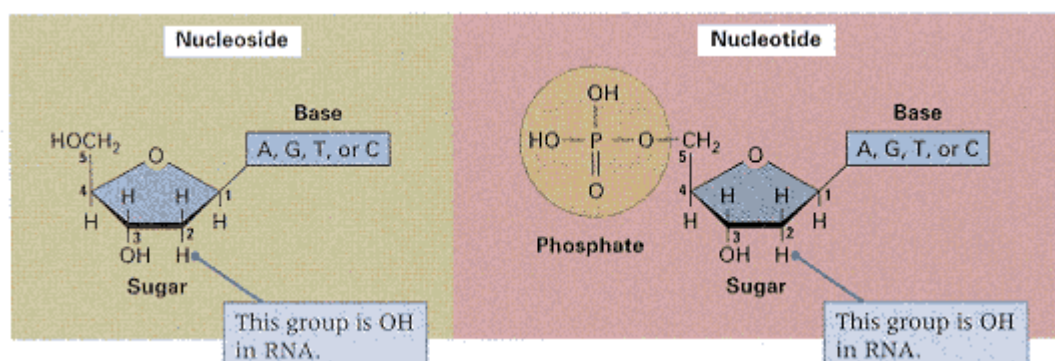


Figure 5.2

A typical nucleotide showing the three major components (phosphate, sugar, and base), the difference between DNA and RNA, and the distinction between a nucleoside (no phosphate group) and a nucleotide (with phosphate). Nucleotides may contain one phosphate unit (monophosphate), two such units (diphosphate) or three (triphosphate).

primed numbers to distinguish them from atoms in the bases.) The nomenclature of the nucleoside and nucleotide derivatives of the DNA bases is somewhat complicated and is summarized in Table 5.1. Most of these terms are not needed in this book; they are included because they are likely to be encountered in further reading.

In nucleic acids, such as DNA and RNA, the nucleotides are joined to form a **polynucleotide chain**, in which the phosphate attached to the 5' carbon of one sugar is linked to the hydroxyl group attached to the 3' carbon of the next sugar in line (Figure 5.3). The chemical bonds by which the sugar components of adjacent nucleotides are linked through the phosphate groups are called **phosphodiester bonds**. The 5'-3'-5'-3' orientation of these linkages continues throughout the chain, which typically consists of millions of nucleotides. Note that the terminal groups of each polynucleotide chain are a 5'-phosphate (**5'-P**) group at one end and a 3'-hydroxyl (**3'-OH**) group at the other. The asymmetry of the ends of a DNA strand implies that each strand has a **polarity** determined by which end bears the 5'-phosphate and which end bears the 3'-hydroxyl.

Three years before Watson and Crick proposed their essentially correct three-dimensional structure of DNA as a double helix, Erwin Chargaff developed a chemical technique to measure the amount of each base present in DNA. As we describe his technique, we will let the molar concentration of any base be represented by the symbol for the base in square brackets; for example, [A] denotes the molar concentration of adenine. Chargaff used his technique to measure the [A], [T], [G], and [C] content of the DNA from a variety of sources. He found that the **base composition** of the DNA, defined as the **percent G + C**, differs among species but is

Table 5.1 DNA nomenclature

Base	Nucleoside	Nucleotide
Adenine (A)	Deoxyadenosine	Deoxyadenosine-5' monophosphate (dAMP) diphosphate (dADP) triphosphate (dATP)
Guanine (G)	Deoxyguanosine	Deoxyguanosine-5' monophosphate (dGMP) diphosphate (dGDP) triphosphate (dGTP)
Thymine (T)	Deoxythymidine	Deoxythymidine-5' monophosphate (dTMP) diphosphate (dTDP) triphosphate (dTTP)
Cytosine (C)	Deoxycytidine	Deoxycytidine-5' monophosphate (dCMP) diphosphate (dCDP) triphosphate (dCTP)

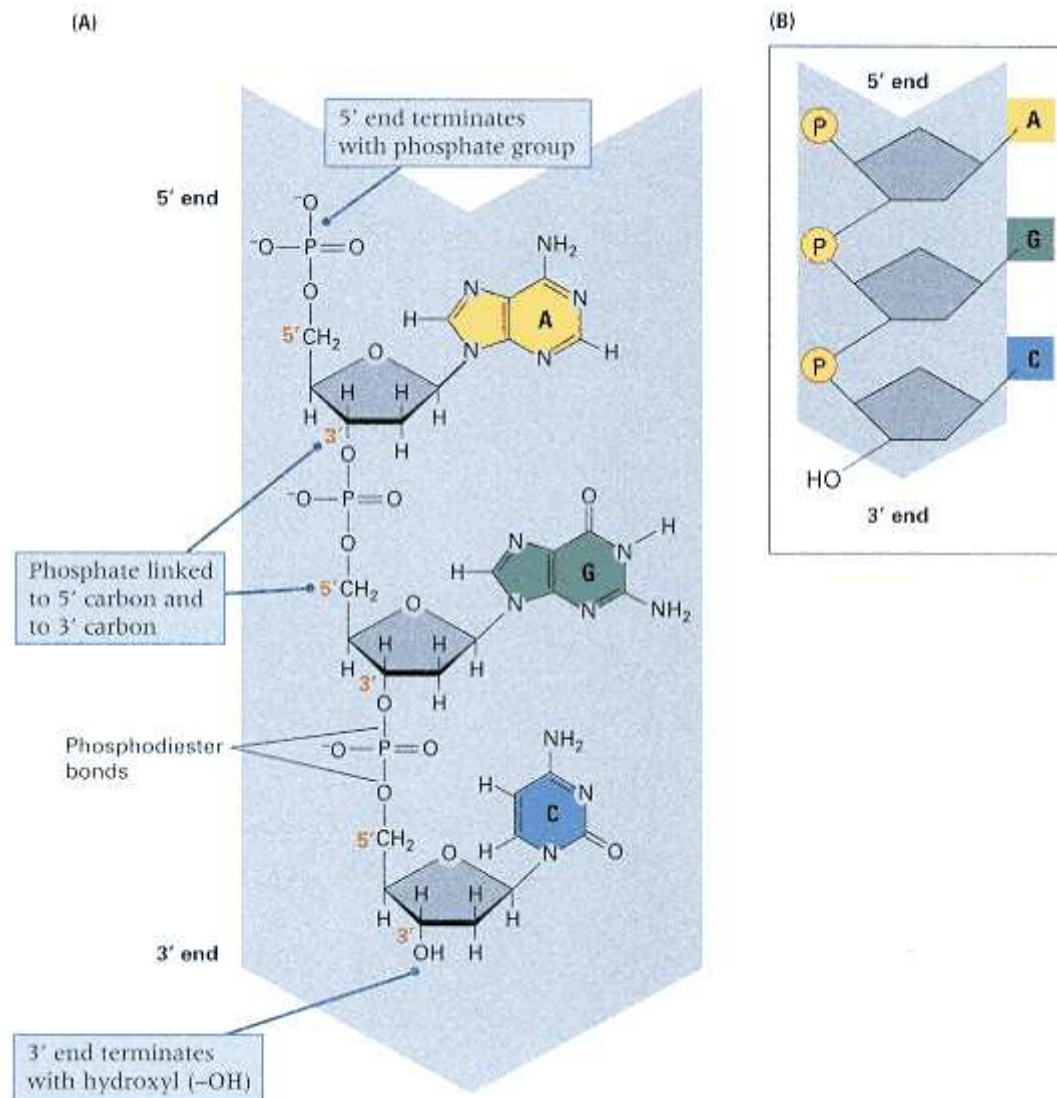


Figure 5.3

Three nucleotides at the 5' end of a single polynucleotide strand. (A) The chemical structure of the sugar-phosphate linkages, showing the 5'-to-3' orientation of the strand (the red numbers are those assigned to the carbon atoms). (B) A common schematic way to depict a polynucleotide strand.

constant in all cells of an organism and within a species. Data on the base composition of DNA from a variety of organisms are given in Table 5.2.

Chargaff also observed certain regular relationships among the molar concentrations of the different bases. These relationships are now called **Chargaff's rules**:

- The amount of adenine equals that of thymine: $[A] = [T]$.
- The amount of guanine equals that of cytosine: $[G] = [C]$
- The amount of purine base equals that of pyrimidine bases:

$$[A] + [G] = [T] + [C]$$

Although the chemical basis of these observations was not known at the time, one of the appealing features of the Watson-Crick structure of paired comple-

Table 5.2 Base composition of DNA from different organisms

Organism	Base (and percentage of total bases)				Base composition (percent G + C)
	Adenine	Thymine	Guanine	Cytosine	
Bacteriophage T7	26.0	26.0	24.0	24.0	48.0
Bacteria					
<i>Clostridium perfringens</i>	36.9	36.3	14.0	12.8	26.8
<i>Streptococcus pneumoniae</i>	30.2	29.5	21.6	18.7	40.3
<i>Escherichia coli</i>	24.7	23.6	26.0	25.7	51.7
<i>Sarcina lutea</i>	13.4	12.4	37.1	37.1	74.2
Fungi					
<i>Saccharomyces cerevisiae</i>	31.7	32.6	18.3	17.4	35.7
<i>Neurospora crassa</i>	23.0	22.3	27.1	27.6	54.7
Higher plants					
Wheat	27.3	27.2	22.7	22.8*	45.5
Maize	26.8	27.2	22.8	23.2*	46.0
Animals					
<i>Drosophila melanogaster</i>	30.8	29.4	19.6	20.2	39.8
Pig	29.4	29.6	20.5	20.5	41.0
Salmon	29.7	29.1	20.8	20.4	41.2
Human being	29.8	31.8	20.2	18.2	38.4

*Includes one-fourth 5-methylcytosine, a modified form of cytosine found in most plants more complex than algae and in many animals

mentary strands was that it explained Chargaff's rules. Because A is always paired with T in double-stranded DNA, it must follow that $[A] = [T]$. Similarly, because G is paired with C, $[G] = [C]$. The third rule follows by addition of the other two: $[A] + [G] = [T] + [C]$. In the next section, we examine the molecular basis of base pairing in more detail.

5.2—

The Physical Structure of the Double Helix

In the three-dimensional structure of the DNA molecule proposed in 1953 by Watson and Crick, the molecule consists of two polynucleotide chains twisted around one another to form a double-stranded helix in which adenine and thymine, and guanine and cytosine, are paired in opposite strands (Figure 5.4). In the standard structure, which is called the **B form** of DNA, each chain makes one complete turn every 34 Å. The helix is right-handed, which means that as you look down the barrel, each chain follows a clockwise path as it progresses. The bases are spaced at 3.4 Å, so there are ten bases per helical turn in each strand and ten base pairs per turn of the double helix. Each base is paired to a complementary base in the other strand by hydrogen bonds, which provide the main force holding the strands together. (A

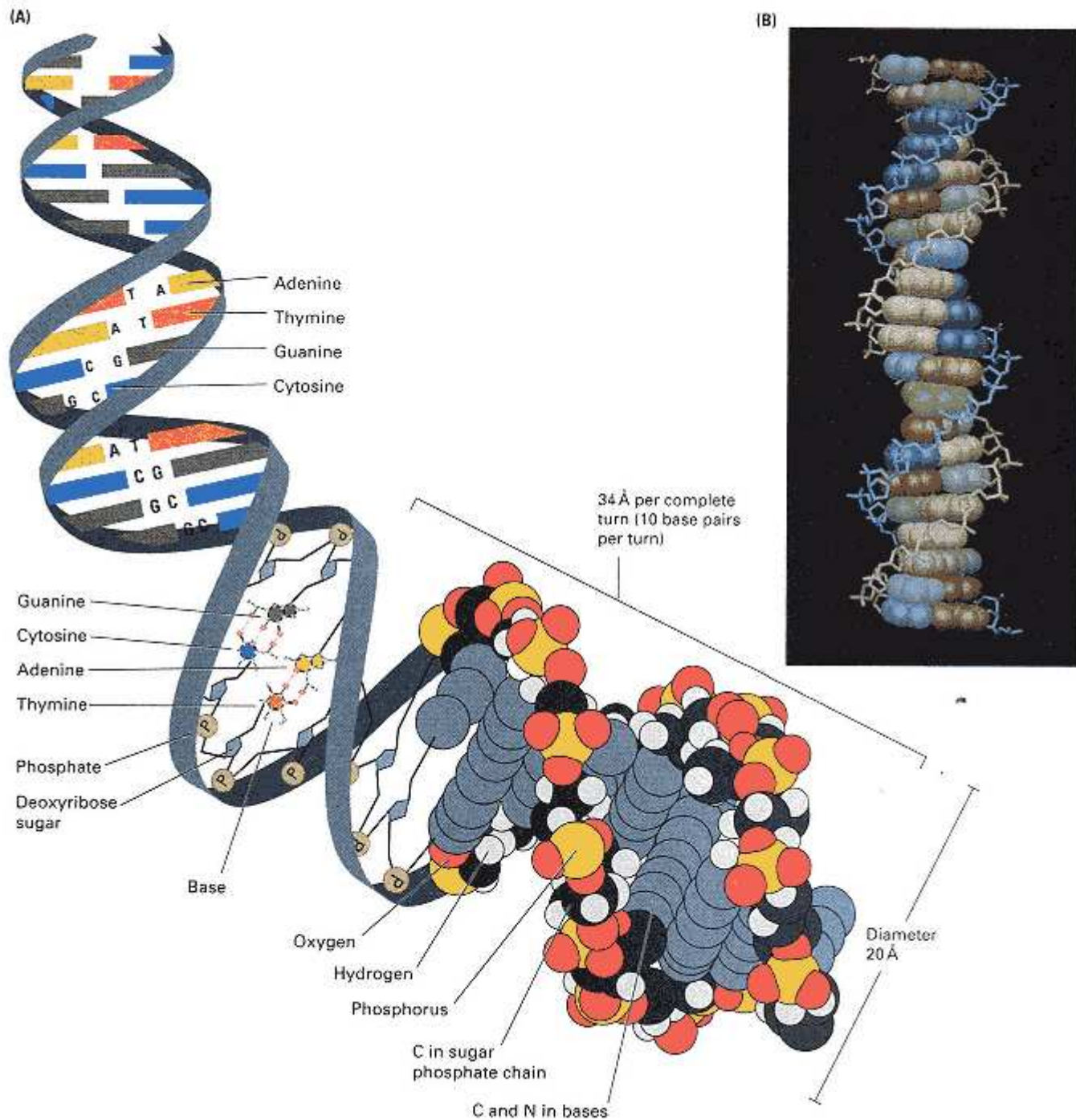


Figure 5.4

Figure 5.4

Two representations of DNA illustrating the three-dimensional structure of the double helix. (A) In a "ribbon diagram" the sugar-phosphate backbones are depicted as bands, with horizontal lines used to represent the base pairs.

(B) A computer model of the B form of a DNA molecule. The stick figures are the sugar-phosphate chains winding around outside the stacked base pairs, forming a major groove and a minor groove.

The color coding for the base pairs is A, red or pink; T, dark green or light green; G, dark brown or beige; C, dark blue or light blue. The bases depicted in dark colors are those attached to the blue sugar-phosphate backbone; bases depicted in light colors are attached to the beige backbone.

[B, courtesy of Antony M. Dean.]

hydrogen bond is a weak bond in which two negatively charged atoms share a hydrogen atom.) The paired bases are planar, parallel to one another, and perpendicular to the long axis of the double helix. When discussing a DNA molecule, molecular biologists frequently refer to the individual strands as single strands or as single-stranded DNA; they refer to the double helix as double-stranded DNA or as a *duplex* molecule. The two grooves spiraling along outside of the double helix are not symmetrical; one groove, called the **major groove**, is larger than the other, which is called the **minor groove**. Proteins that interact with double-stranded DNA often have regions that make contact with the base pairs by fitting into the major groove, into the minor groove, or into both grooves.

The central feature of DNA structure is the pairing of complementary bases, A with T and G with C. The hydrogen bonds that form in the adenine-thymine base pair and in the guanine-cytosine pair are illustrated in Figure 5.5. Note that an A–T pair (Figure 5.5A and B) has two hydrogen bonds and that a G–C pair (Figure 5.5C and D) has three hydrogen bonds. This means that the hydrogen bonding between G and C is

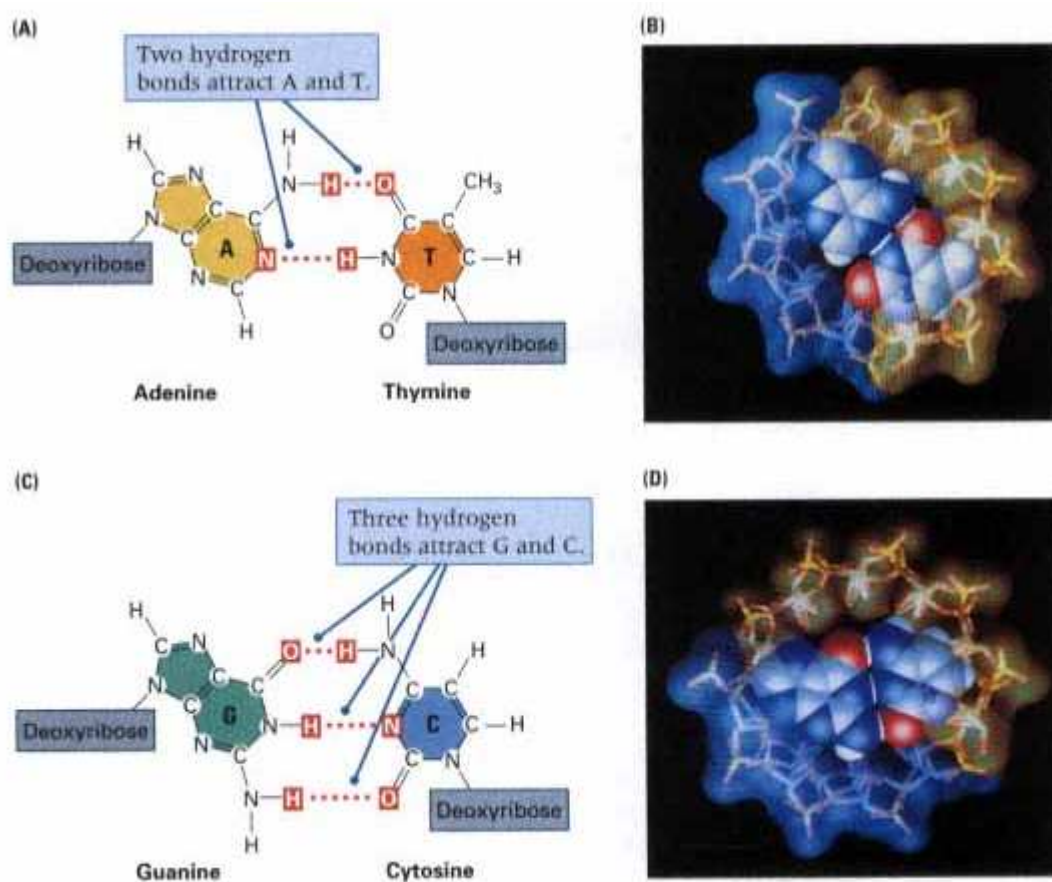


Figure 5.5

Normal base pairs in DNA. On the left, the hydrogen bonds (dotted lines) and the joined atoms are shown in red. (A, B) An A–T base pair. (C, D) A G–C base pair. In the space-filling models (B and D), the colors are C, gray; N, blue; O, red; and H (shown in the bases only), white. Each hydrogen bond is depicted as a white disk squeezed between the atoms sharing the hydrogen. The stick figures on the outside represent the backbones winding around the stacked base pairs.

[Space-filling models courtesy of Antony M. Dean.]

Connection The Double Helix

James D. Watson and Francis H. C. Crick 1953

Cavendish Laboratory
Cambridge, England
A Structure for Deoxyribose Nucleic Acid

This is one of the watershed papers of twentieth-century biology. After its publication, nothing in genetics was the same. Everything that was known, and everything still to be discovered, would now need to be interpreted in terms of the structure and function of DNA. The importance of the paper was recognized immediately, in no small part because of its lucid and concise description of the structure. This excerpt is unusual in that it includes the acknowledgment. Watson and Crick benefited tremendously in knowing that their structure was consistent with the unpublished structural studies of Maurice Wilkins and Rosalind Franklin. The same issue of Nature that included the Watson and Crick paper also included, back to back, a paper from the Wilkins group and one from the Franklin group detailing their data and the consistency of their data with the proposed structure. It has been said that Franklin was poised a mere two half-steps from making the discovery herself, alone. In any event, Watson and Crick and Wilkins were awarded the 1962 Nobel Prize for their discovery of DNA structure. Rosalind Franklin, tragically, died of cancer in 1958 at the age of 38.

We wish to suggest a structure for the salt of deoxyribose nucleic acid (DNA) . . . The structure has two helical chains each coiled round the same axis . . . Both chains follow right-handed helices, but the two chains run in opposite directions . . . The bases are on the inside of the helix and the phosphates on the outside . . . There is a residue on each chain every 3.4 Å and the structure repeats after 10 residues . . . The

If only specific pairs of bases can be formed, it follows that if the sequences of bases on one chain is given, then the sequence on the other chain is automatically determined.

novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fiber axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side. One of the pair must be a purine and the other a pyrimidine for bonding to occur . . . Only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine). In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain is given, then the sequence on the other chain is automatically determined . . . It has not escaped our notice that the specific pairing we have postulated immediately suggests a plausible copying mechanism for the genetic material . . . We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. Maurice H. F. Wilkins, Dr. Rosalind Franklin and their co-workers at King's College, London.

Source: *Nature* 171: 737–738

stronger in the sense that it requires more energy to break; for example, the amount of heat required to separate the paired strands in a DNA duplex increases with the percent of G + C. Because nothing restricts the sequence of bases in a single strand, any sequence could be present along one strand. This explains Chargaff's observation that DNA from different organisms may have different base compositions. However, because the strands in duplex DNA are complementary, Chargaff's rules of $[A] = [T]$ and $[G] = [C]$ are true whatever the base composition.

Each backbone in a double helix consists of deoxyribose sugars alternating with phosphate groups that link the 3' carbon atom of one sugar to the 5' carbon of the next in line (Figure 5.3). The two polynucleotide strands of the

double helix are oriented in opposite directions in the sense that the bases that are paired are attached to >sugars lying above and below the plane of pairing, respectively. The sugars are

offset because the phosphate linkages in the backbones run in opposite directions (Figure 5.6), and the strands are said to be **antiparallel**. This means that each terminus of the double helix possesses one 5'-P group (on one strand) and one 3'-OH group (on the other strand), as shown in Figure 5.6.

The diagrams of the DNA duplexes in Figures 5.4 and 5.6 are static and so some-

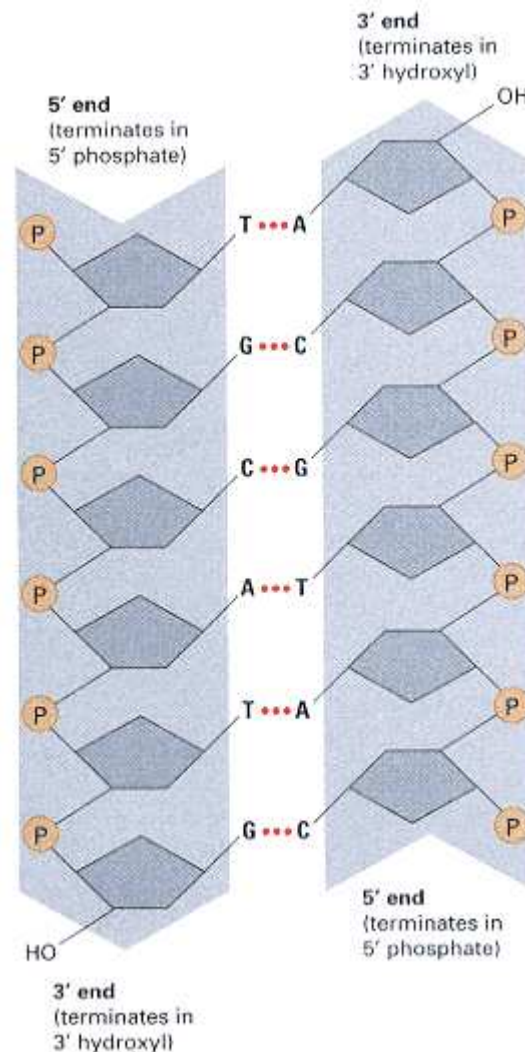


Figure 5.6

A segment of a DNA molecule showing the antiparallel orientation of the complementary strands. The overlying blue arrows indicate the 5'-to-3' direction of each strand. The Phosphates (P) join the 3' carbon atom of one deoxyribose (horizontal line) to the 5' carbon atom of the adjacent deoxyribose.

what misleading. DNA is a dynamic molecule, constantly in motion. In some regions, the strands can separate briefly and then come together again in the same conformation or in a different one. Although the right-handed double helix in Figure 5.4 is the standard form, DNA can form more than 20 slightly different variants of right-handed helices, and some regions can even form helices in which the strands twist to the left (called the **Z form** of DNA). If there are complementary stretches of nucleotides in the same strand, then a single strand, separated from its partner, can fold back upon itself like a hairpin. Even triple helices consisting of three strands can form in regions of DNA that contain suitable base sequences.

5.3—

What a Genetic Material Needs That DNA Supplies

Not every polymer would be useful as genetic material. However, DNA is admirably suited to a genetic function because it satisfies the three essential requirements of a genetic material. First, any genetic material must be able to be replicated accurately, so that the information it contains is precisely replicated and inherited by daughter cells.

The basis for exact duplication of a DNA molecule is the complementarity of the A–T and G–C pairs in the two polynucleotide chains. Unwinding and separation of the chains, with each free chain being copied, results in the formation of two identical double helices (Figure 5.7).

A genetic material must also have the capacity to carry all of the information needed to direct the organization and metabolic activities of the cell. As seen in Chapter 1, the product of most genes is a protein molecule—a polymer composed of molecular units called amino acids. The sequence of amino acids in the protein determines its chemical and physical properties. A gene is expressed when its protein product is synthesized, and one requirement of the genetic material is that it direct the order in which amino acid units are

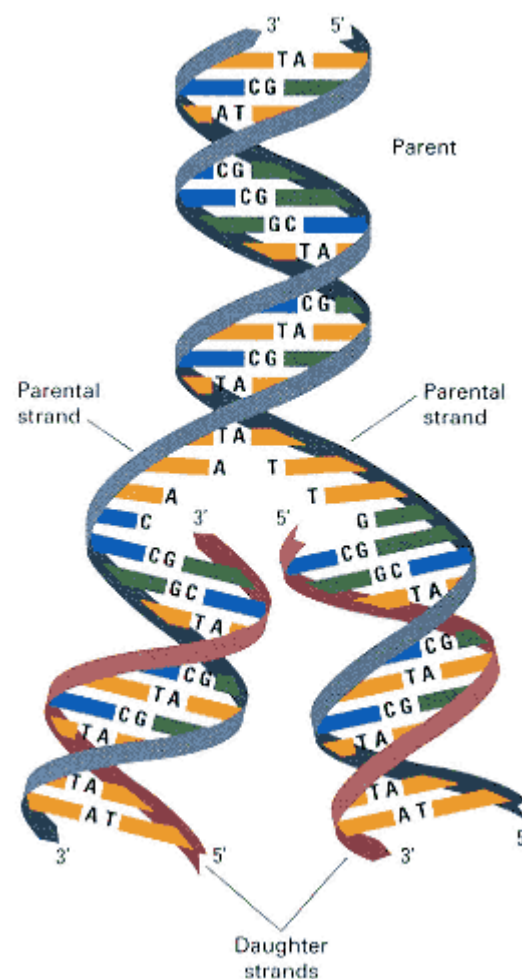


Figure 5.7
Watson-Crick model of DNA replication. The newly synthesized strands are in red. We will see in Section 5–6 that the details of DNA replication are considerably more complex, because any new DNA strand can be synthesized only in the 5'-to-3' direction.

added to the end of a growing protein molecule. In DNA, this is done by means of a genetic code in which groups of three bases specify amino acids. Because the four bases in a DNA molecule can be arranged in any sequence, and because the sequence can vary from one part of the molecule to another and from organism to organism, DNA can contain a great many unique regions, each of which can be a distinct gene. A long DNA chain can direct the synthesis of a variety of different protein molecules.

A genetic material must also be capable of undergoing occasional mutations in which the information it carries is altered. Furthermore, the mutant molecules must be capable of being replicated as faithfully as the parental molecule, so that mutations are heritable. Watson and Crick suggested that heritable mutations might be possible in DNA by rare mispairing of the bases, with the result that an incorrect nucleotide becomes incorporated into a replicating DNA strand.

5.4— The Replication of DNA

The process of replication, in which each strand of the double helix serves as a template for the synthesis of a new strand (Figure 5.7), is simple in principle. It requires only that the hydrogen bonds joining the bases break to allow separation of the chains and that appropriate free nucleotides of the four types pair with the newly accessible bases in each strand. However, it is a complex process with geometric problems requiring a variety of enzymes and other proteins. These processes are examined in this section.

The Basic Rule for the Replication of Nucleic Acids

The primary function of any mode of DNA replication is to reproduce the base sequence of the parent molecule. The specificity of base pairing—adenine with thymine (replaced by uracil in RNA) and guanine with cytosine—provides the mechanism used by all genetic replication systems. Furthermore,

- Nucleotide monomers are added one by one to the end of a growing strand by an enzyme called a DNA polymerase.
- The sequence of bases in each newly replicated strand, or **daughter strand**, is complementary to the base sequence in the old strand, or **parental strand**, being replicated. For example, wherever

an adenine nucleotide is present in the parental strand, a thymine nucleotide will be added to the growing end of the daughter strand.

The following section explains how the two strands of a daughter molecule are physically related to the two strands of the parental molecule.

The Geometry of DNA Replication

The production of daughter DNA molecules from a single parental molecule gives rise to several problems that result from the helical structure and great length of typical DNA molecules and the circularity of many DNA molecules. These problems and their solutions are described here.

Semiconservative Replication of Double-Stranded DNA In the **semiconservative mode** of replication, each parental DNA strand serves as a template for one new strand, and as each new strand is formed, it is hydrogen-bonded to its parental template (Figure 5.7). As replication proceeds, the parental double helix unwinds and then rewinds again into two new double helices, each of which contains one originally parental strand and one newly formed daughter strand.

In theory, DNA could be replicated by a number of mechanisms other than the semiconservative mode. However, the reality of semiconservative replication was demonstrated experimentally by Matthew Meselson and Franklin Stahl in 1958. The experiment made use of a newly developed high-speed centrifuge (an **ultracentrifuge**) that could spin a solution so fast that molecules differing only slightly in density could be separated. In their experiment, the heavy ^{15}N isotope of nitrogen was used for physical separation of parental and daughter DNA molecules. DNA isolated from the bacterium *E. coli* grown in a medium containing ^{15}N as the only available source of nitrogen is denser than DNA from bacteria grown in media with the normal ^{14}N isotope. These DNA molecules can be separated in an ultracentrifuge, because they have about the same density as a very concentrated solution of cesium chloride (CsCl).

When a CsCl solution containing DNA is centrifuged at high speed, the Cs^+ ions gradually sediment toward the bottom of the centrifuge tube. This movement is counteracted by diffusion (the random movement of molecules), which prevents complete sedimentation. At equilibrium, a linear gradient of increasing CsCl concentration—and of density—is present from the top to the bottom of the centrifuge tube. The DNA also moves upward or downward in the tube to a position in the gradient at which the density of the solution is equal to its own density. At equilibrium, a mixture of ^{14}N -containing ("light") and ^{15}N -containing ("heavy") *E. coli* DNA will separate into two distinct zones in a density gradient even though they differ only slightly in density. DNA from *E. coli* containing ^{14}N in the purine and pyrimidine rings has a density of 1.708 g/cm^3 , whereas DNA with ^{15}N in the purine and pyrimidine rings has a density of 1.722 g/cm^3 . These molecules can be separated because a solution of 5.6 molar CsCl has a density of 1.700 g/cm^3 . When spun in a centrifuge, the CsCl solution forms a gradient of density that brackets the densities of the light and heavy DNA molecules. It is for this reason that the separation technique is called **equilibrium density-gradient centrifugation**.

The result of the Meselson-Stahl experiment is shown in Figure 5.8. Prior to the experiment, bacteria were grown for many generations in a ^{15}N -containing medium. Therefore, at the beginning of the experiment, essentially all the DNA was uniformly labeled with ^{15}N and had a heavy density. The cells were then transferred to a ^{14}N -containing medium, and DNA was isolated from samples of cells taken from the culture at intervals and subjected to equilibrium density-gradient centrifugation. Each photograph in Figure 5.8 shows the image of a solution within a centrifuge tube taken in ultraviolet light of wavelength 260 nm (nanometers), which is absorbed by DNA. The positions of the DNA molecules in the density gradient are therefore indicated by the dark bands that absorb the

Connection Replication by Halves

Matthew Meselson and Franklin W. Stahl 1958

California Institute of Technology,
Pasadena, California
The Replication of DNA in Escherichia coli

Replication of DNA by separation of its strands, followed by the synthesis of a new partner strand for each separated parental strand, is so fundamental a biological mechanism that it is easy to forget that it had to be proved experimentally somewhere along the way. Although semiconservative replication is a simple and logical way to proceed, biological systems might have found a different way to replicate their DNA. The Meselson-Stahl experiment that demonstrated semiconservative replication made use of a novel technique, density-gradient centrifugation, which was invented by the authors specifically for the purpose of separating DNA molecules differing slightly in density. The use of cesium chloride was mandated, because in concentrated solution, it has a density similar to that of DNA. The method of cesium chloride density centrifugation is still important in modern molecular biology; it is used in the isolation of DNA molecules.

Studies of bacterial transformation and bacteriophage infection strongly indicate that deoxyribonucleic acid (DNA) can carry and transmit hereditary information and can direct its own replication. Hypotheses for the mechanism of DNA replication differ in the predictions they make concerning the distribution among progeny molecules of atoms derived from parental molecules . . . We anticipated that a label which imparts to the DNA molecule an increased density might permit an analysis of this distribution by sedimentation techniques. To this end, a method was developed for the detection of small density differences among macromolecules . . . A small amount of DNA in a concentrated solution of cesium

The results of the present experiment are in exact accord with the expectation of the Watson-Crick model for DNA duplication.

chloride is centrifuged until equilibrium is closely approached. The opposing processes of sedimentation and diffusion have then produced a stable concentration gradient of the cesium chloride, with a continuous increase in density along the direction of centrifugal force. The macromolecules of DNA present in this density gradient are driven by the centrifugal field into the region where the solution density is equal to their own buoyant density . . . Bacteria uniformly labeled with N^{15} were abruptly changed into N^{14} medium . . . Samples were withdrawn from the culture immediately and afterward at intervals for several generations . . . Until one generation time has elapsed, half-labeled molecules accumulate. One generation after the switch to N^{14} , these half-labeled or "hybrid" molecules alone are observed. Subsequently, only half-labeled DNA and completely unlabeled DNA are found. When two generation times have elapsed, half-labeled and unlabeled DNA are present in equal amounts. These results permit the following conclusions to be drawn: 1. *The nitrogen of a DNA molecule is divided equally between two subunits which remain intact through many generations . . .* 2. *Following replication, each daughter molecule has received one parental subunit . . .* 3. *The replicative act results in a molecular doubling . . .* A molecular structure for DNA has been proposed by Watson and Crick that . . . suggested to them a definite and structurally plausible hypothesis for the duplication of the DNA molecule . . . The results of the present experiment are in exact accord with the expectation of the Watson-Crick model for DNA duplication . . . The results presented here direct our attention to [other important problems]. What are the molecular structures of the subunits of DNA which are passed on intact to each daughter molecule? What is the relationship of these subunits to each other in a DNA molecule? What is the mechanism of the synthesis and dissociation of the subunits in vivo?

Source: *Proceedings of the National Academy of Sciences of the USA* 44:671–682

light. Each photograph is oriented such that the bottom of the tube is at the right and the top is at the left. To the right of each photograph is a graph showing the absorption of the ultraviolet light from the top of the centrifuge tube to the bottom. In each trace, the peaks correspond to the positions of the bands in the photographs, but the height and width

of each peak make it possible to quantify the DNA in each band.

At the start of the experiment (time 0), all of the DNA was heavy (^{15}N). After the transfer to ^{14}N , a band of lighter density began to appear, and it gradually became more prominent as the cells replicated their DNA and divided. After one generation of

growth (one round of replication of the DNA molecules and a doubling of the number of cells), all of the DNA had a "hybrid" density exactly intermediate between the densities of ^{15}N -DNA and ^{14}N -DNA. The finding of molecules with a hybrid density indicates that the replicated molecules contain equal amounts of the two nitrogen isotopes. After a second generation of replication in the ^{14}N medium (1.9 generations in the original experiment), half of the DNA had the density of DNA with ^{14}N in both strands ("light" DNA) and the other half had the hybrid density. After three generations, the ratio of light to hybrid DNA was approximately 3 : 1, and after four generations (4.1 in the original experiments), it was approximately 7 : 1 (Figure 5.8). This distribution of ^{15}N atoms is precisely the result predicted from semiconservative replication of the Watson-Crick structure, as illustrated in Figure 5.9. Subsequent experiments with replicating DNA from numerous viruses, bacteria, and higher organisms have indicated that semiconservative replication is universal.

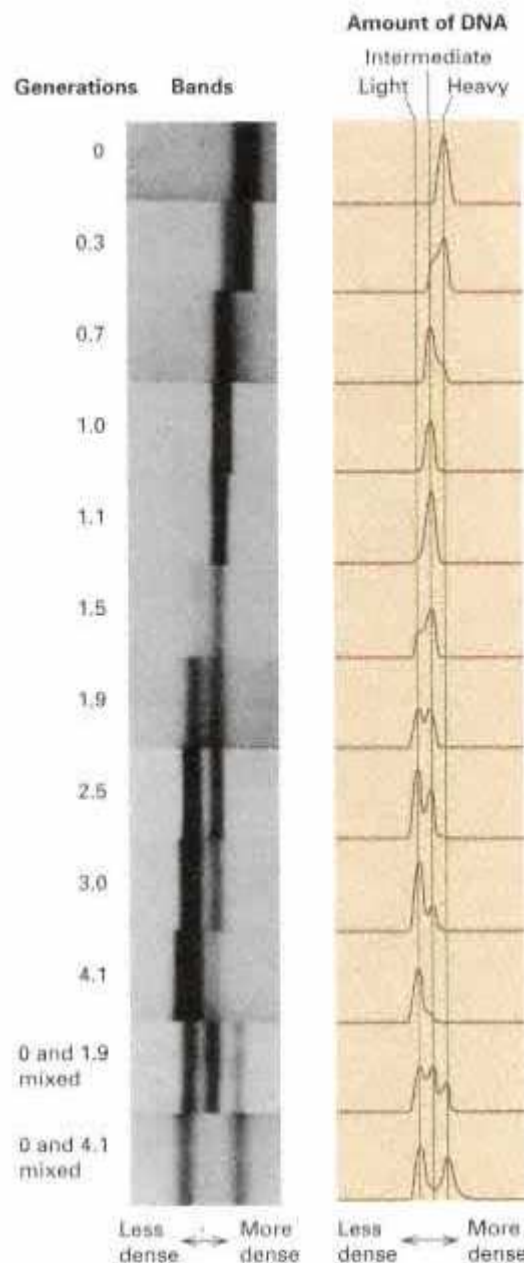


Figure 5.8

The DNA replication experiment of Meselson and Stahl. Cells whose DNA contained heavy nitrogen were transferred into growth medium containing only light nitrogen. Samples were taken at intervals, and the DNA was subjected to equilibrium density-gradient centrifugation in a solution of CsCl . During centrifugation, each type of DNA molecule moves until it comes to rest at a position in the centrifuge tube at which its density equals the density of the CsCl solution at that position. Photographs of

the centrifuge tubes taken with ultraviolet light are shown at the left. The top of the tube is at the left, the bottom at the right. The smooth curves show quantitatively the amount of absorption of the ultraviolet light across the tube.

[Photograph courtesy of Matthew Meselson. From M. Meselson and F. Stahl, Proc. Natl. Acad. Sci. USA 1958, 44:671.]

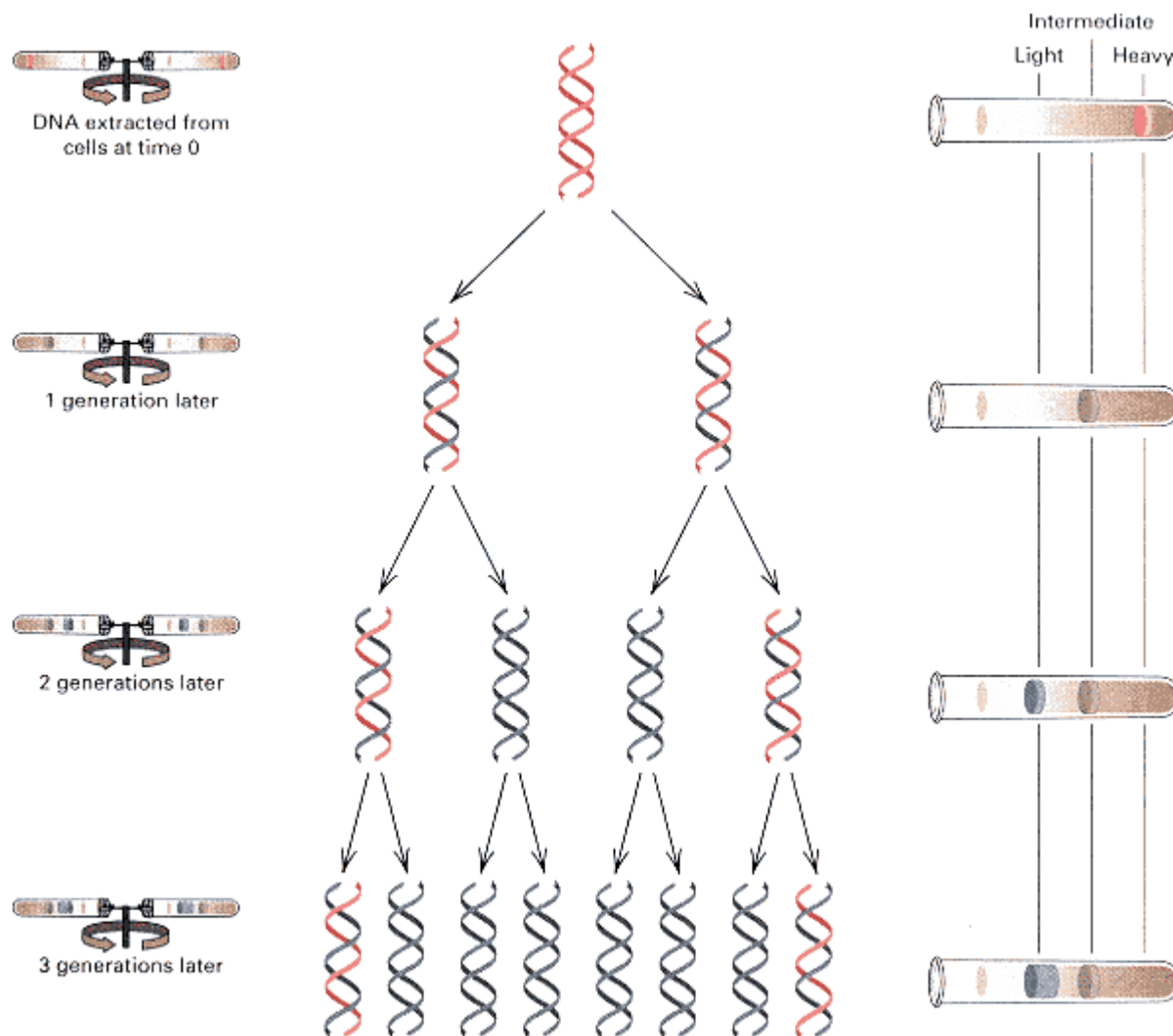


Figure 5.9

Interpretation of the data in Figure 5.8 on the basis of semiconservative replication. The diagrams show the composition of the DNA duplexes after 0, 1, 2 and 3 rounds of replication. DNA strands labeled with ^{15}N are shown in red, those labeled with ^{14}N in blue. The diagrams at the right show the positions at which bands are expected in the CsCl gradient.

Geometry of the Replication of Circular DNA Molecules In the Meselson-Stahl experiment, *E. coli* DNA was extensively fragmented when isolated, so the form of the molecule was unknown. Later, the isolation of unbroken molecules and their examination by two techniques—autoradiography and electron microscopy—showed that the DNA in *E. coli* cells is circular.

The first proof that *E. coli* DNA replicates as a circle came from an autoradiographic experiment. (Genetic mapping experiments to be described in Chapter 8 had already suggested that the bacterial chromosome is circular.) Cells were grown in a medium containing radioactive thymine (^3H -thymine) so that all DNA synthesized would be radioactive. The DNA was isolated without fragmentation and placed on photographic film. Each radioactive decay caused a tiny black spot to appear in the film, and after several months there were enough spots to visualize the DNA with a

microscope; the pattern of black spots on the film located the molecule. One of the now-famous autoradiograms from this experiment is shown in Figure 5.10. The actual length of the *E. coli* chromosome is 1.6 mm (4.7 million base pairs). On a smaller scale, Figure 5.11 shows a replicating circular molecule of a plasmid found inside certain bacterial cells. The total length of the molecule is only 0.01 mm (3000 base pairs). The replicating circle is schematically like the Greek letter θ (theta), and so this mode of replication is usually called **θ replication**.

The circularity of the replicating molecules in Figures 5.10 and 5.11 brings out an important geometric feature of semi-conservative replication. There are about 400,000 turns in an *E. coli* double helix, and because the two chains of a replicating molecule must make a full rotation to unwind each of these gyres, some kind of swivel must exist to avoid tangling the entire structure (Figure 5.12). The axis of rotation for unwinding is provided by nicks made in the backbone of *one* strand of the double helix during replication, nicks that are rapidly repaired after unwinding. Enzymes capable of making such nicks and then rapidly repairing them have been isolated from both bacterial and mammalian cells; they are called **topoisomerases**. How they work is suggested in Figure 5.13, which shows the structure of part of the topoisomerase I enzyme from *E. coli* when it is in contact with duplex DNA. Note that the molecule is wrapped completely around the DNA molecule, which passes through the enzyme like a train through a tunnel.

The position along a molecule at which DNA replication begins is called a **replication origin**, and the region in which parental strands are separating and new strands are being synthesized is called a **replication fork**. The process of generating a new replication fork is **initiation**. In some organisms, replication is initiated at many positions; in others, the origin of replication may be a unique site. In most bacteria, bacteriophage, and viruses, *DNA replication is initiated at a unique origin of replication*. Furthermore, with only a few exceptions, two replication forks move in opposite directions from the origin (Figure

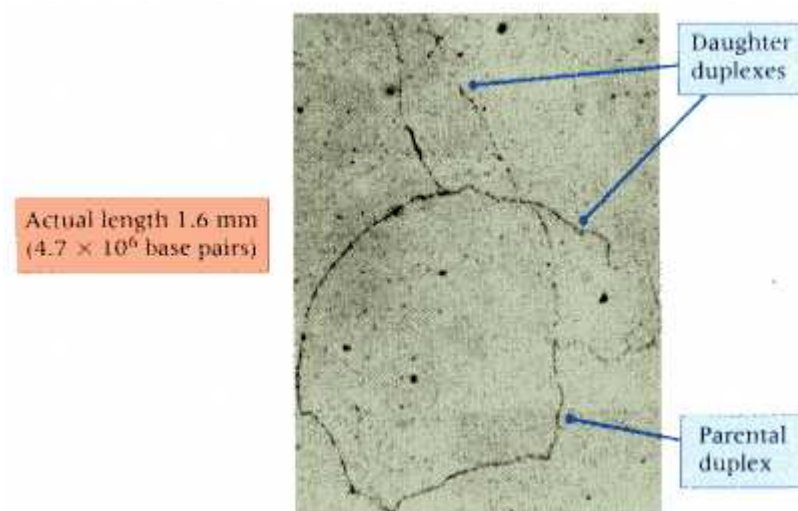


Figure 5.10
 Autoradiogram of the intact replicating chromosome of an *E. coli* cell that has grown in a medium containing ³H-thymine for slightly less than two generations. The continuous lines of dark grains were produced by electrons emitted by decaying ³H atoms in the DNA molecule.
 The pattern is seen by light microscopy
 . [From J. Cairns, *Cold Spring Harbor Symp. Quant. Biol.* 1963. 28:44].

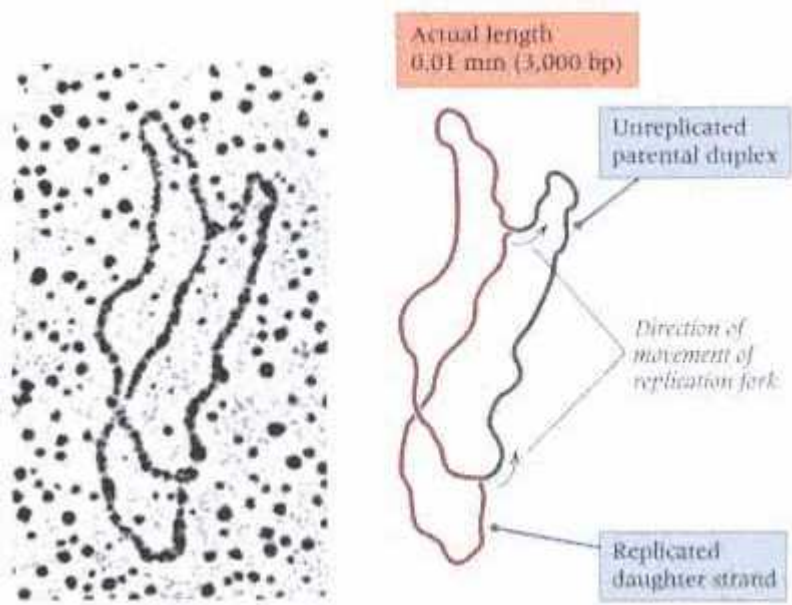


Figure 5.11

Electron micrograph of a small circular DNA molecule replicating by the θ mode.
 The parental and daughter segments are shown in the drawing.
 [Electron micrograph courtesy of Donald Helinski.]

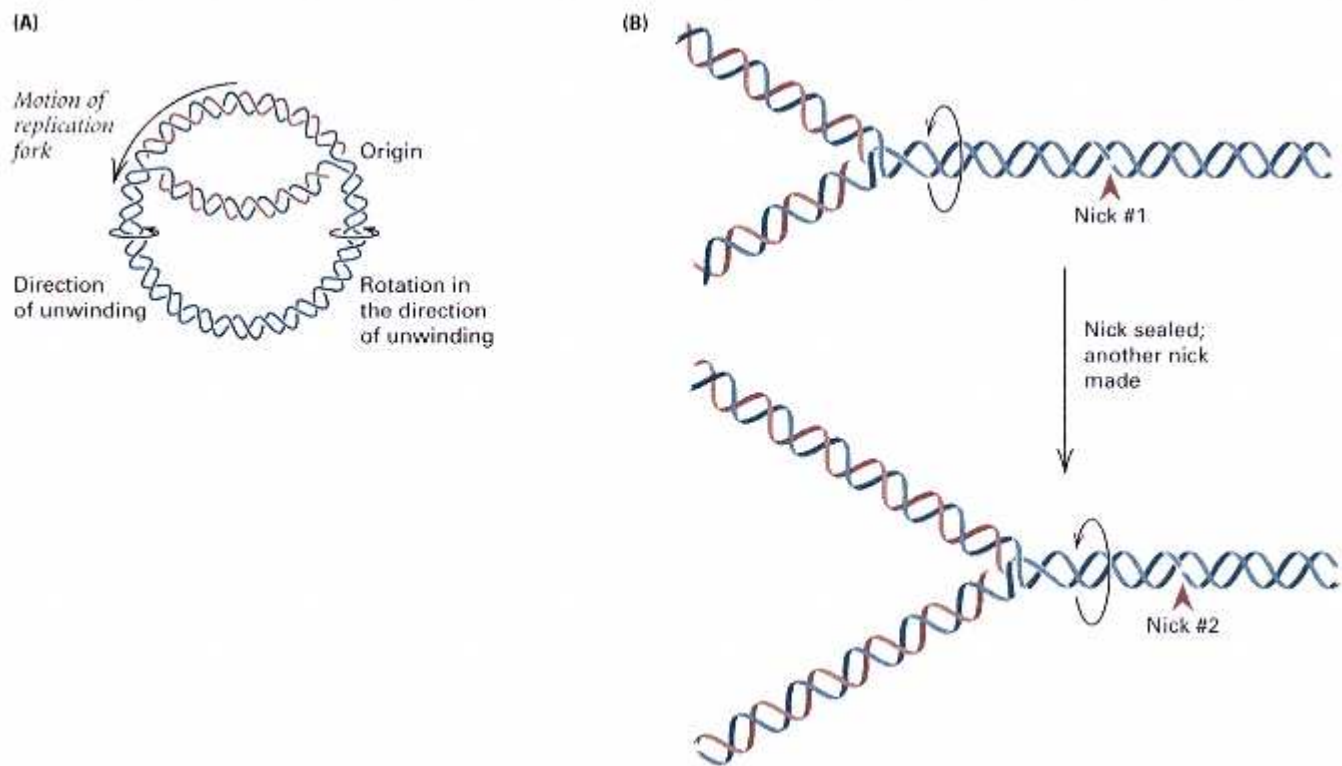


Figure 5.12

Replication of a circular DNA molecule. (A) The unwinding motion of the branches of a replicating circle, without positions at which free rotation can occur, causes overwinding of the unreplicated part. (B) Mechanism by which a single-strand break (a nick) ahead of a replication fork allows rotation.

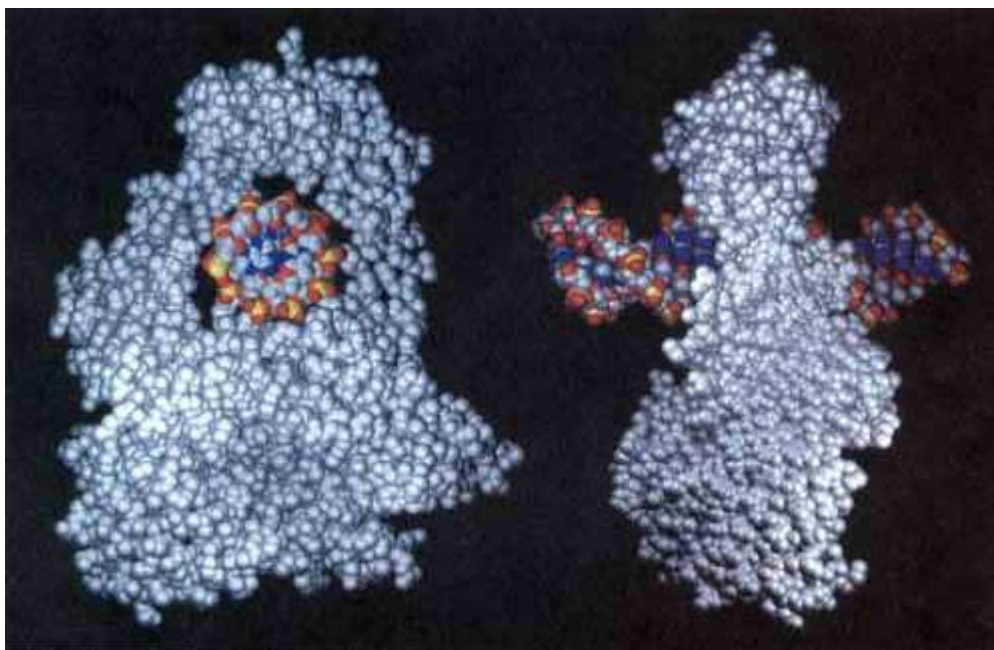


Figure 5.13

Proposed structure of an association between duplex DNA and topoisomerase I in which the DNA passes through a hole formed in the enzyme. The two views are perpendicular to each other. The structure of the enzyme includes a pair of jaw-like projections that can open and close. The intermediate shown here is created when the enzyme closes its jaws around a DNA molecule, completely enclosing it. What happens then is that one DNA strand is cleaved, swiveled, and reconnected, whereupon the jaws open and the DNA molecule is released. [Courtesy of C. D. Lima, J. C. Wang, and A. M3ndragon. From C. D. Lima, J. C. Wang, and A. M3ndragon, *Nature* 1994. 367:138.]

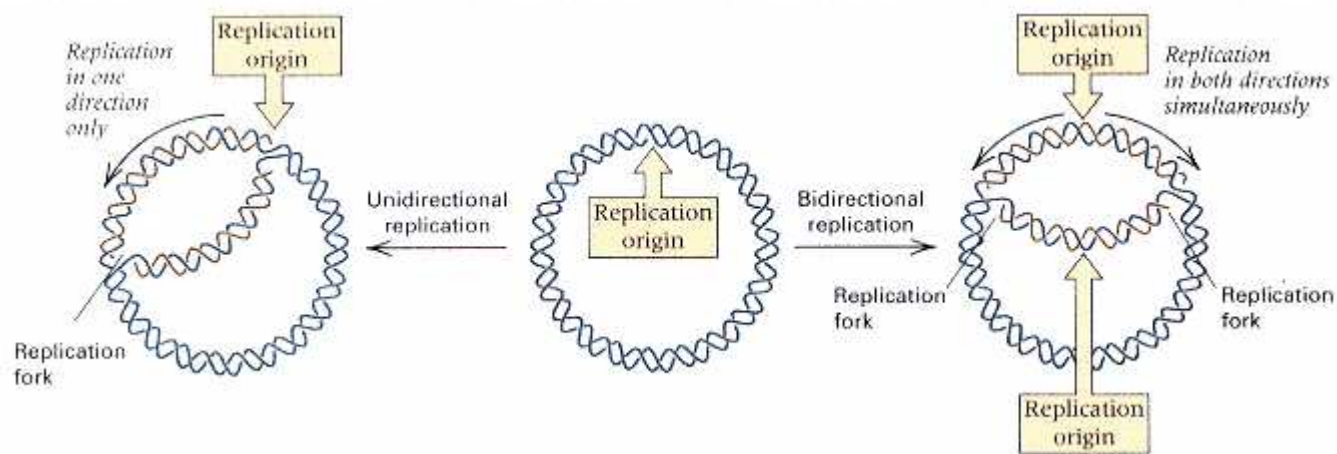


Figure 5.14

The distinction between unidirectional and bidirectional DNA replication. In unidirectional replication, there is only one replication fork; bidirectional replication requires two replication forks. The curved arrows indicate the direction of movement of the forks. Most DNA replicates bidirectionally.

5.14); that is, *DNA almost always replicates bidirectionally in both prokaryotes and eukaryotes.*

Multiple Origins of Replication in Eukaryotes The DNA duplex in a eukaryotic chromosome is linear and replicates **bidirectionally**. Furthermore, replication is initiated at many sites in the DNA. The structures that result from the numerous origins are seen in electron micrographs as multiple loops along a DNA molecule (Figure 5.15). Multiple initiation is a means of reducing the total replication time of a large molecule. In eukaryotic cells, movement of each replication fork proceeds at a rate of approximately 10 to 100 nucleotide pairs per second. For example, in *D. melanogaster*, the rate of replication is about 50 nucleotide pairs per second at 25°C. Because the DNA molecule in the largest chromosome in *Drosophila* contains about 7×10^7 nucleotide pairs, replication from a single bidirectional origin of replication would take about 8 days. Developing *Drosophila* embryos actually use about 8500 replication origins per chromosome, which reduces the replication time to a few minutes. In a typical eukaryotic cell, origins are spaced about 40,000 nucleotide pairs apart, which allows each chromosome to replicate in 15 to 30 minutes. Because chromosomes do not all replicate simultaneously, complete replication of all chromosomes in eukaryotes usually takes from 5 to 10 hours.

DNA replication is much faster in prokaryotes than in eukaryotes. In *E. coli*, for example, replication of a DNA molecule takes place at a rate of approximately 1500 nucleotide pairs per second; similar rates are found in the replication of all phage DNA molecules studied. Complete replication of the *E. coli* DNA molecule from its single bidirectional origin of replication requires approximately 30 minutes because the molecule contains about 4.7×10^6 nucleotide pairs. Yet under optimal culture conditions, an *E. coli* cell can divide every 20 minutes. How is this possible? It is possible because a new round of replication is started at the origin every 20 minutes, even though the previous round of replication has not yet been completed. In a rapidly dividing culture of *E. coli* cells, the DNA molecule passed to each daughter cell in division is already in the process of replicating in preparation for the next division.

Rolling-Circle Replication Some circular DNA molecules, including those of a number of bacterial and eukaryotic viruses, replicate by a process that does not include a θ -shaped intermediate. This replication mode is called **rolling-circle replication**.

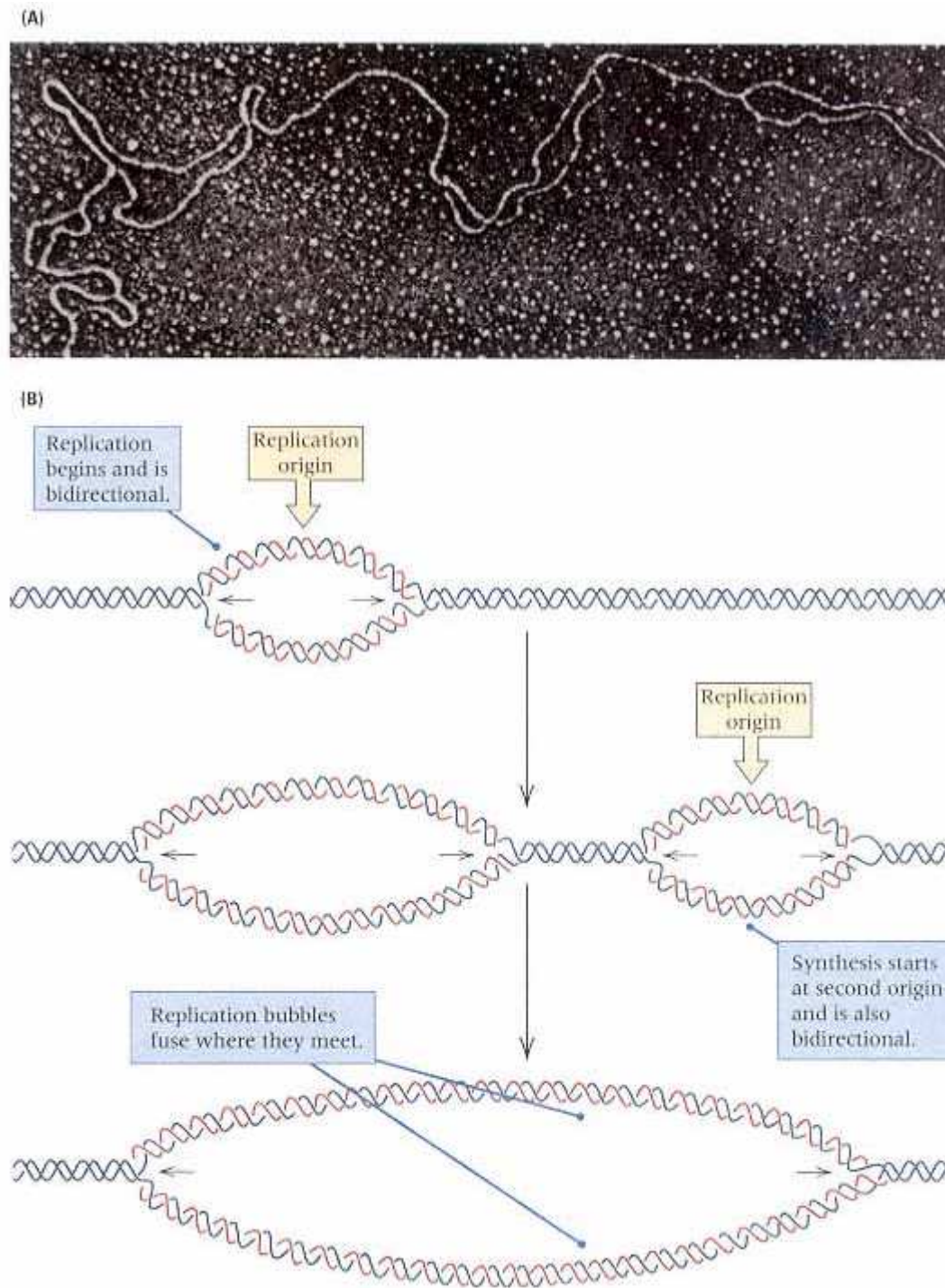


Figure 5.15
Replicating DNA of *Drosophila melanogaster*. (A) An electron micrograph of a 30,000-nucleotide-pair segment showing seven replication loops. (B) An interpretive drawing showing how loops merge. Two replication origins are shown in the drawing. The small arrows indicate the direction of movement of the replication forks.
[Electron micrograph courtesy of David Hogness.]

In this process, replication starts with a cut at a specific sugar-phosphate bond in a double-stranded circle (Figure 5.16). This cut produces two chemically distinct ends: a 3' end (at which the nucleotide has a free 3'-OH group) and a 5' end (at which the nucleotide has a free 5'-P group). The DNA is synthesized by the addition of deoxynucleotides to the 3' end with simultaneous displacement of the 5' end from the circle. As replication proceeds around the circle, the 5' end rolls out as a tail of increasing length.

In most cases, as the tail is extended, a complementary chain is synthesized, which results in a double-stranded DNA

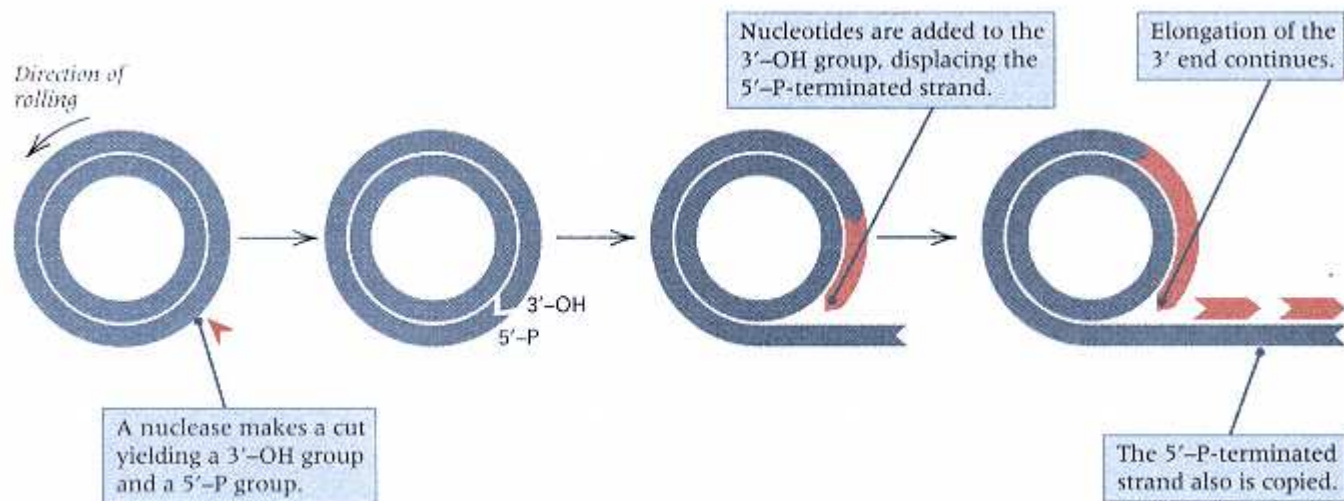


Figure 5.16

Rolling-circle replication. Newly synthesized DNA is in red. The displaced strand is replicated in short fragments, as explained in Section 5.6.

tail. Because the displaced strand is chemically linked to the newly synthesized DNA in the circle, replication does not terminate, and extension proceeds without interruption, forming a tail that may be many times longer than the circumference of the circle. Rolling-circle replication is a common feature in late stages of replication of double-stranded DNA phages that have circular intermediates. An important example of rolling-circle replication will also be seen in Chapter 8, where matings between donor and recipient *E. coli* cells are described.

So far, we have considered only certain geometrical features of DNA replication. In the next section, the enzymes and protein factors used in DNA replication are described.

5.5—

DNA Synthesis

Nucleic acids are synthesized in chemical reactions controlled by enzymes, as is the case with most metabolic reactions in living cells. An enzyme that forms the sugar-phosphate bond (the phosphodiester bond) between adjacent nucleotides in a nucleic acid chain is called a **DNA polymerase**. A variety of DNA polymerases have been purified, and DNA synthesis has been carried out *in vitro* in a cell-free system prepared by disrupting cells and combining purified components in a test tube under precisely defined conditions.

Three principal requirements must be met for DNA polymerases to catalyze synthesis of DNA.

- *The 5'-triphosphates of the four deoxynucleosides must be present.* These are the compounds denoted in Table 5.1 as dATP, dGTP, dTTP, and dCTP, which contain the bases adenine, guanine, thymine, and cytosine, respectively. Details of the structures of dCTP and dGTP are shown in Figure 5.17, in which the phosphate groups cleaved off during DNA synthesis are indicated. DNA synthesis requires all four nucleoside 5'-triphosphates and does not take place if any of them is omitted.
- *A preexisting single strand of DNA to be replicated must be present.* Such a strand is called a **template** strand.
- *A nucleic acid segment, which may be very short, must be present and must be hydrogen-bonded to the template strand.* This segment is called a **primer**. No known DNA polymerase is able to initiate chains, so the presence of a primer chain with a free 3'-OH group is absolutely essential for the initiation of replication. In living cells, the primer is

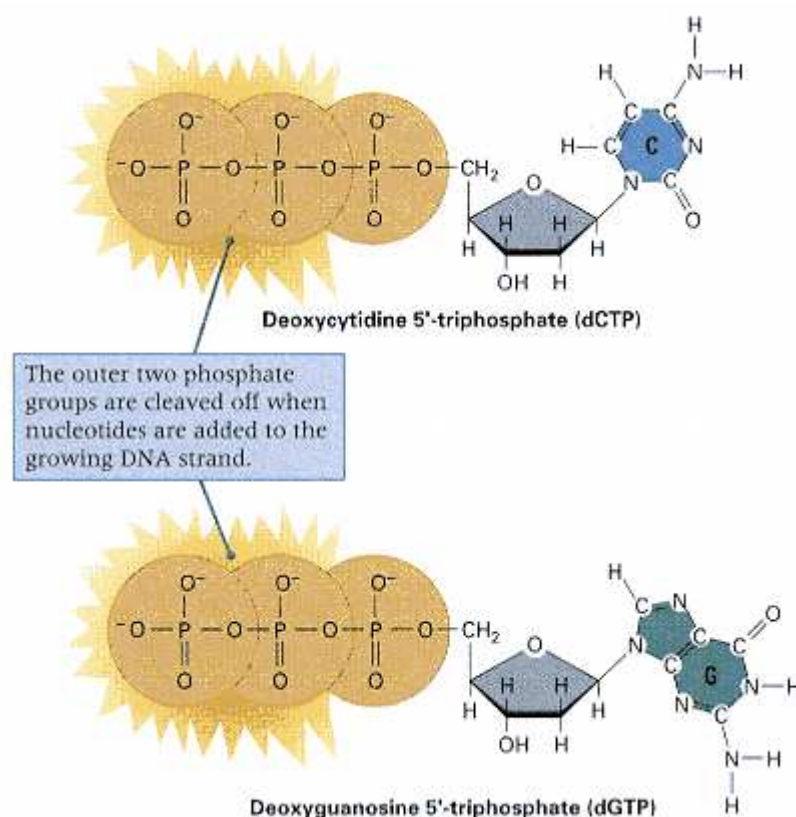


Figure 5.17

Two deoxynucleoside triphosphates used in DNA synthesis. The outer two phosphate groups are removed during synthesis.

a short segment of RNA; in cell-free replication *in vitro*, the primer may be either RNA or DNA.

The reaction catalyzed by the DNA polymerases is the formation of a phosphodiester bond between the free 3'-OH group of the chain being extended and the innermost phosphorus atom of the nucleoside triphosphate being incorporated at the 3' end (Figure 5.18). The result is as follows:

DNA synthesis proceeds by the elongation of primer chains, *always in the 5' to 3' direction*.

Recognition of the appropriate incoming nucleoside triphosphate in replication depends on base pairing with the opposite nucleotide in the template chain. DNA polymerase will usually catalyze the polymerization reaction that incorporates the new nucleotide at the primer terminus only when the correct base pair is present. The same DNA polymerase is used to add each of the four deoxynucleoside phosphates to the 3'-OH terminus of the growing strand.

Two DNA polymerases are needed for DNA replication in *E. coli*: **DNA polymerase I**, which is often written **Pol I**, and **DNA polymerase III (Pol III)**. Polymerase III is the major replication enzyme. In the cell, it exists as a large complex (molecular mass of about 1×10^6 daltons) composed of two Pol III protein subunits plus at least seven other proteins. Although each cell has only 10–20 copies of the Pol III complex, it is responsible not only for the elongation of DNA chains but also for the initiation of the replication fork at origins of replication and the addition of deoxynucleotides to the RNA primers. Polymerase I plays an essential, but secondary, role in replication that will be described in a later section. Eukaryotic cells also contain several DNA polymerases. The enzyme responsible for the replication of chromosomal DNA is called **polymerase α** . Mitochondria have their own DNA polymerase, **polymerase γ** , which replicates the mitochondrial DNA.

In addition to their ability to polymerize nucleotides, most DNA polymerases are capable of **nuclease** activities that break phosphodiester bonds in the sugar-phosphate backbones of nucleic acid chains. The many other enzymes that have nuclease activity, are of two types: (1) **exonucleases** can remove a nucleotide only from the end of a chain, and (2) **endonucleases** break bonds within the chains. DNA polymerases I and III of *E. coli* have an exonuclease activity that acts only at the 3' terminus (a 3'-to-5' exonuclease activity). This exonuclease activity provides a built-in mechanism for correcting rare errors in polymerization. Occasionally, a polymerase adds to the end of the growing chain an incorrect nucleotide, which cannot form a proper base pair with the base in the template strand.

The presence of an unpaired nucleotide activates the 3'-to-5' exonuclease activity, which cleaves the unpaired nucleotide from the 3'-OH end of the growing chain (Figure 5.19). Because it cleaves off an incorrect nucleotide and gives the polymerase another chance to get it right, the 3'-to-5' exonuclease activity of DNA polymerase is also called the proof-

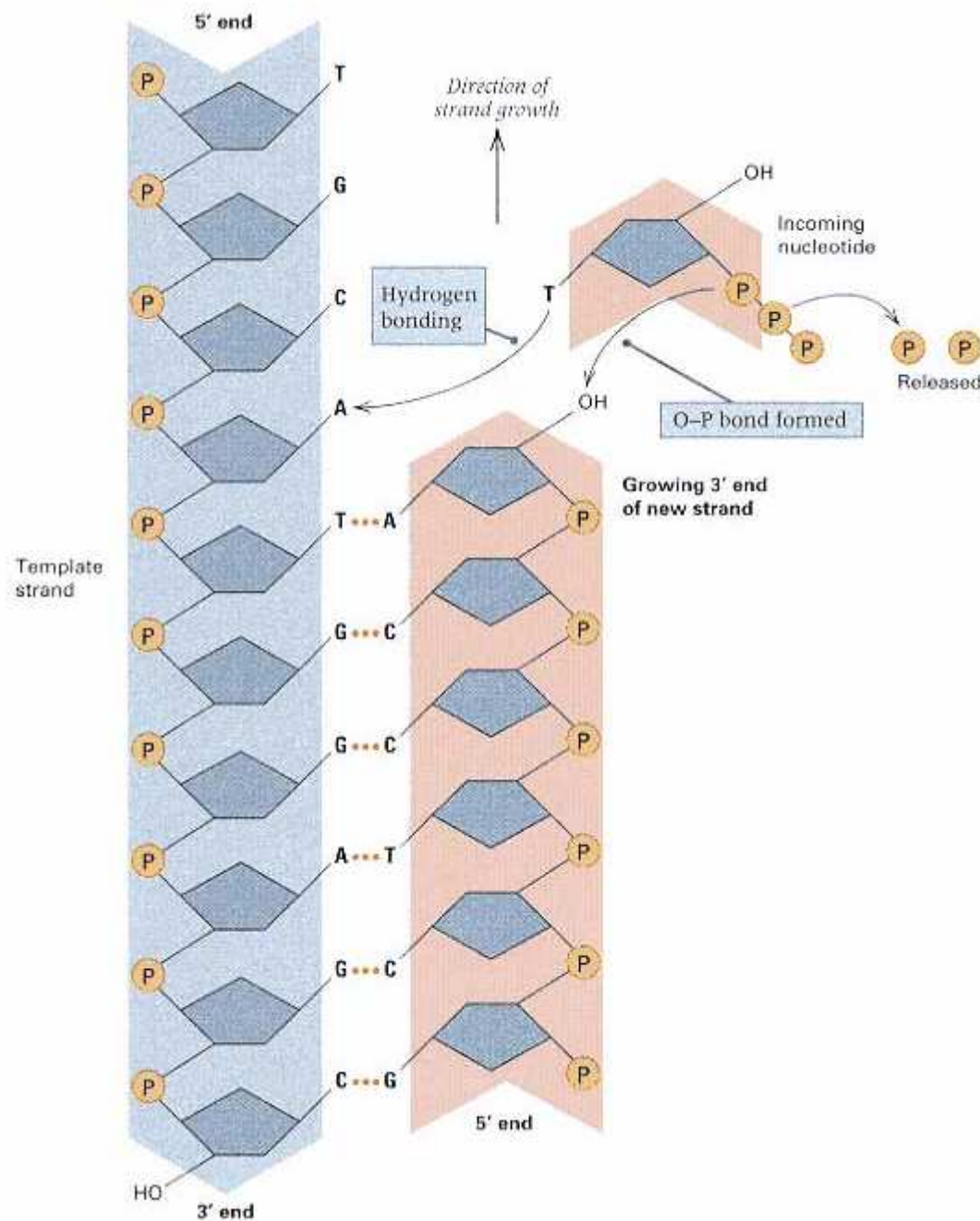


Figure 5.18

Addition of nucleotides to the 3'-OH terminus of a growing strand. The recognition step is shown as the formation of hydrogen bonds between the A and the T. The chemical reaction is between the 3'-OH group of the 3' end of the growing chain that attacks the innermost phosphate group of the incoming trinucleotide.

reading or editing function. The proof-reading function can "look back" only one base (the one added last). Nevertheless,

The genetic significance of the proofreading function is that it is an error-correcting mechanism that serves to reduce the frequency of mutation resulting from the incorporation of incorrect nucleotides in DNA replication.

Two unexpected features of DNA replication result from functional constraints that are present in all known DNA polymerases. One constraint is that a polymerase can elongate a newly synthesized DNA strand only at its 3' end (Figure 5.20). Hence the polymerase can move along the template strand only in the 3'-to-5' direction. The second constraint is that DNA polymerase is unable to initiate new chains but rather requires a preexisting primer. How the process of DNA replication deals with these constraints is described in the next section.

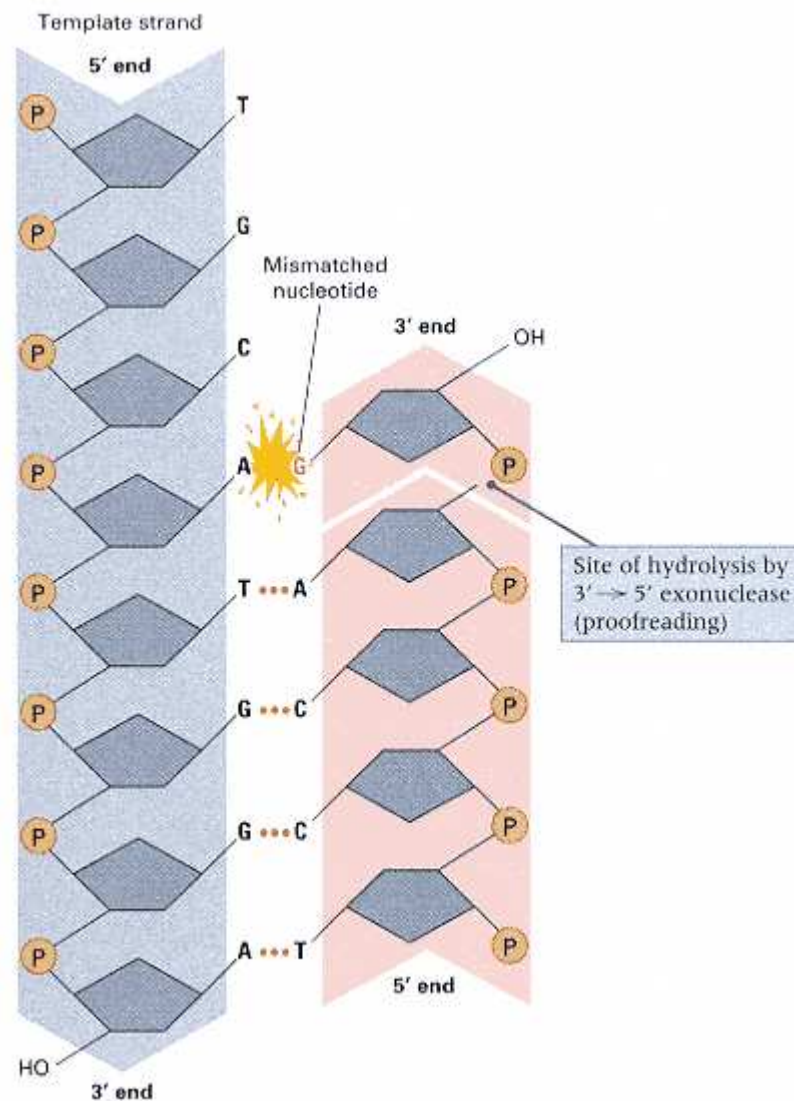


Figure 5.19

The 3'-to-5' exonuclease activity of the proofreading function. The growing strand is cleaved to release a nucleotide containing the base G, which does not pair with the base A in the template strand.

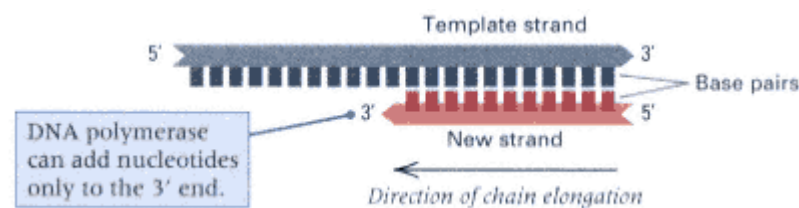


Figure 5.20

The geometry of DNA replication. The new strand (red) is elongated by the addition of successive nucleotides to the 3' end as the polymerase moves along the template strand in the 3'-to-5' direction.

5.6— Discontinuous Replication

In the model of replication suggested by Watson and Crick, which is illustrated in Figure 5.7, both daughter strands were supposed to be replicated as continuous units. However, no known DNA molecule replicates in this way. Because DNA polymerase can elongate a newly synthesized DNA strand only at its 3' end, *one of the daughter strands is made in short fragments, which are then joined together*. The reason for this mechanism and the properties of these fragments are described next.

Fragments in the Replication Fork

As we have seen, all known DNA polymerases can add nucleotides only to a 3'-OH group. Thus if both daughter strands grew in the same overall direction, then each growing strand would need a 3'-OH terminus. However, the two strands of DNA are antiparallel, so only one of the growing strands can terminate in a free 3'-OH group; the other must terminate in a free 5' end. The solution to this topological problem is that within a single replication fork, both strands grow in the 5'-to-3' orientation, which requires that they grow in opposite directions along the parental strands. One strand of the newly made DNA is synthesized continuously (in the lower fork in Figure 5.21). The other strand (in the upper fork in Figure 5.21) is made in small **precursor fragments**. The precursor fragments are also known as **Okazaki fragments**, after their discoverer. The size of the precursor fragments is from 1000 to 2000 base pairs in prokaryotic cells and from 100 to 200 base pairs in eukaryotic cells. Because synthesis of the discontinuous strand is initiated only at intervals, at least one single-stranded region of the parental strand is always present on one side of the replication fork (the upper side in Figure 5.21). Single-stranded regions have been seen in high-resolution electron micrographs of replicating DNA molecules, as indicated by the arrows in Figure 5.22. Another implication of discontinuous replication of one strand is that the

3'-OH terminus of the continuously replicating strand is always ahead of the 5'-P terminus of the discontinuous strand; this is the physical basis of the terms **leading strand** and **lagging strand** that are used for the continuously and discontinuously replicating strands, respectively.

Next, we examine how synthesis of a precursor fragment is initiated.

Initiation by an RNA Primer

As emphasized earlier, DNA polymerases cannot initiate the synthesis of a new strand, so a free 3'-OH is needed. In most organisms, initiation is accomplished by a special type of RNA polymerase. RNA is usually a single-stranded nucleic acid consisting of four types of nucleotides joined together by 3'-to-5' phosphodiester bonds (the same chemical bonds as those in DNA). Two chemical differences distinguish RNA from DNA (Figure 5.23). The first difference is in the sugar component. RNA contains **ribose**, which is identical to the deoxyribose of DNA except for the

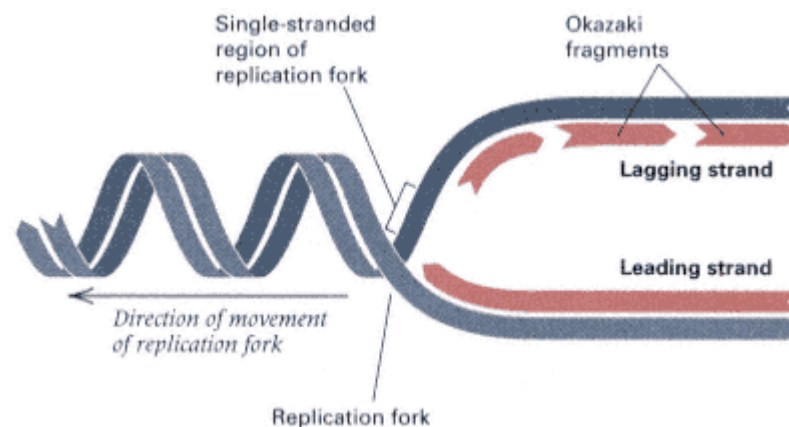


Figure 5.21

Short fragments in the replication fork. For each tract of base pairs, the lagging strand is synthesized later than the leading strand.

presence of an -OH group on the 2' carbon atom. The second difference is in one of the four bases: The thymine found in DNA is replaced by the closely related pyrimidine *uracil* (*U*) in RNA. In RNA synthesis, a DNA strand is used as a template to form a complementary strand in which the bases

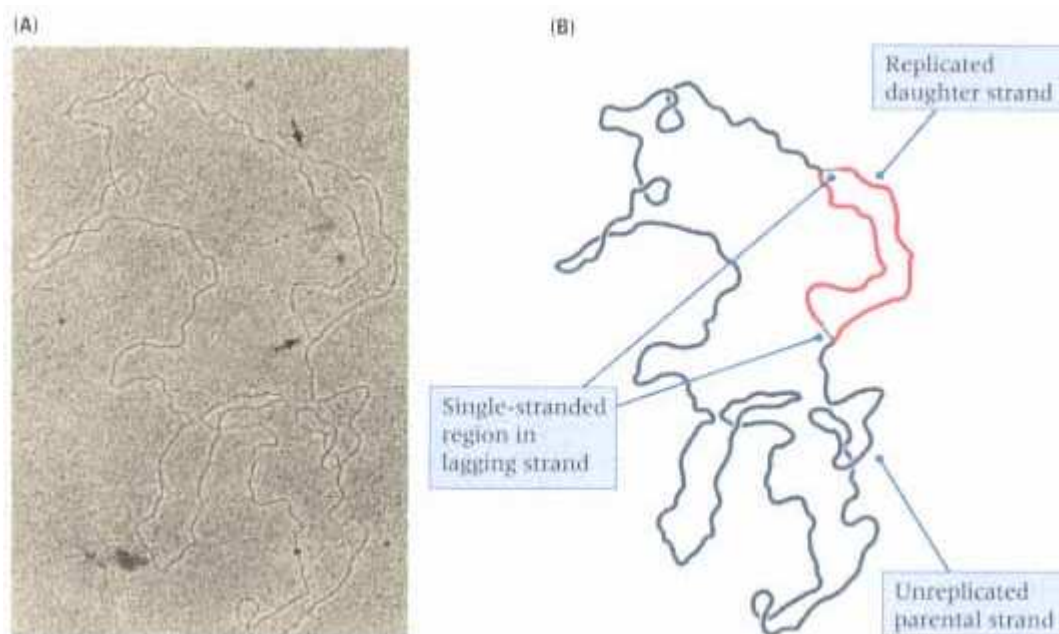


Figure 5.22

Electron micrograph (A) and an interpretive drawing (B) of a replicating θ molecule of phage λ DNA. In the electron micrograph, each arrow points to a short, single-stranded region of unreplicated DNA near the replication fork; the single-stranded regions are somewhat difficult to see, but they are apparent on close inspection.

(Electron micrograph courtesy of Manuel Valenzuela.)

in the DNA are paired with those in the RNA. Synthesis is catalyzed by an enzyme called an **RNA polymerase**. RNA polymerases differ from DNA polymerases in that they can initiate the synthesis of RNA chains without a primer.

DNA synthesis is initiated by using a short stretch of RNA that is base-paired with its DNA template. The size of the primer differs according to the initiation event. In *E. coli*, the length is typically from 2 to 5 nucleotides; in eukaryotic cells, it is usually from 5 to 8 nucleotides. This short stretch of RNA provides a primer onto which a DNA polymerase can add deoxynucleotides (Figure 5.24). The RNA polymerase that produces the primer for DNA synthesis is called **primase**. The primase is usually found in a multienzyme complex composed of 15 to 20 polypeptide chains and called a **primosome**. While it is being synthesized, each precursor fragment in the lagging strand has the structure shown in Figure 5.25.

The Joining of Precursor Fragments

The precursor fragments are ultimately joined to yield a continuous strand of DNA. This strand contains no RNA sequences, so the final stitching together of the lagging strand requires

- Removal of the RNA primer
- Replacement with a DNA sequence
- Joining where adjacent DNA fragments come into contact

In *E. coli*, the first two processes are accomplished by DNA polymerase I, and joining is catalyzed by the enzyme **DNA ligase**, which can link adjacent 3'-OH and 5'-P groups at a nick. How this is done is shown in Figure 5.26. Pol III extends the growing strand until the RNA of the primer of the previously synthesized precursor fragment is reached. Where the DNA and RNA segments meet, there is a single-strand interruption, or **nick**. The *E. coli* DNA ligase cannot seal the nick because a triphosphate is present (it can link only a 3'-OH and a 5'-monophosphate). Here, however, DNA polymerase I takes over. This enzyme has an exonuclease activity that can remove nucleotides from the 5' end of a base-paired fragment. It is effective with both DNA and RNA. The activity is a 5' → 3' **exonuclease** activity. Pol I acts at the nick and displaces it in the 5' → 3' direction by removing RNA nucleotides one by one and adding DNA nucleotides to the 3' end of the DNA strand. When all of the RNA nucleotides have been removed, DNA ligase joins the 3'-OH group to the terminal 5'-P of the precursor fragment. By this sequence of events, the precursor fragment is assimilated into the lagging strand. When the next precursor fragment reaches the RNA primer of the fragment just joined, the sequence begins again. Polymerase I is essential for DNA replication, because the 5'-to-3' exonuclease activity required for removing the RNA primer and joining the precursor fragments is not present in Pol III.

Other Proteins Needed for DNA Replication

The roles of some of the most important components of DNA replication are depicted in Figure 5.27. The replication process includes, in addition to the

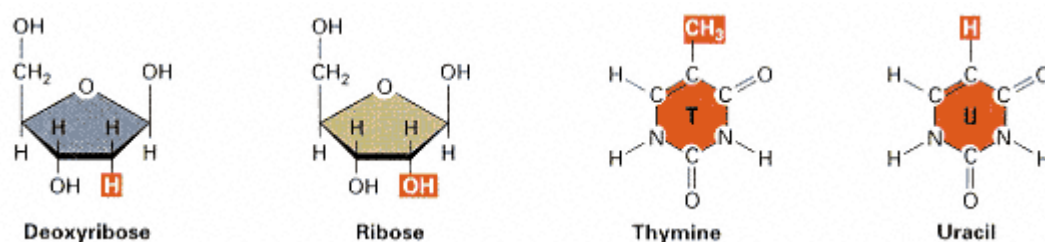


Figure 5.23

Differences between DNA and RNA. The chemical groups that are highlighted are the distinguishing features of deoxyribose and ribose and of thymine and uracil.

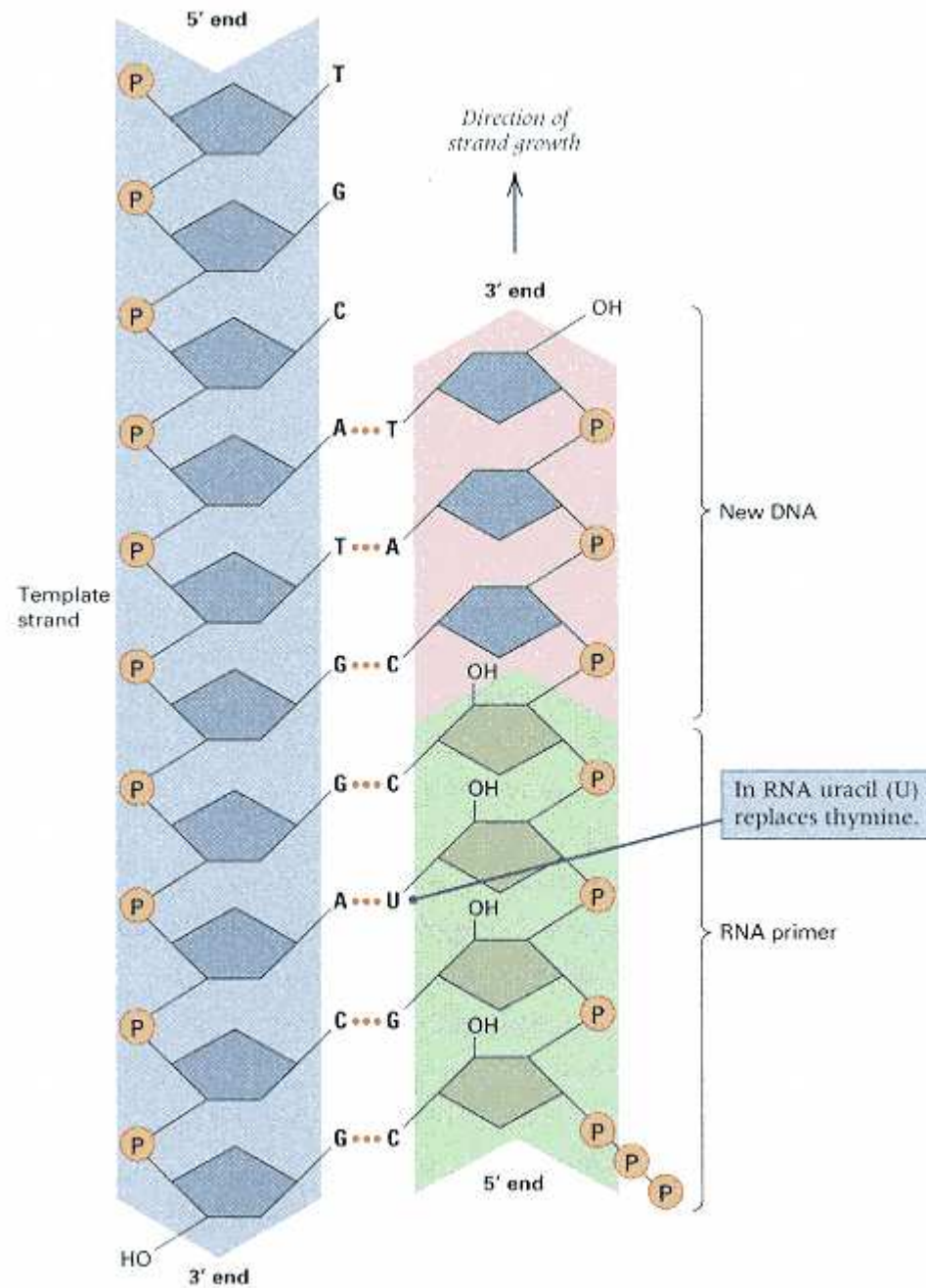


Figure 5.24

Priming of DNA synthesis with an RNA segment. The DNA template strand is shown in blue; newly synthesized DNA is shown in beige. The RNA segment, which was made by an RNA polymerase (primase), is shown in green. In the first step, DNA polymerase adds a deoxyribonucleotide to the 3' end of the primer. Successive deoxyribonucleotides are added to the 3' end of the growing chain.

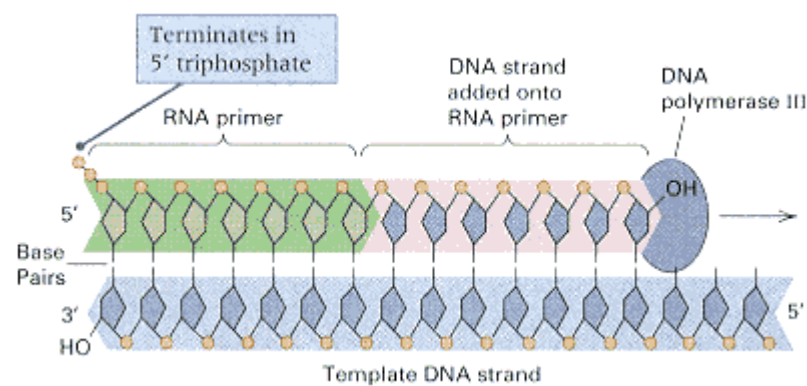


Figure 5.25

Prior to their being joined, each precursor fragment in the lagging strand

has the structure shown here. The short RNA primer is shown in green.

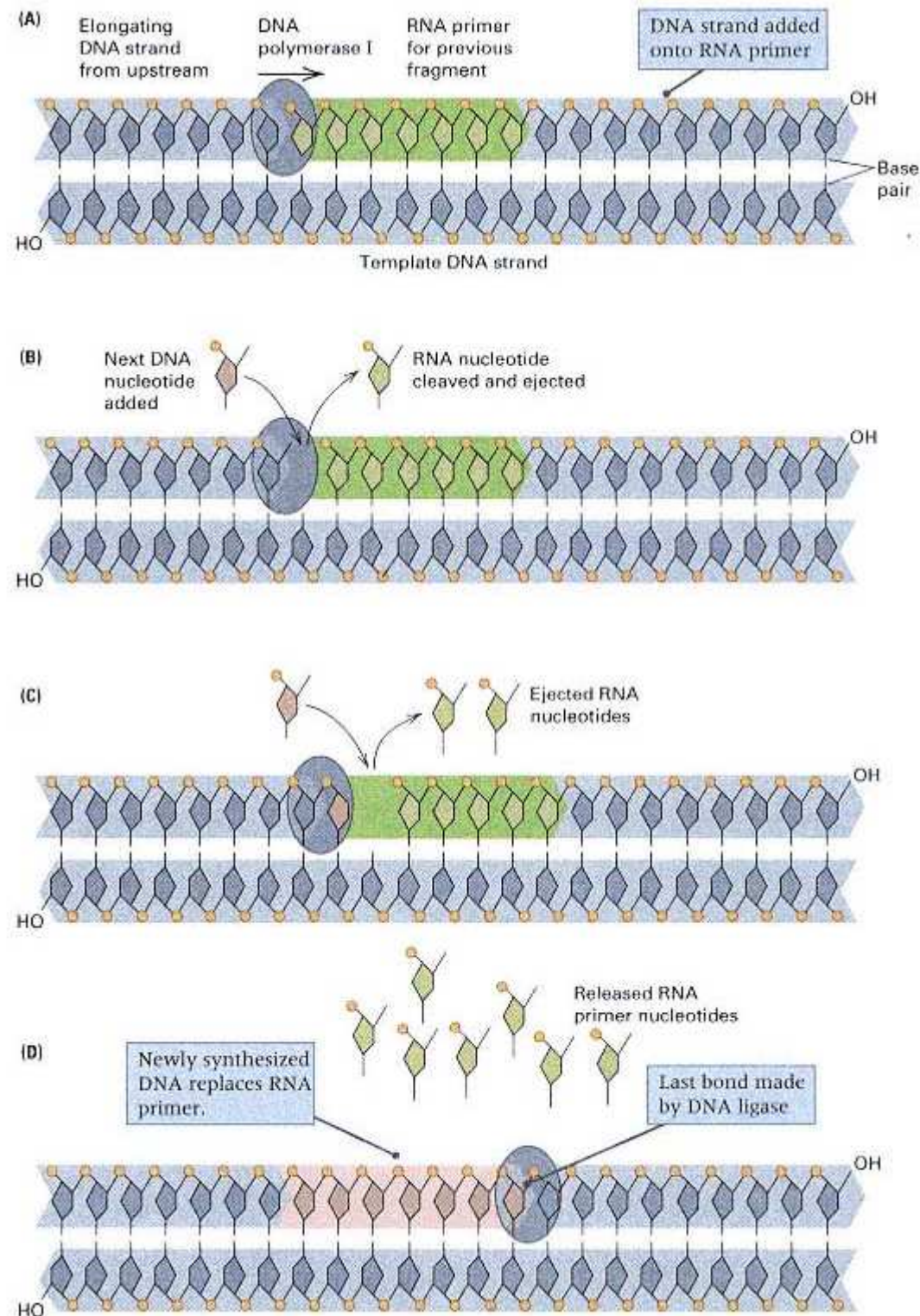


Figure 5.26

Sequence of events in the joining of adjacent precursor fragments. (A) The DNA polymerase of the upstream fragment meets the RNA primer from the next precursor fragment down-stream. (B) The RNA nucleotide immediately adjacent is excised and replaced with a DNA nucleotide. (C) Each RNA nucleotide is cleaved and replaced in turn. (D) When all the RNA nucleotides have been replaced, the adjacent ends of the DNA strand are joined.

polymerase III complex and the RNA primase complex, at least one type of topoisomerase, **helicase** proteins that bind at the replication fork and unwind the double helix, **single-strand binding proteins** that bind with and stabilize the single-stranded DNA at the replication fork, the DNA ligase that joins DNA fragments, and the polymerase I complex (not shown in Figure 5.27) that eliminates the RNA primers from precursor fragments before joining can take place.

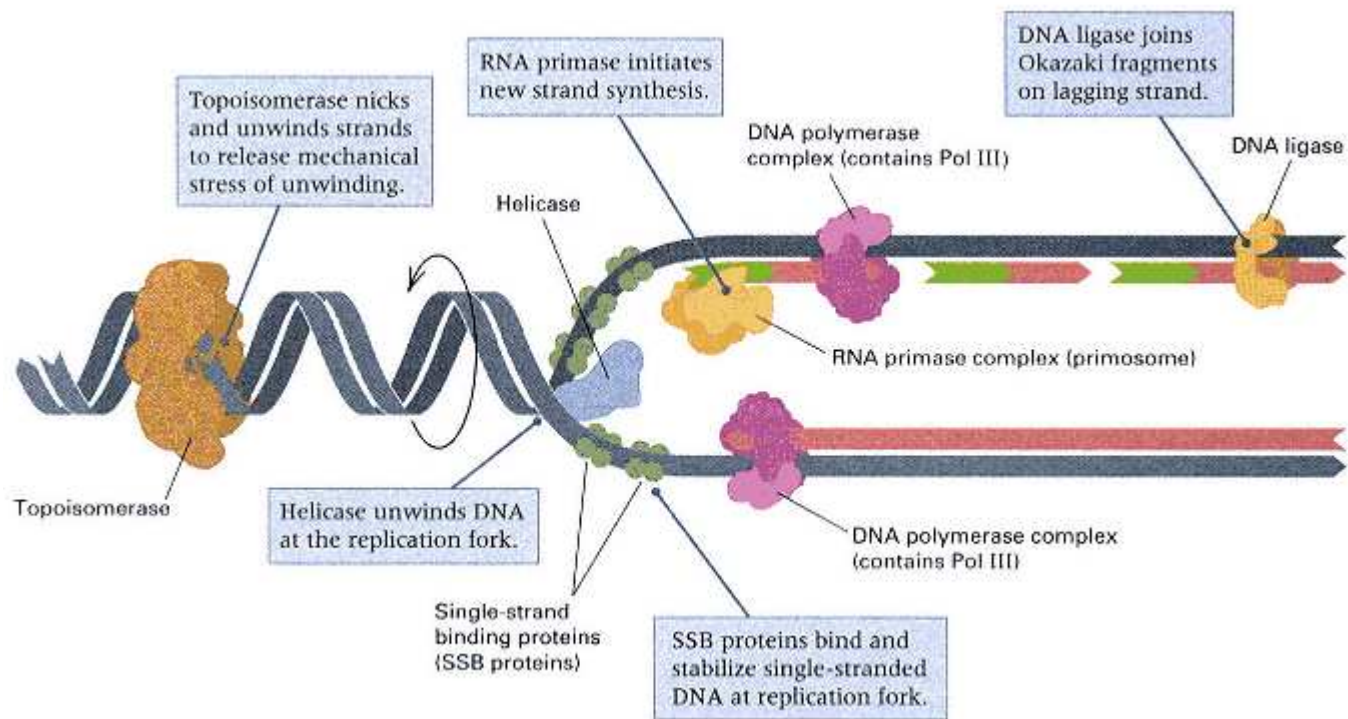


Figure 5.27

Role of some of the key proteins in DNA replication. The DNA polymerase III complex and the primase complex are both composed of multiple different polypeptide subunits. DNA polymerase I, which joins precursor fragments where they meet, is not shown.

5.7—

The Isolation and Characterization of Particular DNA Fragments

This section and the following sections show how our knowledge of DNA structure and replication has been put to practical use in the development of procedures for the isolation and manipulation of DNA. One set of procedures is based on the ability of duplex DNA to be separated into single strands that, under the proper conditions, can form duplexes with other single strands having complementary sequences. This phenomenon is described next.

Denaturation and Renaturation

The double-stranded helical structure of DNA is maintained by forces that include hydrogen bonds between the bases of complementary pairs. When solutions of DNA are exposed to temperatures considerably higher than those normally encountered in most living cells or to excessively high pH, the hydrogen bonds break and the paired strands separate. Unwinding of the helix happens in less than a few minutes, the time depending on the length of the molecule. When the helical structure of DNA is disrupted and the strands are separated, the molecule is said to be **denatured**. A common way to detect denaturation is by measuring the capacity of DNA in solution to absorb ultraviolet light of wavelength 260 nm. The absorption at 260 nm (A_{260}) of a solution of single-stranded molecules is 37 percent higher than the absorption of the double-stranded molecules at the same concentration. When a DNA solution is slowly heated and the value of A_{260} is recorded at various temperatures, a curve called a **melting curve** is obtained. An example is shown in Figure 5.28. The melting transition is usually described in terms of the temperature at which the increase in the value of A_{260} is half complete. This temperature is called the **melting temperature** and is denoted by T_m .

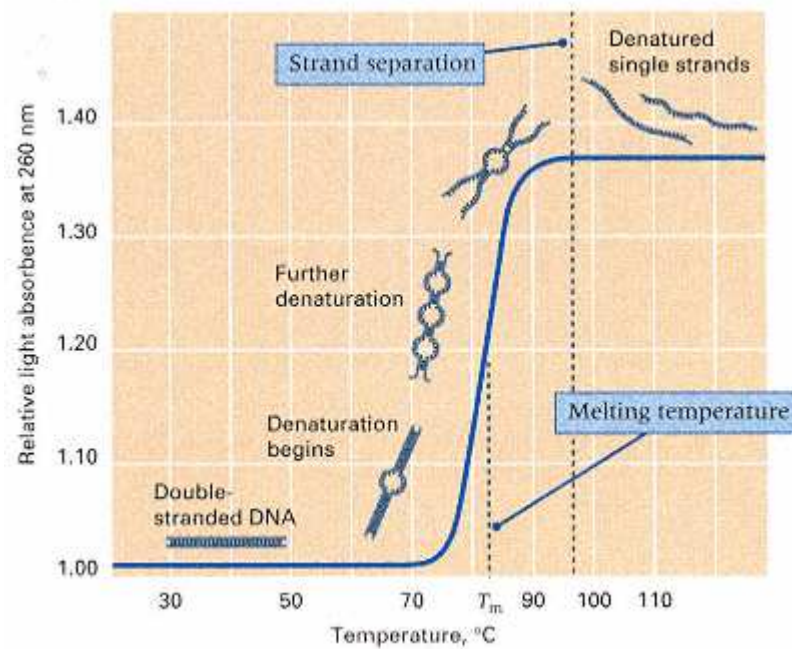


Figure 5.28

A melting curve of DNA showing the T_m and possible shapes of a DNA molecule at various degrees of denaturation.

The value of T_m increases with G + C content, because G - C pairs, joined by three hydrogen bonds, are stronger than A - T pairs, which are joined by two hydrogen bonds.

The single strands in a solution of denatured DNA can, under certain conditions, re-form double-stranded DNA. This process is called **renaturation** or **annealing**. For renaturation to happen; two requirements must be met: (1) The salt concentration must be high ($>0.25\text{ M}$) to neutralize the negative charges of the phosphate groups, which would otherwise cause the complementary strands to repel one another; and (2) the temperature must be high enough to disrupt hydrogen bonds that form at random between short sequences of bases within the same strand, but not so high that stable base pairs between the complementary strands would be disrupted. A temperature about 20°C below T_m is usually optimal. The initial phase of renaturation is a slow process, because its rate is limited by the random chance that a region of two complementary strands will come together to form a short sequence of correct base pairs. This initial pairing step is followed by a rapid pairing of the remaining complementary bases and rewinding of the helix. Rewinding is accomplished in a matter of seconds, and its rate is independent of DNA concentration (because the complementary strands have already found each other). Correct initial base pairing of all molecules in a sample is concentration-dependent and may require several minutes to many hours when standard conditions are used.

An example utilizing a hypothetical DNA molecule will enable us to understand some of the molecular details of renaturation. Consider the double-stranded DNA molecule in Figure 5.29A containing 30,000 base pairs, in which a specific sequence of six base pairs appears only twice. (In general, a six-base sequence would be expected to be found an average of about seven times in a molecule of this length.) This molecule is heat-denatured (Figure 5.29B) and then the temperature is lowered to promote renaturation. In the solution of these denatured molecules, a random collision between noncomplementary base sequences cannot initiate renaturation, but a collision that brings together sequences I and II' or I' and II can result in base pairing (Figure 5.29C). This pairing will be transient at the elevated temperatures used for renaturation, because the paired region is short and the adjacent bases in the two strands are out of register and unable to pair, as is required to form a double-stranded molecule. On the other hand, if a collision results in the pairing of sequence I with I' or II with II'—or any other short complementary sequences—then pairing of the adjacent bases (and all other bases in the strands) will proceed in a zipperlike action (Figure 5.29D). The main point is that only base pairing that brings the complementary sequences into register will cause renaturation.

Nucleic Acid Hybridization

Another important point is that because a solution of denatured DNA usually contains a large number of identical DNA molecules, a double-stranded molecule formed by renaturation is rarely composed of the same two strands

that were paired before

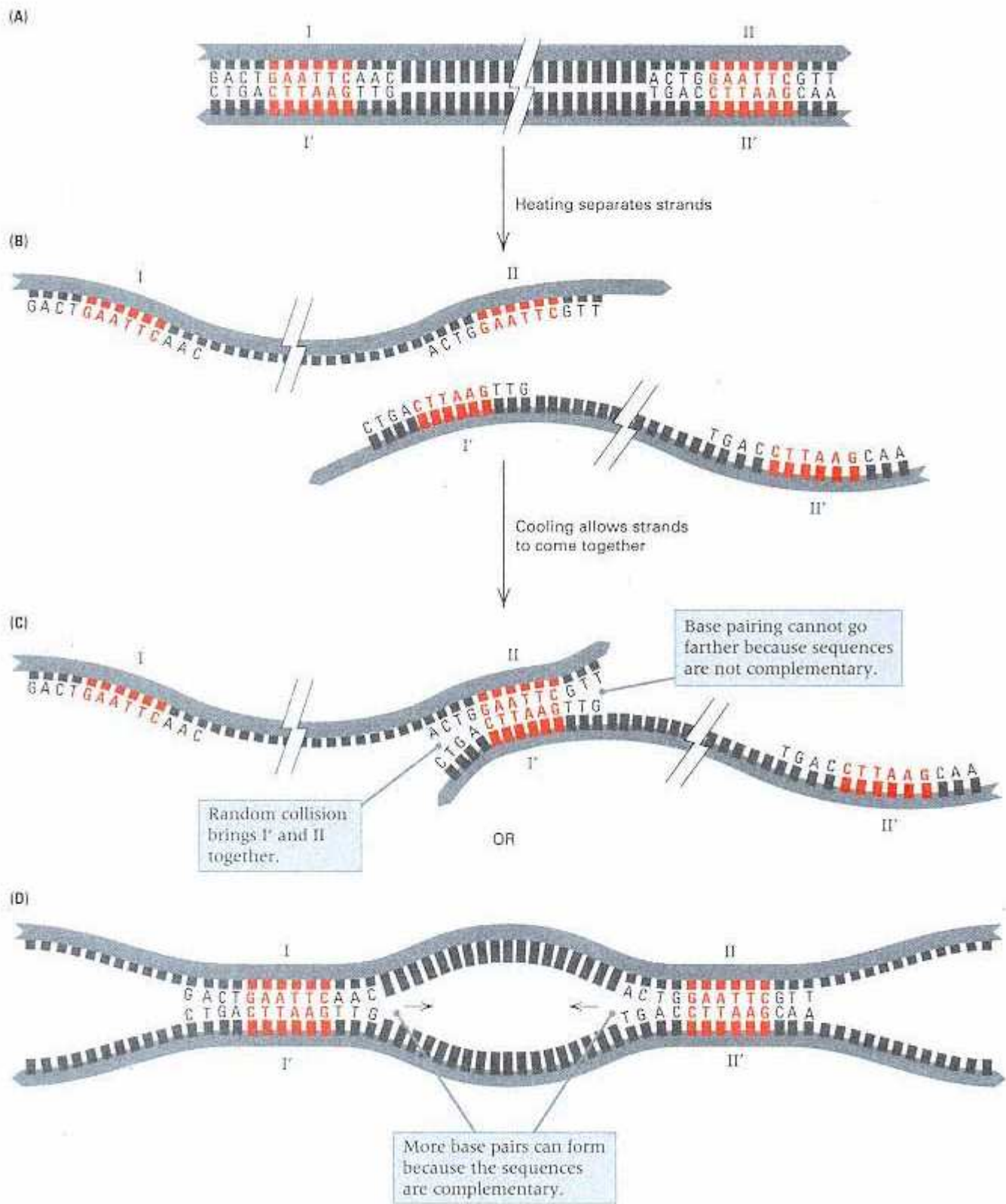


Figure 5.29

Denaturation and renaturation of a DNA duplex. (A) Original duplex. (B) High temperature causes the strands to come apart. Lowering the temperature allows short complementary stretches to come together. (C) If the sequences flanking the paired region are not complementary, then the pairing is unstable and the strands come apart again. (D) If the sequences flanking the paired region are complementary, then further base pairing stabilizes the renatured duplex.

renaturation. This phenomenon is the basis of one of the most useful methods in molecular genetics—**nucleic acid hybridization**. In this context, molecular hybrids are molecules formed when two single nucleic acid strands obtained by denaturing DNA from *different* sources have sufficient sequence complementarity to form a duplex. An example of the use of DNA-DNA hybridization is determination of the fraction of DNA in two different species that have common base sequences. To illustrate one procedure that is used, let us assume that two bacterial species have been grown in media that cause the DNA of one species to contain only the nonradioactive isotope of phosphorus, ^{31}P , and cause the DNA of the second species to contain the radioactive isotope, ^{32}P . The DNA molecules of the two species are isolated, broken into many small fragments, and denatured. The nonradioactive molecules are immobilized on a nitrocellulose filter, and the radioactive DNA is added. The temperature and salt concentration are raised to promote renaturation, and then, after a suitable period of time, the filter is washed. The existence of common base sequences will be detected by the presence of renatured fragments of radioactive ^{32}P -DNA on the filter. The extent of common base sequences determined in this manner in the DNA of two species generally agrees with the evolutionary relatedness of the species as indicated by direct DNA sequencing of a sample of their genes.

Restriction Enzymes and Site-Specific DNA Cleavage

One of the problems with breaking large DNA molecules into smaller fragments by random shearing is that the fragments containing a particular gene, or part of a gene, will all be of different sizes. With random shearing, owing to the random length of each fragment, it is not possible to isolate and identify a *particular* DNA fragment. However, there is an important enzymatic technique, described in this section, that can be used for cleaving DNA molecules at specific sites.

As we saw in Chapter 4, members of a class of enzymes known as **restriction endonucleases** or **restriction enzymes** are able to cleave DNA molecules at the positions at which particular, short sequences of bases are present. For example, the enzyme *Bam*HI recognizes the double-stranded sequence



and cleaves each strand between the G-bearing nucleotides shown in red. Figure 5.30 shows how the regions that make up the active site of *Bam*HI contact the recognition site (blue) just prior to cleavage, and the cleavage reaction is indicated in Figure 5.31.

Table 5.3 lists six of the several hundred restriction enzymes that are known. Most restriction enzymes are isolated from bacteria, and they are named after the species in which they were found. *Bam*HI, for example, was isolated from *Bacillus amyloliquefaciens* strain H, and it is the first (I) restriction enzyme isolated from this organism. Most restriction enzymes recognize only one short base sequence, usually

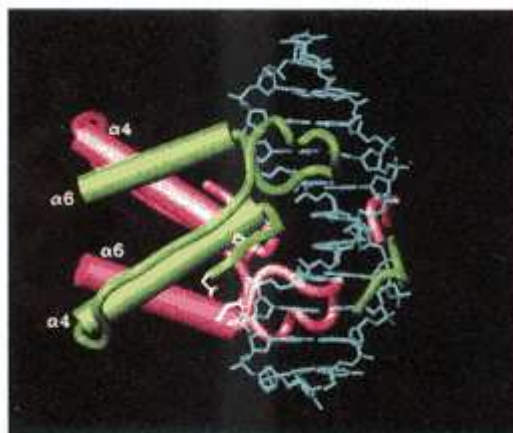


Figure 5.30

Part of the restriction enzyme *Bam*HI in contact with its recognition site in the DNA (blue). The pink and green cylinders represent regions of the enzyme in which the amino acid chain is twisted in the form of a right-handed helix.

[Courtesy of A.A. Aggarwal. From M. Newman, T. Strzelecka, L. F. Dorner, I.

Schildkraut, and A.A. Aggarwal, 1995. *Science*
269:656.]

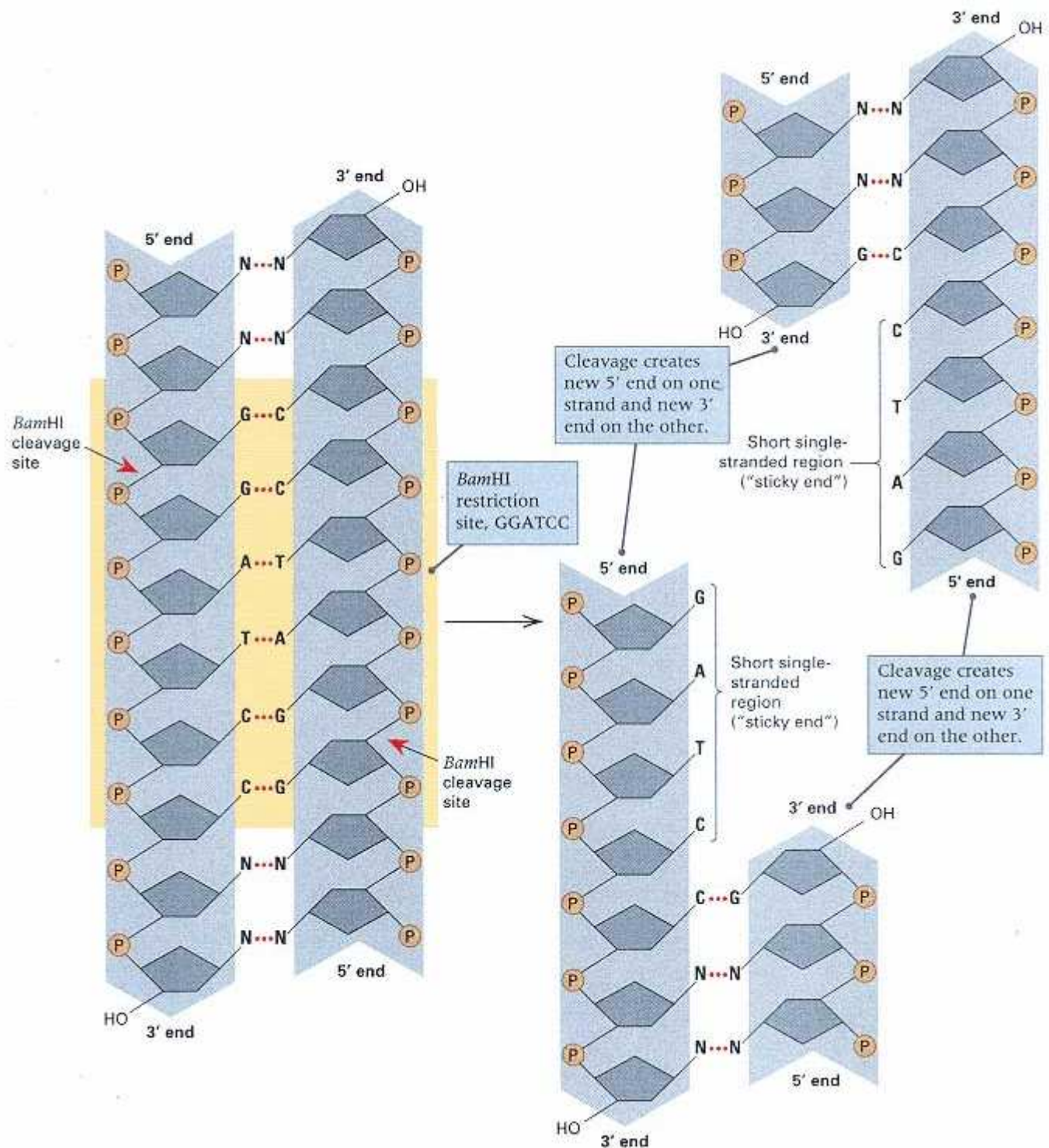


Figure 5.31

Mechanism of DNA cleavage by the restriction enzyme *Bam*HI. The enzyme makes a single cut in the backbone of each DNA strand wherever the duplex contains a *Bam*HI restriction site. Each cut creates a new 3' end and a new 5' end, separating the upper and lower parts of the duplex. In the case of *Bam*HI the cuts are staggered cuts, so the resulting ends terminate in single-stranded regions, each four base pairs in length.

four or six nucleotide pairs. The enzyme binds with the DNA at these sites and makes a break in each strand of the DNA molecule, producing 3'-OH and 5'-P groups at each position. The nucleotide sequence recognized for cleavage by a restriction enzyme is called the **restriction site** of the enzyme. The restriction enzymes in Table 5.3 all cleave their restriction site asymmetrically (at different sites in the two DNA strands), but some restriction enzymes cleave symmetrically (at the same site in both strands). The former leave **sticky ends** because each end of the cleaved site has a small, single-stranded overhang that is complementary in base sequence to the other end (Figure 5.31). In contrast, enzymes that have symmetrical cleavage sites yield DNA fragments that have **blunt ends**. In virtually all cases, the restriction site of a restriction enzyme reads the same on both strands, provided that the opposite polarity of the strands is taken into account; for example, each strand in the restriction site of *Bam*HI reads 5'-GGATCC-3' (Figure 5.31). A DNA sequence with this type of symmetry is called a **palindrome**. (In ordinary English, a palindrome is a word or phrase that reads the same forwards and backwards, such as "madam.")

Restriction enzymes have the following important characteristics:

- Most restriction enzymes recognize a single restriction site.
- The restriction site is recognized without regard to the source of the DNA.
- Because most restriction enzymes recognize a unique restriction site sequence, the number of cuts in the DNA from a particular organism is determined by the number of restriction sites present.

The DNA fragment produced by a pair of adjacent cuts in a DNA molecule is called a

Table 5.3 Some restriction endonucleases, their sources, and their cleavage sites

Enzyme	Microorganism	Target sequence and cleavage sites	Enzyme	Microorganism	Target sequence and cleavage sites
<i>Eco</i> RI	<i>Escherichia coli</i>		<i>Hind</i> III	<i>Haemophilus influenzae</i>	
<i>Bam</i> HI	<i>Bacillus amyloliquefaciens</i> H		<i>Pst</i> I	<i>Providencia stuartii</i>	
<i>Hae</i> II	<i>Haemophilus aegyptus</i>		<i>Taq</i> I	<i>Thermus aquaticus</i>	

Note: The vertical dashed line indicates the axis of symmetry in each sequence. Red arrows indicate the sites of cutting. The enzyme *Taq*I yields cohesive ends consisting of two nucleotides, whereas the cohesive ends produced by the other enzymes contain four nucleotides. Pu and Py refer to any purine and pyrimidine, respectively.

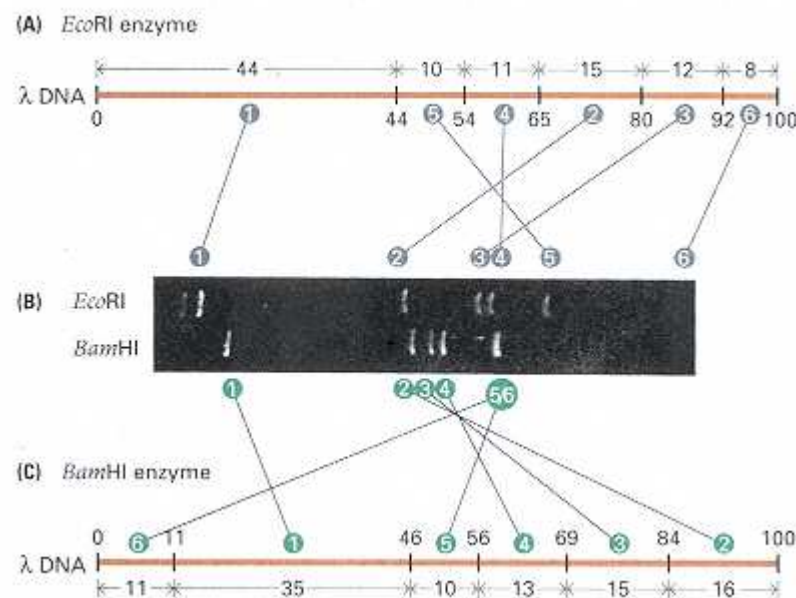


Figure 5.32

Restriction maps of λ DNA for the restriction enzymes. (A) *Eco*RI and (C) *Bam*HI. The vertical bars indicate the sites of cutting. The black numbers indicate the approximate percentage of the total length of λ DNA measured from the end of the molecule that is arbitrarily designated the left end. The numbers within the arrows are the lengths of the fragments, each expressed as percentage of the total length. (B) An electrophoresis gel of *Bam*HI and *Eco*RI enzyme digests of λ DNA. Numbers indicate fragments in order from largest (1) to smallest (6); the circled numbers on the maps correspond to the numbers beside the gel. The DNA has not undergone electrophoresis long enough to separate bands 5 and 6 of the *Bam*HI digest.

restriction fragment. A large DNA molecule will typically be cut into many restriction fragments of different sizes. For example, an *E. coli* DNA molecule, which contains 4.7×10^6 base pairs, is cut into several hundred to several thousand fragments, and mammalian nuclear DNA is cut into more than a million fragments. Although these numbers are large, they are actually quite small relative to the number of sugar-phosphate bonds in the DNA of an organism. Restriction fragments are usually short enough that they can be separated by electrophoresis and manipulated in various ways—for example, using DNA ligase to insert them into self-replicating molecules such as bacteriophage, plasmids, or even small artificial chromosomes. These procedures constitute **DNA cloning** and are the basis of one form of *genetic engineering*, discussed further in Chapter 9.

Because of the sequence specificity, *a particular restriction enzyme produces a unique set of fragments for a particular DNA molecule*. Another enzyme will produce a different set of fragments from the same DNA molecule. Figure 5.32A and 5.32C show the sites of cutting of *E. coli* phage λ DNA by the enzymes *Eco*RI and *Bam*HI. A map showing the unique sites of cutting of the DNA of a particular organism by a single enzyme is called a **restriction map**. The family of fragments produced by a single enzyme can be detected easily by gel electrophoresis of enzyme-treated DNA (Figure 5.32B), and particular DNA fragments can be isolated by cutting out the small region of the gel that contains the fragment and removing the DNA from the gel. Gel electrophoresis for the separation of DNA fragments is described next.

Gel Electrophoresis

The physical basis for the separation of the DNA fragments in Figure 5.32 is that DNA molecules are negatively charged and can move in an electric field. If the terminals of an electrical power source are connected to the opposite ends of a horizontal tube containing a DNA solution, then the molecules will move toward the positive end of the tube, at a rate that depends on the electric field strength and on the shape and size of the molecules. The movement of charged molecules in an electric field is called **electrophoresis**.

The type of electrophoresis most commonly used in genetics is **gel electrophoresis**. An experimental arrangement for gel electrophoresis of DNA is

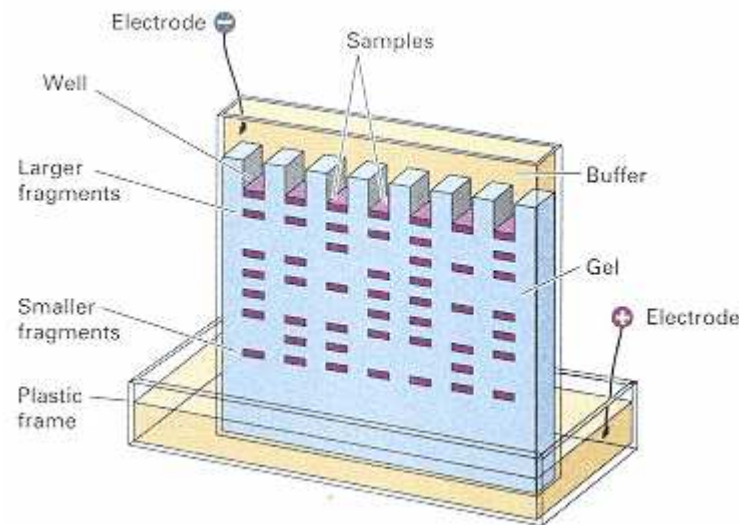


Figure 5.33

Apparatus for gel electrophoresis capable of handling seven samples simultaneously. Liquid gel is allowed to harden in place, with an appropriately shaped mold placed on top of the gel during hardening in order to make "wells" for the samples (purple). After electrophoresis, the samples, located at various positions in the gel, are made visible by removing the plastic frame and immersing the gel in a solution containing a reagent that binds to or reacts with the separated molecules. The separated components of a sample appear as bands, which may be either visibly colored or fluorescent when illuminated with fluorescent light, depending on the particular reagent used. The region of a gel in which the components of one sample can move is called a *lane*. Thus, this gel has seven lanes.

shown in Figure 5.33. A thin slab of a gel, usually agarose or acrylamide, is prepared containing small slots (called wells) into which samples are placed. An electric field is applied, and the negatively charged DNA molecules penetrate and move through the gel. A gel is a complex molecular network that contains narrow, tortuous passages, so smaller DNA molecules pass through more easily; hence the rate of movement increases as the molecular weight decreases. For each molecule, the rate of movement depends primarily on the molecular size, provided the molecule is linear and not too large. Figure 5.34 shows the result of electrophoresis of a collection of double-stranded DNA molecules in an agarose gel. Each discrete region containing DNA is called a **band**.

The Southern Blot

Several techniques enable a researcher to locate a particular DNA fragment in a gel. One of the most generally applicable procedures is the **Southern blot**. In this procedure, a gel in which DNA molecules have been separated by electrophoresis is treated with alkali to denature the DNA and render it single-stranded. Then the DNA is transferred to a sheet of nitrocellulose in such a way that the relative positions of the DNA bands are maintained (Figure 5.35). The nitrocellulose, to which the single-stranded DNA binds tightly, is then exposed to denatured radioactive complementary RNA or

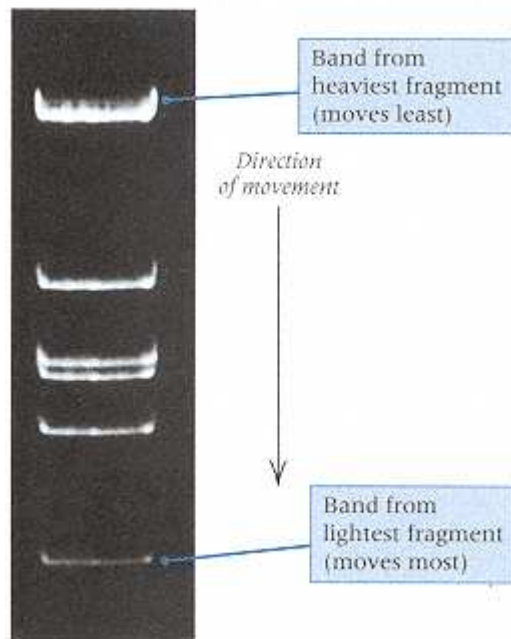


Figure 5.34
Gel electrophoresis of DNA. Molecules of different sizes were mixed and placed in a well. Electrophoresis was in the vertical direction. The DNA has been made visible by the addition of a dye (ethidium bromide) that binds only to DNA and that fluoresces when the gel is illuminated with short-wavelength ultraviolet light.

DNA (the **probe**) in a way that leads the complementary strands to anneal to form duplex molecules. Radioactivity becomes stably bound (resistant to removal by washing) to the DNA only at positions at

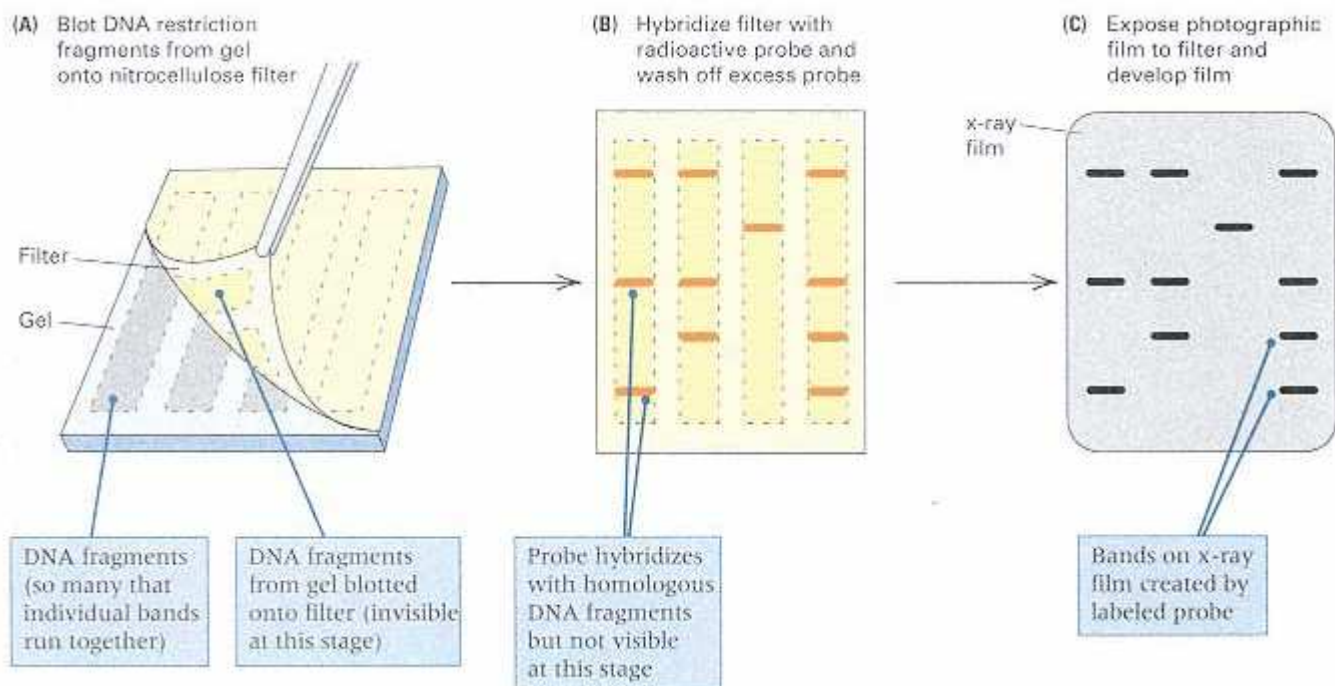


Figure 5.35

Southern blot. (A) DNA restriction fragments are separated by electrophoresis, blotted from the gel onto a nitrocellulose or nylon filter, and chemically attached by the use of ultraviolet light. (B) The strands are denatured and mixed with radioactive probe DNA, which binds with complementary sequences present on the filter. The bound probe remains, whereas unbound probe washes off. (C) Bound probe is revealed by darkening of photographic film placed over the filter. The positions of the bands indicate which restriction fragments contain DNA sequences homologous with those in the probe.

which base sequences complementary to the radioactive molecules are present, so that duplex molecules can form. The radioactivity is located by placing the paper in contact with x-ray film; after development of the film, blackened regions indicate positions of radioactivity. For example, a cloned DNA fragment from one species may be used as probe DNA in a Southern blot with DNA from another species; the probe will hybridize only with restriction fragments containing DNA sequences that are complementary enough to allow stable duplexes to form.

5.8—

The Polymerase Chain Reaction

It is also possible to obtain large quantities of a particular DNA sequence merely by selective replication. The method for selective replication is called the **polymerase chain reaction (PCR)**, and it uses DNA polymerase and a pair of short, synthetic **oligonucleotide primers**, usually about 20 nucleotides in length, that are complementary in sequence to the ends of any DNA sequence of interest. Starting with a mixture containing as little as one molecule of the fragment of interest, repeated rounds of DNA replication increase the number of molecules exponentially. For example, starting with a single molecule, 25 rounds of DNA replication will result in $2^{25} = 3.4 \times 10^7$ molecules. This number of molecules of the amplified fragment is so much greater than that of the other unamplified molecules in the original mixture that the amplified DNA can often be used without further purification. For example, a single fragment of 3000 base pairs in *E. coli* accounts for only 0.06 percent of the total DNA in this

organism. However, if this single fragment were replicated through 25 rounds of replication, then 99.995 percent of the resulting mixture would consist of the amplified sequence.

An outline of the polymerase chain reaction is shown in Figure 5.36. The DNA sequence to be amplified and the oligonucleotide sequences are shown in contrasting colors. The oligonucleotides act as primers for DNA replication, because they anneal to the ends of the sequence to be amplified and become the substrates for chain elongation by DNA polymerase. In the first cycle of PCR amplification, the DNA is denatured to separate the strands. The denaturation temperature is usually around 95°C. Then the temperature is decreased to allow annealing in the presence of a vast excess of the primer oligonucleotides. The annealing temperature is typically in the range from 50°C to 60°C, depending largely on the G + C content of the oligonucleotide primers. The temperature is raised slightly, to about 70°C, for the elongation of each primer. The first cycle in PCR produces two copies of each molecule containing sequences complementary to the primers. The second cycle of PCR is similar to the first. The DNA is denatured, then renatured in the presence of an excess of primer oligonucleotides, and the primers are elongated by DNA polymerase; after this cycle, there are four copies of each molecule present in the original mixture. The steps of denaturation, renaturation, and replication are repeated from 20–30 times, and in each cycle, the number of molecules of the amplified sequence is doubled. The theoretical result of 25 rounds of amplification is 2^{25} copies of each template molecule present in the original mixture.

Implementation of PCR with conventional DNA polymerase is not practical, because at the high temperature necessary for denaturation, the polymerase is itself irreversibly unfolded and becomes inactive. However, DNA polymerase isolated from certain archaebacteria is heat stable because the organisms normally live in hot springs at temperatures well above 90°C, such as are found in Yellowstone National Park. Such organisms are said to be **thermophiles**. The most widely used heat-stable DNA polymerase is called *Taq* polymerase, because it was originally isolated from the thermophilic archaebacterium *Thermus aquaticus*.

Figure 5.37 presents a closer look at the primer oligonucleotide sequences and their relationship to the DNA sequence to be amplified. The primers are designed to anneal to opposite DNA strands at the extreme ends of the region to be amplified. The primers are oriented with their 3' ends facing the sequence to be amplified so that, in DNA synthesis, the new strands grow toward each other but along complementary template strands. In this way, each newly synthesized strand terminates in a sequence that can anneal with the complementary primer and so can be used for further amplification.

PCR amplification is very useful for generating large quantities of a specific DNA sequence. The principal limitation of the technique is that the DNA sequences at the ends of the region to be amplified must be known so that primer oligonucleotides can be synthesized. In addition, sequences longer than about 5000 base pairs cannot be replicated efficiently by conventional PCR procedures. On the other hand, there are many applications in which PCR amplification is useful. PCR can be employed to study many different mutant alleles of a gene whose wildtype sequence is known in order to identify the molecular basis of the mutations. Similarly, DNA sequence variation among alleles present in natural populations can easily be determined by using PCR. The PCR procedure has also come into widespread use in clinical laboratories for diagnosis. To take just one very important example, the presence of the human immunodeficiency virus (HIV), which causes acquired immune deficiency syndrome (AIDS), can be detected in trace quantities in blood banks via PCR by using primers complementary to sequences in the viral genetic material. These and other applications of PCR are facilitated by the fact that the procedure lends itself to automation—for example, by the use of mechanical robots to set up the reactions.

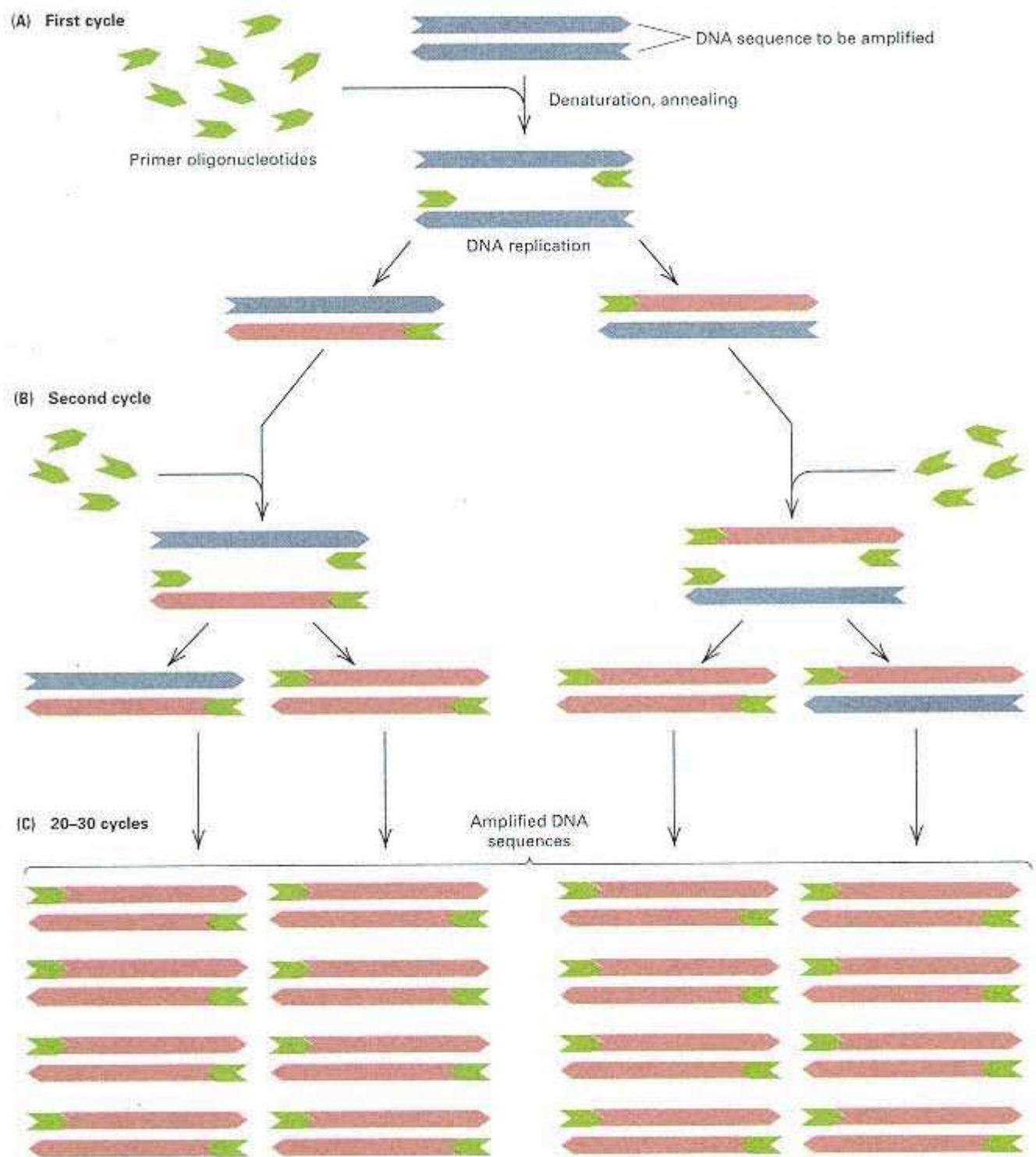


Figure 5.36

Polymerase chain reaction (PCR) for amplification of particular DNA sequences. Oligonucleotide primers (green) that are complementary to the ends of the sequence of interest (blue) are used in repeated rounds of denaturation, annealing, and DNA replication. Copies of the target sequence are shown in pink. The number of copies of the target sequence doubles in each round of replication, eventually overwhelming any other sequences that may be present.

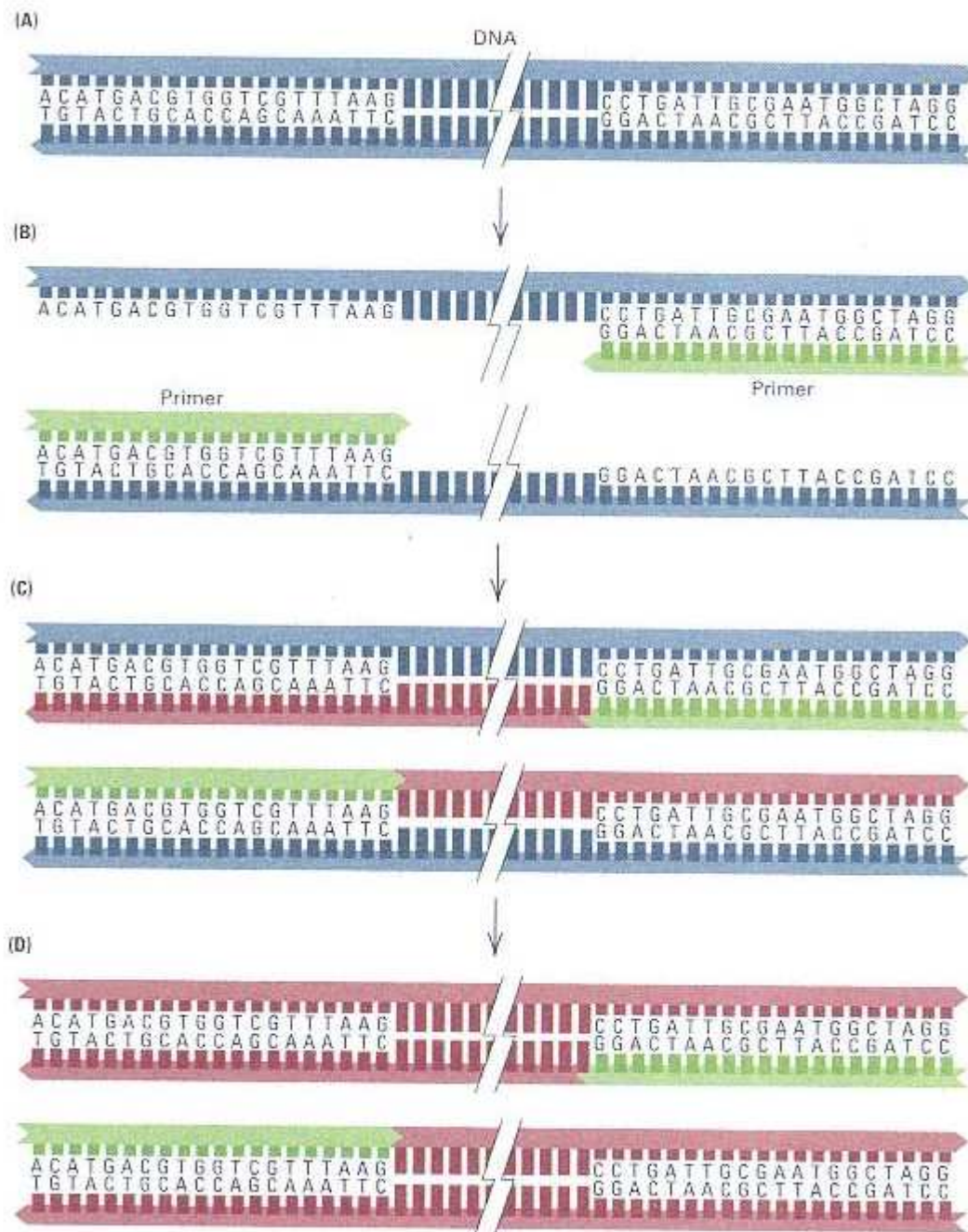


Figure 5.37

Example of primer sequences used in PCR amplification Newly synthesized DNA is shown in pink (A) Original DNA duplex (blue) and primers (green). (B) The original DNA duplex is denatured and the primers annealed. (C) After one round of binding of the primers and elongation. Each primer has been extended at the 3' end. (D) After another round of replication Each primer is again extended at the 3' end, but the elongation terminates at the 5' end of the primer oligonucleotide on the other strand After the first round of replication, each newly synthesized DNA molecule forms a new template. so the number of copies increases exponentially.

5.9—

Determination of the Sequence of Bases in DNA

A great deal of information about gene structure and gene expression can be obtained by direct determination of the sequence of bases in a DNA molecule. Several techniques are available for base sequencing; the most widely used method is described in this section. No technique can determine the sequence of bases in an entire chromosome in a single experiment, so chromosomes are first cut into fragments a few hundred base pairs long, a size that can be sequenced easily. To obtain the sequence of a long stretch of DNA, a set of overlapping fragments is prepared, the sequence of each is determined, and all sequences are then combined. The procedures are straightforward, and DNA sequences have accumulated at such a

rapid rate that large computer databases are necessary to manage the hundreds of millions of nucleotides of DNA sequence already determined for a variety of genes from many organisms. A still larger database would be required to handle the DNA in human sperm or eggs, which consists of 3×10^9 nucleotide pairs. Although the sequencing technique described here is the manual method still in use in many research laboratories, in Chapter 9 we will see how DNA sequencing can be automated and how DNA-sequencing machines work.

The **dideoxy sequencing method** employs DNA synthesis in the presence of small amounts of nucleotides that contain the sugar **dideoxyribose** instead of deoxyribose (Figure 5.38). Dideoxyribose lacks the 3'-OH group, which is essential for attachment of the next nucleotide in a growing DNA strand, so incorporation of a dideoxynucleotide instead of a deoxynucleotide immediately terminates further synthesis of the strand. To sequence a DNA strand, four DNA synthesis reactions are carried out. Each reaction contains the single-stranded DNA template to be sequenced, a single oligonucleotide primer complementary to a stretch of the template strand, all four deoxyribonucleoside triphosphates, and a small amount of *one* of the nucleoside triphosphates in the dideoxy form. Each reaction produces a set of fragments that terminates at the point at which a dideoxynucleotide was randomly incorporated in place of the normal deoxynucleotide. Therefore, in each of the four reactions, the lengths of the fragments are determined by the positions in the daughter strand at which the particular dideoxynucleotide present in that reaction was incorporated. The sizes of the fragments produced by chain termination are determined by gel electrophoresis, and the base sequence is then determined by the following rule:

If a fragment containing n nucleotides is generated in the reaction containing a particular dideoxynucleotide, then position n in the *daughter strand* is occupied by the base present in the dideoxynucleotide. The numbering is from the 5' nucleotide of the primer.

For example, if a 93-base fragment is present in the reaction containing the dideoxy

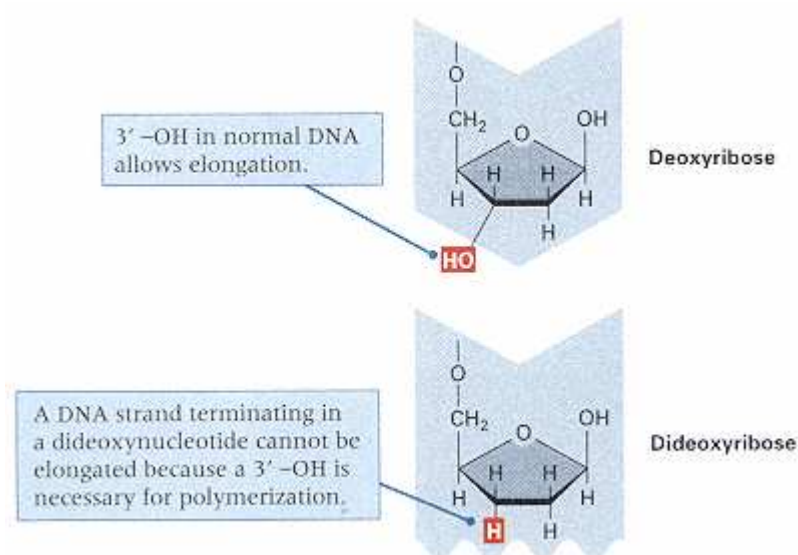


Figure 5.38

Structures of normal deoxyribose and the dideoxyribose sugar used in DNA sequencing. The dideoxyribose has a hydrogen atom (red) attached to the 3' carbon, in contrast with the hydroxyl group (red) at this position in deoxyribose. Because the 3' hydroxyl group is essential for the attachment of the next nucleotide in line in a growing DNA strand, the incorporation of a dideoxynucleotide immediately terminates synthesis.

form of dATP, then the 93rd base in the daughter strand produced by DNA synthesis must be an adenine (A). Because most native duplex DNA molecules consist of complementary strands, it does not matter whether the sequence of the template strand or that of the daughter strand is determined. The sequence of the template strand can be deduced from the daughter strand because their nucleotide sequences are complementary. However, in practice, both strands of a molecule are usually sequenced independently and compared in order to eliminate most of the errors that can be made by misreading the gels.

With short, single-stranded fragments, molecules that differ in size by a single base can be separated by electrophoresis in an acrylamide gel. For example, if a reaction mixture contains a set of DNA fragments consisting of 20, 21, 22, . . . , 100 nucleotides, then electrophoresis will yield separate bands—one containing the fragment of

size 20, the next the fragment of size 21, and so forth. This extraordinary sensitivity to size is the basis of the DNA-sequencing procedure.

The Sequencing Procedure

The procedure for sequencing a DNA fragment is diagrammed in Figure 5.39. In this example, the primer length is 20 nucleotides. Termination at G produces fragments of 21, 24, 27, and 29 nucleotides; termination at A produces fragments of 22 and 30 nucleotides; termination at T produces a fragment of 26 nucleotides; and termination at C produces fragments of 23, 25, and 28 nucleotides. Typically, either the primer or one of the deoxynucleotides used contains a radioactive atom, so that after electrophoresis to separate the fragments by size, the fragments can be visualized as bands on photographic film placed over the gel. After electrophoresis and autoradiography, the positions of the bases in the daughter strand can be read directly from the gel, starting with position 21, as GACGCTGCGA. This corresponds to the template strand sequence CTGCGACGCT.

Figure 5.40 is an autoradiogram of an actual sequencing gel. The shortest fragments are those that move the fastest and farthest. Each fragment contains the primer fragment at the 5' end of the daughter strand. The sequence can be read

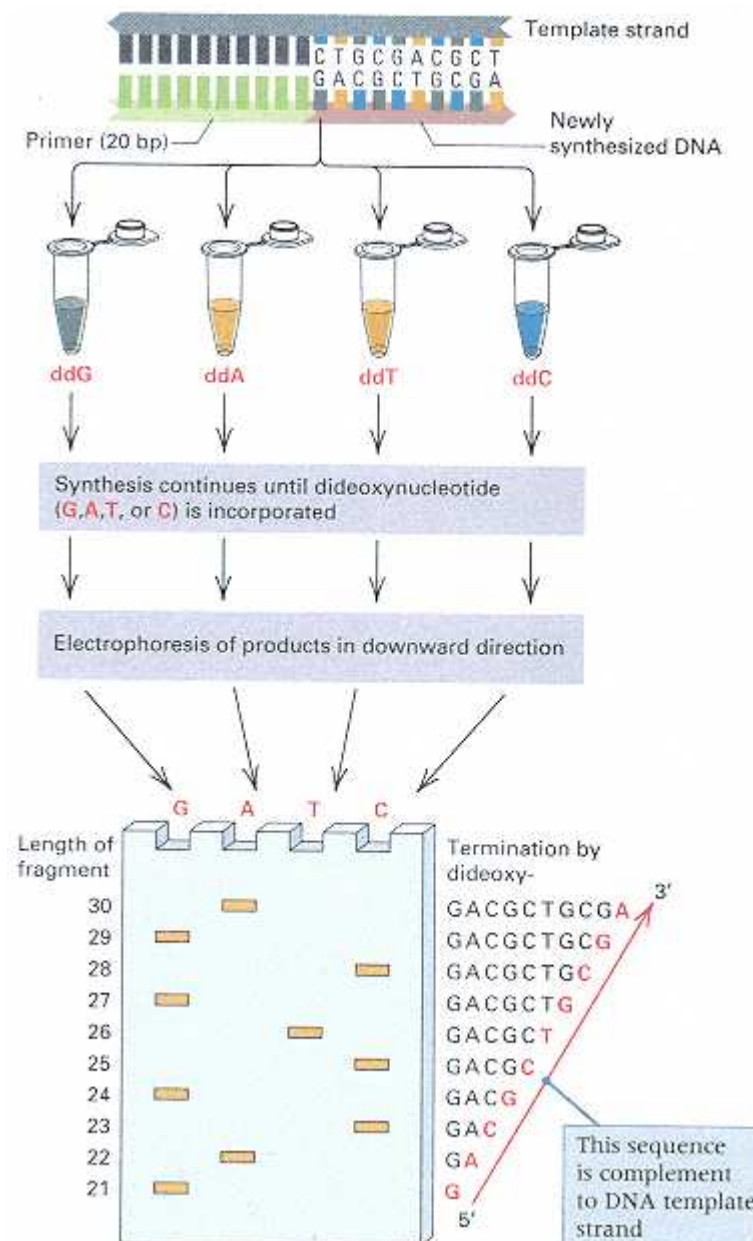


Figure 5.39

Dideoxy method of DNA sequencing. Four DNA synthesis reactions are carried out in the presence of all normal nucleotides plus a small amount of one of the dideoxynucleotides containing G, A, T, C. Synthesis continues along the template strand until a dideoxynucleotide is incorporated. The products that result from termination at each dideoxynucleotide are indicated at the right. The fragments are separated by size by electrophoresis, and the positions of the nucleotides are determined directly from the gel. In this example, the length of the primer needed

to initiate DNA synthesis is 20 nucleotides. Lengths of the DNA fragments are shown at the left of the gel. The sequence of the daughter strand is read from the bottom of the gel as 5'-GACGCTGCGA-3'

directly from the bottom to the top of the gel. The sequence of the first 16 bases in the segment in this gel, read from bottom to top, is

5' - CACTGCCTGCGCCCAG - 3'

and so forth.

Clinical Use of Dideoxynucleoside Analogs

Our knowledge of DNA structure and replication has applications not only in procedures for the manipulation of DNA but also in the development of drugs for clinical use. An example is in one approach to the treatment of AIDS. A number of dideoxynucleoside analogs are effective in inhibiting replication of the viral genetic material. A few of these are illustrated in Figure 5.41. Recall that a *nucleoside* is a base attached to a sugar without a phosphate. A nucleoside **analog** is a molecule similar, but not identical, in structure to a nucleoside. In Figure 5.41, ddC is the normal dideoxyribocytidine nucleoside. It is effective against AIDS, as are the dideoxynucleoside analogs AZT, D4T, and ddI (and other such analogs). The nucleoside, rather than the nucleotide, is used in therapy because the nucleotide, having a highly charged phosphate group, cannot cross the cell membrane as easily. The emergence of these drugs from our basic knowledge of DNA structure and replication is one of the prime examples of the fact that "pure" science may have many unforeseen practical

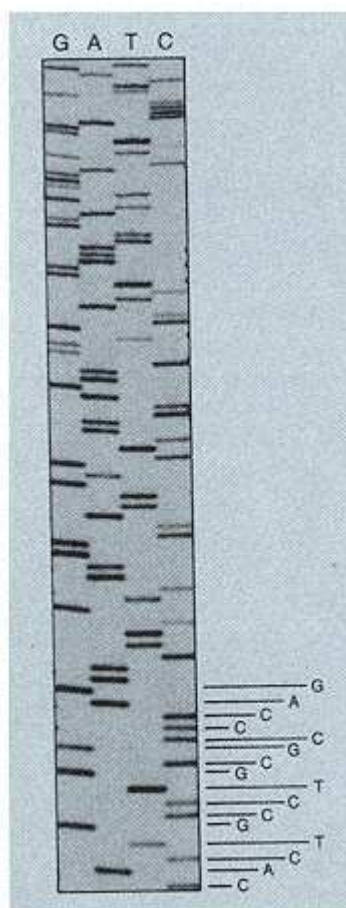


Figure 5.40

A section of a dideoxy sequencing gel. The sequence is read from the bottom to the top. Each horizontal row represents a single nucleotide position in the DNA strand synthesized from the template. The vertical columns result from termination by the dideoxy forms of G, A, T, or C. The sequence from the lower part of the gel is indicated.

applications. The basic experiments on DNA were carried out long before the recognition of AIDS as a distinct infectious disease and the discovery that HIV is the causative agent.

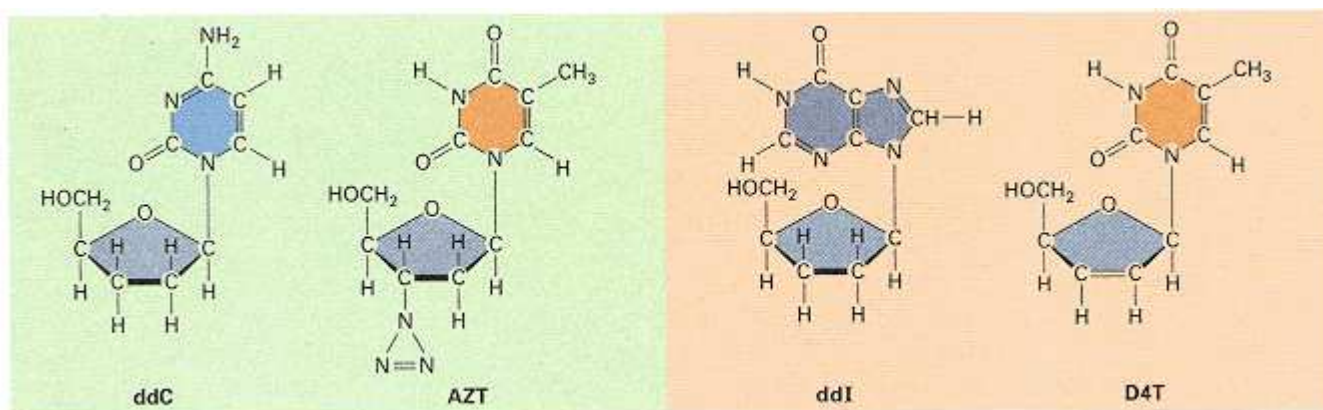


Figure 5.41

A few of the drugs that have been found to be effective in the treatment of AIDS by interfering with the replication of HIV virus. The technical names of the substances are as follows: ddI is 2',3'-dideoxyinosine; ddC is 2',3'-dideoxycytidine; AZT is 3'-azido-2',3'-dideoxythymidine; and D4T is 2',3'-didehydro-2',3'-dideoxythymidine.

Chapter Summary

DNA is a double-stranded polymer consisting of deoxyribonucleotides. A nucleotide has three components: a base, a sugar (deoxyribose in DNA, ribose in RNA), and a phosphate. Sugars and phosphates alternate in forming a single polynucleotide chain with one terminal 3'-OH group and one terminal 5'-P group. In double-stranded (duplex) DNA, the two strands are antiparallel: Each end of the double helix carries a terminal 3'-OH group in one strand and a terminal 5'-P group in the other strand. Four bases are found in DNA: adenine (A) and guanine (G), which are purines; and cytosine (C) and thymine (T), which are pyrimidines. Equal numbers of purines and pyrimidines are found in double-stranded DNA, because the bases are paired as A–T pairs and G–C pairs (Chargaff's rules). This pairing holds the two polynucleotide strands together in a double helix. The base composition of DNA varies from one organism to the next. The information content of a DNA molecule resides in the sequence of bases along the chain, and each gene consists of a unique sequence.

The double helix replicates by using enzymes called DNA polymerases, but many other proteins also are needed. Replication is semiconservative in that each parental single strand, called a template strand, is found in one of the double-stranded progeny molecules. Semiconservative replication was first demonstrated in the Meselson-Stahl experiment, which used equilibrium density-gradient centrifugation to separate DNA molecules containing two ¹⁵N-labeled strands, two ¹⁴N-labeled strands, or one of each. Replication proceeds by a DNA polymerase (1) bringing in a nucleotide triphosphate with a base capable of hydrogenbonding with the corresponding base in the template strand and (2) joining the 5'-P group of the nucleotide to the free 3'-OH group of the growing strand. (The terminal P–P from the nucleotide triphosphate is cleaved off and released.) Because double-stranded DNA is antiparallel, only one strand (the leading strand) grows in the direction of movement of the replication fork. The other strand (the lagging strand) is synthesized in the opposite direction as short fragments (Okazaki fragments) that are subsequently joined together. DNA polymerases cannot initiate synthesis, so a primer is always needed. The primer is an RNA fragment made by an RNA-polymerizing enzyme called primase; the RNA primer is removed at later stages of replication. DNA molecules of prokaryotes usually have a single replication origin; eukaryotic DNA molecules usually have many origins.

Restriction enzymes cleave DNA molecules at the positions of specific sequences (restriction sites) of usually four or six nucleotides. Each restriction enzyme produces a unique set of fragments for any particular DNA molecule. These fragments can be separated by electrophoresis and used for purposes such as DNA sequencing. The positions of particular restriction fragments in a gel can be visualized by means of a Southern blot, in which radioactive probe DNA is mixed with denatured DNA made up of single-stranded restriction fragments that have been transferred to a filter membrane after electrophoresis. The probe DNA will form stable duplexes (anneal or renature) with whatever fragments contain sufficiently complementary base sequences, and the positions of these duplexes can be determined by autoradiography of the filter. Particular DNA sequences can also be amplified without cloning by means of the polymerase chain reaction (PCR), in which short, synthetic oligonucleotides are used as primers to replicate repeatedly and amplify the sequence between them.

The base sequence of a DNA molecule can be determined by dideoxynucleotide sequencing. In this method, the DNA is isolated in discrete fragments containing several hundred nucleotide pairs. Complementary strands of each fragment are sequenced, and the sequences of overlapping fragments are combined to yield the complete sequence. The dideoxy sequencing method uses dideoxynucleotides to terminate daughter strand synthesis and reveal the identity of the base present in the daughter strand at the site of termination.

Key Terms

analog	DNA polymerase	major groove
annealing	DNA polymerase I (Pol I)	melting curve
antiparallel	DNA polymerase III (Pol III)	melting temperature
B form	editing function	minor groove
band	endonuclease	nick

base composition	equilibrium density-gradient	nuclease
bidirectional replication	centrifugation	nucleic acid hybridization
blunt ends	exonuclease	nucleoside
Chargaff's rules	5'-P group	nucleotide
daughter strand	5' → 3' exonuclease	Okazaki fragments
denaturation	gel electrophoresis	oligonucleotide primers
dideoxyribose	helicase	palindrome
dideoxy sequencing method	initiation	parent strand
DNA cloning	lagging strand	percent G + C
DNA ligase	leading strand	phosphodiester bond

polarity	replication fork	template
polymerase α	replication origin	thermophiles
polymerase chain reaction (PCR)	restriction endonuclease	θ replication
polymerase γ	restriction enzyme	3'-OH group
polynucleotide chain	restriction fragment	topoisomerase
precursor fragment	restriction map	ultracentrifuge
primase	restriction site	Z-form DNA
primer	ribose	
primosome	RNA polymerase	
probe DNA	rolling-circle replication	
proofreading function	semiconservative replication	
purine	single-strand binding proteins	
pyrimidine	Southern blot	
renaturation	sticky ends	

Review the Basics

- What are the four bases commonly found in DNA? Which form base pairs? What five-carbon sugar is found in DNA? What is the difference between a nucleoside and a nucleotide?
- How many phosphate groups are there per base in DNA, and how many phosphates are there in each precursor for DNA synthesis?
- Which chemical groups are at the ends of a single polynucleotide strand?
- What is the relationship between the amount of DNA in a somatic cell and the amount in a gamete?
- Name four requirements for initiation of DNA synthesis. To what chemical group in a DNA chain is an incoming nucleotide added, and what group in the nucleotide reacts with the DNA terminus?
- In what sense are the two strands of DNA antiparallel?
- How does the polymerase chain reaction work? What is it used for? What information about the target sequence must be known in advance?

Guide to Problem Solving

Problem 1: A technique is used for determining the base composition of double-stranded DNA. Rather than giving the relative amounts of each of the four bases—[A], [T], [G], and [C]—it yields the value of the ratio [A] [C]. If

this ratio is 1/3, what are the relative amounts of each of the four bases?

Answer: In double-stranded DNA, $[A] = [T]$ and $[G] = [C]$ because of the base pairing between complementary strands. Therefore, if $[A]/[C] = 1/3$, then $[C] = [G] = 3 \times [A]$. Because $[A] + [T] + [G] + [C] = 1$, everything can be put in terms of $[A]$ as

$$[A] + [A] + 3[A] + 3[A] = 1$$

or

$$[A] = [T] = 0.125$$

This makes

$$[C] = [G] = (1 - 0.25)/2 = 0.375$$

In other words, the DNA is 12.5 percent A, 12.5 percent T, 37.5 percent G, and 37.5 percent C.

Problem 2: The restriction enzyme *EcoRI* cleaves double-stranded DNA at the sequence 5'-GAATTC-3', and the restriction enzyme *HindIII* cleaves at 5'-AAGCTT-3'. A 20-kilobase (kb) circular plasmid is digested with each enzyme individually and then in combination, and the resulting fragment sizes are determined by means of electrophoresis. The results are as follows:

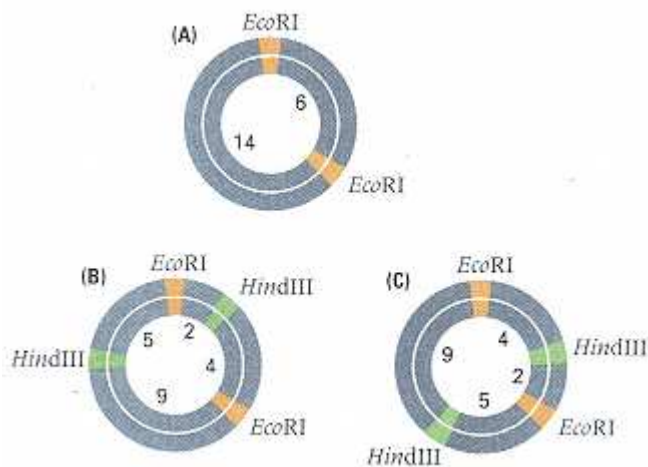
EcoRI alone: fragments of 6 kb and 14 kb

HindIII alone: fragments of 7 kb and 13 kb

EcoRI and *HindIII*: fragments of 2 kb, 4 kb, 5 kb and 9 kb

How many possible restriction maps are compatible with these data? For each possible restriction map, make a diagram of the circular molecule and indicate the relative positions of the *EcoRI* and *HindIII* restriction sites.

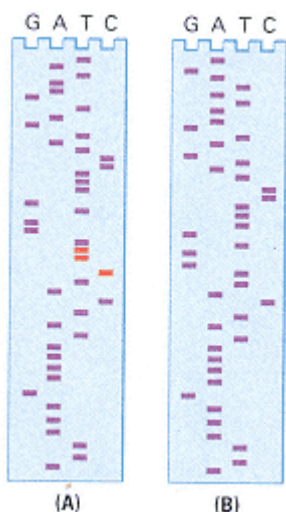
Answer: Because the single-enzyme digests give two bands each, there must be two restriction sites for each enzyme in the molecule. Furthermore, because digestion with *Hind*III makes both the 6-kb and the 14-kb restriction fragments disappear, each of these fragments must contain one *Hind*III site. Considering the sizes of the fragments in the double digest, the 6-kb *Eco*RI fragment must be cleaved into 2-kb and 4-kb fragments, and the 14-kb *Eco*RI fragment must be cleaved into 5-kb and 9-kb fragments. Two restriction maps are compatible with the data, depending on which end of the 6-kb *Eco*RI fragment the *Hind*III site is nearest. The position of the remaining *Hind*III site is determined by the fact that the 2-kb and 5-kb fragments in the double digest must be adjacent in the intact molecule in order for a 13-kb fragment to be produced by *Hind*III digestion alone.



The accompanying figure shows the relative positions of the *Eco*RI sites (part A). Parts B and C are the two possible restriction maps, which differ according to whether the *Eco*RI site at the top generates the 2-kb or the 4-kb fragment in the double digest.

Problem 3: The DNA sequencing gels diagrammed in parts A and B of the accompanying illustration were obtained from human DNA by the dideoxy sequencing method. Part A comes from a person homozygous for the normal allele of the cystic fibrosis gene, part B from an affected person homozygous for a mutant allele. Use the bands in the gel to answer the following questions.

- Why are there discrete bands in the gel?
- Which end of the gel (top or bottom) corresponds to the 5' end of the DNA sequence?
- What nucleotide sequence is indicated by the bands in the gels?
- What is the sequence of the complementary strand in duplex DNA?
- Compare the sequence in parts A and B. What is special about the three nucleotides that give the bands indicated in red?



Answer:

(a) Each band results from a fragment of DNA whose replication was terminated by the incorporation of a dideoxy nucleotide. The fragments differ by one nucleotide in length, and they are arranged by size with the smallest fragments at the bottom, the largest at the top.

(b) Because DNA replication elongates at the 3' end, each fragment of increasing length has an additional 3' nucleotide. Therefore, the DNA sequence is oriented with the 5' end at the bottom of the gel and the 3' end at the top.

(c) The sequences, read directly from the gel from bottom to top, are (part A)

5' -ATTAAAGAAAATATCAT**CTTT**GGTGTTCCTATGATGAATAT -3'

and (part B),

5' -ATTAAAGAAAATATCATTGGTGTTCCTATGATGAATATAGA -3'

(d) The complementary strands in duplex DNA are antiparallel and have A paired with T and G paired with C. For part A, the complementary strand is

3' -TAATTTCTTTTATAGTA**GAA**ACCACAAAGGATACTACTTATA -5'

and, for part B, the complementary strand is

3' -TAATTTCTTTTATAGTAACCACAAAGGATACTACTTATATCT -5'

(e) The three nucleotides in red in the wildtype gene are missing in the mutant gene. This three-base deletion is found in about 70 percent of the mutations in cystic fibrosis patients, and it results in a missing amino acid at position 508 in the corresponding protein.

Chapter 5 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

<http://WWW.jbpub.com/genetics>

Select the link to *Genetics: Principles and Analysis* and then choose the link to *GeNETics on the web*. You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. The keyword **DNA** will connect you with a site that has an excellent collection of DNA structural models, including some animations. Browse through some of these and you will get a better intuitive feel for the structure and the manner in which the two strands are paired. If assigned to do so, take a close look at DNA model Number 24. There is no legend, but you should be able to deduce the color coding of the spheres. What type of atom corresponds to red? To green? To blue? What is the color of the phosphorous atoms?

2. One of the principal tools of the molecular geneticist is the polymerase chain reaction, which is unique in its ability to bring about exponential amplification of a particular DNA sequence present in a large background of other sequences. The keyword site summarizes the development of this key method. If assigned to do so, write a paragraph describing two innovations that were instrumental in the maturation of the method, and explain why each was important.

3. In DNA sequencing, there is sometimes uncertainty in the identification of a particular base because the gel pattern for that base is not completely unambiguous. **An ambiguous base** is denoted in a sequence by any one of a standard set of abbreviations. For example, the letter M stands for the possibilities "either A or C," V stands for "either A or C or G," and N (or X) stands for "unknown base." The keyword will access the complete set of abbreviations. If assigned to do so, organize the abbreviations in a systematic way of your choosing.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 5, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 5.



Analysis and Applications

5.1 What is the base sequence of a DNA strand that is complementary to the hexanucleotide 3'-AGGCTC-5'? Label the termini as to 5' or 3'.

5.2 What is a nuclease enzyme, and how do endonucleases and exonucleases differ?

5.3 In what direction ($5' \rightarrow 3'$ or $3' \rightarrow 5'$) does a DNA polymerase move along the template strand? How do organisms solve the problem that all DNA polymerases move in the same direction along a template strand, yet double-stranded DNA is antiparallel?

5.4 What is the chemical difference between the groups joined by DNA polymerase and DNA ligase?

5.5 What are three enzymatic activities of DNA polymerase I?

5.6 Why can RNA polymerases and primases initiate DNA replication, whereas DNA polymerases cannot?

5.7 In gel electrophoresis, do smaller double-stranded molecules move more slowly or more rapidly than larger molecules?

5.8 Consider a hypothetical phage whose DNA replicates exclusively by rolling-circle replication. A phage with radioactive DNA in both strands infects a bacterium and is allowed to replicate in a nonradioactive medium. Assume that only daughter DNA from the elongating branch ever gets packaged into progeny phage particles.

(a) What fraction of the parental radioactivity will appear in progeny phage?

(b) How many progeny phage will contain radioactive DNA, and how will the number be affected by recombination?

5.9 What is the fundamental difference between the initiation of θ replication and that of rolling-circle replication?

5.10 When the base composition of DNA from the bacterium *Mycobacterium tuberculosis* was determined, 18 percent of the bases were found to be adenine.

(a) What is the percentage of cytosine?

(b) What is the entire base composition of the DNA and the $[G] + [C]$ content?

5.11 The double-stranded DNA molecule of a newly discovered virus was found by electron microscopy to have a length of 34 μm .

(a) How many nucleotide pairs are present in one of these molecules?

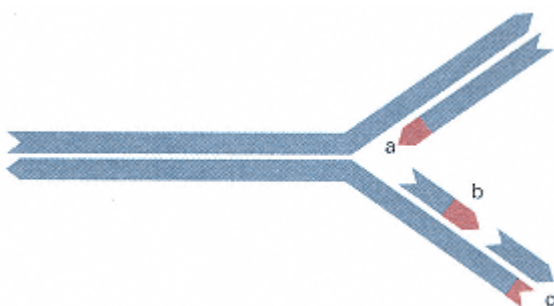
(b) How many complete turns of the two polynucleotide chains are present in such a double helix?

5.12 An elegant combined chemical and enzymatic technique enables one to identify "nearest neighbors" of bases (adjacent bases in a DNA strand). For example, if the single-stranded tetranucleotide 5'-AGTC-3' were treated in this way, the nearest neighbors would be AG, GT, and TC (they are always written with the 5' terminus at the left). Before techniques were available for determining the complete base sequence of DNA, nearest-neighbor analysis was used to determine sequence relations. Nearest-neighbor analysis also indicated that complementary DNA strands are antiparallel, a phenomenon that you are asked to examine in this problem by predicting some nearest-neighbor frequencies. Assume that you have determined the frequencies of the following nearest neighbors: AG, 0.15; GT, 0.03; GA, 0.08; TT, 0.10.

(a) What are the nearest-neighbor frequencies of CT, AC, TC, and AA?

(b) If DNA had a parallel (rather than an antiparallel) structure, what nearest-neighbor frequencies could you deduce from the observed values?

5.13 What is the chemical group (3'-P, 5'-P, 3'-OH, or 5'-OH) at the sites indicated at the positions labeled a, b, and c in the accompanying figure?



5.14 Identify the chemical group that is at the indicated terminus of the daughter strand of the extended branch of the rolling circle in the accompanying diagram.



Challenge Problems

5.15 The following sequence of bases is present along one chain of a DNA duplex that has opened up at a replication fork, and synthesis of an RNA primer on this template begins by copying the base in red.

3' - . . . TCTGATATCAGTACG . . . -5'

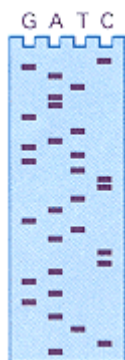
- (a) If the RNA primer consists of eight nucleotides, what is its base sequence?
- (b) In the intact RNA primer, which nucleotide has a free hydroxyl (-OH) terminus and what is the chemical group on the nucleotide at the other end of the primer?
- (c) If replication of the other strand of the original DNA duplex proceeds continuously (with few or no intervening RNA primers), is the replication fork more likely to move in a left-to-right or a right-to-left direction?

5.16 A DNA fragment produced by the restriction enzyme *SalI* is inserted into a unique *SalI* cloning site in a vector molecule. Digestion with restriction enzymes produces the following fragment sizes that originate from the inserted DNA:

- (a) *SalI*: 20 kb
- (b) *SalI* + *EcoRI*: 7 kb, 13 kb
- (c) *SalI* + *HindIII*: 4 kb, 5 kb, 11 kb
- (d) *SalI* + *EcoRI* + *HindIII*: 3 kb, 4 kb, 5 kb, 8 kb

What restriction map of the insert is consistent with these fragment sizes?

5.17 The illustrated DNA sequencing gel was obtained by the dideoxy sequencing method. What is the nucleotide sequence of the strand synthesized in the sequencing reaction? What is the nucleotide sequence of the complementary template strand? Label each end as 3' or 5'.



5.18 For the following restriction endonucleases, calculate the average distance between restriction sites in an organism whose DNA has a random sequence and equal proportions of all four nucleotides. The symbol R means any purine (A or G), and Y means any pyrimidine (T or C), but an R–Y pair must be either A–T or G–C.

- TaqI* 5' - TCGA - 3'
3' - AGCT - 5'
- BamHI* 5' - GGATCC - 3'
3' - CCTAGG - 5'
- HaeII* 5' RGC GCY - 3'
3' YCG CGR - 5'

Further Reading

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Variegated kernel color in corn due to interaction between the transposable elements *Ac* and *Ds* discovered by Barbara McClintock. These kernels contain two copies of *Ds* located in chromosome 9 proximal (toward the centromere) to the locus of *Cl*, which is responsible for the purple anthocyanin pigment. The homologous chromosome carries an inactive mutant allele *cl*. The element *Ac* is present elsewhere in the genome. When *Ac* breaks chromosome 9 at the position of either *Ds* element, the tip of chromosome 9 containing the dominant *Cl* allele is lost, and the portion of the kernel that develops from such a cell is colorless. The colorless patches are large or small depending on whether the breakage occurred early or late in development.

[Courtesy of Clifford Weil and Susan Wessler. From C. F. Weil and S. R. Wessler. 1993. *The Plant Cell* 5:515.]

Chapter 6— The Molecular Organization of Chromosomes

CHAPTER OUTLINE

6-1 Genome Size and Evolutionary Complexity

6-2 The Supercoiling of DNA

Topoisomerase Enzymes

6-3 The Structure of the Bacterial Chromosome

6-4 The Structure of Eukaryotic Chromosomes

The Nucleosome Is the Basic Structural Unit of Chromatin

Nucleosome Core Particles

The Arrangement of Chromatin Fibers in a Chromosome

6-5 Polytene Chromosomes

6-6 Repetitive Nucleotide Sequences in Eukaryotic Genomes

Kinetics of DNA Renaturation

Analysis of Genome Size and Repetitive Sequences by Renaturation

6-7 Nucleotide Sequence Composition of Eukaryotic Genomes

Unique Sequences

Highly Repetitive Sequences

Middle-Repetitive Sequences

6-8 Transposable Elements

6-9 Centromere and Telomere Structure

Molecular Structure of the Centromere

Molecular Structure of the Telomere

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Challenge Problem

Further Reading

GeNETics on the web

PRINCIPLES

- Prokaryotes and lower eukaryotes have smaller genomes (less DNA) than higher eukaryotes.
- Chromosomes consist largely of DNA combined with histone proteins; each chromosome contains a single, usually very long, DNA molecule.
- Most eukaryotic genomes, in addition to the "unique" DNA sequences that make up the majority of genes, also contain DNA sequences that are highly repetitive or moderately repetitive.
- Transposable elements are DNA sequences able to change their location within a chromosome or to move between chromosomes.
- The centromere is a specialized DNA structure that functions as the center of chromosome movement in cell division.
- The telomere is another specialized DNA structure that serves to stabilize the chromosome tips from shortening through progressive loss of DNA; the telomere is elongated by a special enzyme, telomerase.

CONNECTIONS

CONNECTION: Her Feeling for the Organism

Barbara McClintock 1950
The origin and behavior of mutable loci in maize

CONNECTION: Telomeres—The Beginning of the End

Carol W. Greider and Elizabeth H. Blackburn 1987
The telomere terminal transferase of Tetrahymena is a ribonucleoprotein enzyme with two kinds of primer specificity

To understand genetic processes requires a knowledge of the organization of the genetic material at the level of chromosomes. In this chapter, we will see that chromosomes are diverse in size and structural properties and that their DNA differs in the composition and arrangement of nucleotide sequences. The most pronounced differences in structure and genetic organization are between the chromosomes of eukaryotes and those of prokaryotes. Some viral chromosomes are especially noteworthy in that they consist of one single-stranded (rather than double-stranded) DNA molecule and, in a small number of viruses, of one or more molecules of RNA instead of DNA. Also, eukaryotic cells contain several chromosomes, each of which contains one intricately coiled DNA molecule, whereas prokaryotes contain a single major chromosome (and, occasionally, several copies of one or more small, usually circular, DNA molecules called plasmids).

The genetic complement of a cell or virus constitutes its **genome**. In eukaryotes, this term is commonly used to refer to one complete haploid set of chromosomes, such as that found in a sperm or egg.

6.1—

Genome Size and Evolutionary Complexity

Measurement of the nucleic acid content of the genomes of viruses, bacteria, and lower and higher eukaryotes has led to the following generalization:

Genome size increases roughly with evolutionary complexity.

This generalization is based on the observations that the single nucleic acid molecule of a typical virus is smaller than the DNA molecule in a bacterial chromosome; that unicellular eukaryotes, such as the yeasts, contain more DNA than a typical bacterium; and that multicellular eukaryotes have the greatest amount of DNA per genome. However, among the multicellular eukaryotes, no correlation exists between evolutionary complexity and amount of DNA. In higher eukaryotes, DNA content is not directly proportional to number of genes.

A summary of genome size in a sample of organisms is shown in Table 6.1. Bacteriophage MS2 is one of the smallest viruses; it has only four genes in a single-stranded RNA molecule containing 3569 nucleotides. SV40 virus, which infects monkey and human cells, has a genetic complement of five genes in a circular double-stranded DNA molecule consisting of about 5000 nucleotide pairs. Large DNA molecules are measured in **kilobase pairs (kb)**, or thousands of base pairs. The genome of SV40 is about 5 kb. The more complex phages and animal viruses have as many as 250 genes and DNA molecules ranging from 50 to 300 kb. Bacterial genomes are substantially larger. For example, the chromosome of *E. coli* contains about 4000 genes in a DNA molecule composed of about 4700 kb.

Although the genomes of prokaryotes are composed of DNA, their DNA is not packaged into chromosomes. True chromosomes are found only in eukaryotes. The number of chromosomes is characteristic of the particular species, as we saw in Chapter 3. In moving up the evolutionary scale of animals or plants, the DNA content per haploid genome generally tends to increase, but there are many individual exceptions. The number of chromosomes shows no pattern. One of the smallest genomes in a multicellular animal is that of the nematode worm *Caenorhabditis elegans*, with a DNA content about 20 times that of the *E. coli* genome. The *D. melanogaster* and the human genomes have about 40 and 700 times as much DNA, respectively, as the *E. coli* genome. The genomes of some amphibians and fish are very large—many times the size of mammalian genomes. Such large genomes are measured in **megabase pairs (Mb)**, or millions of base pairs. The human genome is large, but it is by no means the largest among animals or higher plants. At 3000 Mb, the human genome is only 67 percent the size of that in corn and only 4 percent the size of that in the salamander *Amphiuma* (Table 6.1).

Among higher animals and plants, a large genome size does not imply a large number of genes. For example, the 3000-Mb human genome is large enough to con-

Table 6.1 Genome size of some representative viral, bacterial, and eukaryotic genomes

Genome	Approximate length in thousands of nucleotides	Form
Virus		
MS2	4	Single-stranded RNA
SV40	5	Circular double-stranded DNA
φX174	5 6	Circular single-stranded DNA; double-stranded replicative form
M13		
λ	50 152 165 267	Linear double-stranded DNA
Herpes simplex		
T2,T4,T6		
Smallpox		
Bacteria		
<i>Mycoplasma hominis</i>	760	Circular double-stranded DNA
<i>Escherichia coli</i>	4700	
Eukaryotes		Haploid chromosome number
<i>Saccharomyces cerevisiae</i> (yeast)	13,000	16
<i>Caenorhabditis elegans</i> (nematode)	100,000	6
<i>Arabidopsis thaliana</i> (wall cress)	100,000	5
<i>Drosophila melanogaster</i> (fruit fly)	165,000	4
<i>Homo sapiens</i> (human being)	3,000,000	23
<i>Zea Mays</i> (maize)	4,500,000	10
<i>Amphiuma sp.</i> (salamander)	76,500,000	14

tain perhaps 10^6 genes; however, various lines of evidence suggest that the number of genes is no greater than approximately 10^5 . Considering the number and size of proteins produced in human cells, it appears that no more than about 4 percent of the human genome actually codes for proteins. Similarly, although closely related species of salamanders are thought to have about the same number of genes, their genome size can differ by 30-fold in the total amount of DNA. Therefore, in higher animals and plants, the actual number of genes is much less than the theoretical maximum. The reason for the discrepancy is that in higher organisms, most of the DNA has functions other than coding for the amino acid sequence of proteins. (This issue is discussed further in Chapter 11).

A remarkable feature of the genetic apparatus of eukaryotes is that the enormous amount of genetic material contained in the nucleus of each cell is precisely divided in each cell division. A haploid human genome contained in a gamete has a DNA content equivalent to a linear DNA molecule 1 meter ($10^6 \mu\text{m}$) in length. The largest of the 23 chromosomes in the human genome contains a DNA molecule that is 82 mm ($8.2 \times 10^4 \mu\text{m}$) long. However, at metaphase of a mitotic division, the DNA molecule is condensed into a compact structure about $10 \mu\text{m}$ long and less than $1 \mu\text{m}$ in diameter. An analogy may be helpful in appreciating the prodigious feat of packaging that such chromosome condensation represents. If the DNA molecule in human chromosome 1 (the

longest chromosome) were a cooked spaghetti noodle 1 mm in diameter, it would stretch for 25 miles; in chromosome condensation, this noodle is gathered together, coil upon coil, until at metaphase it is a canoe-sized tangle of spaghetti 16 feet long and 2 feet wide. After cell division, the noodle is unwound again. Although the genomes of prokaryotes and viruses are much smaller than those of eukaryotes, they also are very compact. For example, an *E. coli* chromosome, which contains a DNA molecule about 1500 μm long, is contained in a cell about 2 μm long and 1 μm in diameter.

6.2—

The Supercoiling of DNA

The DNA of prokaryotic and eukaryotic chromosomes is **supercoiled**, which means that segments of double-stranded DNA are twisted around one another, analogous to the manner in which a telephone cord can be twisted around itself. The geometry of supercoiling can be illustrated by a simple example. Consider first a linear duplex DNA molecule whose ends are joined in such a way that each strand forms a continuous circle. Such a DNA molecule is called a **covalent circle**, and it is said to be **relaxed** if no twisting is present other than the helical twisting (Figure 6.1A). The individual polynucleotide strands of a relaxed circle form the usual right-handed (positive) helical structure with ten nucleotide pairs per turn of the helix. Suppose you were to cut one strand in a relaxed circle and unwind it one complete rotation of 360° so as to undo one complete turn of the double helix. When the ends were rejoined again, the result would be a circular helix that is "underwound." Because a DNA molecule has a strong tendency to maintain its standard helical form with ten nucleotide pairs per turn, the circular molecule would respond to the underwinding in one of two ways: (1) by forming regions with "bubbles" in which the bases are unpaired (Figure 6.1B) or (2) by twisting the circular molecule in the opposite sense from the direction of under-

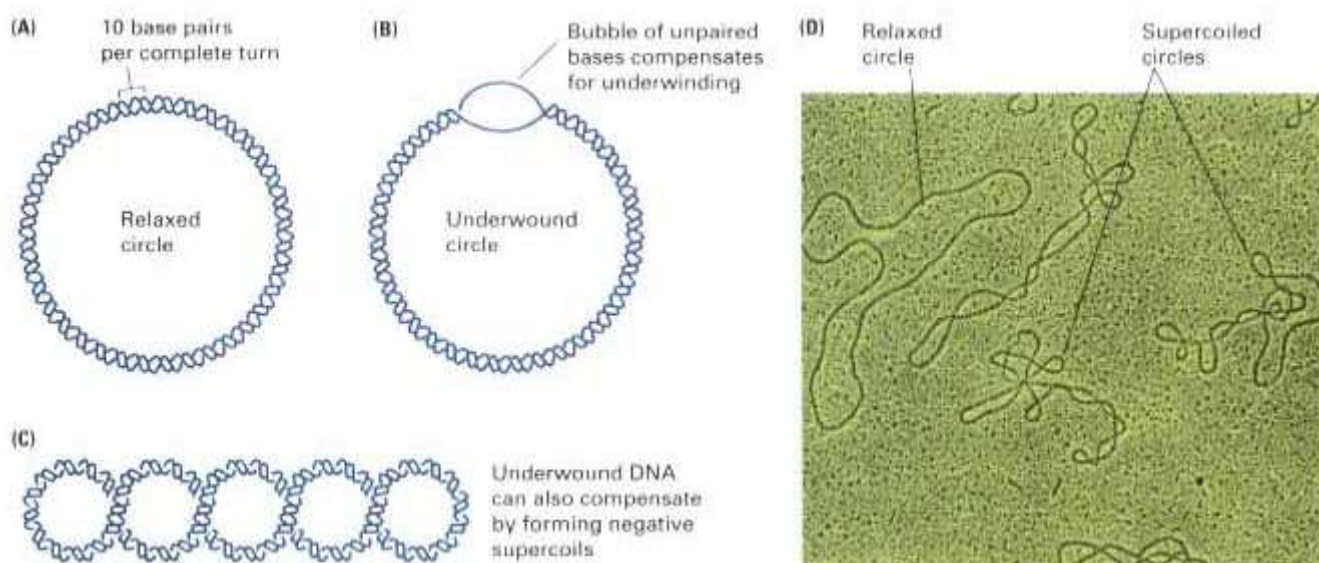


Figure 6.1

Different states of a covalent circle. (A) A nonsupercoiled (relaxed) covalent circle with 36 helical turns. (B) An underwound covalent circle with only 32 helical turns. (C) The molecule in part B, but with four twists to eliminate the underwinding. (D) Electron micrograph showing nicked circular and supercoiled DNA of phage PM2. Note that no bases are unpaired in part C. In solution, parts B and C would be in equilibrium [Electron micrograph courtesy of K. G. Murti.]

winding (Figure 6.1C). This twisting is called **supercoiling**, and a molecule with this sense of twisting is **negatively supercoiled**. Examples of supercoiled molecules are shown in Figure 6.1C and D. The two responses to underwinding are not independent, and underwinding is usually accommodated by a combination of the two processes: An underwound molecule contains some bubbles of unpaired bases and some supercoiling, with the supercoiling predominating. Although supercoiling occasionally plays a role in the expression of some genes, the overall biological function of supercoiling is unknown.

Topoisomerase Enzymes

The negative supercoiling of natural DNA molecules is produced by DNA topoisomerase enzymes (Chapter 5). The class of enzymes denoted **topoisomerase I** cause a single-stranded nick by breaking a phosphodiester bond in the backbone of one of the DNA strands; the nick produces a gap through which the intact DNA strand is passed, and then the nick is resealed. Depending on conditions, topoisomerase I enzymes can either increase or decrease the amount of supercoiling. As we saw in Section 5.4, topoisomerase I enzymes are essential to relieve the coiling produced by separation of the DNA strands in DNA replication.

A second class of topoisomerase enzymes is called **topoisomerase II**. These enzymes work by producing a double-stranded gap in one molecule through which another double-stranded molecule is passed. Therefore, topoisomerase II enzymes are able to pass one DNA duplex entirely through another or to separate two circular DNA molecules that are interlocked. The mechanism of action of topoisomerase II from the yeast *Saccharomyces cerevisiae* is illustrated in Figure 6.2. The molecular structure of the enzyme includes two sets of "jaws" set approximately at right angles (Figure 6.2A). The inner set clamps one of the duplexes, the outer set the other (Figure 6.2B and C). To allow the outer DNA molecule to pass completely through the inner one, the inner duplex is first cleaved, and then the outer duplex is passed through the gap (Figure 6.2D and E). After passage, the gap is repaired and both molecules are released (Figure 6.2F and G).

In a supercoiled DNA molecule free of proteins that maintain the supercoiling, any nick eliminates all supercoiling because the strain of underwinding is relaxed by free rotation of the intact strand about the sugar-phosphate bond opposite the break. Therefore, any treatment that nicks DNA relaxes the supercoiling. Single-stranded nicks can be produced by any of a variety of enzymes, such as **deoxyribonuclease (DNase)**, that cleaves sugar-phosphate bonds.

6.3—

The Structure of the Bacterial Chromosome

The chromosome of *E. coli* is a condensed unit called a **nucleoid** or **folded chromosome**, which is composed of a single circular DNA molecule and associated proteins. The term chromosome is a misnomer for this structure, because it is not a true "chromosome" in the sense of a eukaryotic chromosome. The most striking feature of the bacterial nucleoid is that the DNA is organized into a set of looped domains (Figure 6.3). As isolated from bacterial cells, the nucleoid contains, in addition to DNA, small amounts of several proteins, which are thought to be responsible in some way for the multiply looped arrangement of the DNA. The degree of condensation of the isolated nucleoid (that is, its physical dimensions) is affected by a variety of factors, and some controversy exists about the state of the nucleoid within the cell.

Figure 6.3 also shows that loops of the DNA of the *E. coli* chromosome are supercoiled. Note that some loops are not supercoiled; this is a result of the action of DNases during isolation, and it indicates that the loops are in some way independent of one another. In the preceding section, we stated that supercoiling is generally eliminated in a DNA molecule by one single-strand break. However, such a break in the *E. coli* chromosome does not

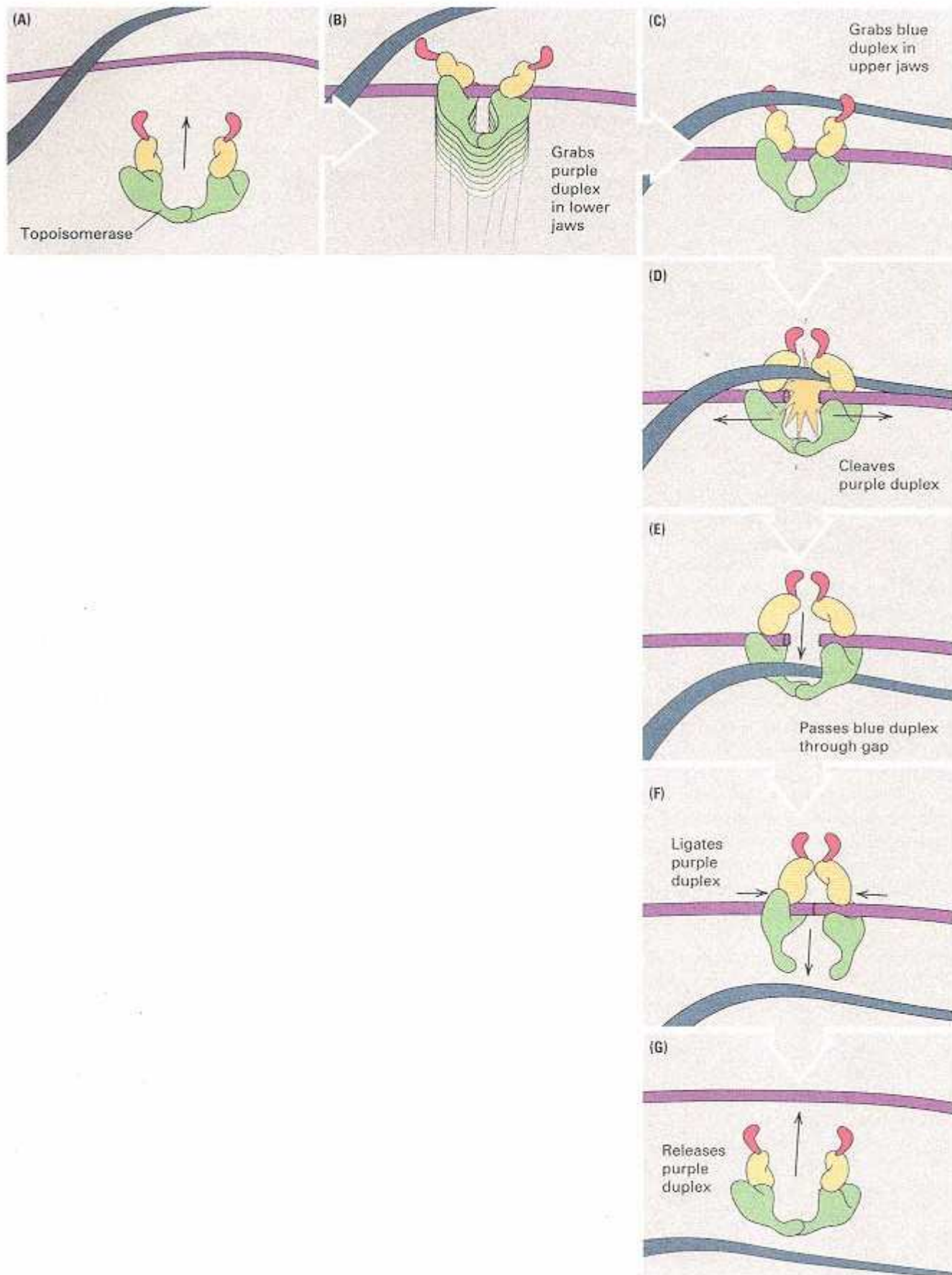


Figure 6.2

Topoisomerase II untangles a pair of DNA molecules by cleaving one DNA duplex and passing the other duplex through the gap. The enzyme illustrated here, from the yeast *Saccharomyces cerevisiae*, has two sets of "jaws." (A through C) The inner jaws (green) trap one duplex. The outer jaws (red) trap the other duplex. (D and E) The duplex in the inner jaws is cleaved, and the second, uncleaved duplex is passed through. (F) The uncleaved duplex is expelled through an opening that forms in the basal part of the enzyme. (G) Then the gap in the cleaved duplex is repaired, and it is released. [After J. M. Berger, S. J. Gamblin, S. C. Harrison and J. C. Wang. 1996. *Nature* 379:225.]

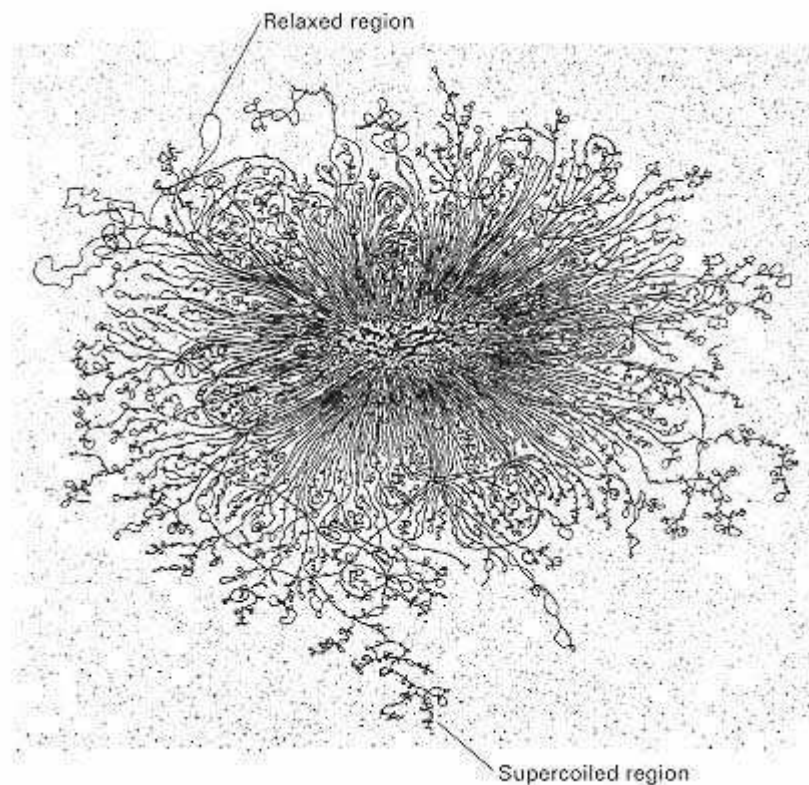


Figure 6.3
An electron micrograph of an *E. coli* chromosome showing the multiple loops emerging from a central region.
[Courtesy of Ruth Kavenoff. Copyright 1983 by Designergenes Posters Ltd.]

eliminate all supercoiling. If nucleoids, all of whose loops are supercoiled, are treated with a DNase and examined at various times after treatment, it is observed that supercoiling is relaxed in one loop at a time, not in all loops at once (Figure 6.4). The loops must be isolated from one another in such a way that rotation in one loop is not transmitted to other loops. The independence is probably the result of proteins that bind to the DNA in a way that prevents rotation of the helix.

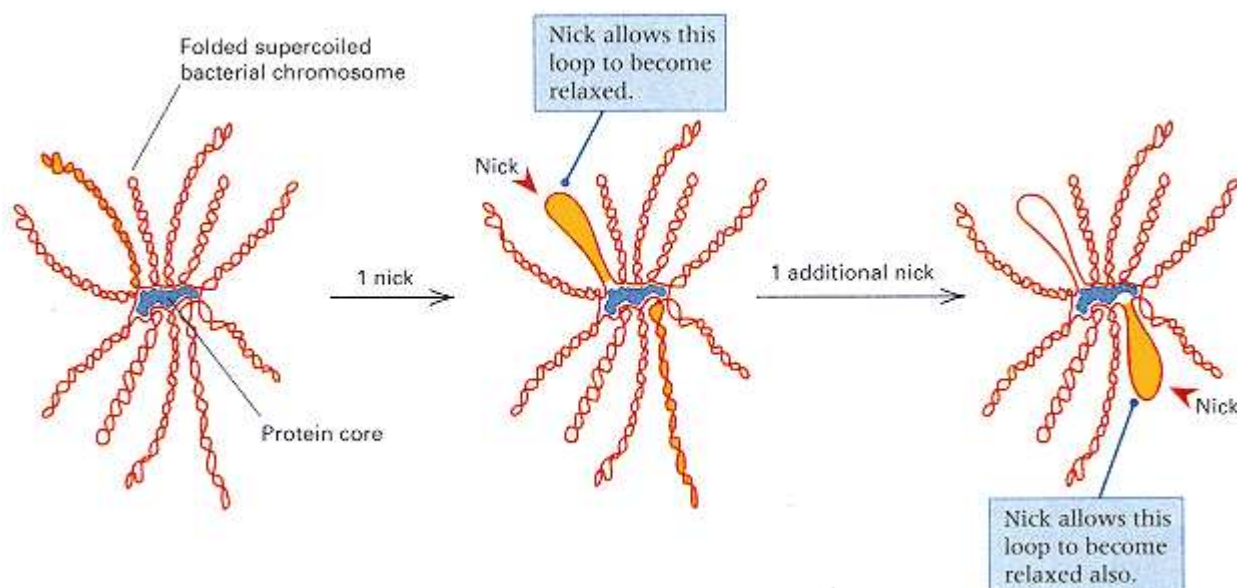


Figure 6.4
A schematic drawing of the folded supercoiled *E. coli* chromosome, showing 11 of the 40 to 50 loops attached to a protein core (blue shaded area) and the opening of loops by nicks.

6.4—

The Structure of Eukaryotic Chromosomes

A eukaryotic chromosome contains a single DNA molecule of enormous length. For example, the largest chromosome in the *D. melanogaster* genome has a DNA content of about 65,000 kb (6.5×10^7 nucleotide pairs), which is equivalent to a continuous linear duplex about 22 mm long. These long molecules usually fracture during isolation, but some fragments that are recovered are still very long. Figure 6.5 is an autoradiograph of radioactively labeled *Drosophila* DNA more than 36,000 kb in length.

In organisms such as baker's yeast, *Saccharomyces cerevisiae*, which have small genomes, the DNA molecules present in the chromosomes can be separated by special types of electrophoretic methods. In conventional electrophoresis, which we examined in Section 5.7, the electric field is maintained in a constant state, usually at constant voltage. The fragments move in

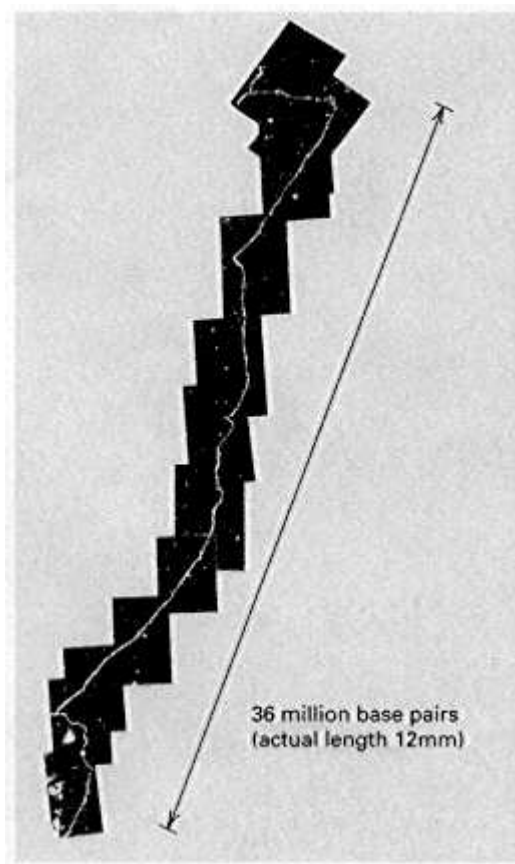


Figure 6.5
 Autoradiogram of a DNA molecule from *D. melanogaster*. The molecule is 12 mm long (approximately 36,000 kb).
 [From R. Kavenoff, L. C. Klotz, and B. H. Zimm. 1974. *Cold Spring Harbor Symp. Quant. Biol.*, 38:4.]

response to the field according to their size, smaller fragments moving faster. However, conventional electrophoresis can separate only molecules smaller than about 20 kb. All molecules larger than about 20 kb have the same electrophoretic mobility under these conditions and so form a single band in the gel. Simple modifications of the electrophoresis result in separation among much larger DNA fragments. The most common modifications alter the geometry of the electric field at periodic intervals during the course of the electrophoretic separation. Some electrophoretic apparatuses alternate the electric field between two sets of electrodes oriented at right angles or among three sets of electrodes forming a hexagon. In one type of large-fragment electrophoresis, called **pulsed-field gel electrophoresis (PFGE)**, the apparatus is similar to that for conventional electrophoresis, but the orientation of the electric field is changed periodically; the improved separation of large DNA molecules apparently results from the additional time it takes for large molecules to reorient themselves when the orientation of the electric field is changed. Figure 6.6 illustrates the separation of the 16 chromosomes from yeast by means of PFGE. The chromosomes range in size from approximately 200 kb to 2.2 Mb. Electrophoretic separation yields a

visual demonstration that each of the chromosomes contains a single DNA molecule that runs continuously throughout its length. Using the methods discussed in Section 5.7, we can determine the identity of each chromosomal band in the gel by hybridization of a Southern blot with probes for genes known to map to a particular chromosome.

The Nucleosome Is the Basic Structural Unit of Chromatin

The DNA of all eukaryotic chromosomes is associated with numerous protein molecules in a stable, ordered aggregate called **chromatin**. Some of the proteins present in chromatin determine chromosome structure and the changes in structure that occur during the division cycle of the cell. Other chromatin proteins appear to play important roles in regulating chromosome functions.

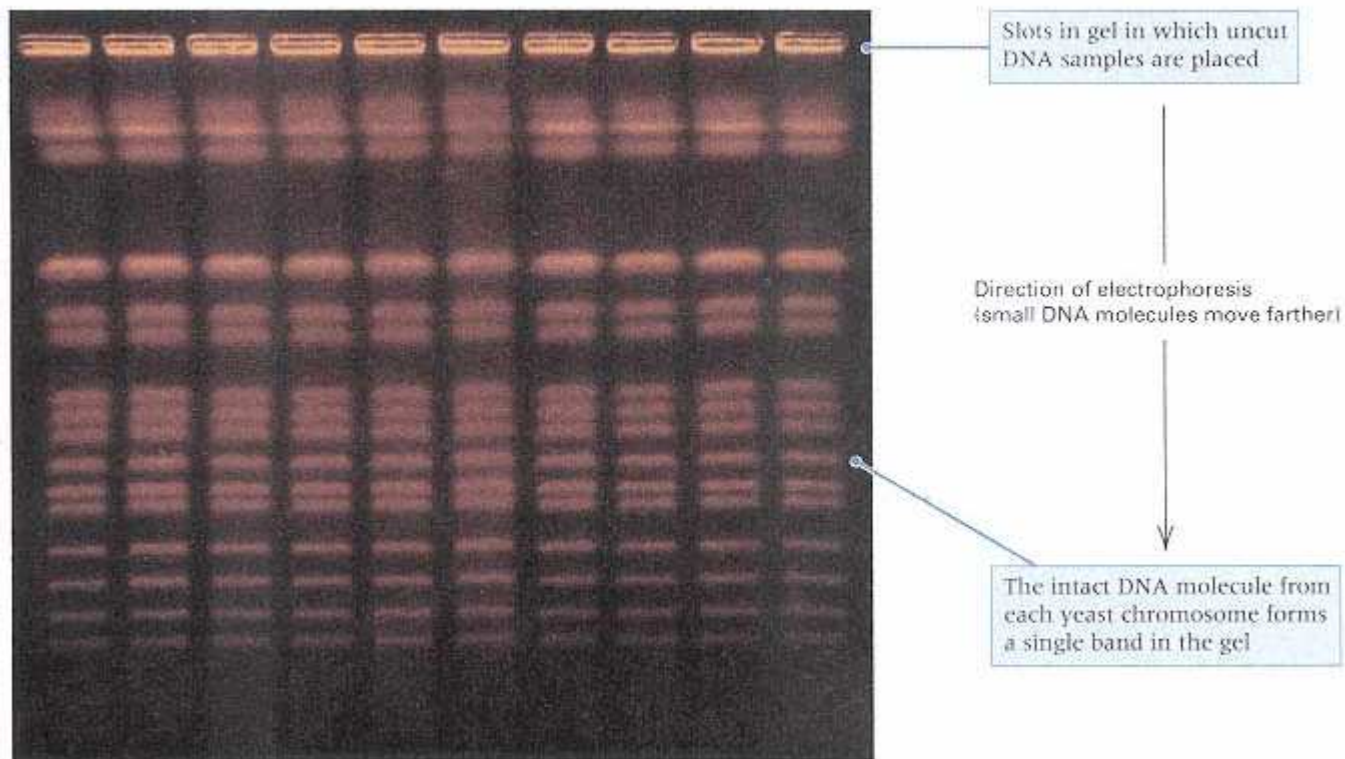


Figure 6.6

Separation of the 16 chromosomes of yeast by pulsed-field gel electrophoresis, in which there is regular change in the orientation of the electric field. Bands for each of the chromosomes are clearly visible. They range in size from 200 kb to 2.2 Mb.
[Courtesy of BioRad Laboratories, Hercules, California.]

Nucleosome Core Particles

The simplest form of chromatin is present in nondividing eukaryotic cells, when chromosomes are not sufficiently condensed to be visible by light microscopy. Chromatin isolated from such cells is a complex aggregate of DNA and proteins. The major class of proteins comprises the **histone** proteins. Histones are largely responsible for the structure of chromatin. Five major types—**H1, H2A, H2B, H3, and H4**—are present in the chromatin of nearly all eukaryotes in amounts about equal in mass to that of the DNA. Histones are small proteins (100–200 amino acids) that differ from most other proteins in that from 20 to 30 percent of the amino acids are lysine and arginine, both of which have a positive charge. (Only a few percent of the amino acids of a typical protein are lysine and arginine.) The positive charges enable histone molecules to bind to DNA, primarily by electrostatic attraction to the negatively charged phosphate groups in the sugar-phosphate back-bone of DNA. Placing chromatin in a solution with a high salt concentration (for example, 2 molar NaCl) to eliminate the electrostatic attraction causes the histones to dissociate from the DNA. Histones also bind tightly to each other; both DNA-histone and histone-histone binding are important for chromatin structure.

The histone molecules from different organisms are remarkably similar, with the exception of H1. In fact, the amino acid sequences of H3 molecules from widely different species are almost identical. For example, the sequences of H3 of cow chromatin and pea chromatin differ by only 4 of 135 amino acids. The H4 proteins of all organisms also are quite similar; cow and pea H4 differ by only 2 of 102 amino acids. There are few other proteins whose amino acid sequences vary so little from one species to the next. When the variation between organisms is very small, we say that the sequence is highly **conserved**. The extraordinary conservation in histone composition through hundreds of millions

of years of evolutionary divergence is consistent with the important role of these proteins in the structural organization of eukaryotic chromosomes.

In the electron microscope, chromatin resembles a regularly beaded thread (Figure 6.7). The bead-like units in chromatin are called **nucleosomes**. The organization of the nucleosomes in chromatin is illustrated in Figure 6.8A. Each unit has a definite composition, consisting of two molecules each of H2A, H2B, H3, and H4, a segment of DNA containing about 200 nucleotide pairs, and one molecule of histone H1. The complex of two subunits each of H2A, H2B, H3, and H4, as well as part of the DNA, forms each "bead," and the remaining DNA bridges between the beads. Histone H1 also appears to play a role in bridging between the beads, but it is not shown in Figure 6.8A.

Brief treatment of chromatin with certain DNases (for example, **micrococcal nuclease** from the bacterium *Staphylococcus aureus*) yields a collection of small particles of quite uniform size consisting only of histones and DNA (Figure 6.9). The DNA fragments in these particles are of lengths equal to about 200 nucleotide pairs or small multiples of that unit size (the precise size varies with species and tissue). These particles result from cleavage of the linker DNA segments between the beads (Figure 6.8B). More extensive treatment with DNase results in loss of the H1 histone and digestion of all the DNA except that protected by the histones in the bead. The resulting structure, called a **core particle**, consists of an octamer of pairs of H2A, H2B, H3, and H4, around which the remaining DNA, approximately 145 base pairs, is wound in about one and three-fourths turns (Figure 6.8B). Each nucleosome is composed of a core particle, additional DNA called **linker DNA** that links adjacent core particles (the linker DNA is removed by extensive nuclease digestion), and one molecule of H1; the H1 binds to the histone octamer and to the linker DNA, causing the linkers extending from both sides of the core particle to cross and draw nearer to the octamer, though some of the linker DNA does not come into contact with any histones. The size of the linker ranges from 20 to 100 nucleotide pairs for different species and even in different cell types in the same organism (200 - 145 = 55 nucleotide pairs is usually considered an average size). Little is known about the structure of the linker DNA or about whether it has a special genetic function, and the cause of the variation in its length is also unknown.

The Arrangement of Chromatin Fibers in a Chromosome

The DNA molecule of a chromosome is folded and folded again in such a way that it is convenient to think of chromosomes as having several hierarchical levels of organization, each responsible for a particular degree of shortening of the enormously long strand (Figure 6.10). Assembly of DNA and histones can be considered the first level—namely, a sevenfold reduction in length of the DNA and the formation of a beaded flexible fiber 110 Å (11 nm) wide (Figure 6.10B), roughly five times the width of free DNA (Figure 6.10A). The structure of chromatin varies with the concentration of salts, and the 110 Å fiber is present only when the salt concentration is quite low. If the salt concentration is increased slightly, then the fiber becomes shortened somewhat by forming a zigzag arrangement of closely spaced beads between which the linking DNA is no longer visible in electron micrographs. If



Figure 6.7
Dark-field electron micrograph of chromatin showing the beaded structure at low salt concentration. The beads have diameters of about 100 Å.
[Courtesy of Ada Olins.]

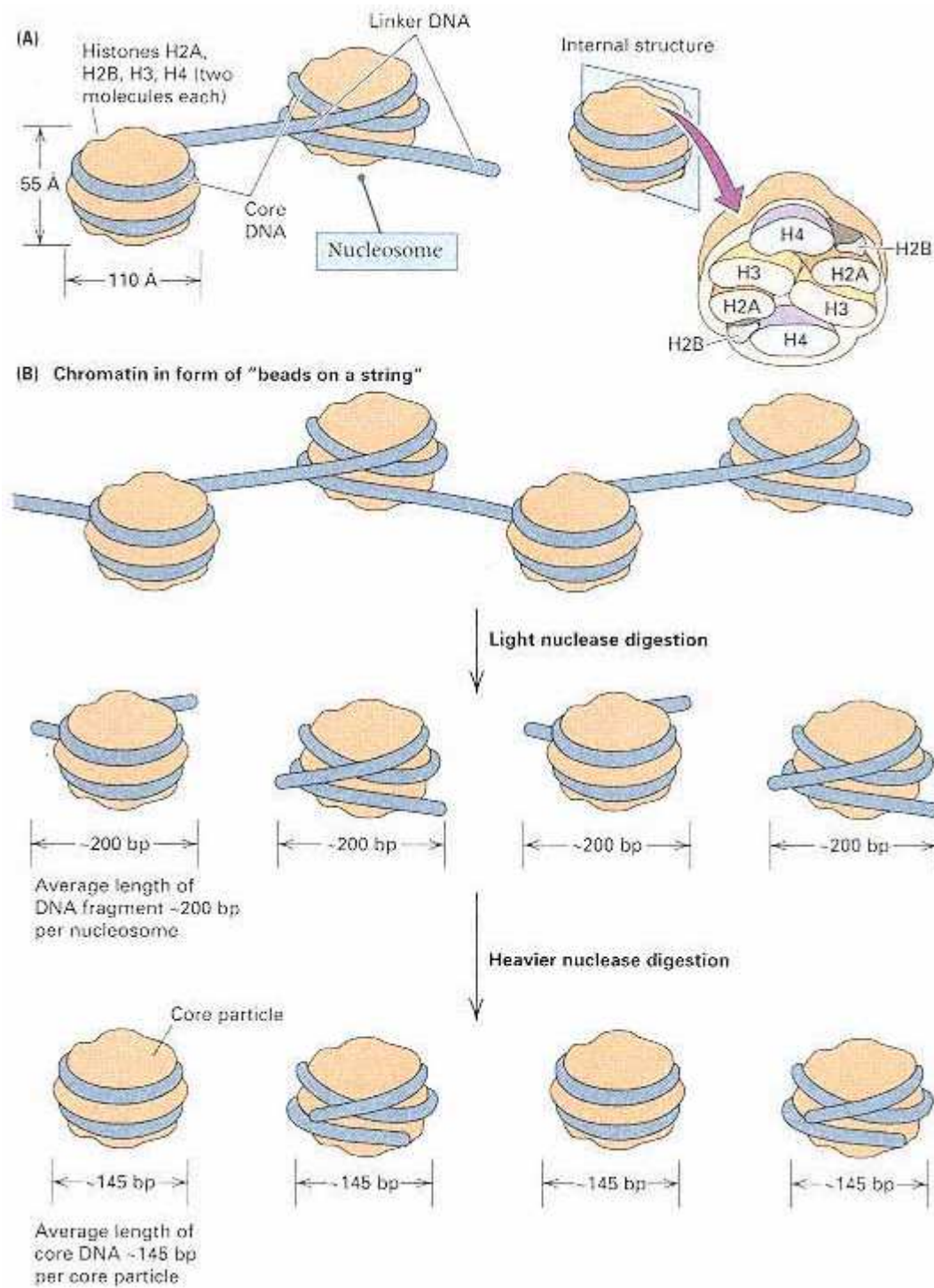


Figure 6.8

(A) Organization of nucleosomes. The DNA molecule is wound one and three-fourths turns around a histone octamer called the core particle. If H1 were present, it would bind to the octamer surface and to the linkers, causing the linkers to cross. (B) Effect of treatment with micrococcal nuclease. Brief treatment cleaves the DNA between the nucleosomes and results in core particles associated with histone H1 and approximately 200 base pairs of DNA. More extensive treatment results in loss of H1 and digestion of all but 145 base pairs of DNA in intimate contact with each core particle.



Figure 6.9

Electron micrograph of nucleosome
monomers.
[Courtesy of Ada Olins.]

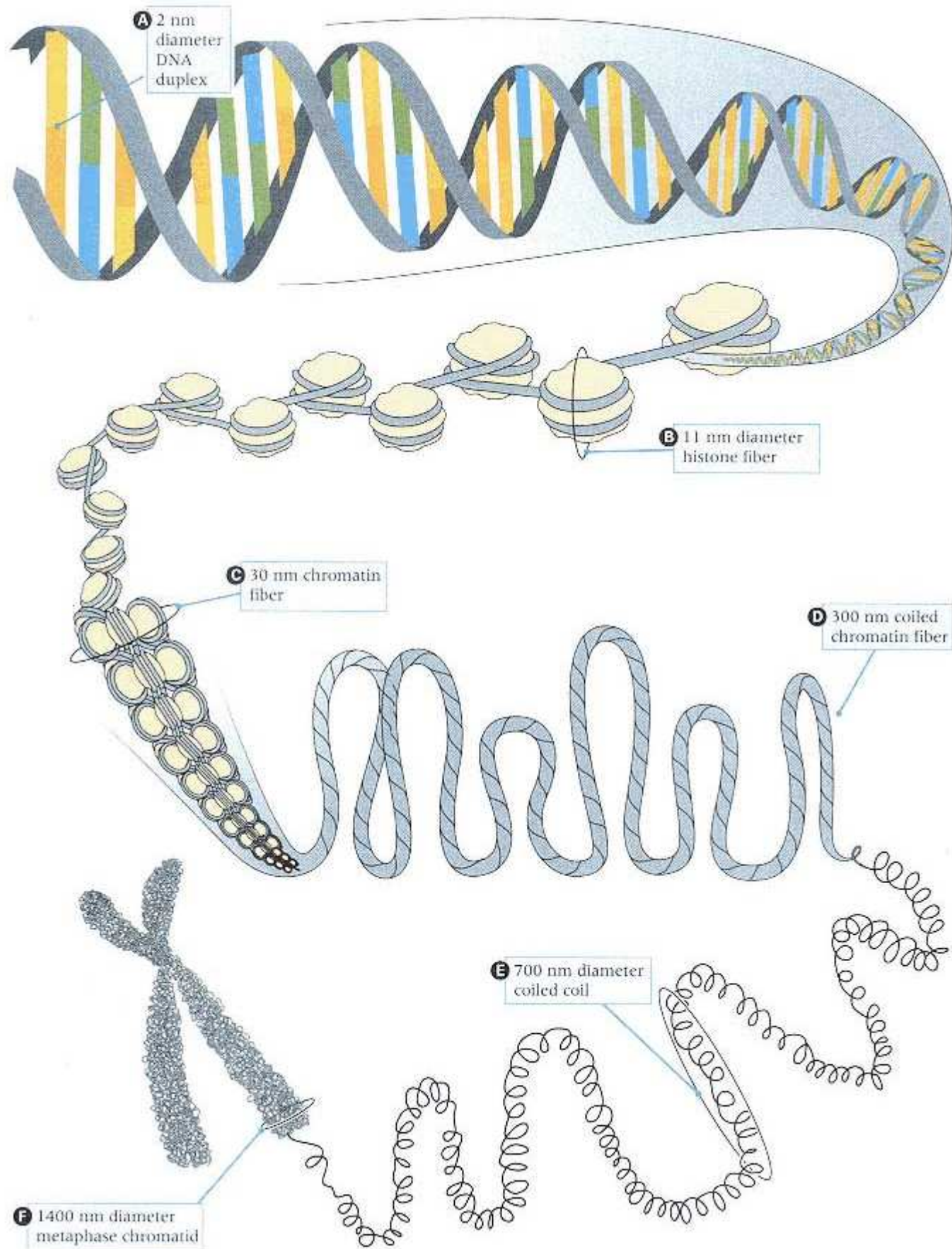


figure 6.10

Various stages in the condensation of DNA (A) and chromatin (B through E) in forming a metaphase chromosome (F). The dimensions indicate known sizes of intermediates, but the detailed structures are hypothetical.

the salt concentration is further increased to that present in living cells, then a second level of compaction occurs: the organization of the 110 Å nucleosome fiber into a shorter, thicker fiber with an average diameter ranging from 300 to 350 Å, called the **30 nm fiber** (Figure 6.10C). In forming this structure, the 110 Å fiber apparently coils in a somewhat irregular left-handed superhelix or solenoidal supercoil with six nucleosomes per turn (Figure 6.11). It is believed that most intracellular chromatin has the solenoidal supercoiled configuration.

The final level of organization is that in which the 30 nm fiber condenses into a chromatid of the compact metaphase chromosome (Figure 6.10D through F). Little is known about this process other than that it seems to proceed in stages. In electron micrographs of isolated metaphase chromosomes from which histones have been removed, the partly unfolded DNA has the form of an enormous number of loops that seem to extend from a central core, or **scaffold**, composed of nonhistone chromosomal proteins (Figure 6.12). Electron microscopic studies of chromosome condensation in mitosis and meiosis suggest that the scaffold extends along the chromatid and that the 30 nm fiber becomes arranged into a helix of loops radiating from the scaffold. Details are not known about the additional folding that is required of the fiber in each loop to produce the fully condensed metaphase chromosome.

The genetic significance of the compaction of DNA and protein into chromatin and ultimately into the chromosome is that it greatly facilitates the movement of the genetic material during nuclear division. Relative to a fully extended DNA molecule, the length of a metaphase chromosome is reduced by a factor of approximately 10^4 as a result of chromosome condensation. Without chromosome condensation, the chromosomes would become so entangled

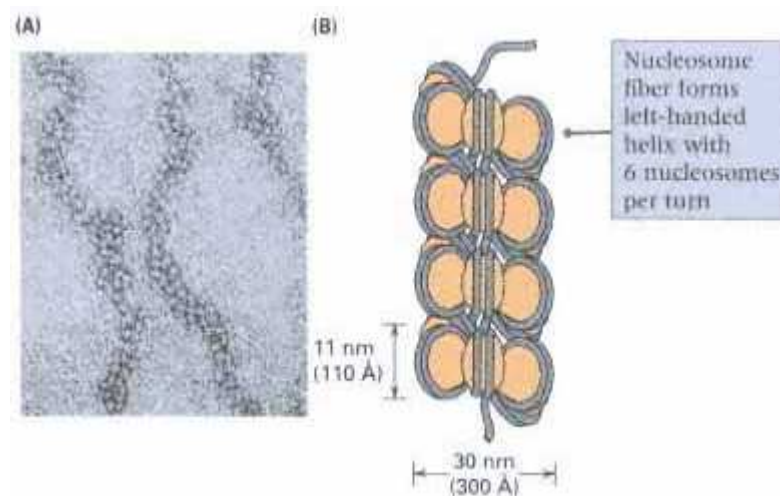


Figure 6.11

(A) Electron micrograph of the 30 nm component of mouse metaphase chromosomes. (B) A proposed solenoidal model of chromatin. The DNA (blue-gray) is wound around each nucleosome. It is unlikely that the real structure is so regular.

[Electron micrograph courtesy of Barbara Hamkalo; drawing after J. T. Finch and A. Klug, 1976. *Proc. Nat. Acad. Sci. USA*, 73: 1900.]

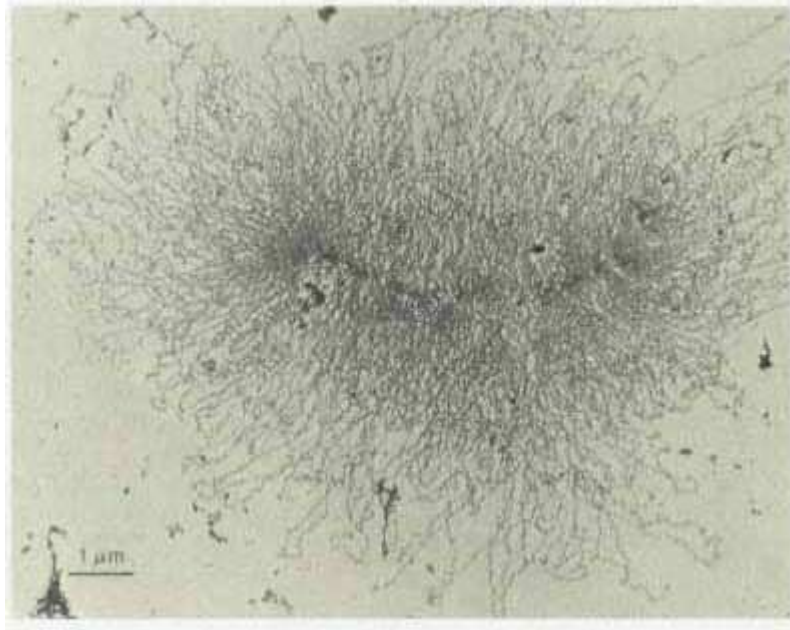


Figure 6.12

Electron micrograph of a partially disrupted anaphase chromosome of the milkweed bug *Oncopeltus fasciatus*, showing multiple loops of 30-nm chromatin at the periphery.

[From V. Foe, H. Forrest, L. Wilkinson, and C. Laird. 1982. *Insect Ultrastructure*, 1:222.]

that there would be many more abnormalities in the distribution of genetic material into daughter cells.

6.5—

Polytene Chromosomes

A typical eukaryotic chromosome contains only a single DNA molecule. However, in the nuclei of cells of the salivary glands and certain other tissues of the larvae of *Drosophila* and other two-winged (dipteran) flies, there are giant chromosomes, called **polytene chromosomes**, that contain about 1000 DNA molecules laterally aligned. Each of these chromosomes has a length and cross-sectional diameter many times greater than those of the corresponding chromosome at mitotic metaphase in ordinary somatic cells, as well as a constant and distinctive pattern of transverse banding (Figure 6.13). The polytene structures are formed by repeated replication of the DNA in a closely synapsed pair of homologous chromosomes without separation of the replicated chromatin strands or of the two chromosomes. Polytene chromosomes are atypical chromosomes and are formed in "terminal" cells; that is, the larval cells containing them do not divide and are eliminated in the formation of the pupa. Although they do not contribute to the tissues in the adult fly, the polytene tissues of larvae have been especially valuable in the genetics of *Drosophila*, as will become apparent in Chapter 7.

In polytene nuclei of *D. melanogaster* and other dipteran species, large blocks of heterochromatin adjacent to the centromeres are aggregated into a single compact mass called the **chromocenter**. Because the two largest chromosomes in *Drosophila* (chromosomes numbered 2 and 3) have centrally located centromeres, the chromosomes appear in the configuration shown in Figure 6.14: The paired X chromosomes (in a female), the left and right arms of chromosomes 2 and 3, and a short chromosome (chromosome 4) project from the chromocenter. In a male, the Y chromosome, which consists almost entirely of heterochromatin, is incorporated in the chromocenter.

The darkly staining transverse bands in polytene chromosomes have about a tenfold range in width. These bands result from the side-by-side alignment of tightly folded regions of the individual chromatin strands that are often visible in mitotic and meiotic prophase chromosomes as chromomeres. More DNA is present within the bands than in the interband (lightly stained) regions. About 5000 bands have been identified in the *D. melanogaster* polytene chromosomes. This linear array of bands, which has a pattern that is constant and characteristic for each species, provides a finely detailed **cytological map** of the chromosomes. The banding pattern is such that observers with sufficient training and experience can identify short regions in any of the chromosomes (Figure 6.14).

Because of their large size and finely detailed morphology, polytene chromosomes are exceedingly useful for a process called *in situ* nucleic acid hybridization. In the procedure of *in situ* hybridization, nuclei

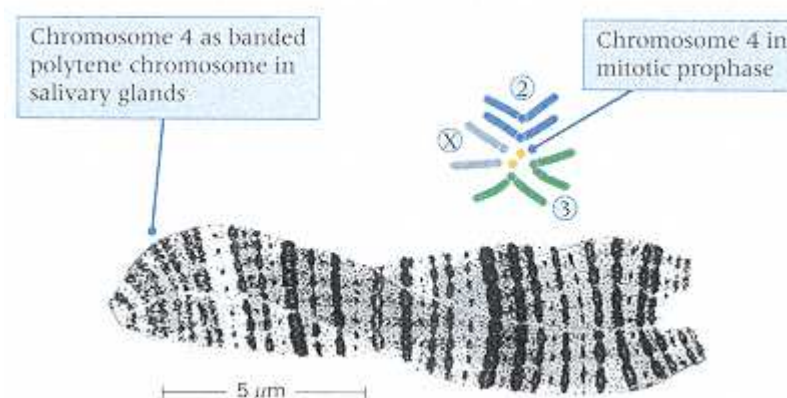


Figure 6.13

The polytene fourth chromosome of *Drosophila melanogaster* adhering to the chromocenter (positioned to the left, not shown). The somatic chromosomes of *Drosophila*, drawn to scale with respect to the polytene fourth chromosome, are shown at the upper right as they appear in mitotic prophase.

[From C. Bridges. 1935. *J. Heredity*, 26: 60.]

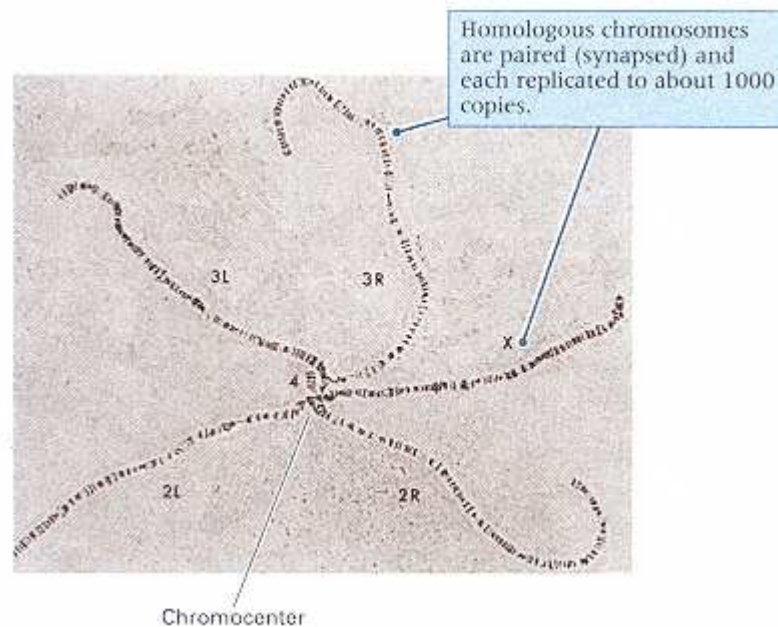


Figure 6.14
Polytene chromosomes from a larval salivary gland cell of *Drosophila melanogaster*. The chromocenter is the central region in which the centromeric regions of all chromosomes are united.

[Courtesy of George Lefevre.]

containing polytene chromosomes are squashed and the chromosomal DNA denatured, after which labeled probe DNA or RNA is added under conditions that favor renaturation. After washing, the only probe that remains in the chromosomes has formed hybrid duplexes with chromosomal DNA, and its position can be identified cytologically (Figure 6.15).

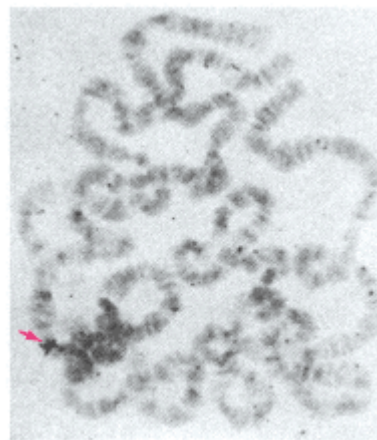


Figure 6.15
Autoradiogram of *Drosophila melanogaster* polytene chromosomes hybridized in situ with radioactively labeled RNA copied from the histone genes, showing hybridization to a particular region (arrow). This region identifies the position of the histone genes in the chromosomes.
[Courtesy of Mary Lou Pardue.]

6.6— Repetitive Nucleotide Sequences in Eukaryotic Genomes

In bacteria, the variation in average base composition from one part of the genome to another is quite small. However, in eukaryotes, some components of the genome can be detected because their base composition is quite

different from the average of the rest of the genome. For example, one component of crab DNA is only 3 percent G + C, compared to an average of approximately 50 percent G + C, in the rest of the genome. The components with unusually low or unusually high G + C contents are called **satellite DNA**. In the mouse, satellite DNA accounts for about 10 percent of the genome. A striking feature of satellite DNA is that it consists of fairly short nucleotide sequences that may be repeated *as many as a million times in a haploid genome*. Other **repetitive sequences** also are present in eukaryotic DNA. Because repetitive DNA consists of many highly similar or identical sequences, fragments of repetitive DNA renature more readily than fragments of nonrepetitive DNA. Information about the size of repeated sequences and the number of copies of a particular sequence can be obtained through studies of the rate of renaturation. The quantitative analysis of DNA renaturation is considered next.

Kinetics of DNA Renaturation

As we saw in Chapter 5, the rate-limiting step in the renaturation of separated DNA strands is the initial collision between two complementary single strands. Because the chance of initial collision is concentration-dependent, the rate of reassociation increases with DNA concentration (Figure 6.16A). Any increase in DNA concentration results in a corresponding increase in the number of potential pairing partners for a given strand. The study of DNA renaturation has contributed greatly to our understanding of the number and types of DNA sequences present in various genomes.

To illustrate the analysis of reassociation, we begin by comparing the renaturation rates of solutions of DNA molecules from the bacteriophages T7 and T4, which have no common base sequences and different molecular weights, and the T7 genome is smaller than that of T4. In solutions in which the number of grams of DNA per milliliter is the same, the molar concentration of T7 is greater than that of T4. Thus if each solution is separately denatured and renatured, the molecules of T7 will renature more rapidly than those of T4. If the two solutions are instead mixed, the T7 and T4 will renature independently of one another (because they are not homologous), and a curve such as that in Figure 6.16B will be obtained. Note that the curve consists of two steps, one for the more rapidly renaturing T7 molecules and the other for the T4 molecules. Each step in the curve accounts for half of the change in the absorption of the solution at 260 nanometers (A_{260}) because the initial concentration (in μg of DNA/ml, which is proportional to A_{260}) of each type of molecule was the same.

As is apparent in Figure 6.16A, the renaturation curve of a molecule that contains no repeating base sequences consists of only a single step, and the rate of renaturation is a function of the size of the molecule. If such molecules are fragmented into many components of equal size, then the molar concentration of each component will be the same as that of the unbroken

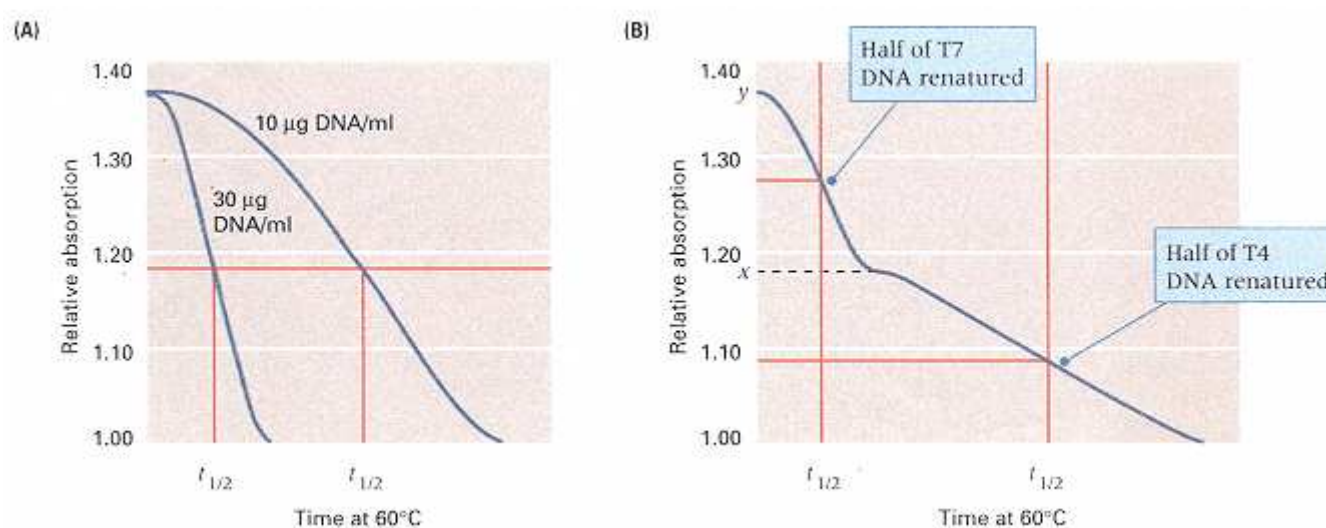


Figure 6.16

- (A) Dependence of renaturation time on the concentration of phage T7 DNA. After a period at 90°C to separate the strands, the DNA was cooled to 60°C. Renaturation is complete when the relative absorption reaches 1. (B) Renaturation of a mixture of T4 and T7 DNA, each at the same temperature. Extrapolation (black dashed line) yields the early portion of the T4 curve; the ratio of the absorptions at points x and y yields the fraction of the total DNA that is T4 DNA. The times required for half-completion of renaturation, $t_{1/2}$, are obtained by drawing the red horizontal lines, which divide each curve equally in the vertical direction, and then extending the red vertical lines to the time axis.

molecule, so the renaturation rate should be unchanged by fragmentation. The rate actually decreases somewhat, but the important point is that *if a DNA molecule contains only a single nonrepetitive sequence of base pairs, then breakage of the molecules does not yield a renaturation curve consisting of steps*. In contrast, if a molecule with an overall sequence that includes both a unique component and several copies of different repeated sequences is fragmented, *fragments containing the more numerous repeated sequences will renature more rapidly than the fragments containing portions of the unique sequence*. For example, consider a molecule containing 50,000 base pairs and consisting of 100 copies of a tandemly repeated sequence of 500 base pairs. If the molecules are broken into about 100 fragments of roughly equal size, each fragment will be about 500 base pairs. Although the renaturation curve for the fragments will have a single step, the renaturation rate will be characteristic of molecules 500 nucleotides in length. If a genome contains multiple families of repeated sequences whose abundances differ, the renaturation curve will have steps—one step for each repeating sequence. This principle is the basis of the analysis of renaturation kinetics.

Renaturation kinetics can be described in a simple mathematical form, because the reaction is one in which the ratelimiting step is the initial collision of two molecules. In such a case, the fraction of single strands remaining denatured at a time t after the start of renaturation is given by the expression

$$\frac{C}{C_0} = \frac{1}{1 + kC_0t}$$

in which C is the concentration of single-stranded DNA in moles of nucleotide per liter, C_0 is the initial concentration, and k is a constant. The expression C_0t is commonly called **cot**, and a plot of C/C_0 versus C_0t is called a **cot curve**. When renaturation is half completed, $C/C_0 = 1/2$ and

$$C_0t_{1/2} = 1/k$$

The value of $1/k$ depends on experimental conditions, but for a particular set of conditions, the value is proportional to the number of bases in the renaturing sequences. The longer the sequence, the greater will be the time to achieve half-complete renaturation for a particular starting concentration (because the number of molecules will be smaller). The equation just stated applies to which we will return. If a molecule consists of several subsequences, then one needs to know C_0 for each subsequence, and a set of values of $1/k$ will be obtained (one for each step in the renaturation curve), each value depending on the length of the subsequence. What is meant by the length of the sequence that determines the rate is best described by example. A DNA molecule containing only adenine in one strand (and thymine in the other) has a repeating length of 1. The repeating tetranucleotide . . . GACTGACT . . . has a repeating length of 4. A nonrepeating DNA molecule containing n nucleotide pairs has a unique length of n .

Experimentally, the number of bases per repeating unit is not determined directly. Generally, renaturation curves for a series of molecules of known molecular weight *with no repeating elements in their sequences* (and hence yielding one-step renaturation curves) serve as standards. Molecules composed of short repeating sequences are also occasionally used. A set of curves of this kind is shown in Figure 6.17. Note that two of these simple curves represent the entire genomes of *E. coli* and phage T4. With the standard conditions for Cot analysis used to obtain this set of curves, the sequence length N (in base pairs) that yields a particular value of $C_0t_{1/2}$ is

$$N = (5 \times 10^5) C_0t_{1/2}$$

in which t is in seconds, C_0 is in nucleotides per liter, and 5×10^5 is a constant dependent on the conditions of renaturation. Through this formula, the experimentally determined value of Cot ($C_0t_{1/2}$) yields an estimate of the repeat length, N , of a repetitive sequence. Note again that C_0 is not the overall DNA concentration but the concentration of the individual sequence producing a particular step in a curve. How one obtains the necessary value of C_0 will become clear when we analyze a Cot curve. Such an analysis begins by first noting the number of steps in the curve (each step of which represents a sequence or class of sequences of a particular length)

and the fraction of the material represented by each step. The observed value of $C_0t_{1/2}$ for each step must be corrected by first inferring the value of C_0 for each sequence class. The lengths of the sequences are then determined from these corrected values by comparison to standards, and the sequence lengths and sequence abundances, as a proportion of the total, are compared to obtain the number of copies of each sequence. This analysis is best understood by looking at an example, which we will do next.

Analysis of Genome Size and Repetitive Sequences by Renaturation

Figure 6.18 shows a Cot curve typical of those obtained in analyses of the renaturation kinetics of eukaryotic genomes. Three discrete steps are evident: 50 percent of the DNA has $C_0t_{1/2} = 10^3$, 30 percent has $C_0t_{1/2} = 10^0 = 1$, and 20 percent has $C_0t_{1/2} = 10^{-2}$. The scale at the top of the figure was obtained from Cot analysis of molecules that have unique sequences of known lengths, as in Figure 6.17. The sequence sizes cannot be determined directly from the observed $C_0t_{1/2}$ values, because each value of C_0 used in plotting the horizontal axis in the body of the figure is the total DNA concentration in the renaturation mixture. Multiplying each $C_0t_{1/2}$ value by the fraction of the total DNA that it represents yields the necessary corrected $C_0t_{1/2}$ values (that is, 0.50×10^3 , 0.30×1 , and 0.20×10^{-2}). From the size scale at the top of Figure 6.18, the corresponding sequence sizes (repeat lengths) are approximately 3.0×10^8 , 2.2×10^5 , and 1×10^3 base pairs, respectively.

To determine the number of copies of each sequence, we make use of the fact that the number of copies of a sequence having a particular renaturation rate is inversely proportional to $t_{1/2}$ and hence to the observed (uncorrected) $C_0t_{1/2}$ values for each class. Thus if the haploid genome contains only one copy of the longest sequence (3.0×10^8 base pairs), it contains 10^3 copies of the sequence (or sequences) of length 2.2×10^5 base pairs and 10^5 copies of the sequence (or sequences) of length 1×10^3 base pairs. An estimate of

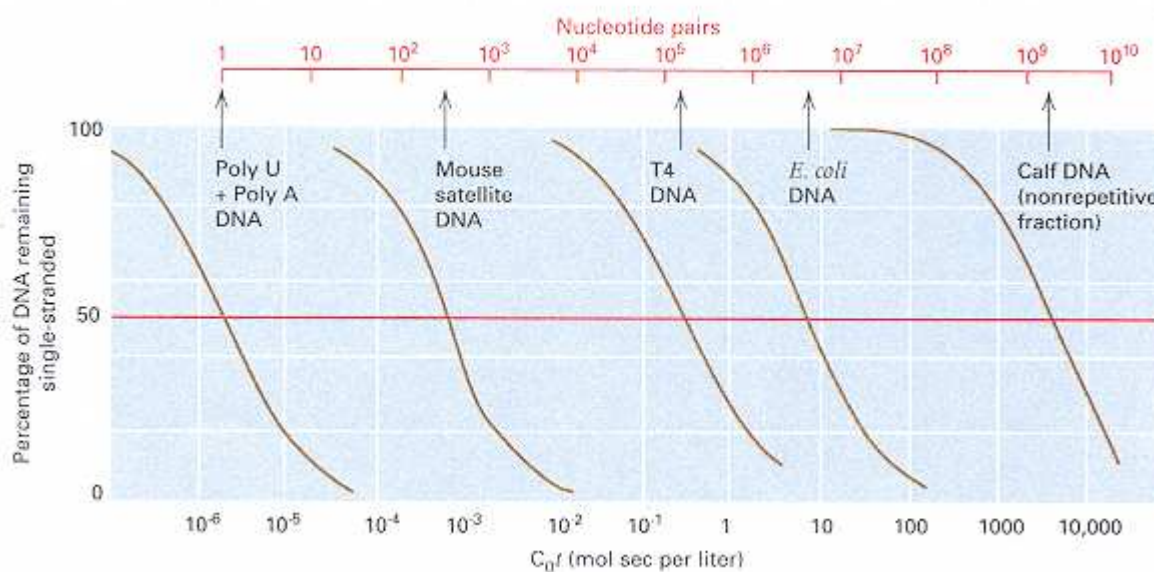


Figure 6.17

A set of Cot curves for various DNA samples. The black arrows pointing up to the red scale indicate the number of nucleotide pairs for each sample; they align with the intersection of each curve with the horizontal red line (the point of half-renaturation, or $C_0t_{1/2}$). The y axis on this graph can be related to that on Figure 6.16 by noting that maximum absorption represents totally single-stranded DNA and minimum absorption represents totally double-stranded DNA.

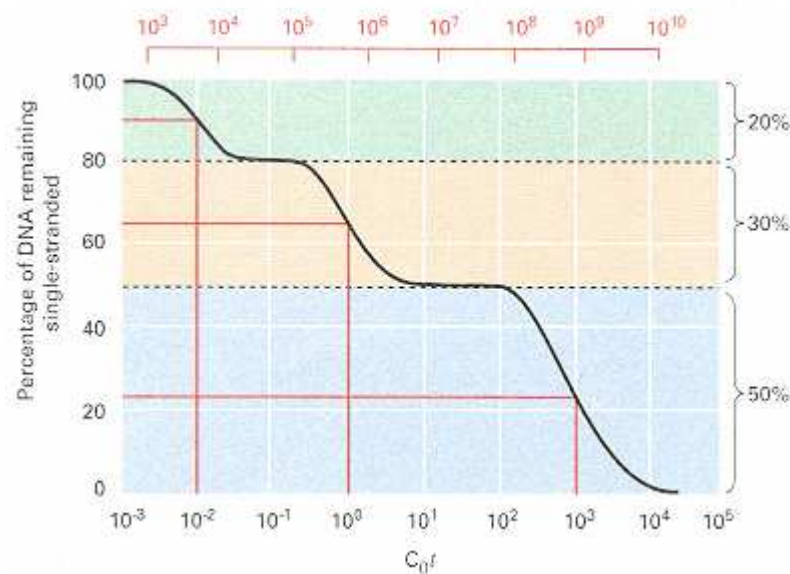


Figure 6.18

The C_{0t} curve analyzed in the text. The scale at the top is the same as that in Figure 6.17. The black dashed lines indicate the fractional contribution of each class of molecules to the total DNA.

the total number of base pairs per genome would in this case be

$$(3.0 \times 10^8) + (10^3 \times 2.2 \times 10^5) + (10^5 \times 1 \times 10^3) = 6.2 \times 10^8 \text{ base pairs}$$

The different sequence components of a eukaryotic DNA molecule can be isolated by procedures that recover the double-stranded molecules formed at different times during a renaturation reaction. The method used is to allow renaturation to proceed only to a particular $C_{0t_{1/2}}$ value. The reassociated molecules present at that point are then separated from the remaining single stranded molecules, usually by passing the solution through a tube filled with a form of calcium phosphate crystal (hydroxylapatite) that preferentially binds double-stranded DNA.

6.7—

Nucleotide Sequence Composition of Eukaryotic Genomes

The nucleotide sequence composition of many eukaryotic genomes has been examined via analysis of the renaturation kinetics of DNA. The principal finding is that eukaryotic organisms differ widely in the proportion of the genome that consists of repetitive DNA sequences and in the types of repetitive sequences that are present. In most eukaryotic genomes, the DNA consists of three major components:

- *Unique, or single-copy, sequences* This is usually the major component and typically comprises from 30 to 75 percent of the chromosomal DNA in most organisms.
- *Highly repetitive sequences* This component constitutes from 5 to 45 percent of the genome, depending on the species. Some of these sequences are the satellite DNA referred to earlier. The sequences in this class are typically from 5 to 300 base pairs per repeat and are duplicated as many as 10^5 times per genome.
- *Middle-repetitive sequences* This component constitutes from 1 to 30 percent of a eukaryotic genome and includes sequences that are repeated from a few times to 10^5 times per genome.

These different components can be identified not only by the kinetics of DNA reassociation but also by the number of bands that appear in Southern blots with the use of appropriate probes and by other methods. It should be clear from the preceding discussion of DNA reassociation that the dividing line between many middle-repetitive sequences is arbitrary.

Unique Sequences

Most gene sequences and the adjacent nucleotide sequences required for their expression are contained in the unique-sequence component. With minor exceptions (for example, the repetition of one or a few genes), the genomes of viruses and prokaryotes are composed entirely of single-copy sequences; in contrast, such sequences constitute only 38 percent of the total genome in some sea urchin species, a little more than 50 percent of the human genome, and about 70 percent of the *D. melanogaster* genome.

Highly Repetitive Sequences

Many highly repetitive sequences are localized in blocks of tandem repeats, whereas others are dispersed throughout the genome. An example of the dispersed type is a family of related sequences in the human genome called the **Alu** family because the sequences contain a characteristic restriction site for the enzyme *AluI* (Section 5.7). The *Alu* sequences are about 300 base pairs in length and are present in approximately 500,000 copies in the human genome; this repetitive DNA family alone accounts for about 5 percent of human DNA.

Among the localized highly repetitive sequences, most are fairly short. Sequences of this type make up about 6 percent of the human genome and 18 percent of the *D. melanogaster* genome, but they account for 45 percent of the DNA of *D. virilis*. One of the simplest possible repetitive sequences is composed of an alternating . . . ATAT . . . sequence with about 3 percent G + C interspersed, which makes up 25 percent of the genomes of certain species of land crabs. In the *D. virilis* genome, the major components of the highly repetitive class are three different but related sequences of seven base pairs rich in A–T:

5' -ACAAACT-3'
 5' -ATAAACT-3'
 5' -ACAAATT-3'

Blocks of satellite (highly repetitive) sequences in the genomes of several organisms have been located by *in situ* hybridization with metaphase chromosomes (Figure 6.19). The satellite sequences located by this method have been found to be in the regions of the chromosomes called **heterochromatin**. These are regions that condense earlier in prophase than the rest of the chromosome and are darkly stainable by many standard dyes used to make chromosomes visible (Figure 6.20); sometimes

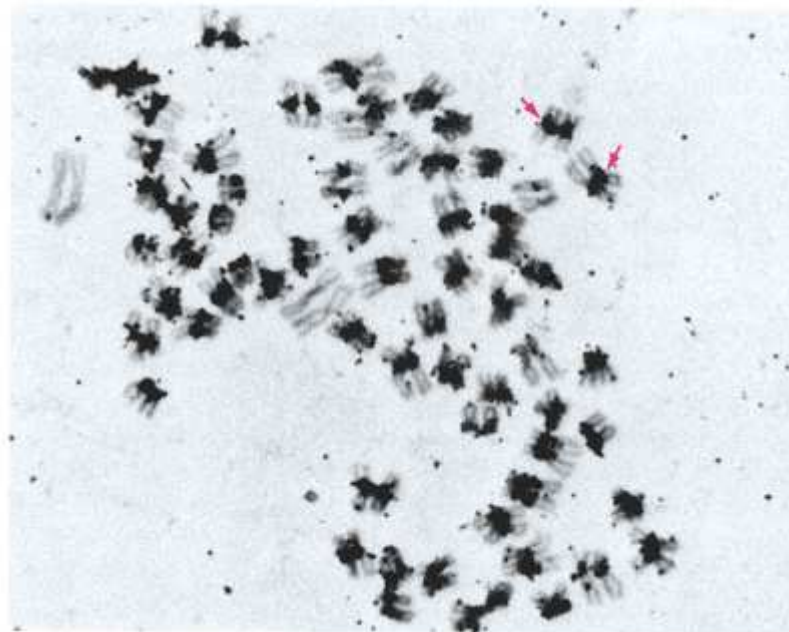


Figure 6.19

Autoradiogram of metaphase chromosomes of the kangaroo rat *Dipodomys ordii*; radioactive RNA copied from purified satellite DNA sequences has been hybridized to the chromosomes to show the localization of the satellite DNA. Hybridization is principally in the regions adjacent to the centromeres (arrows). Note that some chromosomes are apparently free of this satellite DNA. They contain a different satellite DNA not examined in this experiment. This cell is unusual in that each chromosome has undergone two rounds

of chromosome duplication without an intervening mitosis; this is why each chromosome in the normal diploid set is present as a pair.
[Courtesy of David Prescott.]

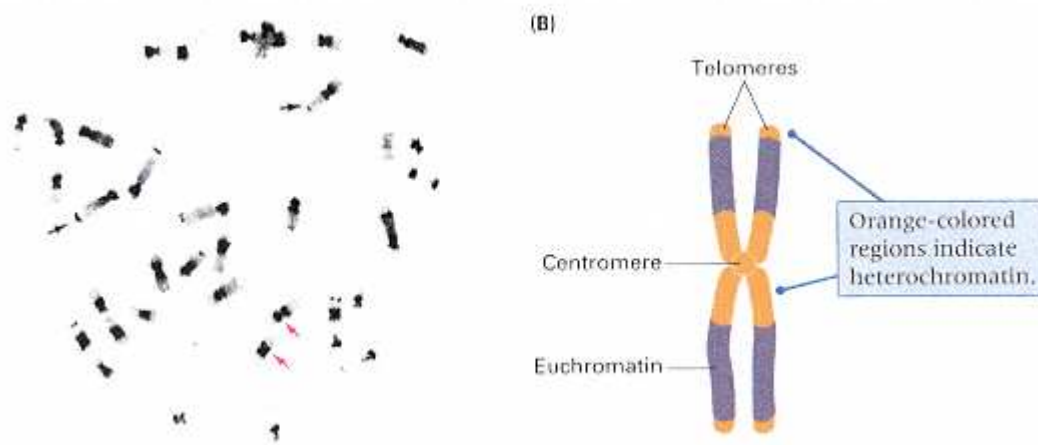


Figure 6.20

(A) Metaphase chromosomes of the ground squirrel *Ammospermophilus harrissi*, stained to show the heterochromatic regions near the centromere of most chromosomes (red arrows) and the telomeres of some chromosomes (black arrows). (B) An interpretive drawing.
[Micrograph courtesy of T.C. Hsu.]

the heterochromatin remains highly condensed throughout the cell cycle. The **euchromatin**, which makes up most of the genome, is visible only in the mitotic cycle. The major heterochromatic regions are adjacent to the centromere; smaller blocks are present at the ends of the chromosome arms (the telomeres) and interspersed with the euchromatin. In many species, an entire chromosome, such as the Y chromosome in *D. melanogaster* and in mammals, is almost completely heterochromatic. Different highly repetitive sequences have been purified from *D. melanogaster*, and in situ hybridization has shown that each sequence has its own distinctive distribution among the chromosomes.

The genetic content of heterochromatin is summarized in the following generalization:

The number of genes located in heterochromatin is small relative to the number in euchromatin.

The relatively small number of genes means that many large blocks of heterochromatin are genetically almost inert, or devoid of function. Indeed, heterochromatic blocks can often be rearranged in the genome, duplicated, or even deleted without major phenotypic consequences.

Middle-Repetitive Sequences

Middle-repetitive sequences constitute about 12 percent of the *D. melanogaster* genome and 40 percent or more of the human and other eukaryotic genomes. These sequences differ greatly in the number of copies and their distribution within a genome. They comprise many families of related sequences and include several groups of genes. For example, the genes for the RNA components of the ribosomes (the particles on which proteins are synthesized; Chapter 10) and the genes for tRNA molecules (which also participate in protein synthesis) are repeated in the genomes of all organisms. The two major ribosomal RNA molecules come from a tandem pair of genes that is repeated several hundred times in most eukaryotic genomes. The genomes of all eukaryotes also contain multiple copies of the histone genes. Each histone gene is repeated about 10 times per genome in chickens, 20 times in mammals, about 100 times in *Drosophila*, and as many as 600 times in certain sea urchin species.

The dispersed middle-repetitive DNA of the *D. melanogaster* genome consists of

about 50 families of related sequences, and from 20 to 60 copies of each family are widely scattered throughout the chromosomes. The positions of these sequences differ from one fly to the next except in completely homozygous laboratory strains. There is variability in position, because many of these sequences are able to move from one location to another in a chromosome and between chromosomes; they are said to be **transposable elements**. Analogous types of sequences are also found in the genomes of yeast, maize, and bacteria (Chapter 8) and are probably present in all organisms. An important dimension has been added to our understanding of the genome as a structural and functional unit by the discovery of these mobile genetic elements, because they can, in some cases, cause chromosome breakage chromosome rearrangements, modification of the expression of genes, and novel types of mutations.

6.8—

Transposable Elements

In the 1940s, in a study of the genetics of kernel mottling in maize (Figure 6.21), Barbara McClintock discovered an element that not only regulated the mottling but also caused breakage of the chromosome carrying the genes for color and consistency of the kernels. The element was called Dissociation (*Ds*). Mapping data showed that the chromosome breakage always occurs at or very near the location of *Ds*. McClintock's critical observation was



Figure 6.21

Sectors of purple and yellow tissue in the endosperm of maize kernels resulting from the presence of the transposable elements *Ds* and *Ac*. The heavier sectoring in some ears results from dosage effects of *Ac*. The least speckled ear has one copy of *Ac*; that in the middle has two (*Ac Ac*); and the most speckled ear has three (*Ac Ac Ac*).

[Courtesy of Jerry L. Kermicle.]

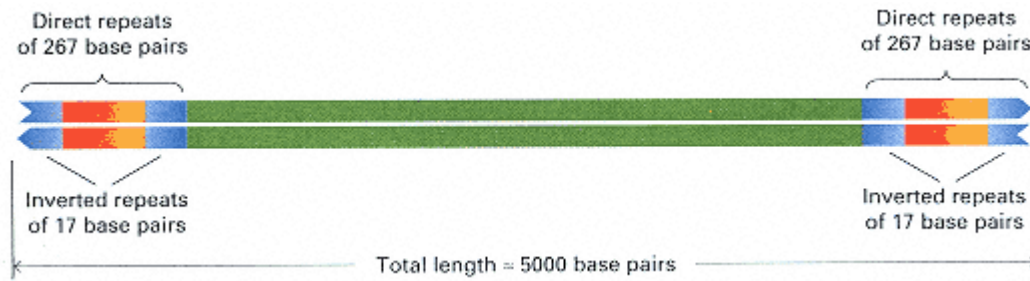


Figure 6.22

Sequence organization of a copia transposable element of *Drosophila melanogaster*.

that *Ds* does not have a constant location but occasionally moves to a new position (**transposition**), causing chromosome breakage at the new site. Furthermore, *Ds* moves only if a second element, called Activator (*Ac*), is also present. In addition, *Ac* itself moves within the genome and can cause, in the expression of genes at or near its insertion site, alterations similar to the modifications resulting from the presence of *Ds*.

Other **transposable elements** with characteristics and genetic effects similar to those of *Ac* and *Ds* are known in maize. Much of the color variegation seen in the kernels of varieties used for decorative purposes are attributable to the presence of one or more of these elements.

Since McClintock's discovery, transposable nucleotide sequences have been observed to be widespread in eukaryotes and prokaryotes. In *D. melanogaster*, they constitute from 5 to 10 percent of the genome and comprise about 50 distinct families of sequences. One well-studied family of closely related, but not identical, sequences is called **copia**. This element is present in about 30 copies per genome. The copia element (Figure 6.22) contains about 5000 base pairs with two identical sequences of 267 base pairs located terminally and in the same orientation. Repeated DNA sequences with the same orientation are called **direct repeats** (Figure 6.23A). The ends of each of these copia direct repeats contain two segments of 17 base pairs, whose sequences are also nearly identical; these shorter segments have opposite orientations. Repeated DNA sequences with opposite orientations are called **inverted repeats** (Figure 6.23B).

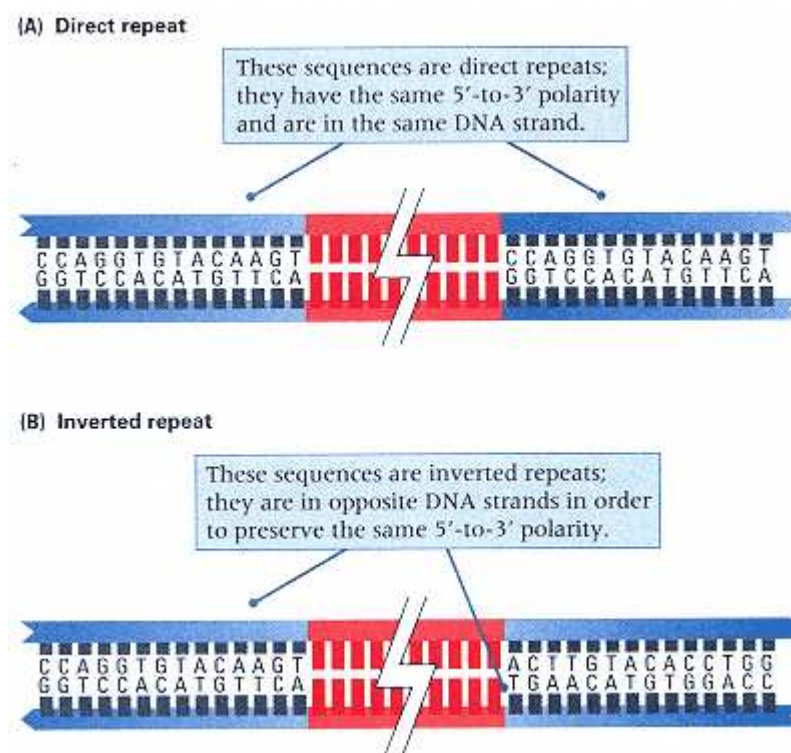


Figure 6.23

(A) In a direct repeat, a DNA sequence is repeated in the same left-to-right orientation. (B) In an inverted repeat, the sequence is repeated in the reverse left-to-right orientation in the opposite strand. The opposite strand is stipulated in order to maintain the correct 5'-to-3' polarity.

Other transposable elements have a similar organization with direct or inverted terminal repeats, as do many such

elements in other organisms, such as the transposable elements in *E. coli* described in Chapter 8.

The molecular processes responsible for the movement of transposable elements

are not well understood (some information will be presented in Chapter 8). A common feature is that transposition of the element is usually accompanied by the duplication of a small number of base pairs originally present at the insertion site, with the result that a copy of this short chromosomal sequence is found immediately adjacent to both ends of the inserted element (Figure 6.24). The length of the duplicated segment

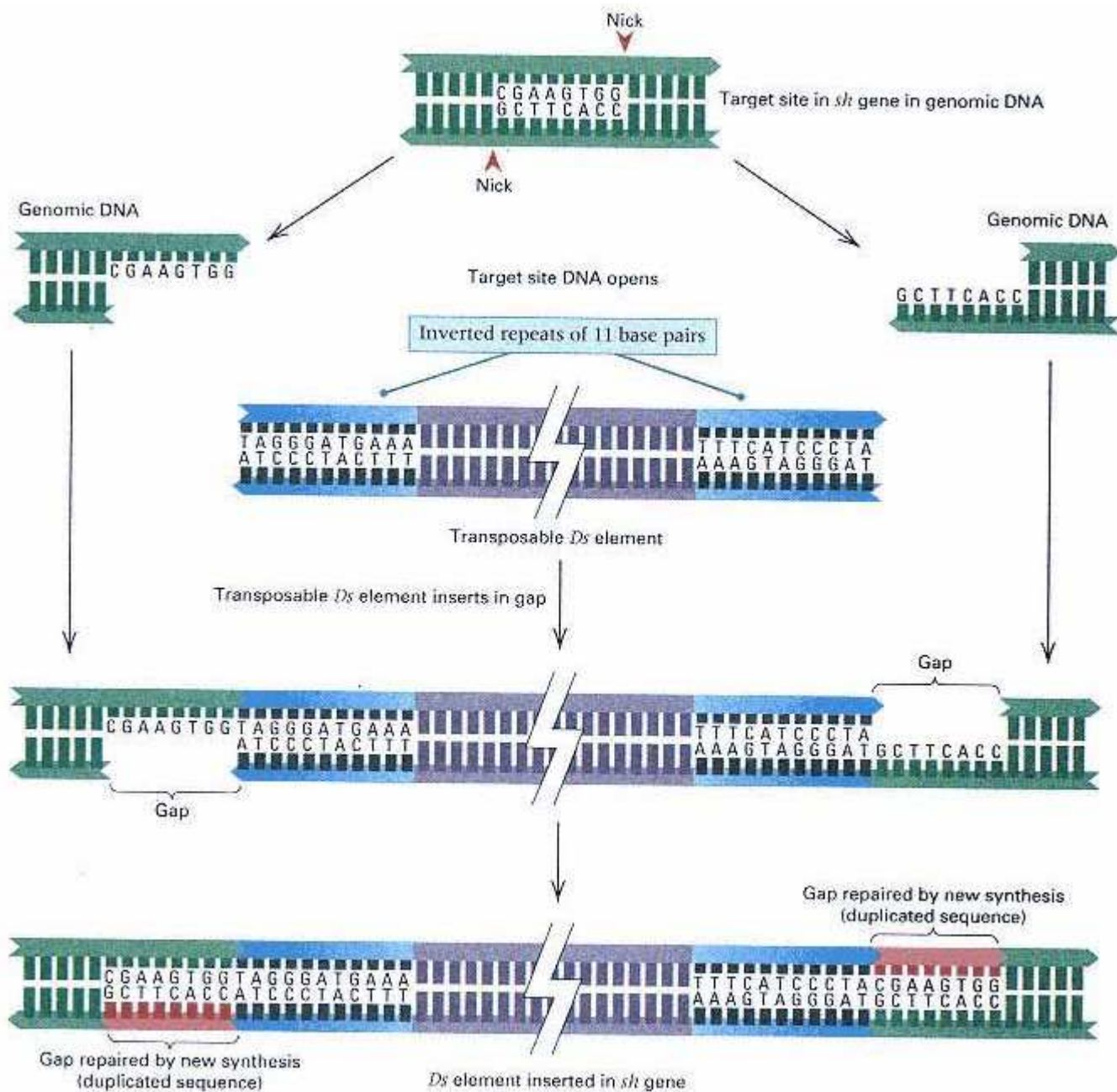


Figure 6.24

The sequence arrangement of one type of transposable element (in this case, *Ds* of maize) and the changes that take place when it inserts into the genome. *Ds* is inserted into the maize *sh* gene at the position indicated. In the insertion process, a sequence of eight base pairs next to the site of insertion is duplicated and flanks the *Ds* element.

Connection Her Feeling for the Organism

Barbara McClintock 1950

Cold Spring Harbor Laboratory,
Cold Spring Harbor, New York
*The Origin and Behavior of Mutable
Loci in Maize*

Many geneticists regard McClintock's papers on transposable elements as difficult. Her discoveries were completely novel. Genes that could move from one place to another in the genome were unheard of. She had no terminology with which to discuss such things, so she had to adapt the conventional terminology to describe a unique situation. What we now call a transposable element, and believe to be universal among organisms, McClintock calls a "chromatin element." McClintock was a superb geneticist and cytogeneticist, perhaps the best of her generation. In tracking down transposable elements by genetic means, McClintock had to use considerable ingenuity in designing crosses that were sometimes quite complex. Her writing style is also uniquely McClintock. The excerpt that follows is a good example. It is a relentless marshaling of observation and hypothesis, experiment and result, interpretation and deduction. McClintock was awarded a Nobel Prize in 1983. The following passage deals with the discovery of Ds (Dissociation) and Ac (Activator). In modern terminology, we call Ac an autonomous transposable element because it codes for all the proteins it needs for its own transposition; Ds is a nonautonomous transposable element because it requires Ac to provide the proteins necessary for it to move.

In the course of an experiment designed to reveal the genic composition of the short arm of chromosome 9, a phenomenon of rare occurrence (or recognition) in maize began to appear with remarkably high frequencies in the cultures. The terms mutable genes, unstable genes, variegation, mosaicism, or mutable loci have been applied to this phenomenon. Its occurrence in a wide variety of organisms has been recognized . . . A fortunate discovery was made early in the study of the mutable loci which proved to be of singular importance in showing the kinds of events that are associated with their origin and behavior. A locus was found in the short arm of chromosome 9 at which breaks were occurring in somatic cells. The time and frequency of the breakage events occurring at this Ds (Dissociation) locus appeared to be the same as the time and frequency of the mutation-producing mutable loci. An extensive study of the Ds locus

Transposition of Ac takes place from one position in the chromosomal complement to another—very often from one chromosome to another.

has indicated the reason for this relationship and has produced the information required to interpret the events occurring at mutable loci. It has been concluded that the changed phenotypic expression of such loci is related to changes in a chromatin element other than that composing the genes themselves, and that mutable loci arise when such chromatin is inserted adjacent to the genes that are affected. . . . The Ds locus is composed of this kind of material. Various types of alterations are observed as a consequence of events occurring at the Ds locus. . . . They involve chromosome breakage and fusion. The breaks are related, however, to events occurring at this one specific locus in the chromosome—the Ds locus. . . . [Among the known events is] transposition of Ds activity from one position to another in the chromosome complement with or without an associated gross chromosomal rearrangement. . . . It is from the transpositions of Ds that some of the new mutable loci may arise. . . . In one case the transposed Ds locus appeared in a single gamete of a plant carrying chromosome 9 with a dominant C (colored aleurone) allele. . . . The new position of Ds corresponded to the known location of C. . . . Significantly, the appearance of Ds activity at this new position was correlated with the disappearance of the normal action of the C locus. The resulting phenotype was the same as that produced by the known recessive allele c. . . . That the c phenotype in this case was associated with the appearance of Ds at the C locus was made evident because mutations from c to C occurred [in which] Ds action concomitantly disappeared. . . . The origin and behavior of this mutation at the c locus have been interpreted as follows: Insertion of the chromatin composed of Ds adjacent to the C locus is responsible for complete inhibition of the action of C. Removal of this foreign chromatin can occur. In many cases, the mechanism associated with the removal results in restoration of former genic organization and action . . . The mutation-producing mechanisms involve only Ds. No gene mutations occur at the C locus; the

restoration of its action is due to the removal of the inhibiting *Ds* chromatin . . . [The movement] of *Ds* requires an activator. This activator has been designated *Ac*. . . . *Ac* shows a very important characteristic not exhibited in studies of the usual genetic factors. This characteristic is the same as that shown by *Ds*. Transposition of *Ac* takes place from one position in the chromosomal complement to another—very often from one chromosome to another. Again, as in *Ds*, changes in state may occur at the *Ac* locus . . . It should be emphasized that when no *Ac* is present in a nucleus, no mutation-producing events occur at *Ds*.

Source: *Proceedings of the National Academy of Sciences of the USA* 36: 344–355

ranges from 2 to 12 base pairs, depending on the particular transposable element. Insertion of a transposable element is not a sequence-specific process in that the element is flanked by a different duplicated sequence at each location in the genome; however, the *number* of duplicated base pairs is usually the same at each location in the genome and is characteristic of a particular transposable element. Experimental deletion or mutation of part of the terminal base sequences of several different elements has shown that the short terminal inverted repeats are essential for transposition, probably because they are necessary for binding an enzyme called a **transposase** that is required for transposition. Many transposable elements code for their own transposase by means of a gene located in the central region between the terminal repeats, so these elements are able to promote their own transposition. Elements in which the transposase gene has been lost or inactivated by mutation are transposable only if a related element is present in the genome to provide this activity. The inability of the maize *Ds* element to transpose without *Ac* results from the absence of a functional transposase gene in *Ds*. Transposable elements are sometimes referred to as **selfish DNA**, because each type of element maintains itself in the genome as a result of its ability to replicate and transpose.

Transposable elements are responsible for many visible mutations. A transposable element is even responsible for the wrinkled-seed mutation in peas studied by Gregor Mendel. The wildtype allele of the gene codes for starch-branching enzyme I (SBEI), which is used in the synthesis of amylopectin (starch with branched chains). In the wrinkled mutation, a transposable element inserted into the gene renders the enzyme nonfunctional. The pea transposable element has terminal inverted repeats that are very similar to those in the maize *Ac* element, and the insertion site in the *wrinkled* allele is flanked by a duplication of eight base pairs of the SBEI coding sequence. This particular insertion appears to be genetically quite stable: The transposable element does not seem to have been excised in the long history of wrinkled peas.

6.9—

Centromere and Telomere Structure

Eukaryotic chromosomes contain regions specialized for maneuvering the chromosomes in cell division and for capping the ends. These regions are discussed next.

Molecular Structure of the Centromere

The centromere is a specific region of the eukaryotic chromosome that becomes visible as a distinct morphological entity along the chromosome during condensation. It serves as a central component of the kinetochore, the complex of DNA and proteins to which the spindle fibers attach as they move the chromosomes in both mitosis and meiosis. The kinetochore is also the site at which the spindle fibers shorten, causing the chromosomes to move toward the poles. Electron microscopic analysis has shown that in some organisms—for example, the yeast *Saccharomyces cerevisiae*—a single spindle-protein fiber is attached to centromeric chromatin. Most other organisms have multiple spindle fibers attached to each centromeric region.

Centromeres exhibit considerable structural variation among species. At one extreme are **holocentric chromosomes**, which appear to have centromeric sequences spread throughout their length (constituting what is called a **diffuse centromere**). The nematode *Caenorhabditis elegans*, which is widely used in genetic research, has holocentric chromosomes. In a holocentric chromosome, microtubules attach along the entire length of each chromatid. If a holocentric chromosome is broken into fragments by x rays, each fragment behaves as a separate, smaller holocentric chromosome that moves normally in cell division. Diffuse centromeres are very poorly understood and will not be discussed further.

The conventional type of centromere is the **localized centromere**, in which microtubules attach to a single region of the chromosome (the kinetochore). Localized centromeres fall into two types, *point centromeres* and *regional centromeres*.

Point centromeres are found in a number of different yeasts, including *Saccharomyces cerevisiae*, and they are relatively small in terms of their DNA content. Other eukaryotes, including higher eukaryotes, have regional centromeres that may contain hundreds of kilobases of DNA.

The chromatin segment of the centromeres of *Saccharomyces cerevisiae* has a unique structure that is exceedingly resistant to the action of various DNases; it has been isolated as a protein-DNA complex containing from 220 to 250 base pairs. The nucleosomal constitution and DNA base sequences of all of the yeast centromeres have been determined. Several common features of the base sequences are shown in Figure 6.25A. There are four regions, labeled CDE1, CDE2, CDE3, and CDE4. All yeast centromeres have sequences highly similar to those indicated for regions 1, 2, and 3, but the sequence of region CDE4 varies from one centromere to another. Region 2 is noteworthy in that approximately 90 percent of the base pairs are A—T pairs. The centromeric DNA is contained in a structure (the centromeric core particle) that contains more DNA than a typical yeast nucleosome core particle (which contains 160 base pairs) and is larger. This structure is responsible for the resistance of centromeric DNA to DNase. The spindle fiber is believed to be attached directly to this particle (Figure 6.25B).

The base-sequence arrangement of the yeast centromeres is not typical of other eukaryotic centromeres. In higher eukaryotes, the chromosomes are about 100 times as large as yeast chromosomes, and several spindle fibers are usually attached to each region of the centromere. Furthermore, the centromeric regions of the chromosomes of many higher eukaryotes contain large amounts of heterochromatin, consisting of many higher eukaryotes contain large amounts of heterochromatin, consisting of repetitive satellite DNA, as described in Section 6.7. For example, the centromeric regions of human chromosomes contain a tandemly repeated DNA sequence of about 170 base pairs called the **alpha satellite** (Figure 6.26). The number of alpha-satellite copies in the centromeric region ranges from 5000 to 15,000, depending on the chromosome. The DNA sequences needed for spindle fiber attachment may be interspersed among the alpha-satellite sequences, but whether the alpha-satellite sequences themselves contribute to centromere activity is unknown.

Molecular Structure of the Telomere

Each end of a linear chromosome is composed of a special DNA-protein structure called a **telomere** that is essential for chromosome stability. Genetic and microscopic

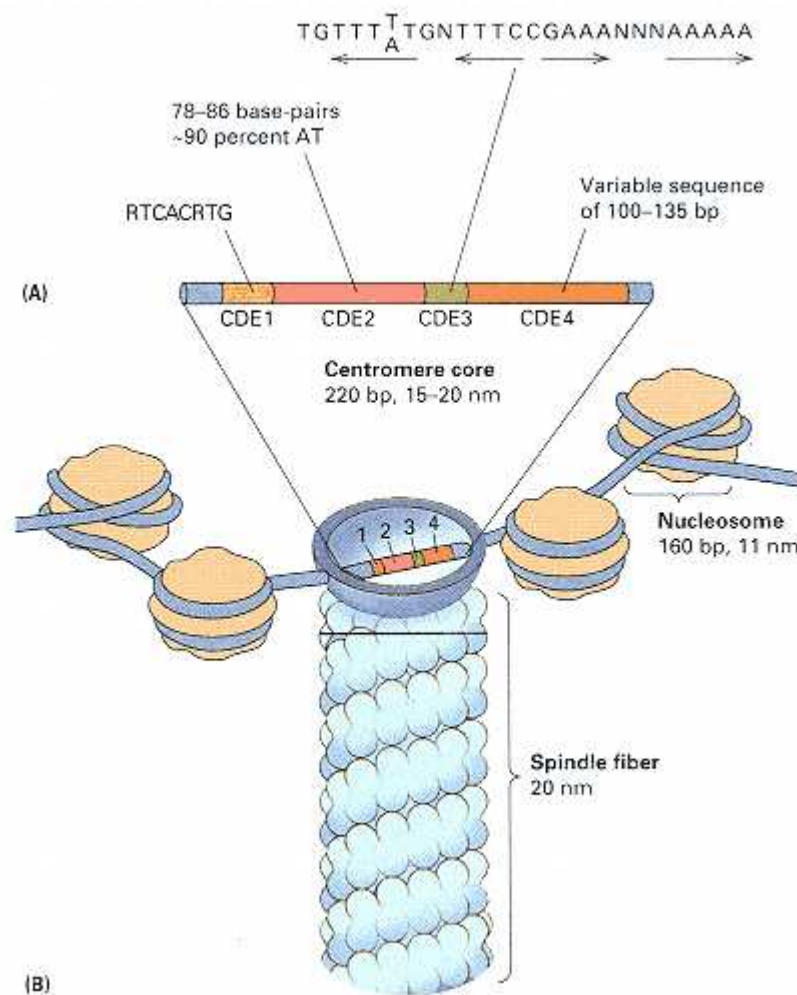


Figure 6.25

A yeast centromere. (A) Diagram of centromeric DNA showing the major regions (CDE1 through CDE4) common to all yeast centromeres. The letter R stands for any purine (A or G), and the letter N indicates any nucleotide. Inverted-repeat segments in region 3 are indicated by arrows. The sequence of region CDE4 varies from one centromere to the next. (B) Positions of the centromere core and the nucleosomes on the DNA. The DNA is wrapped around histones in the nucleosomes, but the detailed organization and composition of the centromere core are unknown.

[After K. S. Bloom, M. Fitzgerald-Hayes, and J. Carbon. 1982. *Cold Spring Harbor Symp. Quant. Biol.*, 47: 1175.]

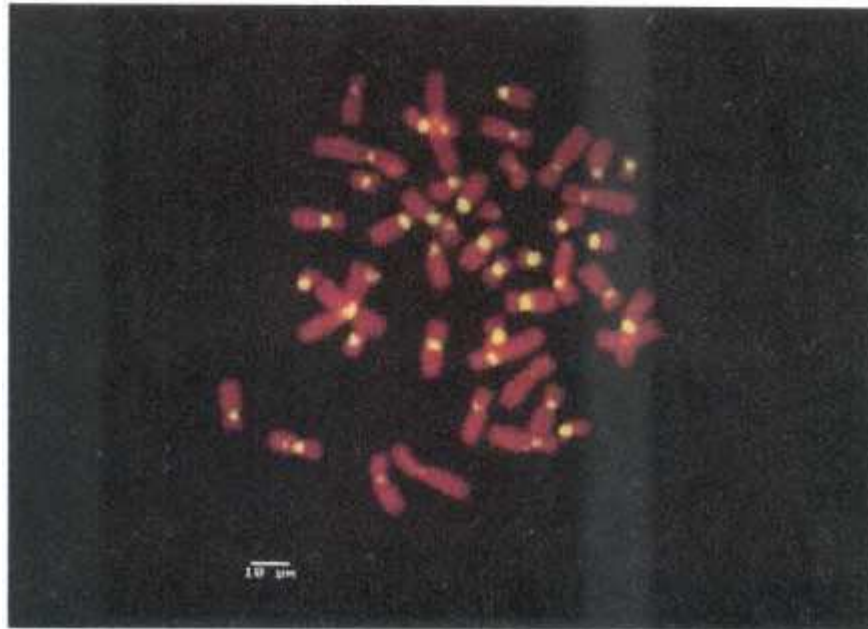


Figure 6.26
Hybridization of human metaphase chromosomes (red) with alpha-satellite DNA. The yellow areas result from hybridization with the labeled DNA. The sites of hybridization of the alpha satellite coincide with the centromeric regions of all 46 chromosomes.
[Courtesy of Paula Coelho and Claudio E. Sunkel.]

observations first indicated that telomeres are special structures. In *Drosophila*, Hermann J. Muller found that chromosomes without ends could not be recovered after chromosomes were broken by treatment with x rays. In maize, Barbara McClintock observed that broken chromosomes frequently fuse with one another and form new chromosomes with abnormal structures (often having two centromeres). As we saw in Chapter 5, DNA polymerases cannot initiate DNA synthesis but instead require an RNA primase, so at least one end of each chromosome must have a short (8 to 12 nucleotides) stretch of single-stranded DNA that remains after the RNA primer at the tip has been removed. It turns out that both ends of linear chromosomes terminate in a stretch of single-stranded DNA.

Telomeric DNA in most organisms consists of tandem repeats of simple sequences such as 5'-TTAGGG-3'. These special telomere DNA sequences are *added* to the ends of eukaryotic chromosomes by an enzyme called **telomerase**. The substrate for the telomerase is a telomere addition sequence consisting of a short repetitive sequence. In mammals, the telomere addition sequence consists of many tandem repeats of 5'-TTAGGG-3'. In the ciliate protozoan *Tetrahymena*, it consists of repeats of the similar sequence 5'-TTGGGG-3'. Relatively few copies of the repeat are necessary to prime the telomerase to add additional copies and form a telomere. Remarkably, the telomerase enzyme incorporates an essential RNA molecule, called a **guide RNA**, that contains sequences complementary to the telomere repeat and that serves as a template for telomere synthesis and elongation. For example, the *Tetrahymena* guide RNA contains the sequence 3'-AACCCCAAC-5'. The guide RNA undergoes base pairing with the telomere repeat and serves as a template for telomere elongation by the addition of more repeating units (Figure 6.27). The complementary DNA strand of the telomere is analogous to the lagging strand at a replication fork and is synthesized by DNA polymerase in the usual manner with the

Connection Telomeres: The Beginning of the End

Carol W. Greider and Elizabeth H. Blackburn 1987

University of California,
Berkeley, California
*The Telomere Terminal Transferase of
Tetrahymena Is a Ribonucleoprotein
enzyme with Two Kinds of Primer Specificity*

What a wonderful surprise that an RNA is a key ingredient in the formation of telomeres! Two limitations of DNA polymerase are that it requires a primer oligonucleotide and that it can elongate a DNA strand only at the 3' end. The limitations imply that the 3' end of a DNA strand should become progressively shorter with each round of replication, owing to the need for a primer at that extremity. (Chromosomes without proper telomeres do, in fact, become progressively shorter.) The organism used in this study, Tetrahymena, is a ciliated protozoan. Each cell has a specialized type of nucleus called a macronucleus that contains many hundreds of small chromosomes. The level of telomerase activity is high because each of these tiny chromosomes needs a pair of telomeres. The convenience of using Tetrahymena for the study of telomere function illustrates a principle that runs through the history of genetics: Breakthroughs often come from choosing just the right organism to study. The authors speculate about the possible presence of a "guide" RNA in the telomerase. They were exactly right.

Precise recognition of nucleic acids is often carried out by enzymes that contain both RNA and protein components. For some of these ribonucleoproteins (RNPs), the RNA components provide specificity to the reaction by base pairing with the substrate. The recognition that RNA can act catalytically has led to the increase in the number of known RNP-catalyzed reactions. We report here that an RNP is involved in synthesizing the telomeric sequences

. . . it is tempting to speculate that the RNA component of the telomerase might be involved in determining the sequence of the telomeric repeats that are synthesized . . .

found at the ends of *Tetrahymena* macronuclear chromosomes. . . . We have previously reported the identification of an activity in *Tetrahymena* cell extracts that adds telomeric repeats onto appropriate telomeric sequence primers in a nontemplated manner. . . . Repeats of the *Tetrahymena* telomeric sequence TTGGGG are added, 1 nucleotide at a time, onto the 3' end of the input primer. . . . We have begun characterizing and purifying the telomerase enzyme in order to investigate the mechanisms controlling the specificity of the reaction. . . . We propose that the RNA component(s) of telomerase may play a role in specifying the sequence of the added telomeric TTGGGG repeats, in recognizing the structure of the G-rich telomeric sequence primers, or both. . . . In the course of purifying the telomerase from crude extracts, we noted a marked sensitivity to salts [in which high] concentrations inactivated the telomeric elongation activity. The salt sensitivity and the large size of the enzyme suggested that the telomerase may be a complex containing a nucleic acid component. To test whether the telomerase contained an essential nucleic acid, we treated active fractions with either micrococcal nuclease or RNAase A. The nuclease activity of each of these enzymes abolished the telomeric elongation activity. . . . These experiments suggest that the telomerase contains an essential RNA component. . . . The RNA of telomerase may simply provide a scaffold for the assembly of proteins in the active enzyme complex; however, . . . it is tempting to speculate that the RNA component of the telomerase might be involved in determining the sequence of the telomeric repeats that are synthesized and/or the specific primer recognition. If the RNA of telomerase contains the sequence CCCCAA, this sequence could act as an internal guide sequence.

Source: *Cell* 51:887–898

help of an RNA primer. In the telomeric regions in most eukaryotes, there are also longer, moderately repetitive DNA sequences just preceding the terminal repeats. These sequences differ among organisms and even among different chromosomes in the same organism.

What limits the length of a telomere? In most organisms the answer is unknown. In yeast, however, a protein called Raplp has been identified that appears to be important in regulating telomere length. The Raplp protein binds to the yeast telomere sequence. Molecules of Raplp bind to the telomere sequence as it is being elongated until about 17 Raplp molecules have been bound. At this point, telomere elongation stops, probably because the accumulation of Raplp inhibits telomerase activity. Because each Raplp molecule binds to approximately 18 base pairs of the telomere, the predicted length of a yeast telomere is $17 \times 18 = 306$ base pairs, which is very close to the value observed. Additional evidence for the role of Raplp comes from mutations in the *RAP 1* gene producing a protein that cannot bind to telomere sequences; in these mutant

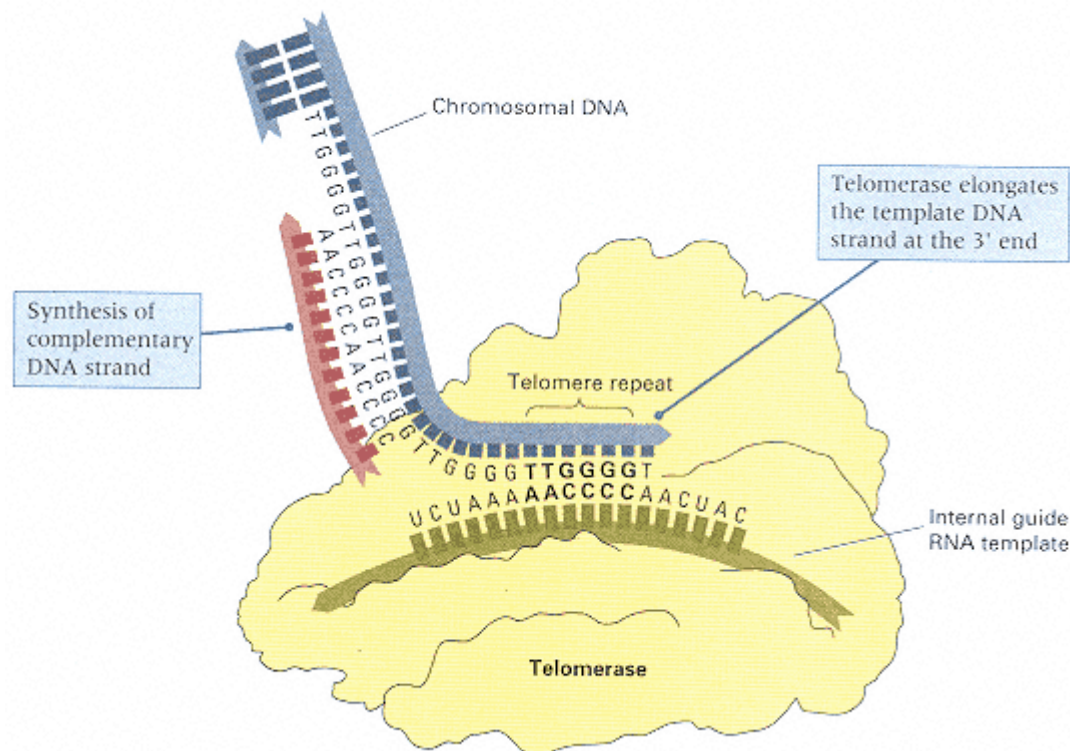
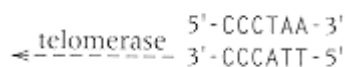


Figure 6.27

Telomere formation in *Tetrahymena*. The telomerase enzyme contains an internal RNA with a sequence complementary to the telomere repeat. The RNA undergoes base pairing with the telomere repeat and serves as a template for telomere elongation. The newly forming DNA strand is produced by DNA polymerase.

strains, massive telomere elongation is observed.

Telomeric sequences found in a variety of organisms are shown in Table 6.2. The table does not include *Drosophila* because its telomeres, rather than consisting of simple repeats as in most other organisms, are composed of specialized transposable elements. Each telomere sequence in Table 6.2 is written as if it were at the left-hand end of a chromosome. Both strands are written in the 5'-to-3' direction. The top strand is the 5' end of the telomere, and the bottom strand is the 3' end of the telomere. It is the 3' end of the bottom strand (in each entry, the 3' end is at the far right) that is elongated by the telomerase and that is single-stranded. Runs of nucleotides are indicated by subscripts; for example, the notation C_3TA_2/T_2AG_3 for vertebrate telomeres means the sequence



In many cases, there is variation in the length of a run; for example, C_{1-8} means variation ranging from as few as one C to as many as eight C's from one repeat to the next. There is also occasional variation in which nucleotide occupies a given position, which is indicated by small stacked letters.

There is a strong tendency for the elongated strand of the telomere repeats in Table 6.2 to be rich in guanine nucleotides. The guanine bases are special in that they have the capacity to hydrogen-bond to one another in a variety of ways. When the elongated 3' strand of the telomere folds back upon itself, the guanine bases can even pair to give the G-quartet structure illustrated in Figure 6.28. The hydrogen bonding in the G-quartet is called **Hoogsteen base pairing** to distinguish it from ordinary Watson-Crick base pairing. A protein has been isolated from the ciliated protozoan *Oxytricha* that binds specifically to the telomeric DNA of linear chromosomes. The β subunit of this telomere-binding protein promotes

Table 6.2 Sequences of telomeric DNAs.

Organism	Sequence
Protozoa	
<i>Tetrahymena</i>	C_4A_2/T_2G_4
<i>Paramecium</i>	$C_3\overset{C}{A}A_2/T_2\overset{C}{T}G_3$
<i>Oxytricha</i>	C_4A_4/T_4G_4
<i>Plasmodium</i>	$C_3T\overset{A}{G}A_2/T_2\overset{T}{C}AG_3$
<i>Trypanosoma</i>	C_3TA_2/T_2AG_3
<i>Giardia</i>	C_3TA/TAG_3
Slime molds	
<i>Physarum</i>	C_3TA_2/T_2AG_3
<i>Didymium</i>	C_3TA_2/T_2AG_3
<i>Dictyostelium</i>	$C_{1-8}T/AG_{1-8}$
Fungi	
<i>Saccharomyces</i>	$C_{2-3}ACA_{1-6}/T_{1-6}GTG_{2-3}$
<i>Kluyveromyces</i>	$ACAC_2ACATAC_2TA_2TCA_3TC_2GA/TCG_2AT_3GAT_2AG_2TATGTG_2TGT$
<i>Candida</i>	$ACAC_2A_2GA_2GT_2AGACATC_2GT/ACG_2ATGTCTA_2CT_2CT_2G_2TGT$
<i>Schizosaccharomyces</i>	$C_{1-6}G_{0-1}T_{0-1}GTA_{1-2}/T_{1-2}ACA_{0-1}C_{0-1}G_{1-6}$
<i>Neurospora</i>	C_3TA_2/TA_2AG_3
<i>Podospora</i>	C_3TA_2/T_2AG_3
<i>Cryptococcus</i>	$A_2C_{3-5}T/AG_{3-5}T_2$
<i>Cladosporium</i>	C_3TA_2/T_2AG_3
Invertebrates	
<i>Caenorhabditis</i>	GC_2TA_2/T_2AG_2C
<i>Ascaris</i>	GC_2TA_2/T_2AG_2C
<i>Parascaris</i>	$TGCA_2/T_2GCA$
<i>Bombyx</i> , other insects	C_2TA_2/T_2AG_2
Vertebrates	
	C_3TA_2/T_2AG_3
Plants	
<i>Chlamydomonas</i>	C_3TA_4/T_4AG_3
<i>Chlorella</i>	C_3TA_3/T_3AG_3
<i>Arabidopsis</i>	C_3TA_3/T_3AG_3
Tomato	$C_3A\overset{T}{A}T_2/A_2\overset{A}{T}TG_3$

Source: From V.A. Zakian. 1995, *Science* 270:1602

G-quartet formation by the telomeric DNA of *Oxytricha*:

. . . 5' -TTTTGGGGTTTGGGGT-3' . . .

or *Tetrahymena*:

. . . 5' -TTGGGGTTGGGGT-3' . . .

It is hypothesized that within telomeres, telomeric DNA may be organized in special three-dimensional conformations that include such G-quartets. Models for *Oxytricha* and *Tetrahymena* telomeric DNA, showing the possible positions of G-quartets, are presented in Figure 6.29.

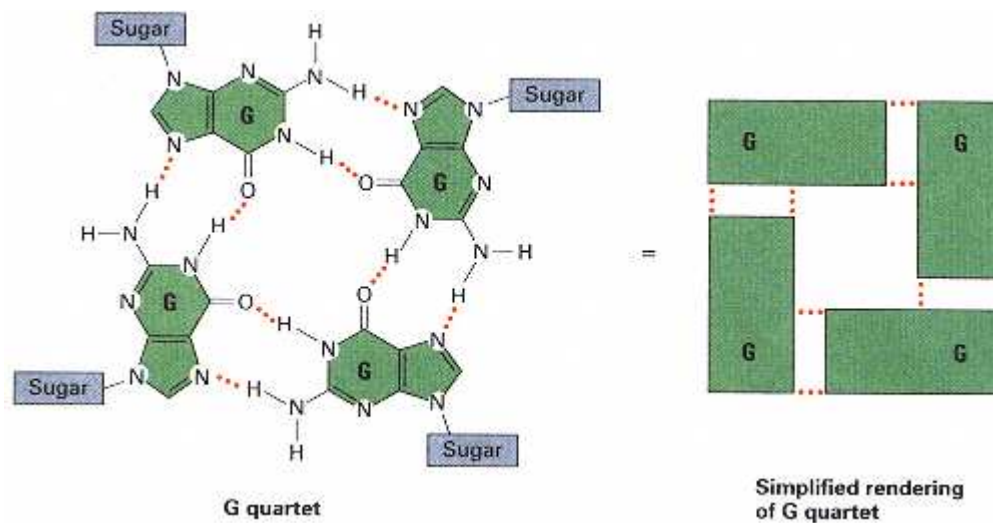


Figure 6.28

G-quartet structure formed by hydrogen bonding between four guanine bases present in a single DNA strand folded back upon itself. The rectangles on the right show how the guanines are oriented, with the deoxyribose sugars attached at the outside corners. On the left, the sugars are indicated by the small blue boxes. There is also a monovalent cation in the center of the quartet (not shown).

[From J. R. Williamson, M. K. Raghuraman and T. R. Cech. 1982 *Cell* 59: 871.]

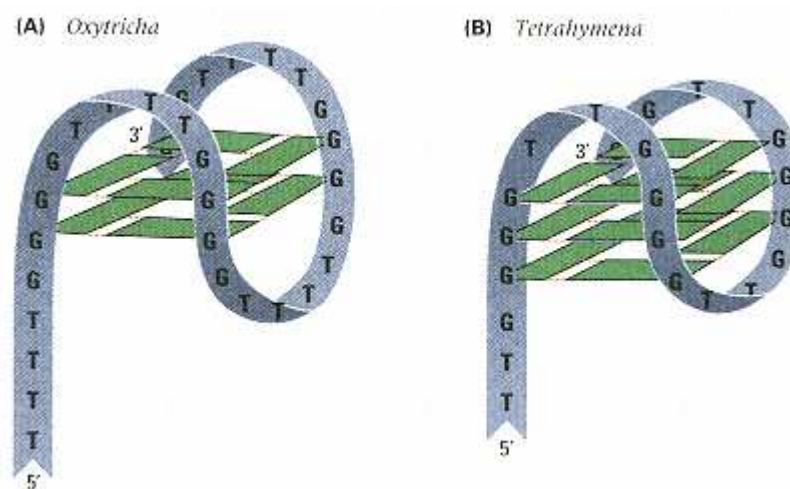


Figure 6.29

Models of telomere structure in *Oxytricha* (A) and *Tetrahymena* (B) that incorporate the G-quartet structure. The arrowhead indicates the 3' end of the DNA strand. The indicated G bases participate in the Hoogsteen base pairing, and the G-quartets are drawn as green squares, as in Figure 6.28.

There are two G-quartets postulated in the *Oxytricha* telomere, three in *Tetrahymena*.

[From J. R. Williamson, M. K. Raghuraman, and T. R. Cech. 1982 *Cell* 59:871.]

Chapter Summary

The DNA content of organisms varies widely. Small viruses exist whose DNA contains only a few thousand nucleotides, and among the higher animals and plants, the DNA content can be as large as 1.5×10^{11} nucleotides. Generally speaking, DNA content increases with the complexity of the organism, but within particular groups of organisms, the DNA content can vary as much as tenfold.

DNA molecules come in a variety of forms. Except for a few of the smallest viruses, whose DNA is single-stranded, and some viruses in which RNA is the genetic material, all organisms contain double-stranded DNA. The chromosomal DNA of higher organisms is almost always linear. Bacterial DNA is circular, as is the DNA of many animal viruses and of some bacteriophages. Circular DNA molecules are invariably supercoiled. The bacterial chromosome consists of independently supercoiled domains.

The DNA of both prokaryotic and eukaryotic cells and of viruses is never in a fully extended state but rather is folded in an intricate way, which reduces its effective volume. In viruses, the DNA is tightly folded but without bound protein molecules. In bacteria, the DNA is folded to form a multiply looped structure called a nucleoid, which includes several proteins that are essential for folding. In eukaryotes, the DNA is compacted into chromosomes, which contain several proteins and which are thick enough to be visible by light microscopy during the mitotic phase of the cell cycle. The DNA-protein complex of eukaryotic chromosomes is called chromatin. The protein component of chromatin consists primarily of five distinct proteins: histones H1, H2A, H2B, H3, and H4. The last four histones aggregate to form an octameric protein that contains two molecules of each. DNA is wrapped around the histone octamer, forming a particle called a nucleosome. This wrapping is the first level of compaction of the DNA in chromosomes. Each nucleosome unit contains about 200 nucleotide pairs, of which about 145 are in contact with the protein. The remaining 55 nucleotide pairs link adjacent nucleosomes. Histone H1 binds to the linker segment and draws the nucleosomes nearer to one another. The DNA in its nucleosome form is further compacted into a helical fiber, the 30-nm fiber. In forming a visible chromosome, this unit undergoes several additional levels of folding, producing a highly compact visible chromosome. The result is that a eukaryotic DNA molecule, whose length and width are about 50,000 and 0.002 μm , respectively, is folded to form a chromosome with a length of about 5 μm and a width of about 0.5 μm .

Polytene chromosomes are found in certain organs in insects. These gigantic chromosomes consist of about 1000 molecules of partly folded chromatin aligned side by side.

Seen by microscopy, they have about 5000 transverse bands. Polytene chromosomes do not replicate further, and cells that contain them do not divide. They are useful to geneticists primarily as morphological markers for particular genes and chromosome segments.

The number of copies of individual base sequences in a DNA molecule can vary tremendously. In prokaryotic DNA, most sequences are unique. However, in eukaryotic DNA, only a fraction of the DNA consists of unique sequences present once per haploid genome. The sequence composition of complex genomes can be studied by DNA renaturation kinetics, or Cot analysis. In eukaryotes, many sequences are present in hundreds to millions of copies. Some highly repetitive sequences are primarily located in the centromeric regions of the chromosomes, whereas others are dispersed. A significant fraction of the DNA, the middle-repetitive DNA, consists of sequences of which from 10 to 1000 copies are present per cell. Much of middle-repetitive DNA in higher eukaryotes consists of transposable elements, which are sequences able to move from one part of the genome to another. A typical transposable element is a sequence ranging in length from one to several thousand nucleotide pairs terminating in short sequences that are repeated, either in the same orientation (direct repeats) or in reverse orientation (inverted repeats). The terminal repeats and a transposase enzyme are necessary for the movement of these elements, a process known as transposition. Many transposable elements contain a gene coding for their own transposase. Insertion of most transposable elements causes duplication of a short chromosomal nucleotide sequence flanking the point of insertion. Insertions of transposable elements are the cause of many visible mutations.

Centromeres and telomeres are regions of eukaryotic chromosomes specialized for spindle fiber attachment and stabilization of the tips, respectively. The centromeres of most higher eukaryotes are associated with localized, highly repeated, satellite DNA sequences. Telomeres are formed by a telomerase enzyme that contains a guide RNA that serves as a template for the addition of nucleotides to the 3' end of a telomerase addition site. The complementary strand is synthesized in a manner analogous to the replication of the lagging strand at an ordinary replication fork. In mammals and other vertebrates, the 3' strand of the telomere terminates in tandem repeats of the simple sequence 5'-TTAGGG-3'. Relatively few copies of this sequence are needed to prime the telomerase. The

special properties of the telomere may be related to the ability of guanine nucleotides to form complex hydrogen-bonded structures—in particular, the G-quartet.

Key Terms

alpha satellites	Cot	diffuse centromere
chromatin	Cot curve	direct repeat
chromocenter	covalent circle	DNase
conserved sequence	cytological map	euchromatin
core particle	deoxyribonuclease	folded chromosome

genome	<i>in situ</i> hybridization	satellite DNA
guide RNA	kilobase pairs (kb)	scaffold
H1 histone	localized centromere	selfish DNA
H2A histone	megabase pairs (Mb)	single-copy sequence
H2B histone	micrococcal nuclease	supercoiled DNA
H3 histone	middle-repetitive sequences	telomerase
H4 histone	negative supercoiling	30-nm fiber
heterochromatin	nucleoid	topoisomerase I
highly repetitive sequences	nucleosome	topoisomerase II
histone	polytene chromosome	transposable element
holocentric chromosome	pulsed-field gel electrophoresis	transposase
Hoogsteen base pairing	relaxed DNA	transposition
inverted repeat	repetitive sequence	unique sequence

Review the Basics

- What are the principal differences between prokaryotes and eukaryotes in the organization of the genetic material?
- In a eukaryotic chromosome, where are the telomeres located? Is it possible to have a chromosome with the centromere exactly at the tip? What shape of chromosome has no telomeres?
- What is a transposable element?
- Are all repetitive DNA sequences transposable elements? If a transposable element were present somewhere in a long stretch of DNA, what features in the nucleotide sequence would help to identify it?
- Histone proteins are positively charged (basic) molecules, and they interact with negatively charged (acidic) molecules. What is the principal type of acid with which histones interact?
- How many polypeptide chains are present in each nucleosome? How many *different* polypeptide chains are present in each nucleosome? Which type of histone is not part of the nucleosome core particle?
- Are the ratios of the different histone types identical in the nucleosomes in all cells of a eukaryotic organism? In all eukaryotic organisms?
- Is it correct to say that no genes are present in heterochromatin? Explain your answer.

Guide to Problem Solving

Problem 1: *Dp(1;f)1187* is a minichromosome induced by x rays in *Drosophila*. It is about 1.3 megabase in length.

What is the approximate physical length of this chromosome in millimeters? (*Note:* One megabase (Mb) equals 1×10^6 base pairs, and $1 \text{ \AA} = 10^{-7} \text{ mm}$.)

Answer: The standard form of duplex DNA contains 10 nucleotide pairs per $34 \text{ \AA}$, or $3.4 \text{ \AA}$ per nucleotide pair. Therefore, a 1.3-Mb molecule has a length of $1.3 \times 10^6 \times 3.4 \text{ \AA} = 4.4 \times 10^6 \text{ \AA} = 0.44 \text{ mm}$; that is, the 1.3-megabase molecule of DNA in *Dp(1;f)1187* has a physical length just a little smaller than half a millimeter.

Problem 2: The genome size of *Drosophila melanogaster* is 165,000 kilobases (kb); approximately 2/3 of the genome is euchromatic and 1/3 is heterochromatic. In salivary gland chromosomes, the euchromatic part of the genome becomes polytene, and the giant chromosomes exhibit approximately 5000 transverse bands used as landmarks for the locations of genes, chromosome breakpoints, and other cytogenetic features. What is the approximate DNA content of an average band in the salivary gland chromosomes?

Answer: The euchromatic part of the *Drosophila* genome consists of approximately $165,000 \text{ kb} \times 2/3 = 110,000 \text{ kb}$ of DNA. This amount of DNA is distributed over 5000 bands, for an average of $110,000 \text{ kb} / 5000 = 22 \text{ kb}$ per band.

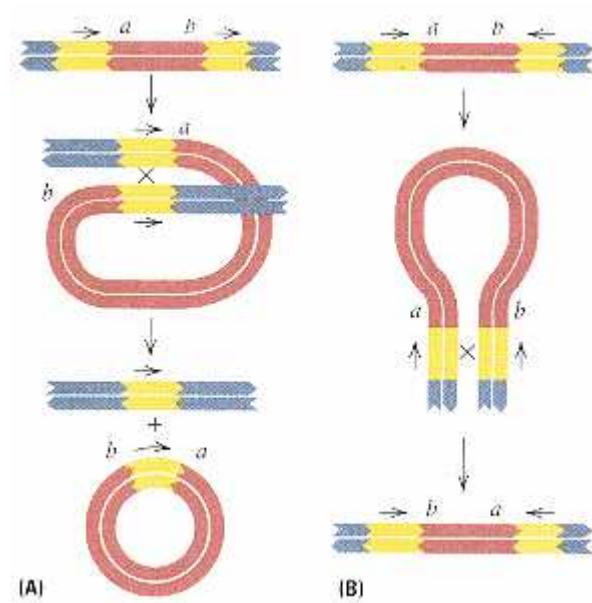
Problem 3: Suppose that the DNA sequences repeated at the ends of a transposable element undergo pairing and recombination. What are the genetic consequences if:

(a) the sequences are direct repeats?

(b) the sequences are inverted repeats?

Draw diagrams to support your answers.

Answer: The diagrams show the consequences.



(a) When direct repeats pair and undergo crossing-over, the result is the deletion of one of the repeats and the DNA segment between them (part A).

(b) When the repeats are inverted repeats, the result is that the transposable element is inverted in the chromosome (part B).

Analysis and Applications

6.1 The w^{pch} mutant allele of the *white* gene in *Drosophila* contains an insertion of 1.3 kb within the controlling region of the gene. The W^{pch} allele undergoes further mutation at a high rate, including reverse mutations to w^+ . The w^+ derivatives are found to lack the 1.3-kb insert. What kind of DNA sequence is the 1.3-kb insert likely to be?

6.2 Some transposable elements in animals are active only in the germ line and not in somatic cells. How might this restriction of activity be advantageous to the persistence of the transposable element?

6.3 Most transposable elements create a short duplication of host sequence when they insert. The duplication is always in a direct, rather than an inverted, orientation. What does this tell you about the process of insertion?

6.4 The *E. coli* chromosome is 4700 kb in length. What is its length in millimeters? The haploid human genome contains 3×10^9 base pairs of DNA. What is its length in millimeters?

6.5 If you had never seen a chromosome in a microscope but had seen nuclei, what feature of DNA structure would tell you that DNA must exist in a highly coiled state within cells?

6.6 What causes the bands in a polytene chromosome? How do polytene chromosomes undergo division?

6.7 Is there an analog of a telomere in the *E. coli* chromosome?

6.8 Many transposable elements contain a pair of repeated sequences. What is their position in the element? Are they in direct or inverted orientation?

6.9 Consider a long linear DNA molecule, one end of which is rotated four times with respect to the other end in the unwinding direction.

(a) If the two ends are joined to keep the molecule in the underwound state, how many base pairs will be broken?

(b) If the underwound molecule is allowed to form a supercoil, how many twists will be present?

6.10 Endonuclease S1 can break single stranded DNA but does not break double-stranded linear DNA. However, S1 can cleave supercoiled DNA, usually making a single break. Why does this occur?

6.11 Circular DNA molecules of the same size migrate at different rates in electrophoretic gels, depending on whether they are supercoiled or relaxed circles. Why?

6.12 Denaturation of DNA refers to breakdown of the double helix and ultimate separation of the individual strands, and it can be induced by heat and other treatments. Because G-C base pairs have three hydrogen bonds and A-T base pairs have only two, the temperature required for denaturation increases with the G-C content and also with the length of continuous G-C tracts in the molecule. Renaturation of DNA refers to the formation of double-stranded DNA from complementary single strands.

(a) Which process—denaturation or renaturation—is dependent on the concentration of DNA?

Chapter 6 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

<http://www.jbpub.com/genetics>

Select the link to **Genetics: Principles and Analysis** and then choose the link to **GeNETics on the web**. You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor

1. The **centromere** consensus sequences for CDE1, CDE2, and CDE3 can be compared among all yeast chromosomes because the complete sequence of the entire genome is available. This exercise concerns only the centromeres of chromosomes IV, IX, XIII, and XVI. Find the CDE1, CDE2, and CDE3 sequences in each of these chromosomes, and compare them with each other and with the consensus sequences given in the chapter. If assigned to do so, copy the CDE1, CDE2, and CDE3 sequences and align them.

2. Another view of the **telomerase** at work can be found at this keyword site. Although the 5'-to-3' polarities of the DNA and RNA strands are not indicated, you should be able to deduce them. If assigned to do so, make a sketch of the telomerase, and indicate the polarity of each of the DNA and RNA strands.

3. You will find up-to-date descriptions of the **transposable element** that Barbara McClintock first described by searching the keyword site for the terms *Activator* and *Dissociation*. Use the hot links to locate the nucleotide sequence of *Ac*, and identify the

(text box continued on next page)

(b) Which of the two illustrated DNA molecules (1 or 2) would have the lower temperature for strand separation? Why?



6.13 DNA from species A, labeled with ^{14}N and randomly fragmented, is renatured with an equal concentration of DNA from species B, labeled with ^{15}N and randomly fragmented, and then centrifuged to equilibrium in CsCl . Five percent of the total renatured DNA has a hybrid density. What fraction of the base sequences is common to the two species?

6.14 What is meant by the terms "direct repeat" and "inverted repeat"? Using the base sequence illustrated, diagram duplex DNA molecule, showing the sequence as a direct repeat and as an inverted repeat.



6.15 Mutations have not been observed that result in non-functional histones. Why should this be expected?

6.16 A sequence of middle-repetitive DNA from *Drosophila* is isolated and purified. It is used as a template in a polymerization reaction with DNA polymerase and radioactive substrates, and highly radioactive probe DNA is prepared. The probe DNA is then used in an *in situ* hybridization experiment with cells containing polytene chromosomes obtained from ten different flies of the same species. Autoradiography indicates that the radioactive material is localized to about 20 sites in the genome, but they are in different sites in each fly examined. What does this observation suggest about the DNA sequence being studied?

Challenge Problem

6.17 A sample of identical DNA molecules, each containing about 3000 base pairs per molecule, is mixed with histone octamers under conditions that allow formation of chromatin. The reconstituted chromatin is then treated with a nuclease, and enzymatic digestion is allowed to occur. The histones are removed, and the positions of the cuts in the DNA are identified by sequencing the fragments. It is found that the breaks have been made at random positions and as

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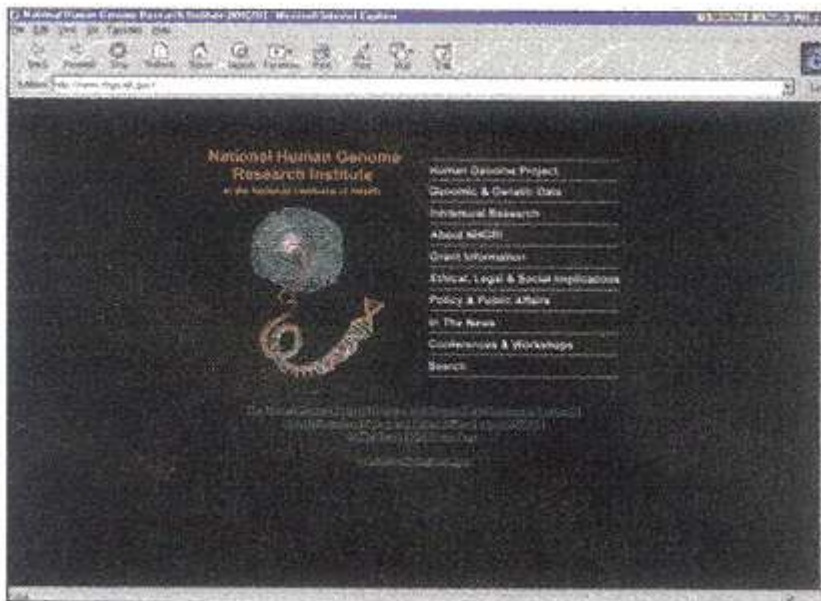
inverted repeats at its ends. If assigned to do so, write a 150-word report on these transposable elements, paying particular attention to the DNA sequence relations between the *Ac* and *Ds* elements and specifying the element that encodes the active transposase.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 6, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 6.



expected, at about 200-base-pair intervals. The experiment is then repeated with a single variation. A protein known bind to DNA is added to the DNA sample before addition of the histone octamers. Again, reconstituted chromatin is formed and digested with nuclease. In this experiment, it is found that the breaks are again at intervals of about 200 base pairs, but they are localized at particular positions in the base sequence. At each position, the site of breakage c vary over only a 2-to-3-base range. However, examination of the base sequences in which the breaks have occurred does not indicate that breakage occurs in a particular sequence; in other words, each 2-to-3-base region in which cutting occurs has a different sequence. Explain the difference between the two experiments.

Further Reading

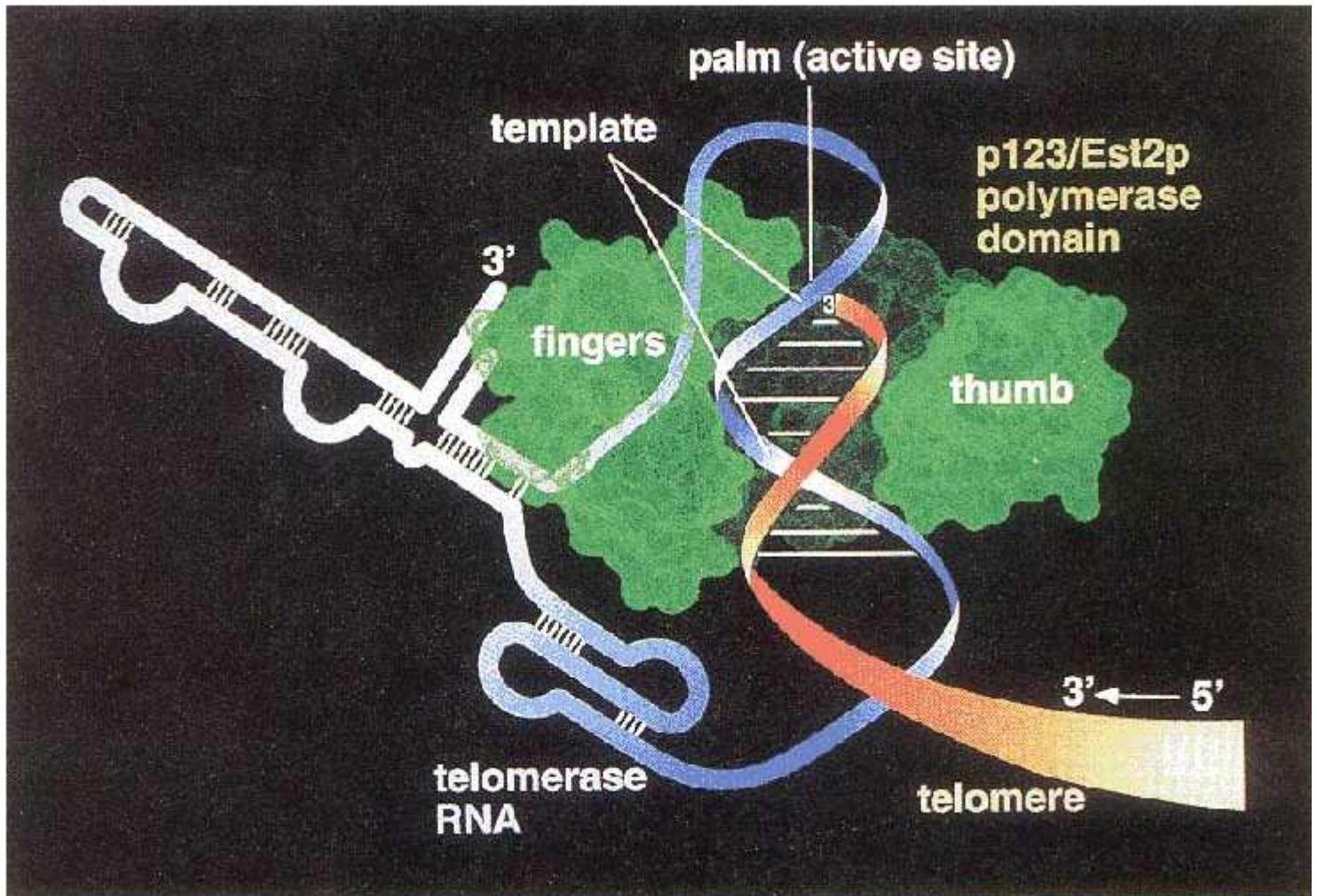
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A model of the telomerase enzyme showing protein components in green and the RNA component in blue.

The "palm" of the protein is the active site of telomere elongation, whereas the "fingers" and "thumb" are protruding. The DNA strand, whose telomere is being elongated, is shown in red:

[Courtesy of Joachim Lingner and Thomas Cech, from J. Lingner, T. R. Hughes, A. Shevchenko, M. Mann, V. Lundblad and T. R. Cech. 1997. *Science* 276: 561.]

Chapter 7— Variation in Chromosome Number and Structure

CHAPTER OUTLINE

7-1 Centromeres and the Genetic Stability of Chromosomes

7-2 Polyploidy

7-3 Monoploid Organisms

7-4 Extra or Missing Chromosomes

7-5 Human Chromosomes

Trisomy in Human Beings

Dosage Compensation

The Calico Cat as Evidence for X-Chromosome Inactivation

Sex-Chromosome Abnormalities

The Fragile-X Syndrome

Chromosome Abnormalities in Spontaneous Abortion

7-6 Abnormalities in Chromosome Structure

Deletions

Duplications

Unequal Crossing-Over in Human Red-Green Color Blindness

Inversions

Reciprocal Translocations

Robertsonian Translocations

7-7 Position Effects on Gene Expression

7-8 Chromosome Abnormalities and Cancer

Retinoblastoma and Tumor-Suppressor Genes

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Challenge Problems

Further Reading

GeNETics on the web

PRINCIPLES

- Duplication of the entire chromosome complement present in a species—or in a hybrid between species—is a major process in the evolution of higher plants.
- The genetic unbalance caused by a single chromosome that is extra or missing may have a more serious phenotypic effect than an entire extra set of chromosomes.
- Chromosome abnormalities are an important cause of human genetic disease and are a major factor in spontaneous abortions.
- Aneuploid (unbalanced) chromosome rearrangements usually have greater phenotypic effects than euploid (balanced) chromosome rearrangements.
- By a process of mispairing and unequal crossing-over, genes that are duplicated in tandem along the chromosome can give rise to chromosomes with even more copies.
- Gene-for-gene pairing between a wildtype chromosome and one that contains an inversion of a segment of genes results in the formation of a loop in one of the chromosomes; crossing-over within the "inversion loop" leads to chromosomal abnormalities.
- Reciprocal translocations result in abnormal gametes because they upset segregation.
- Some types of cancer are associated with particular chromosome rearrangements.

CONNECTIONS

CONNECTION: The First Human Chromosomal Disorder

Jerome Lejeune, Marthe Gautier, And Raymond Turpin 1959

Study of the somatic chromosomes of nine Down syndrome children

CONNECTION: Lyonization of an X Chromosome

Mary F. Lyon 1961

Gene Action In The X Chromosome Of The Mouse (*Mus Musculus* L.)

In all species, an occasional organism is found that has extra chromosomes or that lacks a particular chromosome. Such a deviation from the norm is an abnormality in chromosome *number*. Other organisms, usually rare, are found to have alterations in the arrangement of genes in the genome, such as by having a chromosome with a particular segment missing, reversed in orientation, or attached to a different chromosome. These variations are abnormalities in chromosome *structure*. This chapter will deal with the genetic effects of both numerical and structural chromosome abnormalities. We will see that animals are much less tolerant of such changes than are plants. Furthermore, in

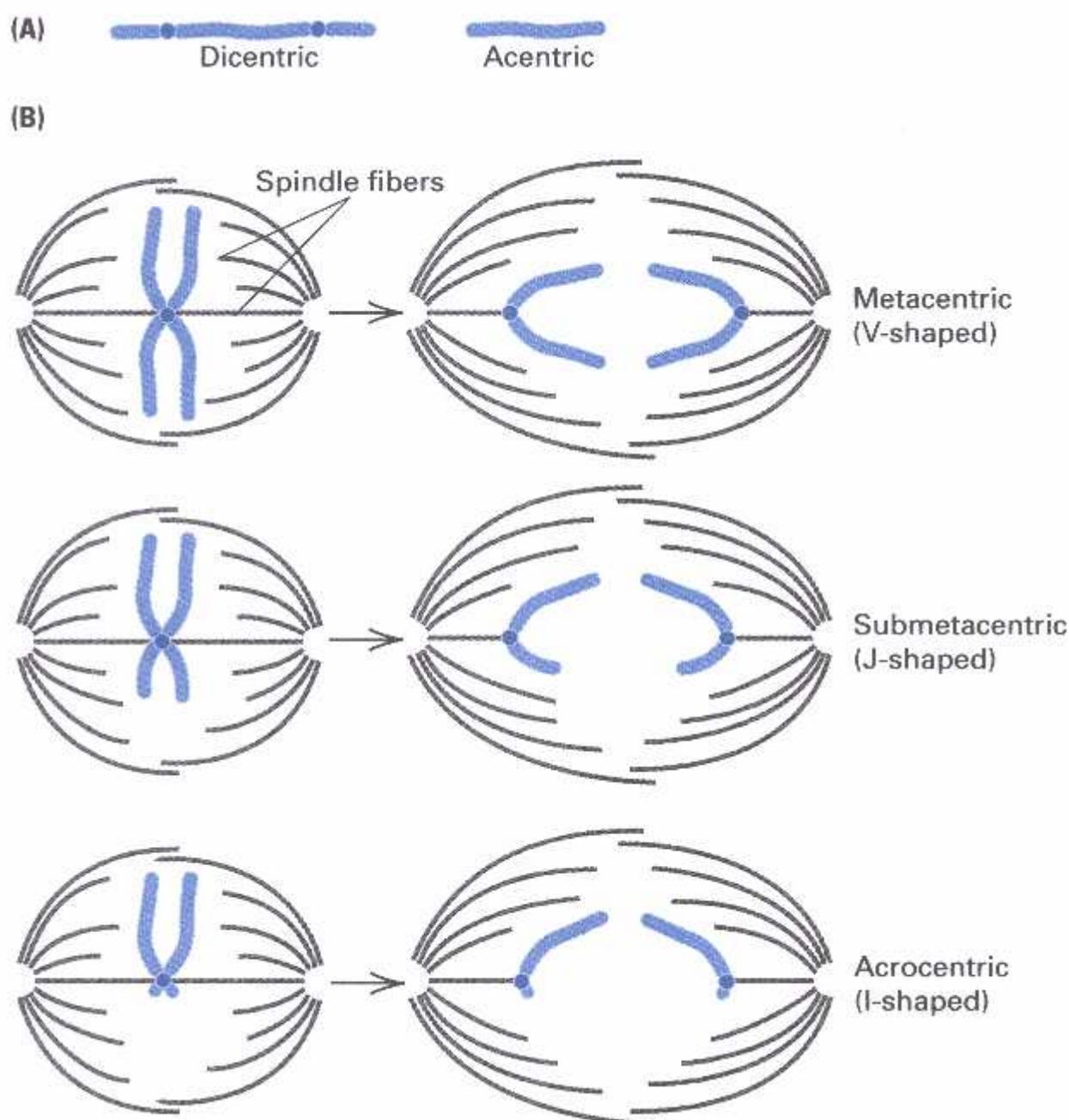


Figure 7.1

(A) Diagram of a chromosome that is dicentric (two centromeres) and one that is acentric (no centromere). Dicentric and acentric chromosomes are frequently lost in cell division, the former because the two centromeres may bridge between the daughter cells, and the latter because the chromosome cannot attach to the spindle fibers. (B) Three possible shapes of monocentric chromosomes in anaphase as determined by the position of the centromere. The centromeres are shown in dark blue.

animals, numerical alterations often produce greater effects on phenotype than do structural alterations.

Centromeres and the Genetic Stability of Chromosomes

Chromosomes that have a single centromere are usually the only ones that are transmitted reliably from parental cells to daughter cells and from parental organisms to their progeny. When a cell divides, spindle fibers attach to the centromere of each chromosome and pull the sister chromatids to opposite poles. Occasionally, a chromosome arises that has an abnormal number of centromeres, as diagrammed in Figure 7.1A. The chromosome on the left has two centromeres and is said to be **dicentric**. A dicentric chromosome is genetically unstable, which means it is not transmitted in a predictable fashion. The dicentric chromosome is frequently lost from a cell when the two centromeres proceed to opposite poles in the course of cell division; in this case, the chromosome is stretched and forms a *bridge* between the daughter cells. This bridge may not be included in either daughter nucleus, or it may break, with the result that each daughter nucleus receives a broken chromosome. The chromosome on the right in Figure 7.1A is an **acentric** chromosome, which lacks a centromere. Acentric chromosomes also are genetically unstable because they cannot be maneuvered properly during cell division and tend to be lost.

In eukaryotic organisms, virtually all chromosomes have a single centromere and are rod-shaped. (Rarely, a **ring** chromosome is found, which results from breakage and loss of the telomere at each end of a rod chromosome and subsequent fusion of the broken ends.) Monocentric rod chromosomes are often classified according to the relative position of their centromeres. A chromosome with its centromere about in the middle is a **metacentric chromosome**; the arms are of approximately equal length and form a V shape at anaphase (Figure 7.1B). When the centromere is somewhat off center, the

chromosome is a **submetacentric chromosome**, and the arms form a J shape at anaphase. A chromosome with the centromere very close to one end appears I-shaped at anaphase because the arms are grossly unequal in length; such a chromosome is **acrocentric**.

The distinction among metacentric, submetacentric, and acrocentric chromosomes is useful because it draws attention to the chromosome arms. In the evolution of chromosomes, often the number of chromosome *arms* is conserved without conservation of the individual *chromosomes*. For example, *Drosophila melanogaster* has two large metacentric autosomes, but many other *Drosophila* species have four acrocentric autosomes instead of the two metacentrics. Detailed comparison of the genetic maps of these species reveals that the acrocentric chromosomes in the other species correspond, arm for arm, with the large metacentrics in *Drosophila melanogaster* (Figure 7.2). Among higher primates, chimpanzees and human beings have 22 pairs of chromosomes that are morphologically similar, but chimpanzees have two pairs of acrocentrics not found in human beings, and human beings have one pair of metacentrics not found in chimpanzees. In this case, the human metacentric chromosome was formed by fusion of the telomeres between the short arms of the chromosomes that, in chimpanzees, remain acrocentrics. The metaphase chromosome resulting from the fusion is human chromosome 2.

7.2—

Polyploidy

The genus *Chrysanthemum* illustrates **polyploidy**, an important phenomenon found frequently in higher plants in which a species has a genome composed of multiple complete sets of chromosomes. One *Chrysanthemum* species, a diploid species, has 18 chromosomes. A closely related species has 36 chromosomes. However, comparison of chromosome morphology indicates that the 36-chromosome species has two complete sets of the chromosomes found in the 18-chromosome species (Figure 7.3). The basic chromosome set in the group, from which all the other genomes are formed, is called the **monoploid** chromosome set. In *Chrysanthemum*, the monoploid chromosome number is 9. The diploid species has two complete copies of the monoploid set, or 18 chromosomes altogether. The 36-chromosome species has four copies of the monoploid set ($4 \times 9 = 36$) and is a **tetraploid**. Other species of *Chrysanthemum* have 54 chromosomes (6×9 , constituting the **hexaploid**), 72 chromosomes (8×9 , constituting the **octoploid**), and 90 chromosomes (10×9 , constituting the **decaploid**).

In meiosis, the chromosomes of all *Chrysanthemum* species synapse normally in pairs to form bivalents (Section 3.3). The 18-chromosome species forms 9 bivalents, the 36-chromosome species forms 18 bivalents, the 54-chromosome species forms 27

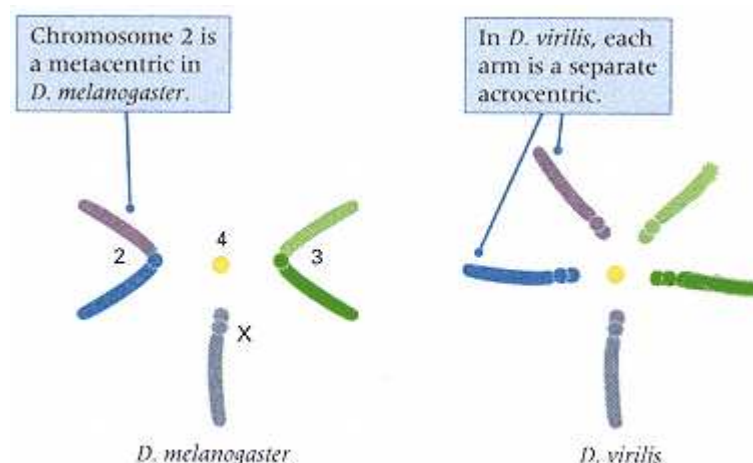


Figure 7.2

The haploid chromosome complement of two species of *Drosophila*. Color indicates homology of chromosome arms. The large metacentric chromosomes of *Drosophila melanogaster* (chromosomes 2 and 3) correspond arm for arm with the four large acrocentric autosomes of *Drosophila virilis*.

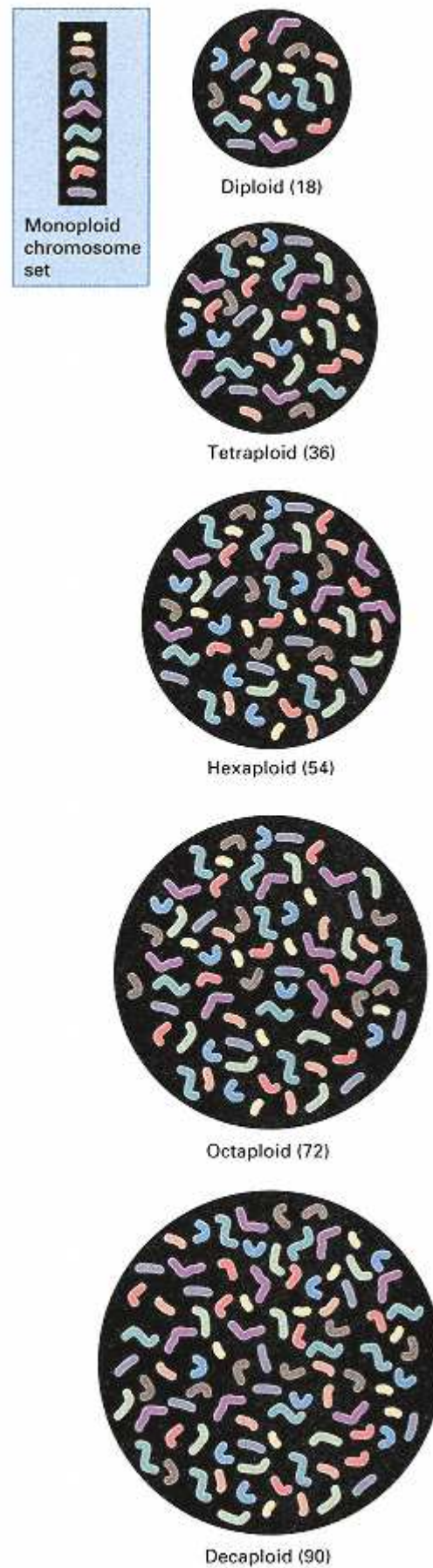


Figure 7.3
Chromosome numbers in
diploid and polyploid species of
Chrysanthemum. Each set of
homologous chromosomes
is depicted in a different
color.

bivalents, so forth. Gametes receive one chromosome from each bivalent, so the number of chromosomes in the gametes of any species is exactly half the number of chromosomes in its somatic cells. The chromosomes present in the gametes of a species constitute the **haploid** set of chromosomes. In the species of *Chrysanthemum* with 90 chromosomes, for example, the haploid chromosome number is 45; in meiosis, 45 bivalents are formed, so each gamete contains 45 chromosomes. When two such gametes come together in fertilization, the complete set of 90 chromosomes in the species is restored. Thus the gametes of a polyploid organism are not always monoploid, as they are in a diploid organism; for example, a tetraploid organism has diploid gametes.

The distinction between the terms *monoploid* and *haploid* is subtle:

- The *monoploid* chromosome set is the basic set of chromosomes that is multiplied in a polyploid series of species, such as *Chrysanthemum*.
- The *haploid* chromosome set is the set of chromosomes present in a gamete, irrespective of the chromosome number in the species.

The potential confusion arises because of diploid organisms, in which the monoploid chromosome set and the haploid chromosome set are the same. Considering the tetraploid helps to clarify the difference: It contains four monoploid chromosome sets, and the haploid gametes are diploid.

Polyploidy is widespread in certain plant groups. Among flowering plants, from 30 to 35 percent of existing species are thought to have originated as some form of polyploid. Valuable agricultural crops that are polyploid include wheat, oats, cotton, potatoes, bananas, coffee, and sugar cane. Polyploidy often leads to an increase in the size of individual cells, and polyploid plants are often larger and more vigorous than their diploid ancestors; however, there are many exceptions to these generalizations. Polyploidy is rare invertebrate animals, but it is found in a few groups of invertebrates. One reason why polyploidy is rare in animals is the difficulty in regular segregation of the sex chromo-

somes. For example, a tetraploid animal with XXXX females and XXYY males would produce XX eggs and XY sperm (if all chromosomes paired to form bivalents), so the progeny would be exclusively XXXY and thus unlike either of the parents.

Polyploid plants found in nature almost always have an even number of sets of chromosomes, because organisms that have an odd number have low fertility. Organisms with three monoploid sets of chromosomes are known as **triploids**. As far as growth is concerned, a triploid is quite normal because the triploid condition does not interfere with mitosis; in mitosis in triploids (or any other type of polyploid), each chromosome replicates and divides just as in a diploid. However, because each chromosome has more than one pairing partner, chromosome segregation is severely upset in meiosis, and most gametes are defective. Unless the organism can perpetuate itself by means of asexual reproduction, it will eventually become extinct.

The infertility of triploids is sometimes of commercial benefit. For example, the seeds in commercial bananas are small and edible because the plant is triploid and most of the seeds fail to develop to full size. In oysters, triploids are produced by treating fertilized diploid eggs with a chemical that causes the second polar body of the egg to be retained. The triploid oysters are sterile and do not spawn, so they remain edible through the hot summer months of June, July, and August (the months that lack the letter *r*), when normal oysters are spawning. In Florida and in certain other states, weed control in waterways is aided by the release of weed-eating fish (the grass carp), which do not become overpopulated because the released fish are sterile triploids.

Tetraploid organisms can be produced in several ways. The simplest mechanism is a failure of chromosome separation in either mitosis or meiosis, which instantly doubles the chromosome number. Chromosome doubling through an abortive cell division is called **endoreduplication**. In a plant species that can undergo self-fertilization, endoreduplication creates a new, genetically stable species, because the chromosomes in the tetraploid can pair two by two in meiosis and therefore segregate regularly, each gamete receiving a full diploid set of chromosomes. Self-fertilization of the tetraploid restores the chromosome number, so the tetraploid condition can be perpetuated. The genetics of tetraploid species, and that of other polyploids, is more complex than that of diploid species because the organism carries more than two alleles of any gene. With two alleles in a diploid, only three genotypes are possible: *AA*, *Aa*, and *aa*. In a tetraploid, by contrast, five genotypes are possible: *AAAA*, *AAAa*, *AAaa*, *Aaaa*, and *aaaa*. Among these genotypes, the middle three represent different types of tetraploid heterozygotes.

An octoploid species (eight sets of chromosomes) can be generated by failure of chromosome separation in mitosis in a tetraploid. If only bivalents form in meiosis, then an octoploid organism can be perpetuated sexually by self-fertilization or through crosses with other octoploids. Furthermore, cross-fertilization between an octoploid and a tetraploid results in a hexaploid (six sets of chromosomes). Repeated episodes of polyploidization and cross-fertilization may ultimately produce an entire polyploid series of closely related organisms that differ in chromosome number, as exemplified in *Chrysanthemum*.

Chrysanthemum represents a type of polyploidy, known as **autopolyploidy**, in which all chromosomes in the polyploid species derive from a single diploid ancestral species. In many cases of polyploidy, the polyploid species have complete sets of chromosomes from two or more *different* ancestral species. Such polyploids are known as **allopolyploids**. They derive from occasional hybridization between different diploid species when pollen from one species germinates on the stigma of another species and sexually fertilizes the ovule, followed by endoreduplication in the zygote to yield a hybrid plant in which each chromosome has a pairing partner in meiosis. The pollen may be carried to the wrong flower by wind, insects, or other pollinators. Figure 7.4 illustrates hybridization between species A and B in which endoreduplication leads to the formation of an allopolyploid (in this case, an *allotetraploid*), which carries a complete diploid genome from each of its two ancestral species. The formation of allopolyploids through hybridization and endoreduplication is an extremely important process in

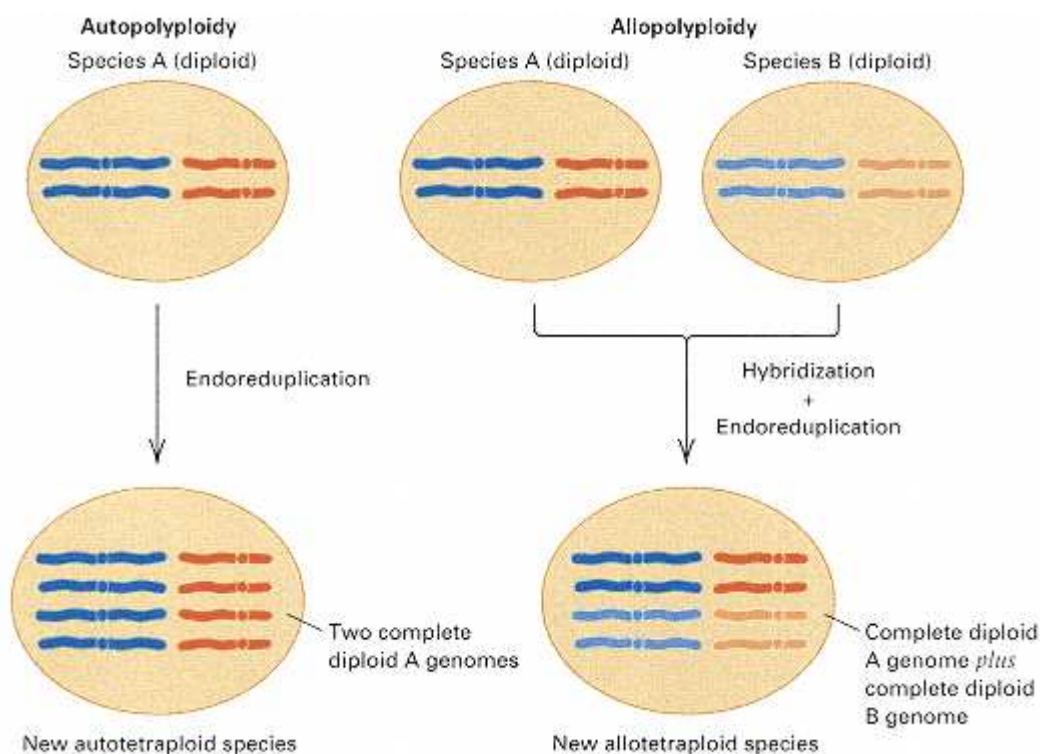


Figure 7.4

Endoreduplication (the doubling of the chromosome complement) plays a key role in the formation of polyploid species. When it takes place in a diploid species, endoreduplication results in the formation of an autotetraploid species. When it takes place in the hybrid formed by cross-fertilization between distinct species, endoreduplication results in the formation of an allotetraploid species that has a complete diploid set of chromosomes from each of the parental species.

plant evolution and plant breeding. At least half of all naturally occurring polyploids are allopolyploids. Cultivated wheat provides an excellent example of allopolyploidy. Cultivated bread wheat is a hexaploid with 42 chromosomes constituting a complete diploid genome of 14 chromosomes from each of three ancestral species. The 42-chromosome allopolyploid is thought to have originated by the series of hybridizations and endoreduplications outlined in Figure 7.5.

The ancestral origin of the chromosome sets in an allopolyploid can often be revealed by the technique of **chromosome painting**, in which chromosomes are "painted" different colors by hybridization with DNA strands labeled with fluorescent dyes. DNA from each of the putative ancestral species is isolated, denatured, and labeled with a different fluorescent dye. Then the labeled single strands are spread on a microscope slide and allowed to renature with homologous strands present in the chromosomes of the allopolyploid species.

An example of chromosome painting is shown in Figure 7.6. The flower is from a variety of crocus called Golden Yellow. Its genome contains seven pairs of chromosomes, which are shown painted in yellow and green. Golden Yellow was thought to be an allopolyploid formed by hybridization of two closely related species followed by endoreduplication of the chromosomes in the hybrid. The putative ancestral species are *Crocus flavus*, which has four pairs of chromosomes, and *Crocus angustifolius*, which has three pairs of chromosomes. To paint the chromosomes of Golden Yellow, DNA from *C. flavus* was isolated and labeled with a fluorescent green dye, and that from *C. angustifolius* was isolated and labeled with a fluorescent yellow

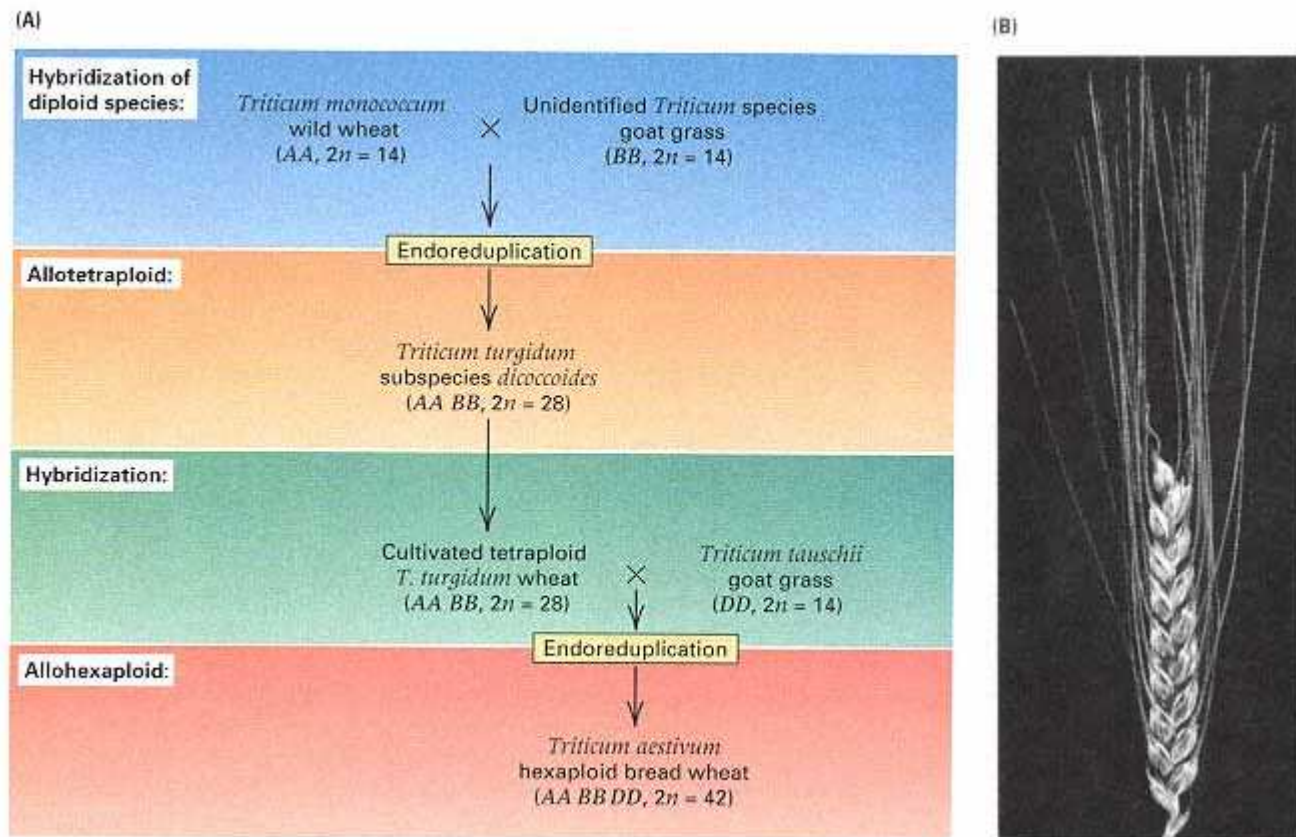


Figure 7.5

(A) Repeated hybridization and endoreduplication in the ancestry of cultivated bread wheat (*Triticum aestivum*), which is an allohexaploid containing complete diploid genomes (AA, BB, DD) from three ancestral species. (B) The spike of *T. turgidum* in the photograph is an allotetraploid. The large grains of this species made it attractive to early hunter-gatherer societies in the Middle East. One of the earliest cultivated wheats, *T. turgidum*, is the progenitor of commercial macaroni wheat [Photograph courtesy of Gordon Kimber.]

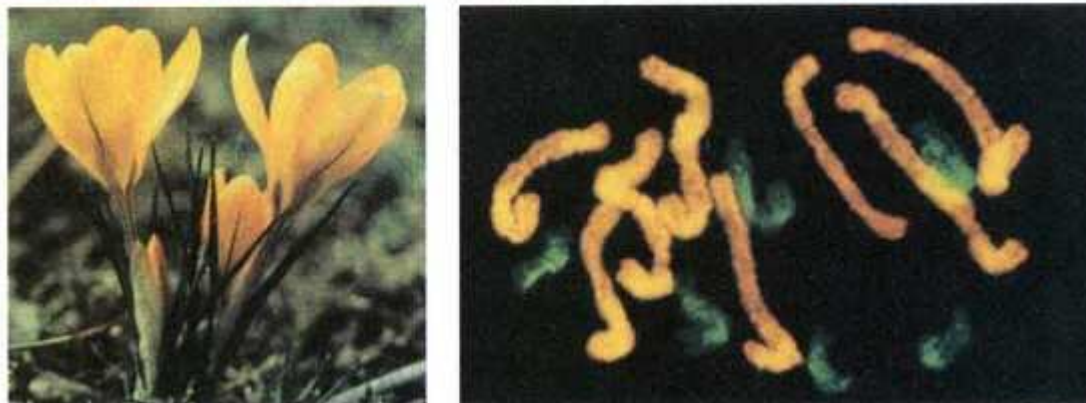


Figure 7.6

Flower of the crocus variety Golden Yellow and chromosome painting that reveals its origin as an allopolyploid. Its seven pairs of chromosomes are shown at the right. The chromosomes in green hybridized with DNA from *C. angustifolius*, which has three pairs of chromosomes, and those in yellow hybridized with DNA from *C. flavus*, which has four pairs of chromosomes. [Courtesy of J. S. Heslop-Harrison, John Innes Centre, Norwich, UK. With permission of the *Annals of Botany*.]

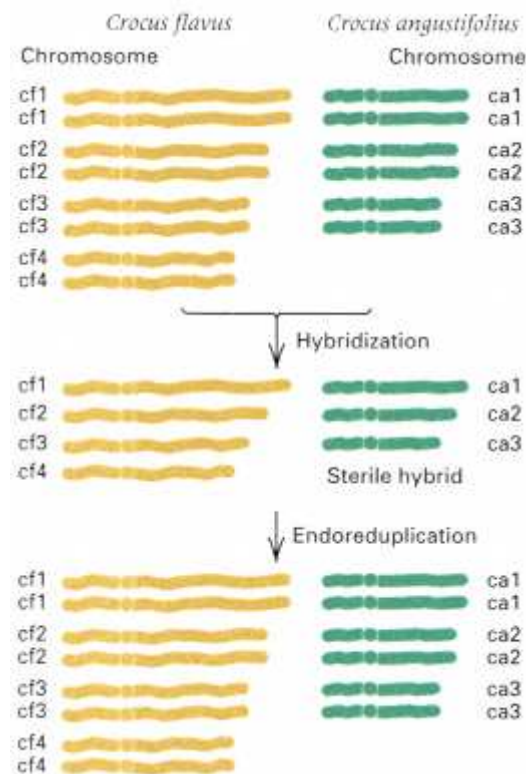


Figure 7.7

Allopolyploidy in the evolutionary origin of the Golden Yellow crocus. The original hybrid between *C. flavus* and *C. angustifolius* produced a sterile monoploid with one copy of each chromosome from each species. Chromosome endoreduplication in the monoploid results in a fertile allotetraploid with a complete diploid genome from each species.

dye. The result of the chromosome painting is very clear: three pairs of chromosomes hybridize with the green-labeled DNA from *C. flavus*, and four pairs of chromosomes hybridize with the yellow-labeled DNA from *C. angustifolius*. This pattern of hybridization strongly supports the hypothesis of the autopolyploid origin of Golden Yellow diagrammed in Figure 7.7.

7.3—

Monoploid Organisms

As noted, the monoploid chromosome set is the set of chromosomes multiplied in polyploid species. An organism is monoploid if it develops from a monoploid cell. Meiosis cannot take place normally in the germ cells of a monoploid, because each chromosome lacks a pairing partner, and hence monoploids are usually sterile. Monoploid organisms are quite rare, but they occur naturally in certain insect species (ants, bees) in which males are derived from unfertilized eggs. These monoploid males are fertile because the gametes are produced by a modified meiosis in which chromosomes do not separate in meiosis I.

Monoploids are important in plant breeding because, in the selection of diploid organisms with desired properties, favorable recessive alleles may be masked by heterozygosity. This problem can be avoided by studying monoploids, provided that their sterility can be overcome. In many plants, the production of monoploids capable of reproducing can be stimulated by conditions that yield aberrant cell divisions. Two techniques make this possible.

With some diploid plants, monoploids can be derived from cells in the anthers (the pollen-bearing structures). Extreme chilling of the anthers causes some of the haploid cells destined to become pollen grains to begin to divide. These cells are monoploid as well as haploid. If the cold-shocked cells are placed on an agar surface containing suitable nutrients and certain plant hormones, then a small dividing mass of cells called an **embryoid** forms. A subsequent change of plant hormones in the growth medium causes the embryoid to form a small plant with roots and leaves that can be potted in soil and allowed to grow normally. Because monoploid cells have only a single set of chromosomes, their genotypes can be identified without regard to the dominance or recessiveness of

individual alleles. A plant breeder can then select a monoploid plant with the desired traits. In some cases, the desired genes are present in the original diploid plant and are merely sorted out and selected in the monoploids. In other cases, the anthers are treated with mutagenic agents in the hope of producing the desired traits.

When a desired mutation is isolated in a monoploid, it is necessary to convert the monoploid into a homozygous diploid because the monoploid plant is sterile and

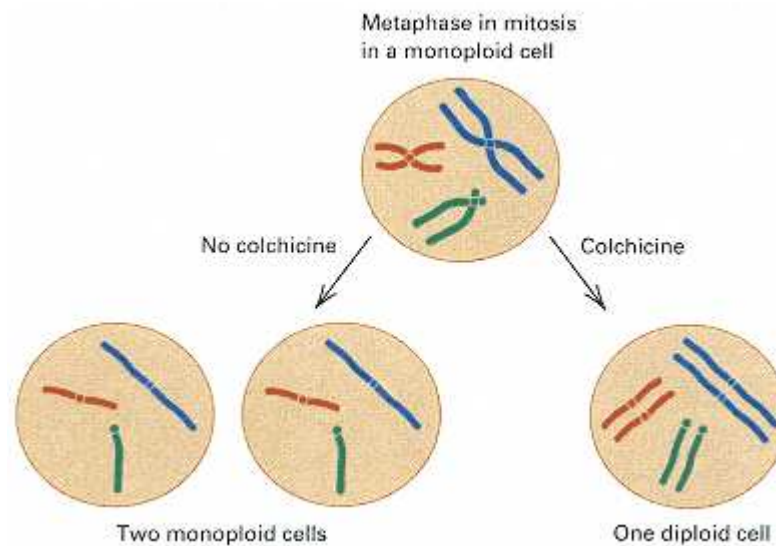


Figure 7.8
Production of a diploid from a monoploid by treatment with colchicine. The colchicine disrupts the spindle and thereby prevents separation of the chromatids after the centromeres (circles) divide.

does not produce seeds. Converting the monoploid into a diploid is possible by treatment of the meristematic tissue (the growing point of a stem or branch) with the substance **colchicine**. This chemical is an inhibitor of the formation of the mitotic spindle. When the treated cells in the monoploid meristem begin mitosis, the chromosomes replicate normally; however, the colchicine blocks metaphase and anaphase, so the result is endoreduplication (doubling of each chromosome in a given cell). Many of the cells are killed by colchicine, but, fortunately for the plant breeder, a few of the monoploid cells are converted into the diploid state (Figure 7.8). The colchicine is removed to allow continued cell multiplication, and many of the now-diploid cells multiply to form a small sector of tissue that can be recognized microscopically. If placed on a nutrient agar surface, this tissue will develop into a complete plant. Such plants, which are completely homozygous, are fertile and produce normal seeds.

7.4—

Extra or Missing Chromosomes

Occasionally, organisms are formed that have extra copies of individual chromosomes, rather than extra entire sets of chromosomes. This situation is called **polysomy**. In contrast to polyploids, which in plants are often healthy and in some cases are more vigorous than the diploid, polysomics are usually less vigorous than the diploid and have abnormal phenotypes. In most plants, a single extra chromosome (or a missing chromosome) has a more severe effect on phenotype than the presence of a complete extra set of chromosomes. Each chromosome that is extra or missing results in a characteristic phenotype. For example, Figure 7.9 shows the seed capsule of the Jimson weed *Datura stramonium*, beneath which is a series of capsules of strains, each having an extra copy of a different chromosome. The seed capsule of each of the strains is distinctive.

An otherwise diploid organism that has an extra copy of an individual chromosome is called a **trisomic**. In a trisomic organism, the segregation of chromosomes in meiosis is upset because the trisomic chromosome has two pairing partners instead of one. The behavior of the chromosomes in meiosis depends on the manner in which the homologous chromosome arms pair and on the chiasmata formed between them. In some cells, the three chromosomes form a **trivalent** in which distinct parts of one chromosome are paired with homologous parts of each of the others (Figure 7.10A). In metaphase, the trivalent is usually oriented with two centromeres pointing toward one pole and the other centromere pointing toward the other. The result is

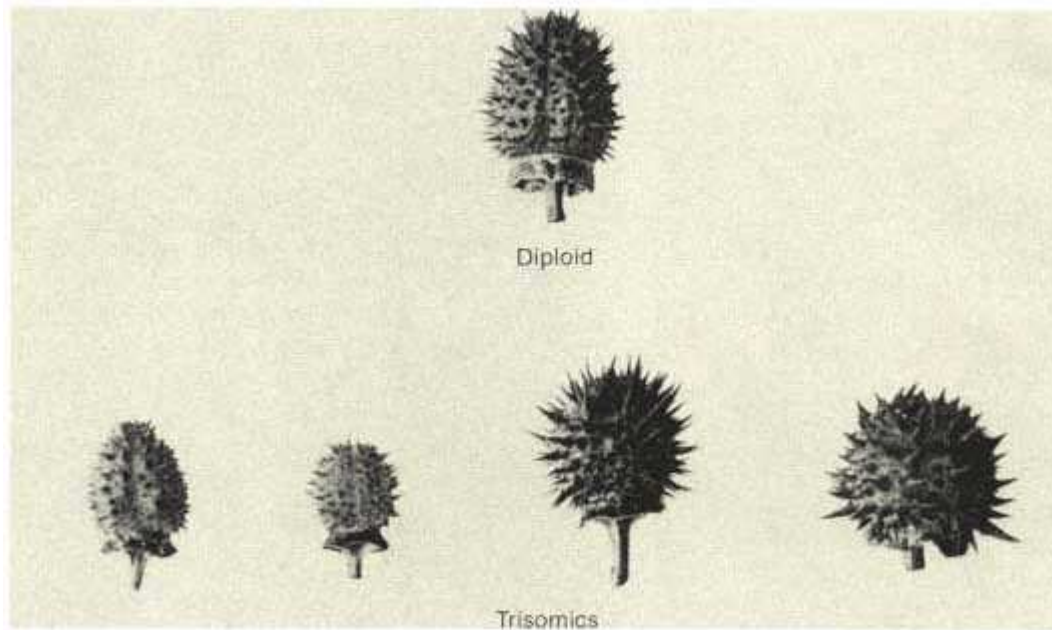


Figure 7.9
Seed capsules of the normal diploid *Datura stramonium* (Jimson weed), which has a haploid number of 12 chromosomes, and 4 of the 12 possible trisomics. The phenotype of the seed capsule in trisomics differs according to the chromosome that is trisomic.

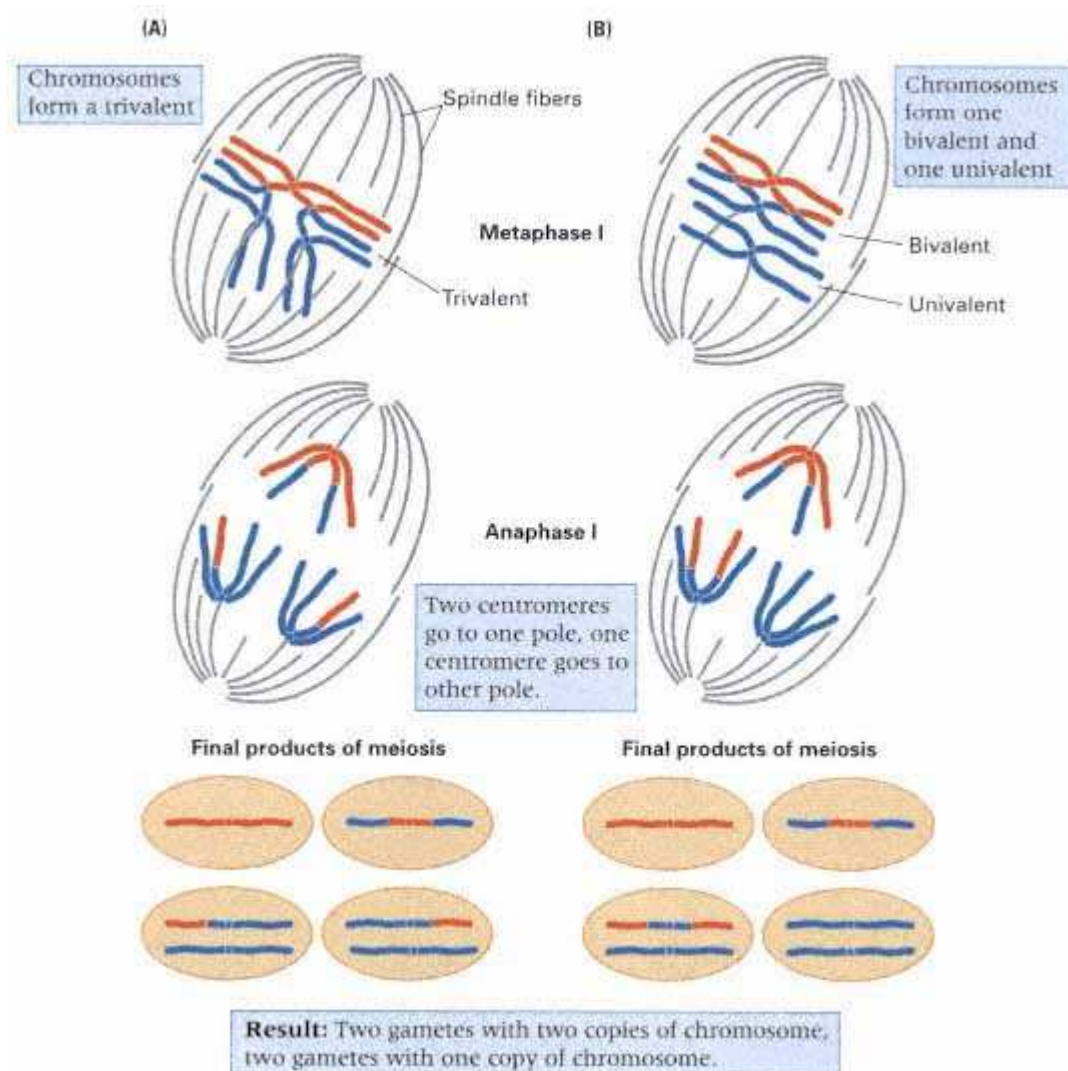


Figure 7.10
Meiotic pairing in a trisomic. (A) Formation of a trivalent. (B) Formation of a bivalent and a univalent. Both types of pairing result in one pair of gametes containing two copies of the trisomic chromosome and the other pair of gametes containing one copy

of the trisomic chromosome

that, at the end of both meiotic divisions, one pair of gametes contains two copies of the trisomic chromosome, and the other pair of gametes contains only a single copy. Alternatively, the trisomic chromosome can form one normal bivalent and one **univalent**, or unpaired chromosome, as shown in Figure 7.10B. In anaphase I, the bivalent disjoins normally and the univalent usually proceeds randomly to one pole or the other. Again, the end result is the formation of two products of meiosis that contain two copies of the trisomic chromosome and two products of meiosis that contain one copy. To state the matter in another way, a trisomic organism with three copies of a chromosome (say, $C C C$) will produce gametes among which half contain $C C$ and half contain C . The unequal segregation of trisomic chromosomes is also the cause of the infertility in triploids, because in a triploid, each chromosome behaves independently as though it were trisomic. For each of the homologous chromosomes in a triploid, a gamete can receive either one copy or two copies.

Polysomy generally results in more severe phenotypic effects than does polyploidy. The usual explanation is that the greater harmful phenotypic effects in trisomics are related to the imbalance in the number of copies of different genes. A polyploid organism has a "balanced" genome in the sense that the ratio of the numbers of copies of any pair of genes is the same as in the diploid. For example, in a tetraploid, each gene is present in twice as many copies as in a diploid, so no gene or group of genes is out of balance with the others. Balanced chromosome abnormalities, which retain equality in the number of copies of each gene, are said to be **euploid**. In contrast, gene equality is upset in a trisomic because three copies of the genes located in the trisomic chromosome are present, whereas two copies of the genes in the other chromosomes are present. Such unbalanced chromosome complements are said to be **aneuploid**. Aneuploid abnormalities are usually more severe than euploid abnormalities. For example, in *Drosophila*, triploid females are viable, fertile, and nearly normal in morphology, whereas trisomy for either of the two large autosomes is invariably lethal (the larvae die at an early stage).

Just as an occasional organism may have an extra chromosome, a chromosome may be missing. Such an individual is said to be **monosomic**. A missing copy of a chromosome results in more harmful effects than an extra copy of the same chromosome, and monosomy is often lethal. If the monosomic organism can survive to sexual maturity, then chromosomal segregation in meiosis is unequal because the monosomic chromosome forms a univalent. Half of the gametes will carry one copy of the monosomic chromosome, and the other half will contain no copies.

7.5—

Human Chromosomes

The chromosome complement of a normal human male is illustrated in Figure 7.11. The chromosomes have been treated with a staining reagent called Giemsa, which causes the chromosomes to exhibit transverse bands that are specific for each pair of homologs. These bands permit the chromosome pairs to be identified individually. By convention, the autosome pairs are arranged and numbered from longest to shortest and, on the basis of size and centromere position, are separated into seven groups designated by the letters A through G. This conventional representation of chromosomes is called a **karyotype**; it is obtained by cutting each chromosome out of a photograph taken at metaphase and pasting it into place. In a karyotype of a normal human female, the autosomes would not differ from those of a male and hence would be identical to those in Figure 7.11 but there would be two X chromosomes instead of an X and a Y. Abnormalities in chromosome number and morphology are made evident by a karyotype.

The nomenclature of the banding patterns in human chromosomes is shown in Figure 7.12. For each chromosome, the short arm is designated with the letter p, which stands for "petite," and the long arm by the letter q, which stands for "not-p." Within each arm the regions are numbered. The first digit is the major region, numbered consecutively proceeding from the centromere toward the telomere;

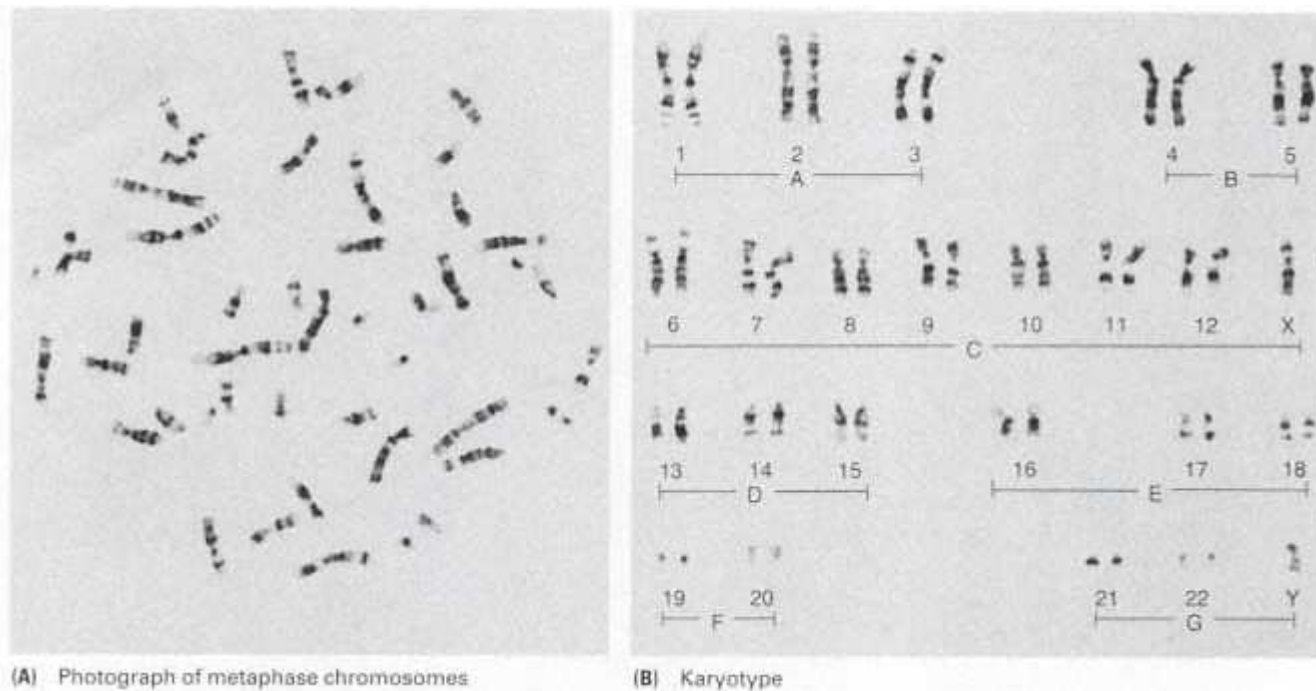


Figure 7.11

A karyotype of a normal human male. Blood cells arrested in metaphase were stained with Giemsa and photographed with a microscope. (A) The chromosomes as seen in the cell by microscopy. (B) The chromosomes have been cut out of the photograph and paired with their homologs. [Courtesy of Patricia Jacobs.]

within each region, the second number indicates the next-smaller division, again numbered outward from the centromere. For example, the designation 1p34 indicates chromosome 1, short arm, division 34. Some divisions can be subdivided still further by the bands and interbands within them, which are numbered consecutively and indicated by a digit placed after a decimal point following the main division; for example, 1p36.2 means the second band in 1p36. Some familiar genetic landmarks in the human genome are the Rh (Rhesus) blood group locus, which is somewhere in the region between 1p34 and 1p36.2; the ABO blood group locus at 9q34; the red-green color-blindness genes at Xq28, and the male-determining gene on the Y chromosome, called *SRY* (*sex-determining region, Y*), at Yp11.3. At the base of each chromosome in Figure 7.12, the red number indicates the number of genes presently assigned a position on the chromosome. These numbers do not include the approximately 5000 DNA markers used in developing the human genetic map. Many of the genes enumerated in Figure 7.12 are associated with inherited diseases. There are a total of 3399 genes assigned a position on one of the autosomes, 251 assigned a position on the X chromosome, and 19 assigned a position on the Y chromosome.

The technique of chromosome painting seen earlier in Figure 7.6 has been applied to human chromosomes to achieve spectacular effects. An example is shown in the metaphase spread and karyotype in Figure 7.13. To produce this effect, chromosomes were hybridized simultaneously with 27 different fluorescent DNA probes, each specific for hybridization with a single chromosome or chromosome arm. The probes were obtained by microdissection of metaphase nuclei to isolate each individual chromosome or chromosome arm, and the minuscule amount of DNA so obtained was amplified by the polymerase chain reaction (Section 5.8). After hybridization, each chromosome was scanned along its length, and at each point, the fluorescent signal

Connection The First Human Chromosomal Disorder

Jerome Lejeune, Marthe Gautier, and Raymond Turpin 1959

National Center for Scientific Research, Paris, France

Study of the Somatic Chromosomes of Nine Down Syndrome Children (original in French)

Down syndrome had been one of the greatest mysteries in human genetics. One of the most common forms of mental retardation, the syndrome did not follow any pattern of Mendelian inheritance. Yet some families had two or more children with Down syndrome. (Many of these cases are now known to be due to a translocation involving chromosome 21.) This paper marked a turning point in human genetics by demonstrating that Down syndrome actually results from the presence of an extra chromosome. It was the first identified chromosomal disorder. The excerpt uses the term telocentric, which means a chromosome that has its centromere very near one end. In the human genome, the smallest chromosomes are three very small telocentric chromosomes. These are chromosomes 21, 22, and the Y. A normal male has five small telocentrics (21, 21, 22, 22, and Y); a normal female has four (21, 21, 22, and 22). (The X is a medium-sized chromosome with its centromere somewhat off center.) In the table that follows, note the variation in chromosome counts in the "doubtful" cells. The methods for counting chromosomes were then very difficult, and many errors were made either by counting two nearby chromosomes as one or by including in the count of one nucleus a chromosome that actually belonged to a nearby nucleus. Lejeune and collaborators wisely chose to ignore these doubtful counts and based their conclusion only on the "perfect" cells. Sometimes good science is a matter of knowing which data to ignore.

The culture of fibroblast cells from nine Down syndrome children reveals the presence of 47 chromosomes, the supernumerary chromosome being a small telocentric one. The hypothesis of the chromosomal determination of Down syndrome is considered . . . The observations made in these nine cases (five boys and four girls) are recorded in the table to the right.

The number of cells counted in each case may seem relatively small. This is due to the fact that only the pictures [of the spread chromosomes] that claim a minimum of interpretation have been retained in this table. The apparent variation in the chromosome number in the "doubtful" cells, that is to say, cells in

Analysis of the chromosome set of the "perfect" cells reveals the presence in Down syndrome boys of 6 small telocentric chromosomes (instead of 5 in the normal man) and 5 small telocentric ones in Down syndrome girls (instead of 4 in the normal woman).

which each chromosome cannot be noted individually with certainty, has been pointed out by several authors. It does not seem to us that this phenomenon represents a cytological reality, but merely reflects the difficulties of a delicate technique. It therefore seems logical to prefer a small number of absolutely certain counts ("perfect" cells in the table) to a mass of doubtful observations, the statistical variance of which rests solely on the lack of precision of the observations. Analysis of the chromosome set of the "perfect" cells reveals the presence in Down syndrome boys of 6 small telocentric chromosomes (instead of 5 in the normal man) and 5 small telocentric ones in Down syndrome girls (instead of 4 in the normal woman) . . . It therefore seems legitimate to conclude that there exists in Down syndrome children a small super-numerary telocentric chromosome,

		Number of chromosomes

		"Doubtful" cells			"Perfect" cell		
		46	47	48	46	47	48
Boys	1	6	10	2	–	11	–
	2	–	2	1	–	9	–
	3	–	1	1	–	7	–
	4	–	3	–	–	1	–
	5	–	–	–	–	8	–
Girls	1	1	6	1	–	5	–
	2	1	2	–	–	8	–
	3	1	2	1	–	4	–
	4	1	1	2	–	4	–

accounting for the abnormal figure of 47. To explain these observations, the hypothesis of nondisjunction of a pair of small telocentric chromosomes at the time of meiosis can be considered . . . It is, however, not possible to say that the supernumerary small telocentric chromosome is indeed a normal chromosome and at the present time the possibility cannot be discarded that a fragment resulting from another type of aberration is involved.

Source: Comptes rendus des séances de l'Académie des Sciences 248: 1721–1722 Translation in S. H. Boyer. 1963. (Papers on Human Genetics). Englewood Cliffs, NJ: Prentice Hall

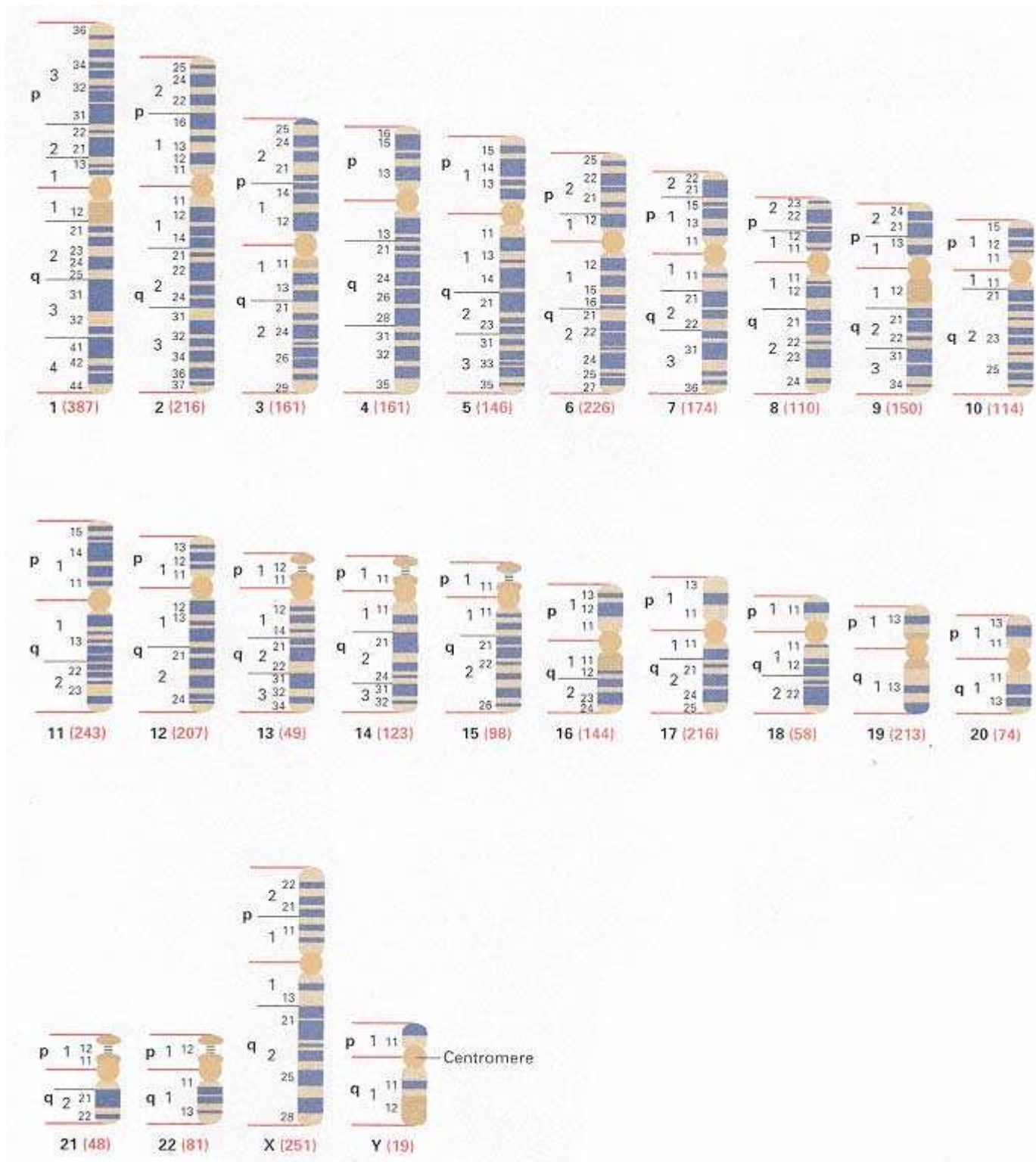
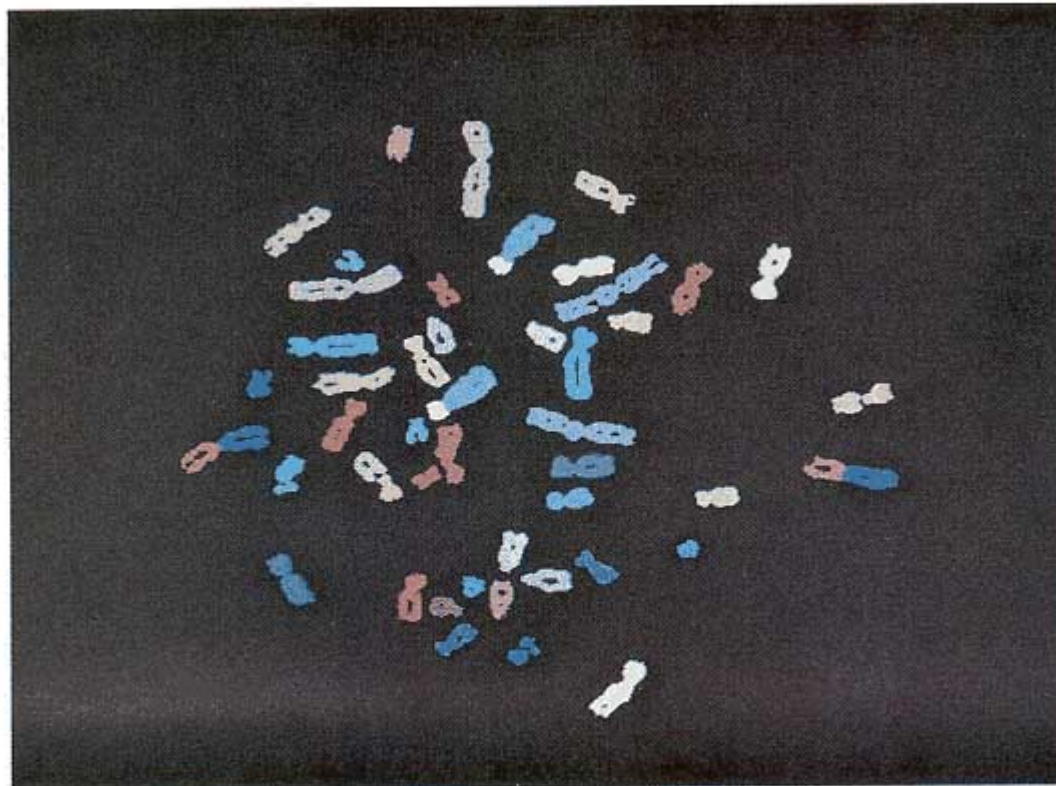


Figure 7.12

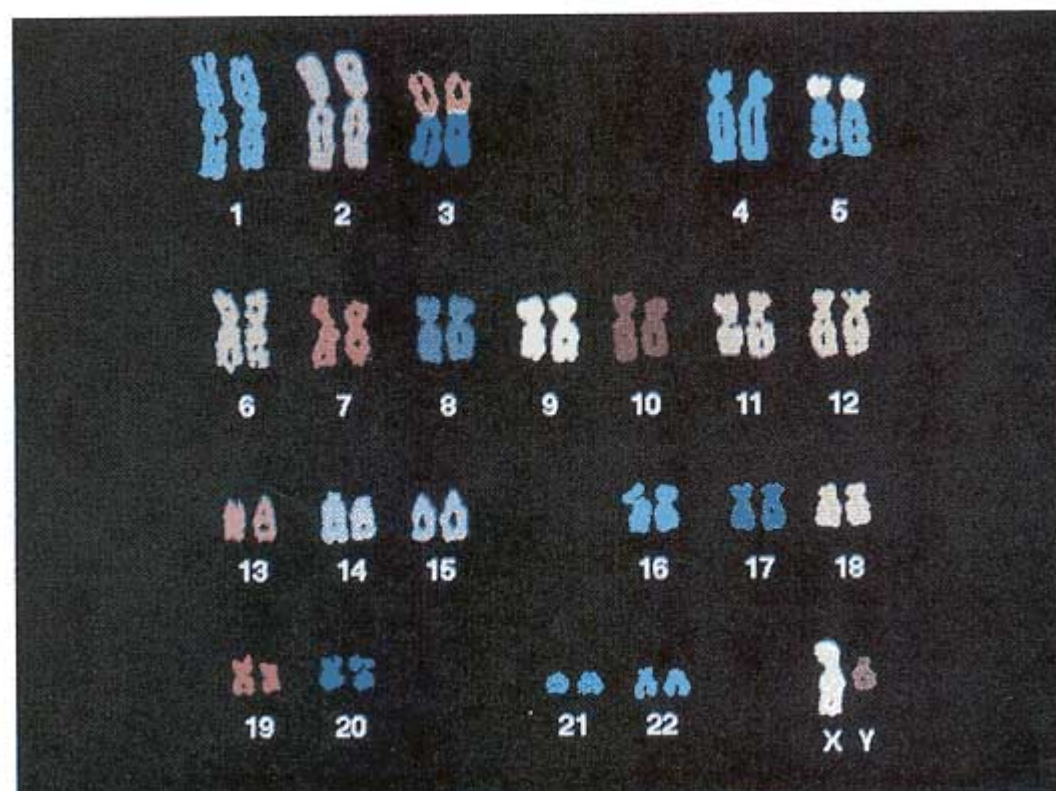
Designations of the bands and interbands in the human karyotype. Below each chromosome is the number of genes presently assigned a location on the chromosome.

[Data from V. A. McKusick. 1988. *Mendelian Inheritance in Man*, 8th ed. Baltimore: Johns Hopkins University Press.]

was converted to a specific color in the visible spectrum. The chromosomes in Figure 7.13 are therefore "painted" in 27 different colors. This technique is of considerable utility in human cytogenetics because even complex chromosome rearrangements can be detected rapidly and easily. The painting technique makes it possible to decipher some chromosome rearrangements, particularly those involving small pieces of chromosome, that are not amenable to analysis by conventional banding procedures.



(A) Metaphase spread



(B) Karyotype

Figure 7.13

(A) Metaphase spread and (B) karyotype in which human chromosomes have been "painted" in 27 different colors according to their hybridization with fluorescent probes specific to individual chromosomes or chromosome arms.

[Courtesy of M. R. Speicher and D. C. Ward. See M. R. Speicher, S. G. Ballard, and D. C. Ward. 1996. *Nature Genetics* 12: 368.]

Trisomy in Human Beings

Monosomy or trisomy of most human autosomes is usually incompatible with life. Most zygotes with missing chromosomes or extra chromosomes either fail to begin embryonic development or undergo spontaneous abortion at an early stage. There are a few exceptions. One exception is **Down syndrome**, which is caused by trisomy of chromosome 21. Down syndrome affects about 1 in 750 live-born children. Its major symptom is mental retardation, but there can be multiple physical abnormalities as well, such as major heart defects.

Most cases of Down syndrome are caused by nondisjunction, which means the failure of homologous chromosomes to separate in meiosis, as explained in Chapter 3. The result of chromosome-21 nondisjunction is one gamete that contains two copies of chromosome 21 and one that contains none. If the gamete with two copies participates in fertilization, then a zygote with trisomy 21 is produced. The gamete with one copy may also participate in fertilization, but zygotes with monosomy 21 do not survive even through the first few days or weeks of pregnancy. For unknown reasons, nondisjunction of chromosome 21 is more likely to happen in oogenesis than in spermatogenesis, and so the abnormal gamete in Down syndrome is usually the egg.

Chromosome 21 is a small chromosome and therefore is somewhat less likely to undergo meiotic crossing-over than a longer one. Noncrossover bivalents sometimes have difficulty aligning at the metaphase plate because they lack a chiasma to hold them together, so there is an increased risk of nondisjunction. Among the events of nondisjunction that result in Down syndrome, about 40 percent are derived from such nonexchange bivalents. (Students already familiar with trisomy 21 may know it as *Down's syndrome*, with an apostrophe s. This text follows current practice in human genetics in avoiding the possessive form of proper names used to designate syndromes.)

The risk of nondisjunction of chromosome 21 increases dramatically with the age of the mother, and the risk of Down syndrome reaches 6 percent in mothers of age 45 and older (Figure 7.14). Thus many physicians recommend that older women who are pregnant have cells from the fetus tested in order to detect Down syndrome prenatally. This can be done from 15 to 16 weeks after fertilization by **amniocentesis**, in which cells of a developing fetus are obtained by insertion of a fine needle through the wall of the uterus and into the sac of fluid (the *amnion*) that contains the fetus, or even earlier in pregnancy by sampling cells from another of the embryonic membranes (the *chorion*). In about 3 percent of families with a Down syndrome child, the risk of another affected child is very high—up to 20 percent of births. This high risk is caused by a chromosome abnormality called a *translocation* in one of the parents, which will be considered in Section 7.6.

Dosage Compensation

Abnormal numbers of sex chromosomes usually produce less severe phenotypic effects than do abnormal numbers of autosomes. For example, the effects of extra Y chromosomes are relatively mild. This is in part because the Y chromosome in mammals is largely heterochromatic. In human beings, there is a region at the tip of the short arm of the Y that is homologous with a corresponding region at the tip of the short arm of the X chromosome. It is in this region of homology that the X and Y chromosomes synapse in spermatogenesis, and an obligatory crossover in the region holds the chromosomes together and ensures proper separation during anaphase I. The crossover is said to be obligatory because it takes place somewhere in this region in every meiotic division. The shared X–Y homology defines the **pseudoautosomal** region. Genes within the pseudoautosomal region show a pattern of inheritance very similar to ordinary autosomal inheritance, because they are not completely linked to either the X chromosome or the Y chromosome but can exchange between the sex chromosomes by crossing-over. Near the pseudoautosomal region, but not within it, the Y chromosome contains the master sex-controller gene *SRY* at position Yp11.3. *SRY* codes for a protein transcription factor, the **testis-determining factor (TDF)**, which triggers male embryonic develop-

ment by inducing the undifferentiated embryonic genital ridge, the precursor of the gonad, to develop as a testis. A transcription factor, as the name implies, is a component that, together with other factors, stimulates transcription of its target genes. Apart from the pseudoautosomal pairing region and *SRY*, the Y chromosome contains very few genes, so extra Y chromosomes are milder in their effects on phenotype than are extra autosomes.

Extra X chromosomes have milder effects than extra autosomes because, in mammals, all X chromosomes except one are genetically inactivated very early in embryonic development. The inactivation tends to minimize the phenotypic effects of extra X chromosomes, but there are still some effects due to a block of genes near the tip of the short arm that are *not* inactivated.

In female mammals, X-chromosome inactivation is a normal process in embryonic development. In human beings, at an early stage of embryonic development, one of the two X chromosomes is inactivated in each somatic cell; different tissues undergo X inactivation at different times. The X chromosome that is inactivated in a particular somatic cell is selected at random, but once the decision is made, the same X chromosome remains inactive in all of the descendants of the cell.

X-chromosome inactivation has two consequences. First, it equalizes the number of active copies of X-linked genes in females and males. Although a female has two X chromosomes and a male has only one, because of inactivation of one X chromosome in each of the somatic cells of the female, the number of active X chromosomes in both sexes is one. In effect, gene dosage is equalized except for the block of genes in the short arm of the X that escapes inactivation. This equalization in dosage of active genes is called **dosage compensation**. The mammalian method of dosage compensation by means of X inactivation was originally proposed by Mary Lyon and is called the **single-active-X principle**.

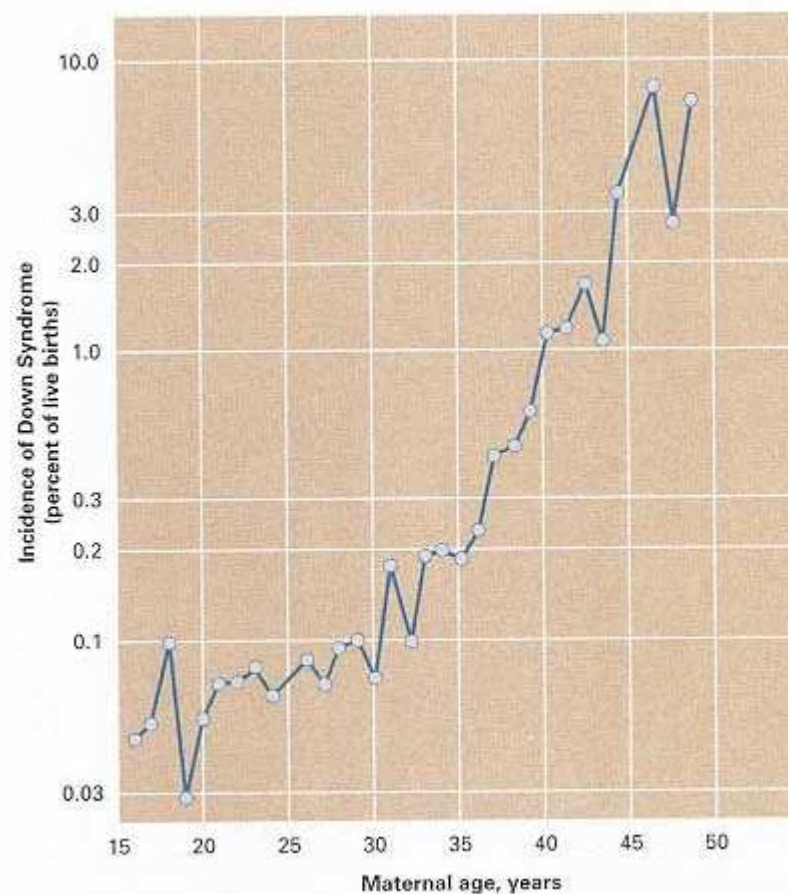


Figure 7.14

Frequency of Down syndrome number of cases per 100 live births) related to age of mother. The graph is based on 438 Down syndrome births (among 330,859 total births) in Sweden in the period 1968 to 1970.

[Data from E. B. Hook and A. Lindsjö. 1978. *Am. J. Human Genet.* 30:19.]

Connection Lyonization of an X Chromosome

Mary F. Lyon 1961

Medical Research Council,
Harwell, England
Gene Action in the X Chromosome of the Mouse (Mus musculus L.)

How do organisms solve the problem that females have two X chromosomes whereas males have only one? Unless there were some type of correction (called dosage compensation), the unequal number would mean that for all the genes in the X chromosome, cells in females would have twice as much gene product as cells in males. It would be difficult for the developing organism to cope with such a large difference in dosage for so many genes. The problem of dosage compensation has been solved by different organisms in different ways. The hypothesis put forward in this paper is that in the mouse (and, by inference, in other mammals), the mechanism is very simple: One of the X chromosomes, chosen at random in each cell lineage early in development, becomes inactivated and remains inactivated in all descendant cells in the lineage. In certain cells, the inactive X chromosome becomes visible in interphase as a deeply staining "sex-chromatin body." We now know that there are a few genes in the short arm of the X chromosome that are not inactivated. There is also good evidence that the inactivation of the rest of the X chromosome takes place sequentially from an "X-inactivation center." We also know that in marsupial mammals, such as the kangaroo, it is always the paternal X chromosome that is inactivated.

It has been suggested that the so-called sex chromatin body is composed of one heteropyknotic [that is, deeply staining during interphase] X chromosome . . . The

The coat of the tortoiseshell cat, being a mosaic of the black and yellow colours of the two homozygous genotypes, fulfills this expectation.

present communication suggests that evidence of mouse genetics indicates: (1) that the heteropyknotic X chromosome can be either paternal or maternal in origin, in different cells of the same animal; (2) that it is genetically inactivated. The evidence has two main parts. First, the normal phenotype of XO females in the mouse shows that only one active X chromosome is necessary for normal development, including sexual development. The second piece of evidence concerns the mosaic phenotype of female mice heterozygous for some sex-linked [X-linked] mutants. All sex-linked mutants so far known affecting coat colour cause a "mottled" or "dappled" phenotype, with patches of normal and mutant colour. . . . It is here suggested that this mosaic phenotype is due to the inactivation of one or other X chromosome early in embryonic development. If this is true, pigment cells descended from the cells in which the chromosome carrying the mutant gene was inactivated will give rise to a normal-coloured patch and those in which the chromosome carrying the normal gene was inactivated will give rise to a mutant-coloured patch. . . . Thus this hypothesis predicts that for all sex-linked genes of the mouse in which the phenotype is due to localized gene action the heterozygote will have a mosaic appearance . . . The genetic evidence does not indicate at what early stage of embryonic development the inactivation of the one X chromosome occurs. . . . The sex-chromatin body is thought to be formed from one X chromosome in the rat and in the opossum. If this should prove to be the case in all mammals, then all female mammals heterozygous for sex-linked mutant genes would be expected to show the same phenomena as those in the mouse. The coat of the tortoiseshell cat, being a mosaic of the black and yellow colours of the two homozygous genotypes, fulfills this expectation.

Source: *Nature* 190:372

The second consequence of X-chromosome inactivation is that a normal female is a **mosaic** for X-linked genes (Figure 7.15A). That is, each somatic cell expresses the genes in only one X chromosome, but the X chromosome that is active genetically differs from one cell to the next. This mosaicism has been observed directly in females that are heterozygous for X-linked alleles that determine different forms of an enzyme, A and B; when cells from the heterozygous female are individually cultured in the laboratory, half of the clones are found to produce only the A

form of the enzyme and the other half to produce only the B form. Mosaicism can be observed directly in women who are heterozygous for an X-linked recessive mutation that results in the absence of sweat glands; these women exhibit patches of skin in which sweat glands are present (these patches are derived from embryonic cells in which the normal X chromosome remained active and the mutant X was inactivated) and other patches of skin in which sweat glands are absent (these patches are derived from embryonic cells in which the normal X chromosome was inactivated and the mutant X remained active.)

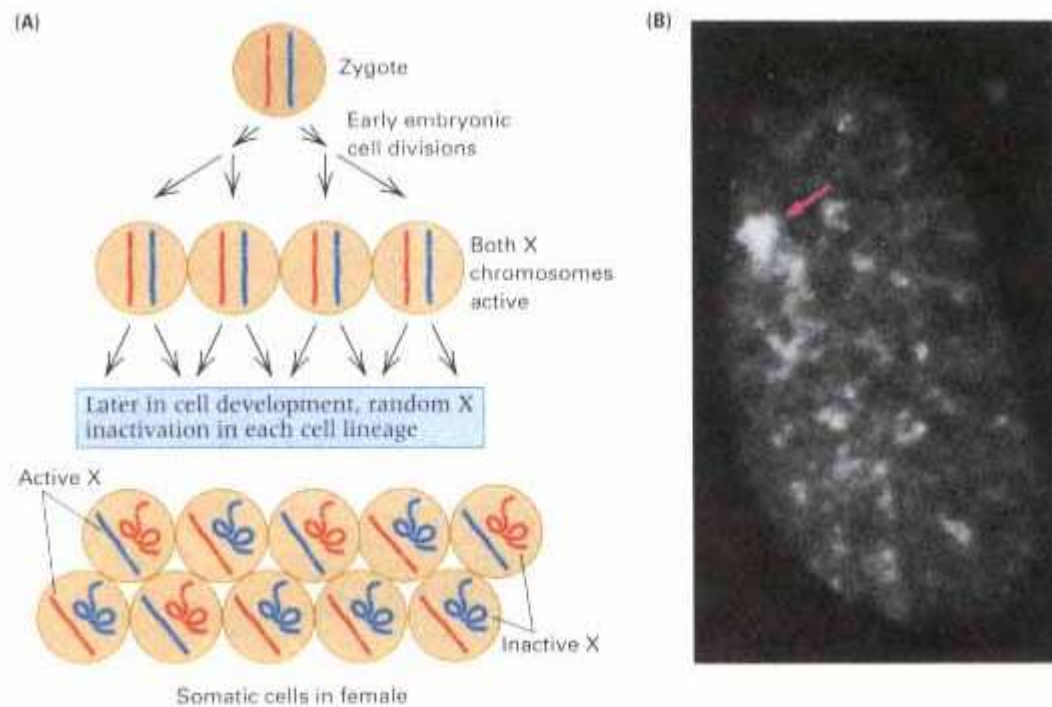


Figure 7.15

(A) Schematic diagram of somatic cells of a normal female showing that the female is a mosaic for X-linked genes. The two X chromosomes are shown in red and blue. An active X is depicted as a straight chromosome, an inactive X as a tangle. Each cell has just one active X, but the particular X that remains active is a matter of chance. In human beings, the inactivation includes all but a few genes in the tip of the short arm. (B) Fluorescence micrograph of a human cell showing a Barr body (bright spot at the upper left; see arrow.) This cell is from a normal human female, and it has one Barr body.

[Micrograph courtesy of A. J. R. de Jonge.]

In certain cell types, the inactive X chromosome in females can be observed microscopically as a densely staining body in the nucleus of interphase cells. This is called a **Barr body** (upper left in Figure 7.15B). Although cells of normal females have one Barr body, cells of normal males have none. Persons with two or more X chromosomes have all but one X chromosome per cell inactivated, and the number of Barr bodies equals the number of inactivated X chromosomes.

Although all mammals use X chromosomal inactivation for dosage compensation in females, the choice of which chromosome to inactivate is not always random. In marsupial mammals, which include the kangaroo, the koala, and the wombat, the X chromosome that is inactivated is always the one contributed by the father. The result is that female marsupials are not genetic mosaics of paternal and maternal X-linked genes. A female marsupial expresses the X-linked genes that she inherited from her mother.

The Calico Cat As Evidence for X-Chromosome Inactivation

In nonmarsupial mammals, the result of random X inactivation in females can sometimes be observed in the external phenotype. One example is the "calico" pattern of coat coloration in female cats. Two alleles affecting coat color are present in the X chromosome in cats. One allele results in an orange coat color (sometimes referred to as "yellow"), the other in a black coat color. Because he has only one X chromosome, a normal male has either the orange or the black allele. A female can be heterozygous for orange and black, and in this case the coat color is "calico"—a mosaic of orange and black patches mixed with patches of white. Figure 7.16 is a photograph of a female cat with the classic calico pattern.



Figure 7.16

A female cat that is heterozygous for the orange and black coat-color alleles and shows the classic "calico" pattern of patches of orange, black, and white fur.

The orange and black patches result from X chromosome inactivation. In cell lineages in which the X chromosome bearing the orange allele is inactivated, the X chromosome with the black allele is active and so the fur is black. In cell lineages in which the X chromosome with the black allele is inactivated, the orange allele in the active X chromosome results in orange fur.

The white patches have a completely different explanation. The white patches are due to an autosomal gene *S* for white spotting, which prevents pigment formation in the cell lineages in which it is expressed. Why the *S* gene is expressed in some cell lineages and not others is not known. Homozygous *SS* cats have more white than heterozygous *Ss* cats. The female in the photograph is homozygous.

Sex-Chromosome Abnormalities

Many types of sex-chromosome abnormalities have been observed. As noted, they are usually less severe in their phenotypic effects than are abnormal numbers of autosomes. The four most-common types are

- **47,XXX** This condition is often called the **trisomy-X syndrome**. The number 47 in the chromosome designation refers to the total number of chromosomes, and XXX indicates that the person has three X chromosomes. People with the karyotype 47,XXX are female. Many are phenotypically normal or nearly normal, though the frequency of mild mental retardation is somewhat greater than among 46,XX females.
- **47,XYY** This condition is often called the **double-Y syndrome**. These people are male and tend to be tall, but they are otherwise phenotypically normal. At one time it was thought that 47,XYY males developed severe personality disorders and were at a high risk of committing crimes of violence, a belief based on an elevated incidence of 47,XYY among violent criminals. Further study indicated that most 47,XYY males have slightly impaired mental function and that, although their rate of criminality is higher than that of normal males, the crimes are mainly nonviolent petty crimes such as theft. The majority of 47,XYY males are phenotypically and psychologically normal, have mental capabilities in the normal range, and have no criminal convictions.

- **XXY** This condition is called **Klinefelter syndrome**. Affected persons are male. They tend to be tall, do not undergo normal sexual maturation, are sterile, and in some cases have enlargement of the breasts. Mild mental impairment is common.
- **45,X** Monosomy of the X chromosome in females is called **Turner syndrome**. Affected persons are phenotypically female but short in stature and without sexual maturation. Mental abilities are typically within the normal range.

The Fragile-X Syndrome

An important form of inherited mental retardation is associated with a class of X chromosomes containing a site at Xq27–Xq28 (toward the end of the long arm) that tends to break in cultured cells that are starved for DNA precursors, such as the nucleotides. The X chromosomes containing this site are called **fragile-X** chromosomes, and the associated form of mental retardation is the **fragile-X syndrome**. The fragile-X syndrome affects about 1 in 2500 children. It accounts for about half of all cases of X-linked mental retardation and is second only to Down syndrome as a cause of inherited mental impairment. The fragile-X syndrome has an unusual pattern of inheritance in which approximately 1 in 5 males with the fragile-X chromosome are phenotypically normal and also have phenotypically normal children. However, the heterozygous daughters of such a "transmitting male" often have affected sons, and about one-third of their heterozygous daughters are also affected. This pattern is illustrated in Figure 7.17. The transmitting male denoted I-2 is not affected, but the X chromosome that he transmits to his daughters (II-2 and II-5) somehow becomes altered in the female germ line in such a way that sons and daughters in the next generation (III) are affected. Affected and normal grand-daughters of the transmitting male sometimes have affected progeny (generation IV). Males who are affected are severely retarded and do not reproduce. Among females, there is substantial variation in severity of expression (variable expressivity). In general, females are less severely affected than males, and, as noted, some heterozygous females are not affected at all (incomplete penetrance).

The molecular basis of the fragile-X chromosome has been traced to a **trinucleotide repeat** of the form (CCG)_n present in the DNA at the site where the breakage takes place. Normal X chromosomes have from 6 to 54 tandem copies of the repeating unit (the average is average is about 30), and affected persons have more than 230 copies of the repeat. Transmitting males have an intermediate number, between 52 and 230 copies, which is called the fragile-X "premutation." The unprecedented feature of the premutation is that, in females, it invariably increases in copy number to reach a level of 230 copies or greater, at which stage the chromosome causes mental retardation. The amplification in the number of copies present in the germ line of daughters of transmitting males is related

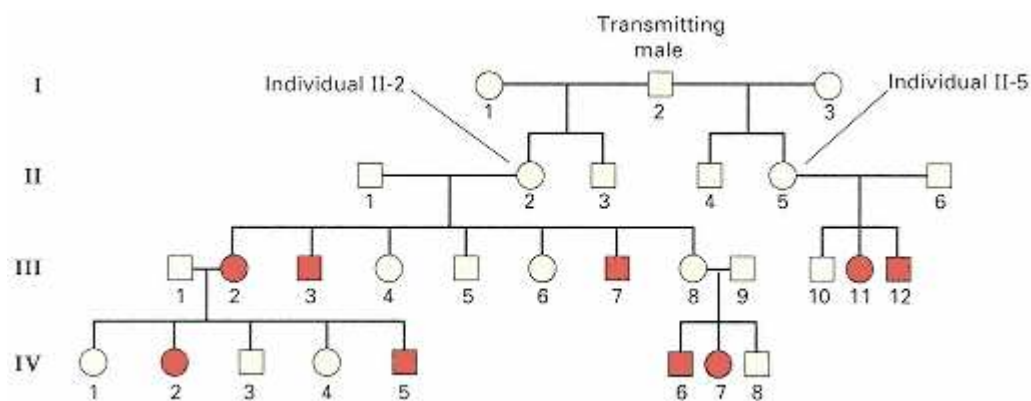


Figure 7.17
Pedigree showing transmission of the fragile-X syndrome. Male I-2 is not affected, but his daughters (II-2 and II-5) have affected children and grand-children.

[After C. D. Laird. 1987. *Genetics*, 117: 587.]

to the cycle of X-chromosome inactivation and reactivation. Chromosomes with the premutation that go through the X-inactivation cycle become permanently altered. In these chromosomes, the inactivation of a gene designated *FMR1* (fragile-site mental retardation-I) is irreversible, and the gene is unable to be reactivated in oogenesis. The X chromosome is said to have undergone a process of **imprinting**. Current evidence suggests that imprinting is associated with the addition of methyl groups ($-\text{CH}_3$) to certain cytosine bases in the DNA. Failure to remove the methyl groups irreversibly inactivates *FMR1* and also triggers amplification of the CCG repeats. The variable expressivity of the fragile-X syndrome in affected females also is related to X-chromosome inactivation: More severely affected females have a higher proportion of cells in which the normal X chromosome is inactivated.

Chromosome Abnormalities in Spontaneous Abortion

Approximately 15 percent of all recognized pregnancies in human beings terminate in spontaneous abortion, and in about half of

Table 7.1 Chromosome abnormalities per 100,000 recognized human pregnancies

	15,000 abort spontaneously 7500 chromosomally abnormal	85,000 live births 550 chromosomally abnormal
Trisomy		
A: $\begin{cases} 1 \\ 2 \\ 3 \end{cases}$	0 159 53	0 0 0
B: $\begin{cases} 4 \\ 5 \end{cases}$	95 0	0 0
C: 6–12	561	0
D: $\begin{cases} 13 \\ 14 \\ 15 \end{cases}$	128 275 318	17 0 0
E: $\begin{cases} 16 \\ 17 \\ 18 \end{cases}$	1229 10 223	0 0 13
F: 19–20	52	0
G: $\begin{cases} 21 \\ 22 \end{cases}$	350 424	113 0
Sex chromosomes		
XYY	4	46
XXY	4	44
XO	1350	8
XXX	21	44
Translocations		
Balanced	14	164
Unbalanced	225	52
Polyploid		
Triploid	1275	0
Tetraploid	450	
Other (mosaics. etc.)	280	49

Total

7500

550

them, the fetus has a major chromosome abnormality. Table 7.1 summarizes the average rates of chromosome abnormality found per 100,000 recognized pregnancies in several studies. Many of the spontaneously aborted fetuses have trisomy of one of the autosomes. Triploids and tetraploids also are common in spontaneous abortions. Note that the majority of trisomy-21 fetuses are spontaneously aborted, as are the vast majority of 45,X fetuses. If all trisomy-21 fetuses survived to birth, the incidence of Down syndrome would rise to 1 in 250, approximately a threefold increase from the incidence observed.

Although many autosomal trisomies are found in spontaneous abortions, autosomal monosomies are not found. Monosomic embryos do exist. They are probably created in greater numbers than the trisomic fetuses because chromosome loss, leading to monosomy, is usually much more frequent than chromosome gain, leading to trisomy. The absence of autosomal monosomies in spontaneously aborted fetuses is undoubtedly due to these embryos being aborted so early in development that the pregnancy goes unrecognized. The spontaneous abortions summarized in Table 7.1, although they represent a huge fetal wastage, serve the important biological function of eliminating many fetuses that are grossly abnormal in their development because of major chromosome abnormalities.

7.6—

Abnormalities in Chromosome Structure

Thus far, abnormalities in chromosome *number* have been described. The remainder of this chapter deals with abnormalities in chromosome *structure*. There are several principal types of structural aberrations, each of which has characteristic genetic effects. Chromosome aberrations were initially discovered through their genetic effects, which, though confusing at first, eventually came to be understood as resulting from abnormal chromosome structure. This interpretation was later confirmed directly by microscopic observations.

Deletions

A chromosome sometimes arises in which a segment is missing. Such a chromosome is said to have a **deletion** or a **deficiency**. Deletions are generally harmful to the organism, and the usual rule is the larger the deletion, the greater the harm. Very large deletions are usually lethal, even when heterozygous with a normal chromosome. Small deletions are often viable when they are heterozygous with a structurally normal homolog, because the normal homolog supplies gene products that are necessary for survival. However, even small deletions are usually homozygous lethal (lethal when both members of a pair of homologous chromosomes carry the deletion).

Deletions can be detected genetically by making use of the fact that a chromosome with a deletion no longer carries the wild-type alleles of the genes that have been eliminated. For example, in *Drosophila*, many *Notch* deletions are large enough to remove the nearby wildtype allele of *white*, also. When these deleted chromosomes are heterozygous with a structurally normal chromosome carrying the recessive *w* allele, the fly has white eyes because the wildtype *w*⁺ allele is no longer present in the deleted *Notch* chromosome. This **uncovering** of the recessive allele implies that the corresponding wildtype allele of *white* has also been deleted. Once a deletion has been identified, its size can be assessed genetically by determining which recessive mutations in the region are uncovered by the deletion. This method is illustrated in Figure 7.18.

With the banded polytene chromosomes in *Drosophila* salivary glands, it is possible to study deletions and other chromosome aberrations physically. For example, all the *Notch* deletions cause particular bands to be missing in the salivary chromosomes. Physical mapping of deletions also allows individual genes, otherwise known only from genetic studies, to be assigned to specific bands or regions in the salivary chromosomes.

Physical mapping of genes in part of the *Drosophila* X chromosome is illustrated in Figure 7.19. The banded chromosome is shown near the top, along with the num-

bering system used to refer to specific bands. Each chromosome is divided into numbered sections (the X chromosome comprises sections 1 through 20), and each of the sections is divided into subdivisions designated A through F. Within each lettered subdivision, the bands are numbered in order, and so, for example, 3A6 is the sixth band in subdivision A of section 3 (Figure 7.19). On the average, each band contains about 20 kb of DNA, but there is considerable variation in DNA content from band to band.

In Figure 7.19, the mutant X chromosomes labeled I through VI have deletions. The deleted part of each chromosome is shown in red. These deletions define regions along the chromosome, some of which correspond to specific bands. For example, the deleted region in both chromosome I and II that is present in all the other chromosomes consists of band 3A3. In crosses, only deletions I and II uncover the mutation *zeste* (*z*), so the *z* gene must be in band 3A3, as indicated at the top. Similarly, the recessive-lethal mutation *zw2* is uncovered by all deletions except VI; therefore, the *zw2* gene must be in band 3A9. As a final example, the *w* mutation is uncovered only by deletions II, III, and IV; thus the *w* gene must be in band 3C2. The *rst* (rough eye texture) and *N* (notched wing margin) genes are not uncovered by any of the deletions. These genes were localized by a similar analysis of overlapping deletions in regions 3C5 to 3C10.

Duplications

Some abnormal chromosomes have a region that is present twice. These chromosomes are said to have a **duplication**.

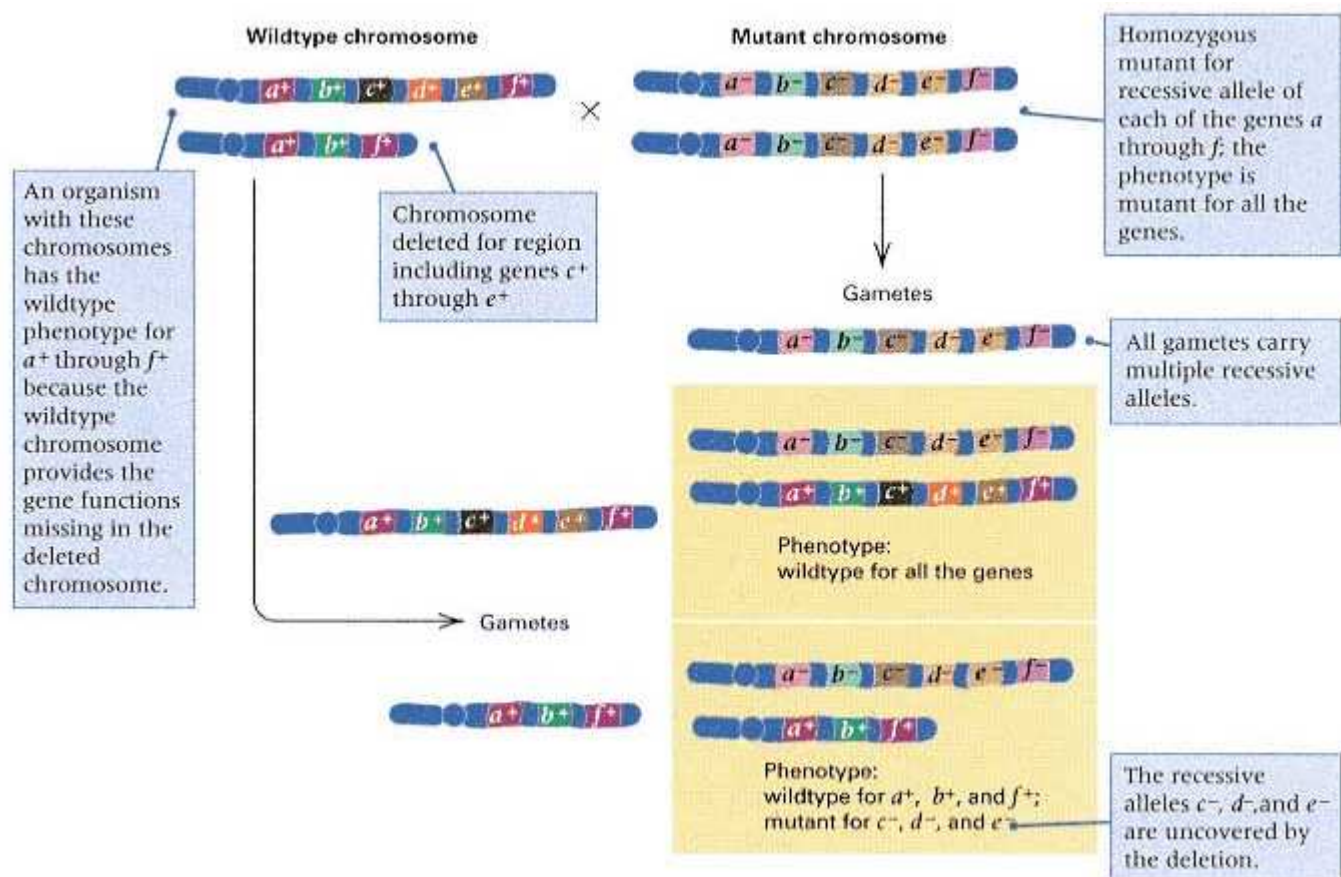


Figure 7.18

Mapping of a deletion by testcrosses. The F₁ heterozygotes with the deletion express the recessive phenotype of all deleted genes. The expressed recessive alleles are said to be uncovered by the deletion.

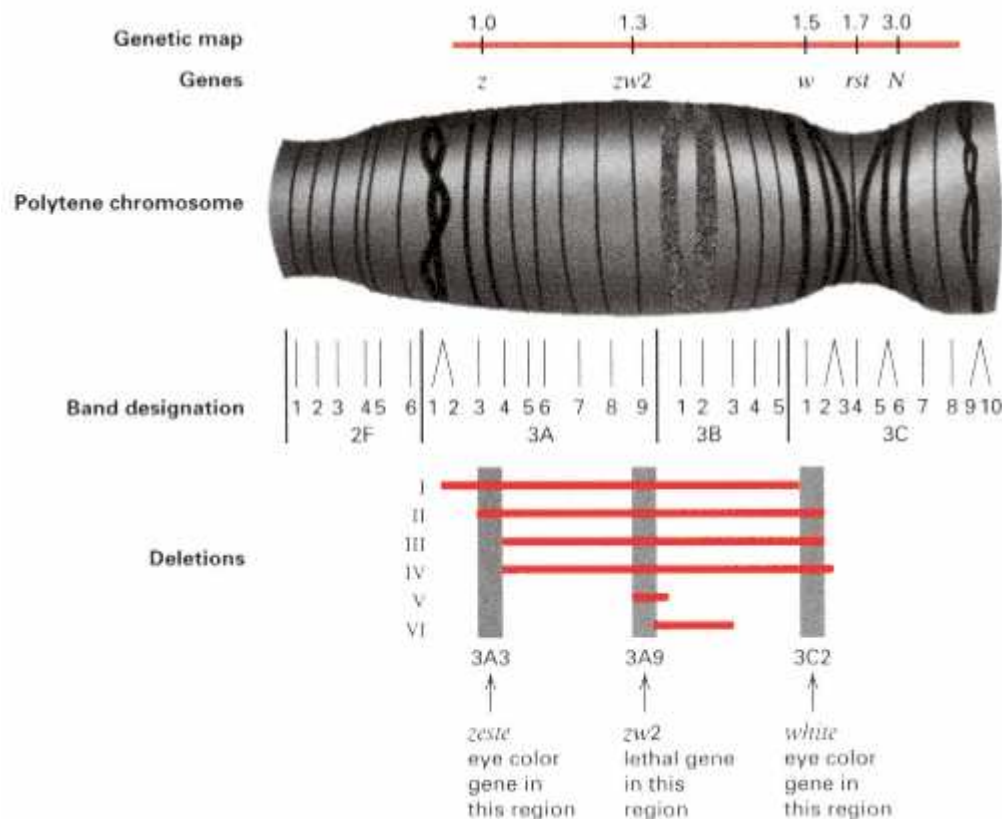


Figure 7.19

Part of the X chromosome in polytene salivary gland nuclei of *Drosophila melanogaster* and the extent of six deletions (I–VI) in a set of chromosomes. Any recessive allele that is uncovered by a deletion must be located inside the boundaries of the deletion. This principle can be used to assign genes to specific bands in the chromosome.

Certain duplications have phenotypic effects of their own. An example is the *Bar* duplication in *Drosophila*, which is a tandem duplication of bands 16A1 through 16A7 in the X chromosome. A **tandem duplication** is one in which the duplicated segment is present in the same orientation immediately adjacent to the normal region in the chromosome. In the case of *Bar*, the tandem duplication produces a dominant phenotype of bar-shaped eyes.

Tandem duplications are able to produce even more copies of the duplicated region by means of a process called **unequal crossing-over**. Figure 7.20A illustrates the chromosomes in meiosis of an organism that is homozygous for a tandem duplication (brown region). When they undergo synapsis, these chromosomes can mispair with each other, as illustrated in Figure 7.20B. A crossover within the mispaired part of the duplication (Figure 7.20C) will thereby produce a chromatid carrying a triplication, and a reciprocal product (labeled "single copy" in Figure 7.20D) that has lost the duplication. For the *Bar* region, the triplication can be recognized because it produces an even greater reduction in eye size than the duplication.

The most frequent effect of a duplication is a reduction in viability (probability of survival); in general, survival decreases with increasing size of the duplication. However, deletions are usually more harmful than duplications of comparable size.

Unequal Crossing-Over in Human Red-Green Color Blindness

Human color vision is mediated by three light-sensitive protein pigments present in the cone cells of the retina. Each of the pigments is related to **rhodopsin**, the pigment

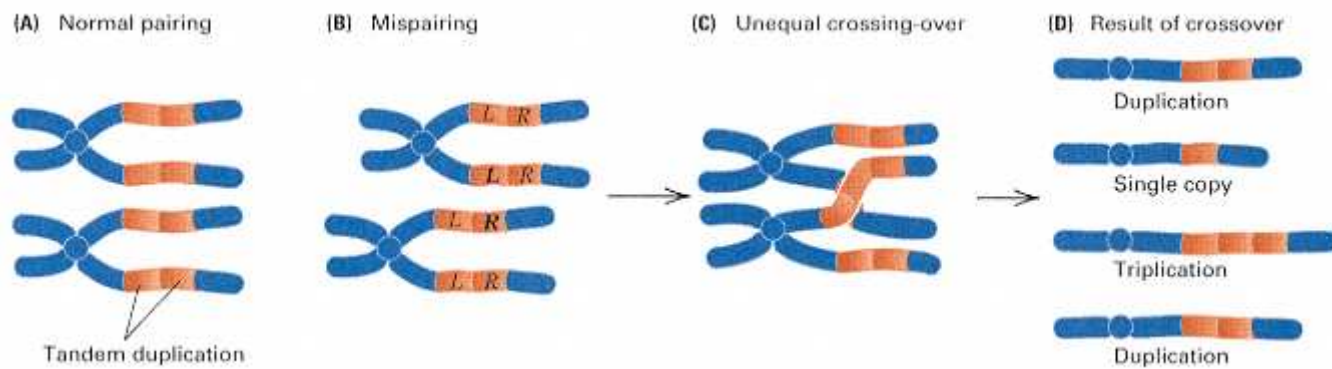


Figure 7.20

An increase in the number of copies of a chromosome segment resulting from unequal crossing-over of tandem duplications (brown). (A) Normal synapsis of chromosomes with a tandem duplication. (B) Mispairing. The right-hand element of the lower chromosome is paired with the left-hand element of the upper chromosome. (C) Crossing-over within the mispaired duplication, which is called unequal crossing-over. (D) The outcome of unequal crossing-over. One product contains a single copy of the duplicated region, another chromosome contains a triplication, and the two strands not participating in the crossover retain the duplication.

that is found in the rod cells and mediates vision in dim light. The light sensitivities of the cone pigments are toward blue, red, and green. These are our primary colors. We perceive all other colors as mixtures of these primaries. The gene for the blue-sensitive pigment resides somewhere in 7q22-qter (which means 7q22 to the terminus), and the genes for the red and green pigments are on the X chromosome at Xq28 separated by less than 5 cM (roughly 5 Mb of DNA). Because the red and green pigments arose from the duplication of a single ancestral pigment gene and are still 96 percent identical in amino acid sequence, the genes are similar enough that they can pair and undergo unequal crossing-over. The process of unequal crossing-over is the genetic basis of red-green color blindness.

Almost everyone is familiar with **red-green color blindness**; it is one of the most common inherited conditions in human beings. Approximately 5 percent of males have some form of red-green color blindness. The preponderance of affected males immediately suggests X-linked inheritance, which is confirmed by pedigree studies (Section 3.4). Affected males have normal sons and carrier daughters, and the carrier daughters have 50 percent affected sons and 50 percent carrier daughters.

Actually, there are several distinct varieties of red-green color blindness. Defects in red vision go by the names of **protanopia**, an inability to perceive red, and **protanomaly**, an impaired ability to perceive red. The comparable defects in green perception are called **deutanopia** and **deutanomaly**, respectively. Isolation of the red and green pigment genes and study of their organization in people with normal and defective color vision have indicated quite clearly how the "-opias" and "-omalies" differ and have also explained why the frequency of color blindness is so relatively high.

The organization of the red and green pigment genes in men with normal vision is illustrated in Figure 7.21A. Unexpectedly, a significant proportion of normal X chromosomes contain two or three green-pigment genes. How these arise by unequal crossing-over is shown in Figure 7.21B. The red-pigment and green-pigment genes pair and the crossover takes place in the region of homology between the genes. The result is a duplication of the green-pigment gene in one chromosome and a deletion of the green-pigment gene in the other.

The recombinational origins of the defects in color vision are illustrated in Figure 7.22. The top chromosome in Figure 7.22A is the result of deletion of the green-

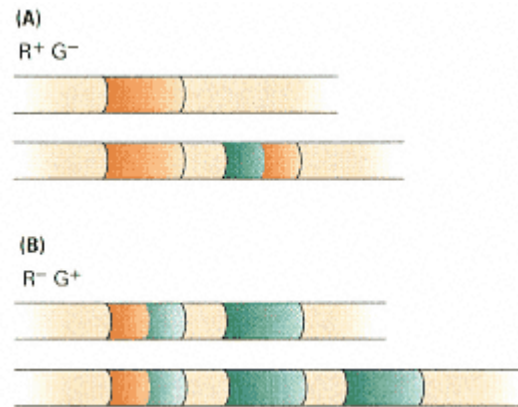


Figure 7.22
 Genetic basis of absent or impaired red-green color vision. (A) Defects in green vision result from unequal crossing-over between mispaired red-pigment and green-pigment genes, yielding a green-red chimeric gene. If the green-pigment gene is missing, or if the chimeric gene is largely "red" in its sequence, then deuteranopia is the result. If the chimeric gene is largely "green" in its sequence, then deuteranomaly is the result. (B) Defects in red vision result from unequal crossing-over between mispaired red-pigment and green-pigment genes, again yielding a green-red chimeric gene. If the chimeric gene is largely "green," then protanopia results; if it is largely "red," then protanomaly results. Note that the red gene cannot be eliminated altogether (as the green gene can), because the red gene is at the end of the region of homology between the chromosomes

pigment gene shown earlier in Figure 7.21B. Males with such an X chromosome have deuteranopia, or "green-blindness." Other types of abnormal pigments result when crossing-over takes place within mispaired red-pigment and green-pigment genes. Crossing-over between the genes yields a **chimeric gene**, which is a composite gene: part of one joined with part of the other. The chimeric gene in Figure 7.22A joins the 5' end of the green-pigment gene with the 3' end of the red-pigment gene. If the crossover point is toward the 5' end (toward the left in the figure), then the resulting chimeric gene is mostly "red" in sequence, so the chromosome causes deuteranopia, or "green-blindness." However, if the crossover point is near the 3' end (toward the right in the figure), then most of the green-pigment gene remains intact, and the chromosome causes deuteranomaly.

Chromosomes associated with defects in red vision are illustrated in Figure 7.22B. The chimeric genes are the reciprocal products of the unequal crossovers that yield defects in green vision. In this case, the chimeric gene consists of the red-pigment gene at the 5' end and the green-pigment gene at the 3' end. If the crossover point is near the 5' end, most of the red-pigment gene is replaced with the green-pigment gene. The result is protanopia, or "red-

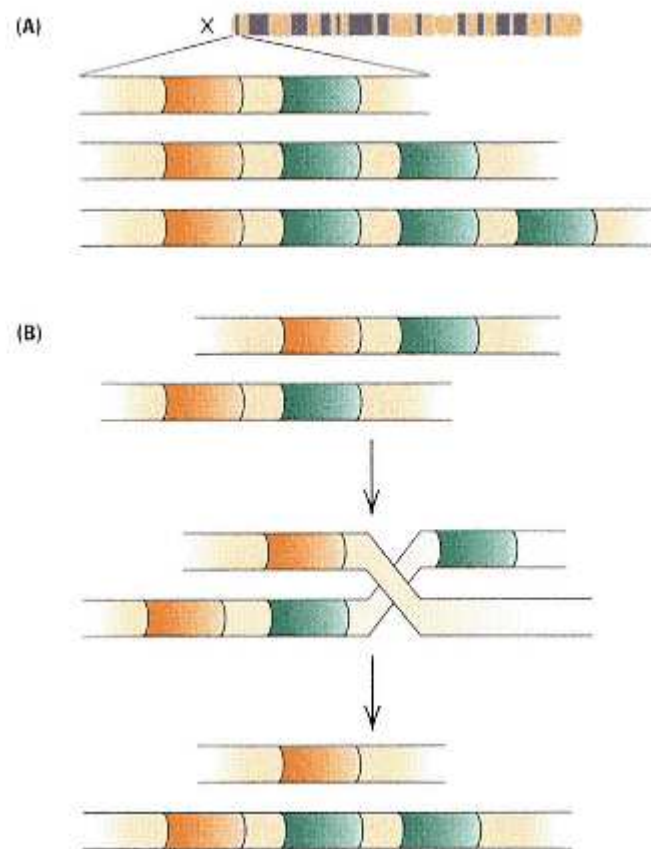


Figure 7.21

(A) Organization of red-pigment and green-pigment genes in normal X chromosomes. Some chromosomes contain one copy of the green-pigment gene, others two, still others three. (B) Origin of multiple green-pigment genes by unequal crossing over in the region of DNA homology between the genes. Note that one product of unequal crossing-over is a chromosome containing a red-pigment gene but no green-pigment gene

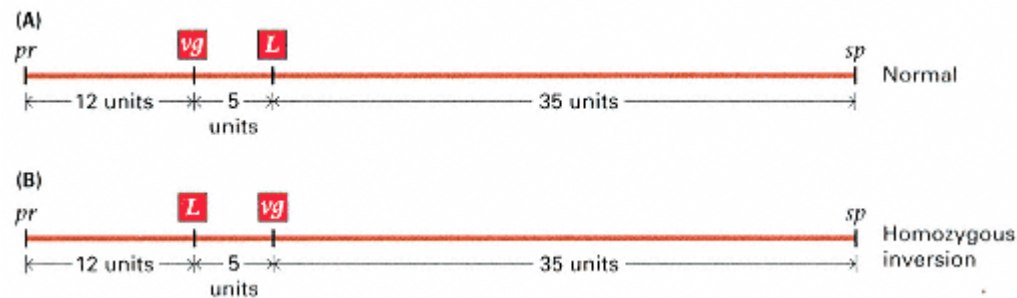


Figure 7.23

Partial genetic map of chromosome 2 of *Drosophila melanogaster*. (A) Normal order. (B) Genetic map of the same region in a chromosome that has undergone an inversion in which the order of *vg* and *L* is reversed.

blindness." The same is true of the other chromosome indicated in Figure 7.22B. However, if the crossover point is near the 3' end, then most of the red-pigment gene remains intact, and the result is protanomaly.

Inversions

Another important type of chromosome abnormality is an **inversion**, a segment of a chromosome in which the order of the genes is the reverse of the normal order. An example is shown in Figure 7.23.

In an organism that is heterozygous for an inversion, one chromosome is structurally normal (wildtype), and the other carries an inversion. These chromosomes pass through mitosis without difficulty because each chromosome duplicates and its chromatids are separated into the daughter cells without regard to the other chromosome. There is a problem in meiosis, however. The problem is that the chromosomes are attracted gene for gene in the process of synapsis, as shown in Figure 7.24. In an inversion heterozygote, in order for gene-for-gene pairing to take place everywhere along the length of the chromosome, one or the other of the chromosomes must twist into a loop in the region in which the gene order is inverted. In

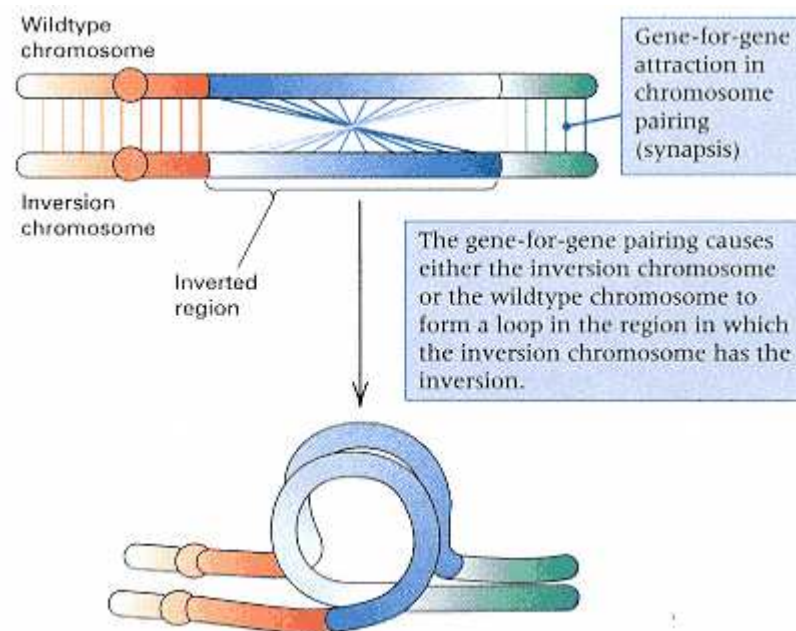


Figure 7.24

In an organism that carries a chromosome that is structurally normal along with a homologous chromosome with an inversion, the gene-for-gene attraction between the chromosomes during synapsis causes one of the chromosomes to form into a loop in the region in which the gene order is inverted. In this example, the structurally normal chromosome forms the loop.

Figure 7.24, it is the structurally normal chromosome that is shown as looped, but in other cells it may be the inverted chromosome that is looped. In either case, the loop is called an **inversion loop**.

The loop itself does not create a problem. The looping apparently takes place without difficulty and can be observed through the microscope. As long as there is no crossing-over within the inversion, the homologous chromosomes can separate normally at anaphase I, as illustrated in Figure 7.25. On the other hand, when there is crossing-over within the inversion loop, the chromatids involved in the crossing-over become physically joined, and the result is the formation of chromosomes containing large duplications and deletions. The products of the crossing-over can be deduced from Figure 7.26A by tracing along the chromatids. The outer chromatids are the ones not participating in the crossover. One of these contains the inverted sequence and the other the normal sequence, as shown in Figure 7.26B. Because of the crossover, the inner chromatids, which did participate in the crossover, are connected. If the centromere is not included in the inversion loop, as is the case here, then the result is a dicentric chromosome. The reciprocal product of the crossover produces an acentric chromosome. Neither the dicentric chromosome nor the acentric chromosome can be included in a normal gamete. The acentric chromosome is usually lost because it lacks a centromere and, in any case, has a deletion of the *a* region and a duplication of the *d* region. The dicentric chromosome is also often lost because it is held on the meiotic spindle by the chromatid bridging between the centromeres; in any case, this chromosome is deleted for the *d* region and duplicated for the *a* region. Hence, when there is a crossover in the inversion loop, the only chromatids that can be recovered in the gametes are the chromatids that did not participate on the crossover. One of these carries the inversion, and the other does not. It is for this reason that inversions prevent the recovery of crossover products. In the early years of genetics, before their identity as inversions was discovered, inversion-bearing chromosomes were known as "crossover suppressors."

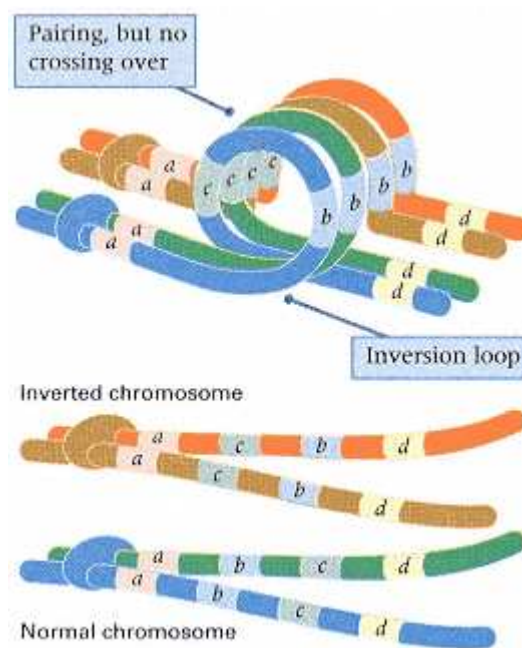


Figure 7.25

In an inversion heterozygote, if there is no crossing over within the inversion loop, then the homologous chromosomes disjoin without problems. Two of the resulting gametes carry the inverted chromosome, and two carry the structurally normal chromosome.

The inversion shown in Figure 7.26, in which the centromere is not included in the inverted region, is known as a **paracentric inversion**, which means inverted "beside" (*para-*) the centromere. As seen in the figure, the products of crossing-over include a dicentric and an acentric chromosome.

When the inversion does include the centromere, it is called a **pericentric inversion**, which means "around" (*peri-*) the centromere. Chromatids with duplications and deficiencies are also created by crossing-over within the inversion loop of a pericentric inversion, but in this case the crossover products are monocentric. The situation is illustrated in Figure 7.27A. The diagram is identical to that in Figure 7.26 except for the position of the centromere. The products of crossing-over can again be deduced by tracing the chromatids. In this case, both products of the crossover are monocentric, but one chromatid carries a duplication of *a* and a deletion of *d*, and the other carries a duplication of *d* and a deletion of *a* (Figure 7.27B). Although

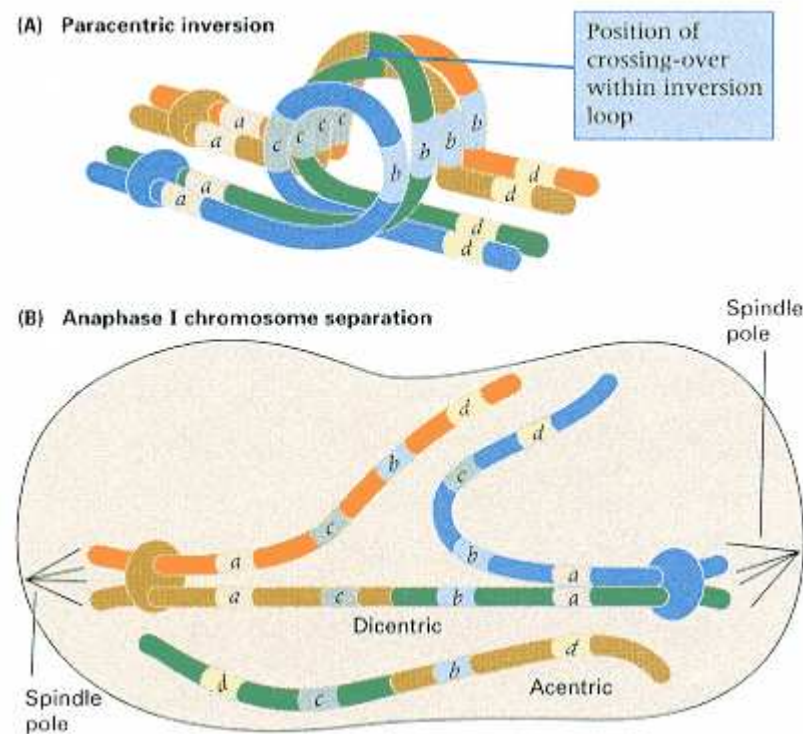


Figure 7.26

(A) Synapsis between homologous chromosomes, one of which contains an inversion. There is a crossing-over within the inversion loop. (B) Anaphase I configuration resulting from the crossover. Because the centromere is not included in the inverted region, one of the crossover products is a dicentric chromosome, and the reciprocal product is an acentric chromosome. Among the two chromatids not involved in the crossover, one carries the inversion and the other the normal gene sequence.

either of these chromosomes could be included in a gamete, the duplication and deficiency usually result in inviability. Thus, as with the paracentric inversion, the products of recombination are not recovered, but for a different reason. Among the chromatids not participating in the crossing-over in Figure 7.27A, one carries the pericentric inversion, and the other has the normal sequence.

The looping of one chromosome in an inversion heterozygote creates a rather complex structure, but it can be depicted in simplified form as in Figure 7.28. In this diagram, only two of the four chromatids are shown (the two that participate in the crossing-over), and the location of the centromere is not indicated. This kind of diagram is convenient because it shows that the essential result of the crossing-over is the production of the duplication and deletion products. These will also be dicentric or acentric if the centromere is located outside of the inverted region (a paracentric inversion).

Reciprocal Translocations

A chromosomal aberration resulting from the interchange of parts between nonhomologous chromosomes is called a **translocation**. In Figure 7.29, organism A is homozygous for two pairs of structurally normal chromosomes. Organism B contains one structurally normal pair of chromosomes and another pair of chromosomes that have undergone an interchange of terminal parts. The organism is said to be *heterozygous* for the translocation. The translocation is properly called a **reciprocal translocation** because it consists of two reciprocally interchanged parts. As indicated in Figure 7.29C, an organism can also be homozygous for a translocation if both pairs of homologous chromosomes undergo an interchange of parts.

An organism that is heterozygous for a reciprocal translocation usually produces only about half as many offspring as normal—a condition that is called **semisterility**. The reason for the semisterility is difficulty in chromosome segregation in meiosis. When meiosis takes place in a translocation heterozygote, the normal and translocated chromosomes must undergo synapsis, as shown in Figure 7.30. Ordinarily, there would also be chiasmata between nonsister chromatids in the arms of the homologous chromosomes, but these are not shown, as though the translocation were present in an organism with no crossing-over, such as a male *Drosophila*. Segregation from this

configuration can take place in any of three ways. In the list that follows, the symbol $1 + 2 \longleftrightarrow 3 + 4$ means that at the first meiotic anaphase, the chromosomes in Figure 7.30 labeled 1 and 2 go to one pole and those labeled 3 and 4 go to the opposite pole. The red

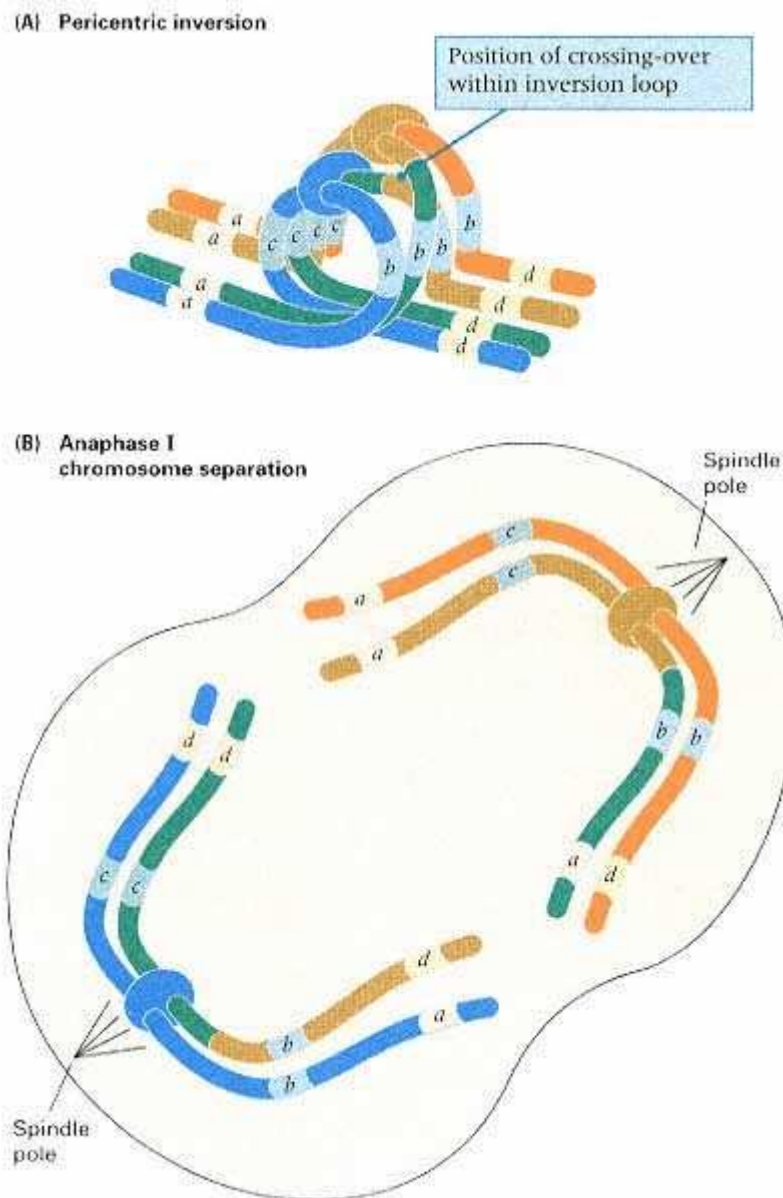


Figure 7.27

(A) Synapsis between homologous chromosomes, one of which carries a pericentric inversion. A crossing-over within the inversion loop is shown. (B) Anaphase I configuration resulting from the crossover. One of the crossover products is duplicated for *a* and deficient for *d*; the other is duplicated for *d* and deficient for *a*. Among the two chromatids not involved in the crossover, one carries the inversion and the other is normal.

numbers indicate the two parts of the reciprocal translocation. The three types of segregation are

- $1 + 2 \leftrightarrow 3 + 4$ This mode is called **adjacent-1** segregation. Homologous centromeres go to opposite poles, but each normal chromosome goes with one part of the reciprocal translocation. All gametes formed from adjacent-1 segregation have a large duplication and deficiency for the distal part of the translocated chromosomes. (The *distal* part of a chromosome is the part farthest from the centromere.) The pair of gametes originating from the $1 + 2$ pole are duplicated for the distal part of the blue chromosome and deficient for the distal part of the red chromosome; the pair of gametes from the $3 + 4$ pole have the reciprocal deficiency and duplication.

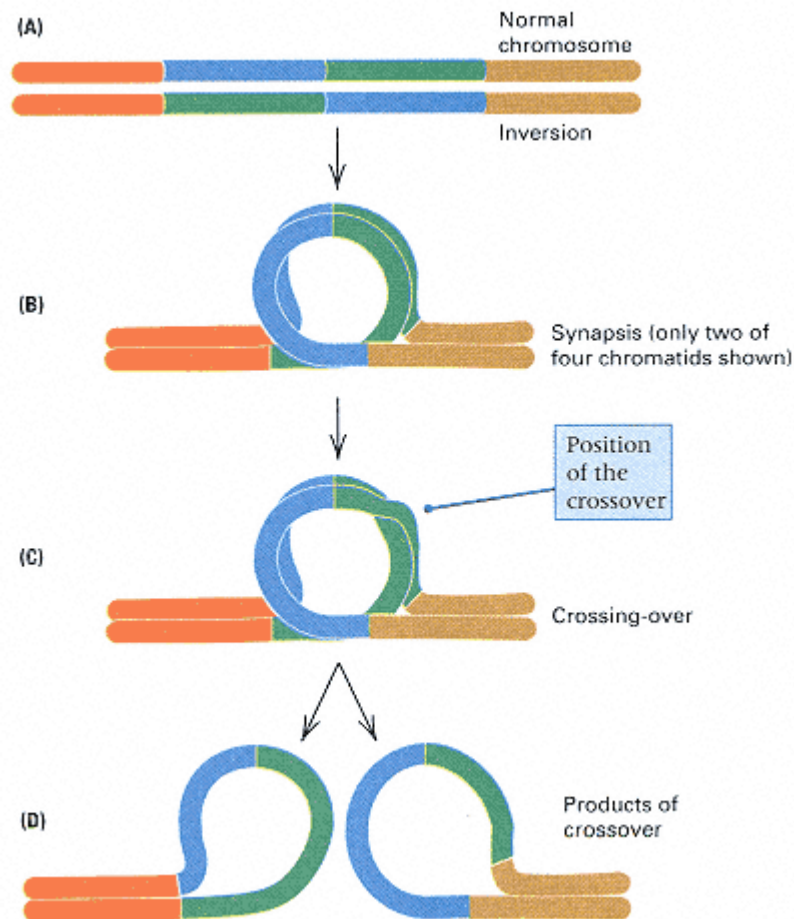


Figure 7.28
Simplified diagram of an inversion loop showing the consequence of crossing-over within the loop. Only the two chromatids participating in the crossover are shown, and the location of the centromeres is not indicated.

- $1 + 3 \longleftrightarrow 2 + 4$ This mode is **adjacent-2** segregation, in which homologous centromeres go to the same pole at anaphase I. In this case, all gametes have a large duplication and deficiency of the proximal part of the translocated chromosome. (The *proximal* part of a chromosome is the part closest to the centromere.) The pair of gametes from the $1 + 3$ pole have a duplication of the proximal part of the red chromosome and a deficiency of the proximal part of the blue chromosome; the pair of gametes from the $2 + 4$ pole have the reciprocal deficiency and duplication.

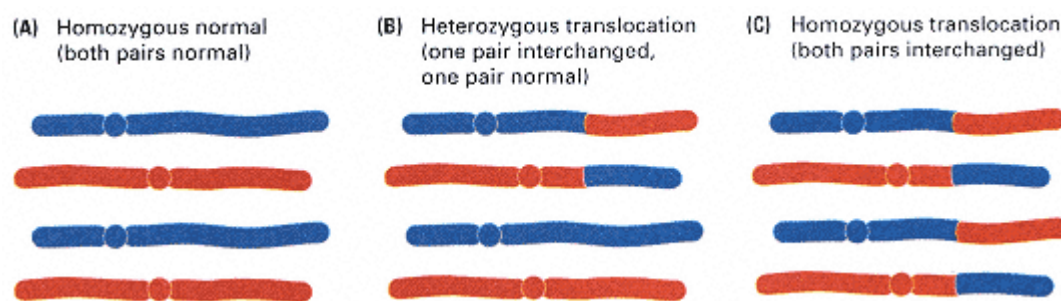


Figure 7.29
(A) Two pairs of nonhomologous chromosomes in a diploid organism. (B) Heterozygous reciprocal translocation, in which two nonhomologous chromosomes (the two at the top) have interchanged terminal segments. (C) Homozygous reciprocal translocation.

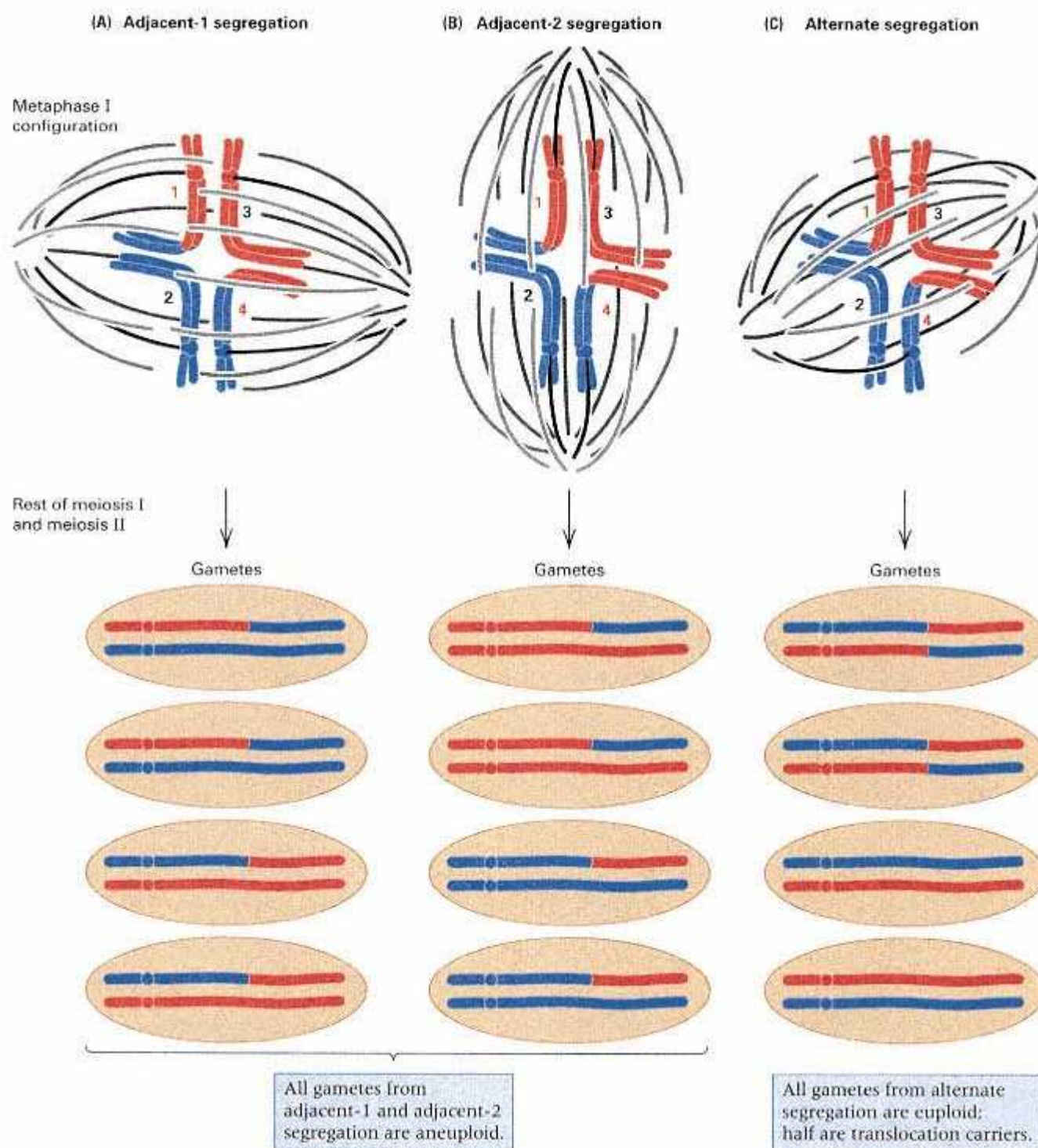


Figure 7.30

Quadrivalent formed in the synapsis of a heterozygous reciprocal translocation. The translocated chromosomes are numbered in red, their normal homologs in black. No chiasmata are shown. (A) Adjacent-1 segregation, in which homologous centromeres separate at anaphase I; all of the resulting gametes have a duplication of one terminal segment and a deficiency of the other. (B) Adjacent-2 segregation, in which homologous centromeres go together at anaphase I; all of the resulting gametes have a duplication of one proximal segment and a deficiency of the other. (C) Alternate segregation, in which half of the gametes receive both parts of the reciprocal translocation and the other half receive both normal chromosomes.

- $1 + 4 \longleftrightarrow 2 + 3$ In this type of segregation, which is called **alternate** segregation, the gametes are all balanced (euploid), which means that none has a duplication or deficiency. The gametes from the $1 + 4$ pole have both parts of the reciprocal translocation; those from the $2 + 3$ pole have both normal chromosomes.

The semisterility of genotypes that are heterozygous for a reciprocal translocation results from lethality due to the duplication and deficiency gametes produced by adjacent-1 and adjacent-2 segregation. The frequency with which these types of segregation take place is strongly influenced by the position of the translocation breakpoints, by the number and distribution of chiasmata in the interstitial region between the centromere and each breakpoint, and by whether the quadrivalent tends to open out into a ring-shaped structure on the metaphase plate. A ring often forms if the breakpoints are in the middle region of the arms and there is no crossing-over in the interstitial regions; in this case, the frequencies of adjacent-1: adjacent-2: alternate segregation are approximately 1: 1: 2. If there is crossing-over in the interstitial regions, or if one breakpoint is near a telomere, then the orientation of the quadrivalent at metaphase tends to discourage adjacent-2 segregation. In some cases, adjacent-2 segregation is eliminated, and the ratio of adjacent-1: alternate segregation is approximately 1: 1. Adjacent-1 segregation is quite common in any event, which means that semisterility is to be expected from virtually all translocation heterozygotes.

Translocation semisterility is manifested in different life-history stages in plants and animals. Plants have an elaborate gametophyte phase of the life cycle, a haploid phase in which complex metabolic and developmental processes are necessary. In plants, large duplications and deficiencies are usually lethal in the gametophyte stage. Because the gametophyte produces the gametes, in higher plants the semisterility is manifested as pollen or seed lethality. In animals, by contrast, only minimal gene activity is necessary in the gametes, which function in spite of very large duplications and deficiencies. In animals, therefore, the semisterility is usually manifested as zygotic lethality.

Certain groups of plants have reciprocal translocations present in natural populations without the semisterility usually expected. Among these are the evening primroses in the genus *Oenothera*, of which there are about 100 wild species native to North America and many cultivated varieties. *Oenothera* escapes the translocation semisterility because segregation is always in the alternate mode. In terms of Figure 7.30, the mode of segregation is always $1 + 4 \longleftrightarrow 2 + 3$. In some species of *Oenothera*, entire sets of chromosomes are interconnected through a chain of reciprocal translocations of the type illustrated in Figure 7.31A. The chromosomes at the top left are all normal; those at the top right are translocated in such a way that each chromosome has exchanged an arm with the next in line. When the chromosomes in the complex translocation heterozygote undergo synapsis, the result is a ring of chromosomes (Figure 7.31B). The astonishing feature of meiosis in such *Oenothera* heterozygotes is that the segregation is exclusively of the alternate type, so the only gametes formed contain either the entire set of normal chromosomes or the entire set of translocated chromosomes. This is not the end of the surprises. In *Oenothera*, one of the gametic types is inviable in the pollen and the other is inviable in the ovule, so fertilization restores the karyotype of the complex translocation heterozygote!

In species in which translocation heterozygotes exhibit semisterility, the semisterility can be used as the phenotype to map the breakpoint of the translocation just as though it were a normal gene. The mapping procedure can be made clear by means of an example. Translocation *TB-10LI* is a translocation with one breakpoint in the long arm of chromosome 10 in maize, and it results in semisterility when heterozygous. A cross is made between a translocation heterozygote and a genotype homozygous for both *zn1* (*zebra necrotic 1*) and *tp2* (*teopod 2*), and semisterile progeny are testcrossed with *zn1 tp2* homozygotes. (The phenotypes of *zn1* and *tp2* are shown in Figure 4.8.) The parental genotype is

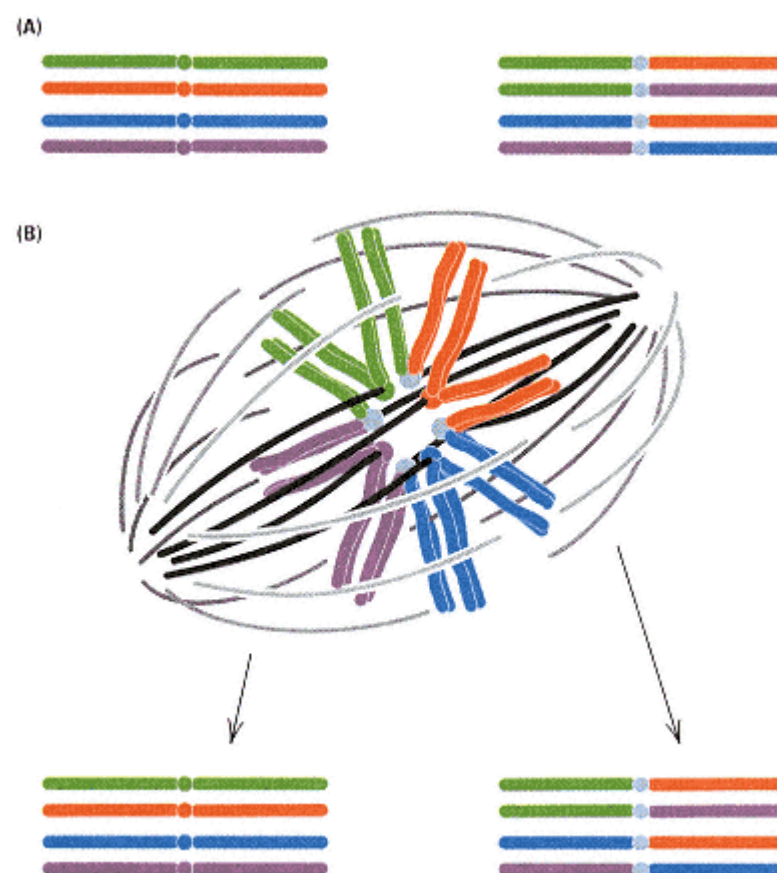


Figure 7.31

(A) Complex translocation heterozygote of the type found in some species of *Oenothera*. The chromosomes at the top left are not rearranged. Those at the top right are connected by a chain of translocations, each chromosome having exchanged an arm with the next chromosome in line. (B) At metaphase I in meiosis, the pairing configuration of the translocation heterozygote is a ring of chromosomes in which each arm is paired with its proper partner; note that each chromosome consists of two chromatids. Alternate segregation from the metaphase ring yields, after the second meiotic division, two types of gametes: those containing all normal chromosomes and those containing all translocated chromosomes.

therefore *Zn1 Tp2 TB-10L1/zn1 tp2 +*, where the + denotes the position of the translocation breakpoint in the homologous chromosome. The progeny phenotypes are as follows:

nonzebrastripe	nonteopod	semisterile	392
nonzebrastripe	nonteopod	fertile	3
nonzebrastripe	teopod	semisterile	42
nonzebrastripe	teopod	fertile	73
zebrastripe	nonteopod	semisterile	83
zebrastripe	nonteopod	fertile	34
zebrastripe	teopod	semisterile	1

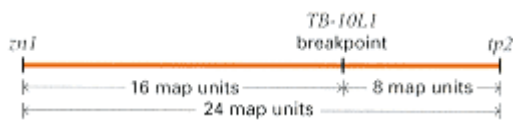
These data are analyzed exactly like the three-point crosses in Section 4.3. The double recombinants are present in the rarest classes of progeny and differ from the parental genotypes in exchanging the gene that is in the middle of the three. The double recombinants are *Zn1 Tp2 +* and *zn1 tp2 TB-10L1*, which means that the translocation breakpoint lies between *zn1* and *tp2*. Hence, the map distance between *zn1* and the breakpoint is

$$(3 + 73 + 83 + 1)/1000 = 16 \text{ cM}$$

and that between the translocation breakpoint and *tp2* is

$$(3 + 42 + 34 + 1)/1000 = 8 \text{ cM}$$

The genetic map of the region containing the *TB-10L1* translocation breakpoint is



Robertsonian Translocations

A special type of *nonreciprocal* translocation is a **Robertsonian translocation**, in which the centromeric regions of two

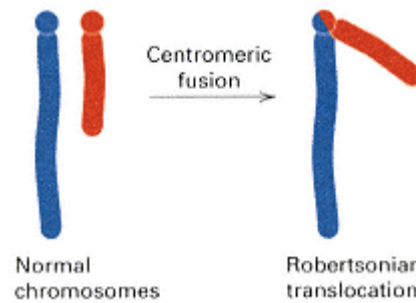


Figure 7.32
Formation of a Robertsonian translocation
by fusion of two acrocentric chromosomes
in the centromeric region.

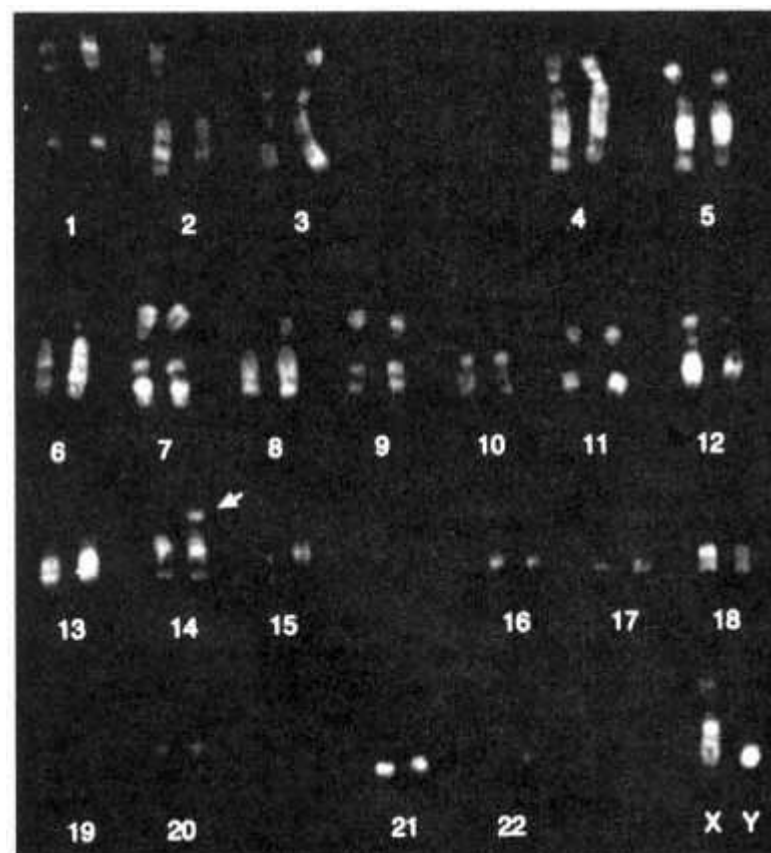


Figure 7.33
A karyotype of a child with Down syndrome, carrying a Robertsonian
translocation of chromosomes 14 and 21 (arrow). Chromosomes 19
and 22 are faint in this photo; this has no significance.
[Courtesy of Irene Uchida.]

nonhomologous acrocentric chromosomes become fused to form a single centromere (Figure 7.32). Robertsonian translocations are important in human genetics, especially as a risk factor to be considered in Down syndrome. When chromosome 21 is one of the acrocentrics in a Robertsonian translocation, the rearrangement leads to a familial type of Down syndrome in which the risk of recurrence is very high. Approximately 3 percent of children with Down syndrome are found to have one parent with such a translocation.

A Robertsonian translocation that joins chromosome 21 with chromosome 14 is shown in Figure 7.33 (arrow). The heterozygous carrier is phenotypically normal, but a high risk of Down syndrome results from aberrant segregation in meiosis. The possible modes of segregation are shown in Figure 7.34. The symbol *rob* refers to the translocation, and the +, or -, preceding a chromosome number designates an extra copy, or a missing copy, of the entire chromosome. Among the several possible types of gametes that can arise, one contains a normal chromosome 21 along with the 14/21 Robertsonian translocation (Figure 7.34A). If this aberrant gamete is used in fertilization, then the fetus will contain two copies of the normal chromosome 21 plus the 14/21 translocation. In effect, the fetus contains three copies of chromosome 21 and hence has Down syndrome. The other abnormal gametes that result from adjacent-1 or adjacent-2 segregation either are missing chromosome 21 or chromosome 14 or contain

effectively two copies of chromosome 14 (Figure 7.34A and B). If these gametes participate in fertilization, the result is monosomy 21, monosomy 14, or trisomy 14, respectively. The monosomic embryos undergo very early spontaneous abortion; the trisomy 14-fetus undergoes spontaneous abortion later in pregnancy. Hence families with a high risk of translocation Down syndrome also have a high risk of spontaneous abortion due to other chromosome abnormalities. Alternate segregation of a Robertsonian translocation yields gametes carrying either the translocation or both normal chromosomes (Figure 7.34C). Because these

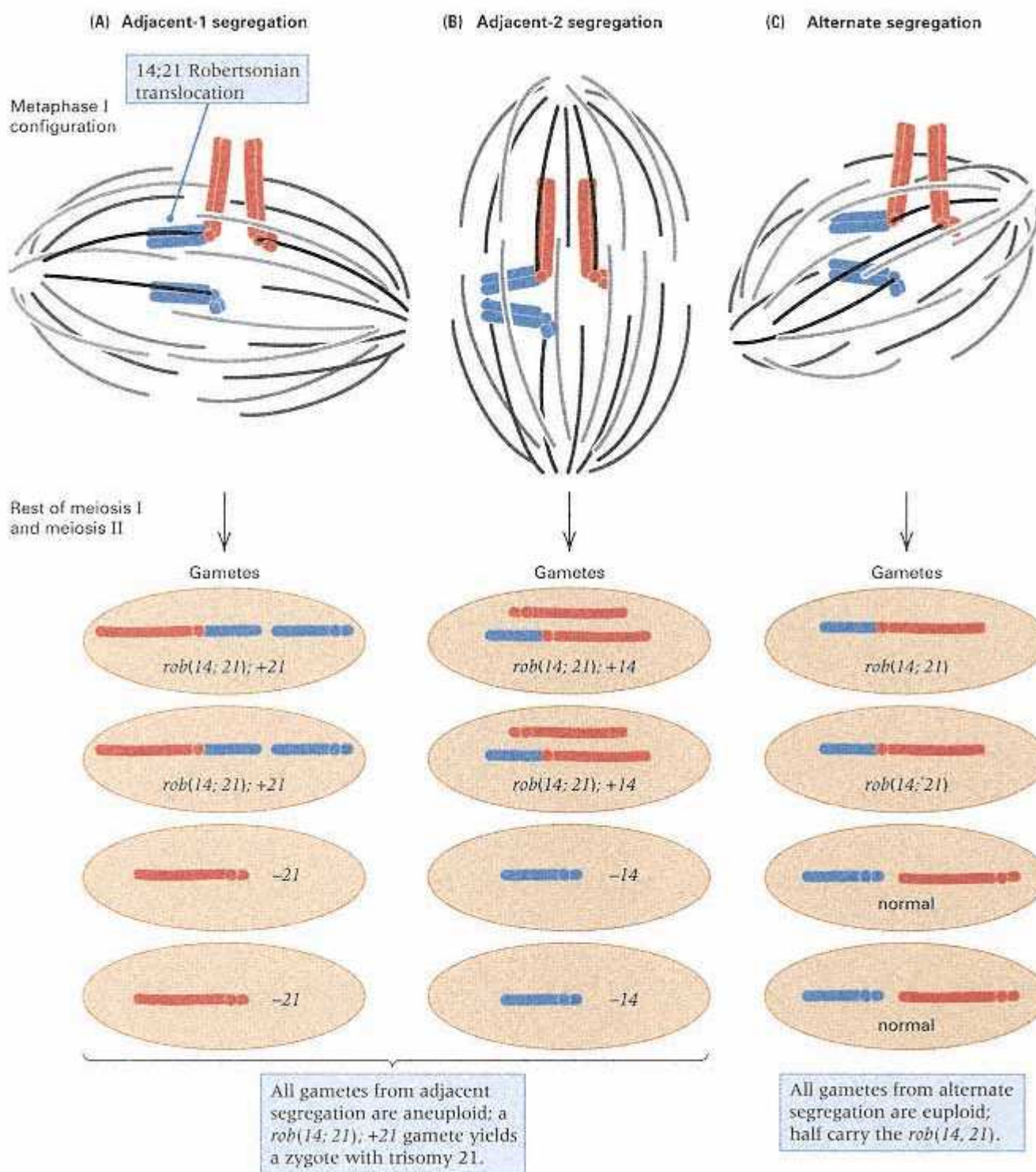


Figure 7.34

Segregation of Robertsonian translocation between chromosomes 14 and 21. (A) Adjacent-1 segregation, in which the gametes formed from the pole at the top have, effectively, an extra copy of chromosome 21. (B) Adjacent-2 segregation. The gametes are either duplicated or deficient for chromosome 14. (C) Alternate segregation. Half of the gametes give rise to phenotypically normal children who are carriers of the Robertsonian translocation.

gametes derive from reciprocal products of meiosis, the nonaffected children have a risk of 1/2 of carrying the translocation.

7.7—

Position Effects on Gene Expression

Genes near the breakpoints of chromosomal rearrangement become repositioned in the genome and flanked by new neighboring genes. In many cases, the repositioning of a gene affects its level of expression or, in some cases, its ability to function; this is called **position effect**. These effects have been studied extensively in *Drosophila* and also in yeast. In *Drosophila*, the most common type of position effect results in a mottled (mosaic) phenotype that is observed as interspersed patches of wildtype cells, in which the wildtype gene is expressed, and mutant cells, in which the wildtype gene is inactivated. The phenotype is said to show **variegation**, and the phenomenon is called **position-effect variegation (PEV)**. In the older literature, PEV is often referred to as variegated or V-type position effect.

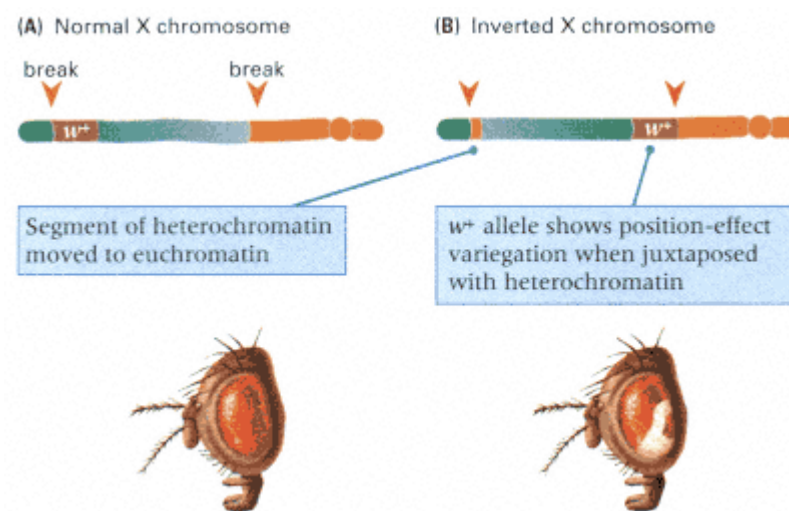


Figure 7.35

Position-effect variegation (PEV) is often observed when an inversion or other chromosome rearrangement repositions a gene normally in euchromatin to a new location in or near heterochromatin. In this example, an inversion in the X chromosome of *Drosophila melanogaster* repositions the wildtype allele of the *white* gene near heterochromatin. PEV of the w^+ allele is observed as mottled red and white eyes.

PEV usually results from a chromosome aberration that moves a wildtype gene from a position in euchromatin to a new position in or near heterochromatin (Figure 7.35). (Euchromatin and heterochromatin are discussed in Chapter 6.) Figure 7.36 illustrates some of the patterns of wildtype (red) and mutant (white) facets that are observed in male flies that carry a rearranged X chromosome in which an inversion repositions the wildtype w^+ allele into heterochromatin. The same types of patterns are found in females heterozygous for the rearranged X chromosome and an X chromosome carrying the w allele. The patterns of w^+ expression coincide with the clonal lineages in the eye; that is, all of the red cells in a particular patch derive from a single ancestral cell in the embryo in which the w^+ allele was activated. In contrast with the pattern shown in Figure 7.36, other chromosome rearrangements with PEV yield a salt-and-pepper pattern of mosaicism, which consists of numerous very small patches of wildtype tissue, and still others yield a combination of many small and a few large patches. These patterns imply that gene activation can be very late in development as well as very early.

Although the mechanism of PEV is not understood in detail, it is thought to result from the unusual chromatin structure of heterochromatin interfering with gene activation. The determination of gene expression or nonexpression is thought to take place when the boundary between condensed heterochromatin and euchromatin is established. Where heterochromatin is juxtaposed with euchromatin, the chromatin condensation characteristic of heterochromatin may spread into the adjacent euchromatin, inactivating euchromatic genes in the cell and all of its descendants. A similar inactivation phenomenon takes place in cells of female mammals when euchromatic genes translocated to the X chromosome become heterochromatic and inactive. The length of the euchromatic region that is inactivated ranges from 1 to 50 bands in the *Drosophila* polytene chromosomes, depending on the particular chromosome abnormality. At the molecular level, this range is approximately 20 to 1000 kb. The term *spreading*,

with its implication of smooth

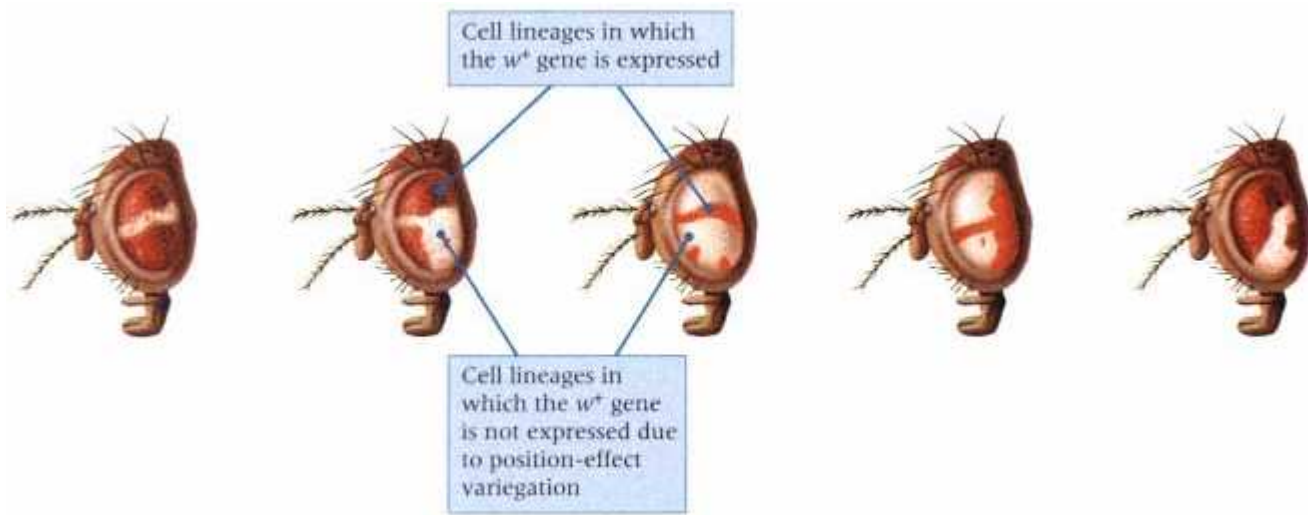


Figure 7.36

Patterns of red and white sectors in the eye of *Drosophila melanogaster* resulting from position-effect variegation. Each group of contiguous facets of the same color derives from a single cell in development. Such large patches of red are atypical. More often, one observes numerous very small patches of red or a mixture of many small and a few large patches.

continuity, should not be taken literally, however, because in both *Drosophila* and mammals, chromosomal regions affected by PEV are known to contain active genes interspersed with inactive ones. It is likely that heterochromatic regions associate to form a relatively compact "compartment" in the nucleus, and the presence of heterochromatin near a euchromatic gene may silence the gene simply by attracting it into the heterochromatic compartment.

The heterochromatin of different chromosomes differs in its ability to induce variegation of specific genes and in its response to environmental or genetic modifiers. Environmental modifiers include temperature; genetic modifiers include the presence of extra blocks of heterochromatin (such as a Y chromosome). Some mutations that affect chromatin structure also suppress PEV, and certain euchromatic DNA sequences, when repositioned into heterochromatin, act as "insulators" protecting nearby genes by binding with chromatin-associated proteins that prevent inactivation. Other than position-effect variegation, genes are rarely shut down completely because of their position in the genome. Most genes can be expressed irrespective of neighboring genes, but the level of expression may vary substantially according to position.

7.8—

Chromosome Abnormalities and Cancer

Cancer is an unrestrained proliferation and migration of cells. In all known cases, cancer cells derive from the repeated division of a single mutant cell whose growth has become unregulated, and so cancer cells initially constitute a clone. With continued proliferation, many cells within such clones develop chromosomal abnormalities, such as extra chromosomes, missing chromosomes, deletions, duplications, or translocations. The chromosomal abnormalities found in cancer cells are diverse, and they may differ among cancer cells in the same person or among people with the same type of cancer. The accumulation of chromosome abnormalities is evidently one accompaniment of unregulated cell division.

Amid the large number of apparently random chromosome abnormalities found in cancer cells, a small number of aberrations are found consistently in certain types of cancer, particularly in blood diseases such as the leukemias. For example, chronic myelogenous leukemia is frequently associated with an apparent deletion of part of the long arm of chromosome 22. The abnormal chromosome 22 in this

disease is called the **Philadelphia chromosome**; this chromosome is actually one part of a reciprocal translocation in which the missing segment of chromosome 22 is attached to either chromosome 8 or chromosome 9. Similarly, a deletion of part of the short arm of chromosome 11 is frequently associated with a kidney tumor called Wilms tumor, which is usually found in children.

For many of these characteristic chromosome abnormalities, the breakpoint in the chromosome is near the chromosomal location of a **cellular oncogene**. An **oncogene** is a gene associated with cancer. Cellular oncogenes, also called *protooncogenes*, are the cellular homologs of **viral oncogenes** contained in certain cancer-causing viruses. The distinction is one of location: Cellular oncogenes are part of the normal genome; viral oncogenes are derived from cellular oncogenes through some rare mechanism in which they become incorporated into virus particles. More than 50 different cellular oncogenes are known. They are apparently normal developmental genes that predispose cells to unregulated division when mutated or abnormally expressed. Many of the genes function in normal cells as growth factors that promote and regulate cell division. When a chromosome rearrangement happens near a cellular oncogene (or when the gene is incorporated into a virus), the gene may become expressed abnormally and result in unrestrained proliferation of the cell that contains it. However, abnormal expression by itself is usually not sufficient to produce cancerous growth. One or more additional mutations in the same cell are also required.

A sample of characteristic chromosome abnormalities found in certain cancers is given in Figure 7.37, along with the locations of known cellular oncogenes on the same chromosomes. The symbols are again *p* for the short arm and *q* for the long arm. The + and - signs are new designations:

- A + or - preceding a chromosome number indicates an extra (or missing) copy of the entire chromosome.
- A + or - following a chromosome designation means extra material (or missing material) corresponding to part of the designated chromosome or arm; for example, 11p- refers to a deletion of part of the short arm of chromosome 11.

The symbol *t* means reciprocal translocation; hence *t*(9;22) refers to a reciprocal translocation between chromosome 9 and chromosome 22.

In Figure 7.37, the location of the cellular oncogene, when known, is indicated by a red arrow at the left, and the chromosomal breakpoint is indicated by a black arrow at the right. In most cases for which sufficient information is available, the correspondence between the breakpoint and the cellular oncogene location is very close, and in some cases, the breakpoint is within the cellular oncogene itself. The significance of this correspondence is that the chromosomal rearrangement disturbs normal cellular oncogene regulation and ultimately leads to the onset of cancer.

Retinoblastoma and Tumor-Suppressor Genes

A second class of genes associated with inherited cancers consists of the **tumor-suppressor genes**. These are genes whose presence is necessary to suppress tumor formation. Absence of both normal alleles, through either mutational inactivation or deletion, results in tumor formation. An example of a tumor-suppressor gene is the human gene *Rb-1*, located in chromosome 13 in band 13q14. When the normal *Rb-1* gene product is absent, malignant tumors form in the retinas, and surgical removal of the eyes becomes necessary. The disease is known as **retinoblastoma**.

Retinoblastoma is unusual in that the predisposition to retinal tumors is dominant in pedigrees but the *Rb-1* mutation is recessive at the cell level; that is, a person who inherits one copy of the *Rb-1* mutation through the germ line is heterozygous, and the penetrance of retinoblastoma in this person is 100 percent. However, the retinal cells that become malignant have the genotype *Rb-1 Rb-1*. The explanation for this apparent paradox is illustrated in Figure 7.38. Part A of Figure 7.38 shows the genotype of an *Rb-1* heterozygote, along with a few other genes in the same chromosome. Parts B through E show four possible ways in which a second genetic event can result

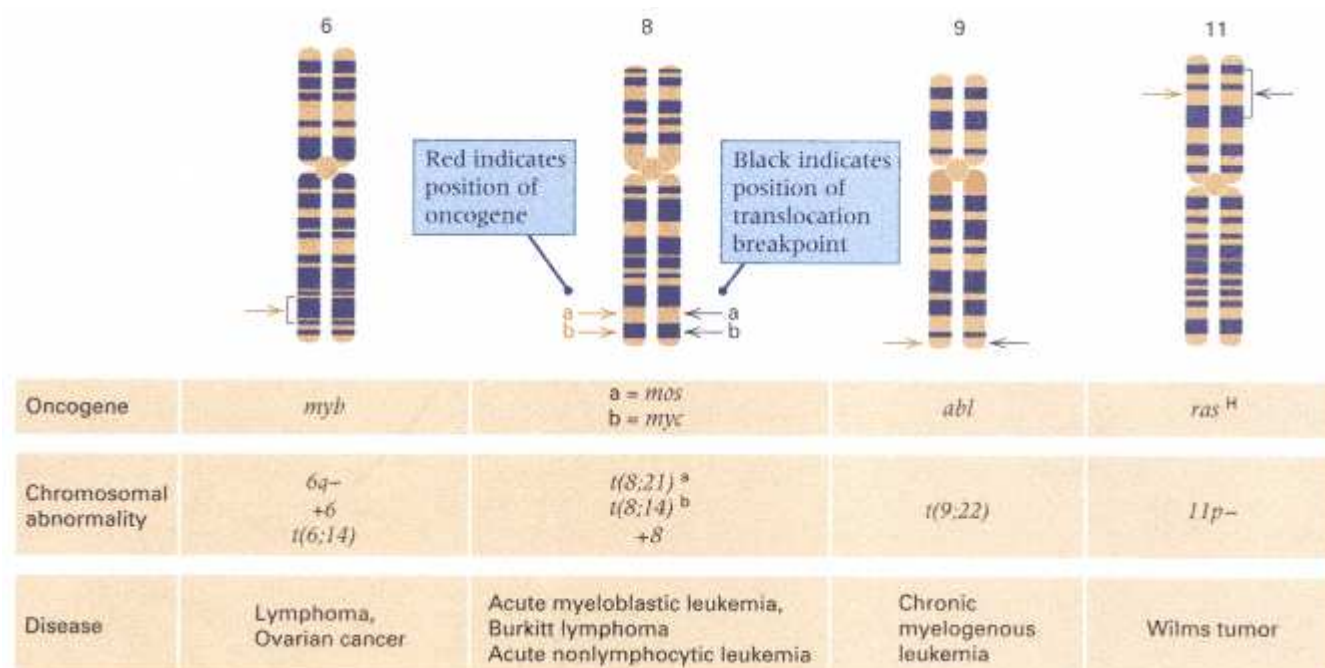


Figure 7.37

Correlation between oncogene positions (red arrows) and chromosome breaks (black arrows) in aberrant human chromosomes frequently found in cancer cells. A number of break-points near *myb* result in a cancer-associated *t(6;14)*.

in *Rb-1Rb-1* cells in the retina. The simplest (part B) is mutation of the wildtype allele in the homolog. The wildtype allele could also be deleted (part C). Part D illustrates a situation in which the normal homolog of the *Rb-1* chromosome is lost and replaced by nondisjunction of the *Rb-1*-bearing chromosome. Mitotic recombination (part E) is yet another possibility for making *Rb-1* homozygous. Although each of these events takes place at a very low rate per cell division, there are so many cells in the retina (approximately 10^8) that the *Rb-1* allele usually becomes homozygous in at least one cell. (In fact, the average number of tumors per retina is three.)

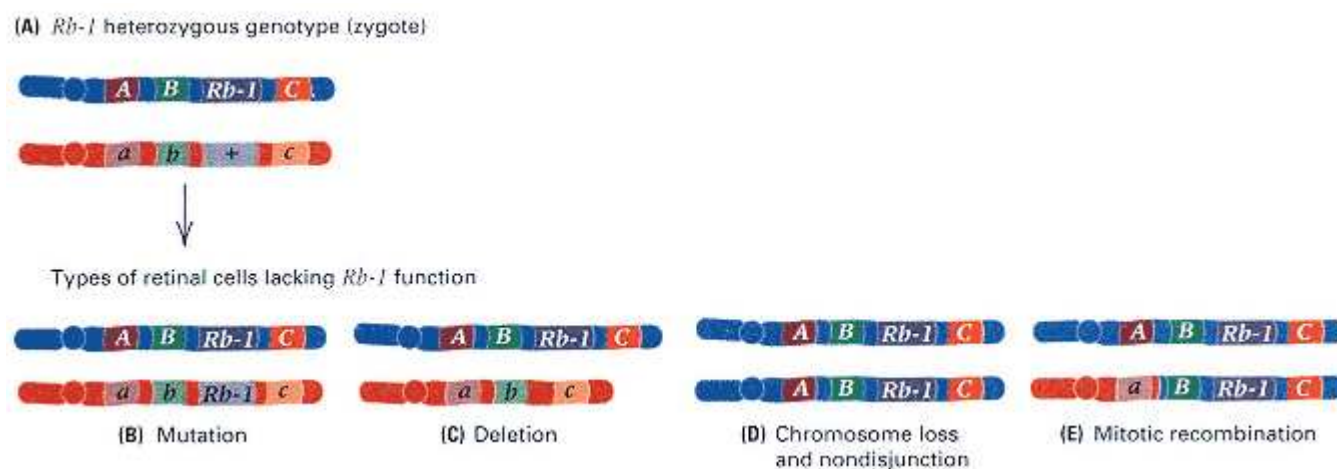


Figure 7.38

Mechanisms by which (A) a single copy of *Rb-1* inherited through the germ line can become homozygous in cells of the retina: (B) new mutation; (C) deletion; (D) loss of the normal homologous chromosome and replacement by nondisjunction of the *Rb-1*-bearing chromosome; (E) mitotic recombination. Each of these events is rare, but there are so many cells in the retina that on average, there are three such events per eye.

Chapter Summary

A typical chromosome contains a single centromere, the position of which determines the shape of the chromosome as it is pulled to the poles of the cell during anaphase. Rare chromosomes with no centromere, and those with two or more centromeres, are usually lost within a few cell generations because of aberrant separation during anaphase. Ring chromosomes are genetically relatively stable but are very rare.

Polyploid organisms contain more than two complete sets of chromosomes. Polyploidy is widespread among higher plants and uncommon otherwise. Between 30 and 35 percent of all species of flowering plants are thought to have originated as some form of polyploid. An autopolyploid organism contains multiple sets of chromosomes from a single ancestral species; allopolyploid organisms contain complete sets of chromosomes from two or more ancestral species. Organisms occasionally arise in which an individual chromosome either is missing or is present in excess; in either case, the number of copies of genes in such a chromosome is incorrect. Departures from normal gene dosage (aneuploidy) often result in reduced viability of the zygote in animals or of the gametophyte in plants. In general, too many copies of genes or chromosomes have less severe effects than too few copies.

The normal human chromosome complement consists of 22 pairs of autosomes, which are assigned numbers 1 through 22 from longest to shortest, and one pair of sex chromosomes (XX in females and XY in males). Fetuses that contain an abnormal number of autosomes usually fail to complete normal embryonic development or die shortly after birth, though people with Down syndrome (trisomy 21) sometimes survive for several decades. Persons with excess sex chromosomes survive, because the Y chromosome contains relatively few genes other than the master sex-controller *SRY*, and because only one X chromosome is genetically active in the cells of females (dosage compensation through the single-active-X principle). The mosaic orange and black pattern of the female "calico" cat results from X inactivation, because these alternative coat color alleles are X-linked in cats.

Most structural abnormalities in chromosomes are duplications, deletions, inversions, or translocations. In a duplication, there are two copies of a chromosomal segment. In a deletion, a chromosomal segment is missing. An organism can often tolerate an imbalance of gene dosage resulting from small duplications or deletions, but large duplications or deletions are almost always harmful. Chromosome rearrangements may affect gene expression through position effects. Although most genes can be expressed whatever their location in the genome, the level of expression may vary substantially according to position. A major effect of gene location on expression is observed in position-effect variegation (PEV), in which a wildtype allele repositioned in or near heterochromatin is unable to be expressed in a fraction of the cell lineages.

A chromosome that contains an inversion has a group of adjacent genes in reverse of the normal order. Expression of the genes is usually unaltered, so inversions rarely affect viability. However, crossing-over between an inverted chromosome and its noninverted homolog in meiosis yields abnormal chromatids. Crossing-over within a heterozygous paracentric inversion yields an acentric chromosome and a dicentric chromosome, both of which also have a duplication and a deficiency. Crossing-over within a heterozygous pericentric inversion yields monocentric chromosomes, but both have a duplication and a deficiency.

Two nonhomologous chromosomes that have undergone an exchange of parts constitute a reciprocal translocation. Organisms that contain a reciprocal translocation, as well as the normal homologous chromosomes of the translocation, produce fewer offspring (this is called semisterility) because of abnormal segregation of the chromosomes in meiosis. The semisterility is caused by the aneuploid gametes produced in adjacent-1 and adjacent-2 segregation. Alternate segregation yields equal numbers of normal and translocation-bearing gametes. In genetic crosses, the semisterility of a heterozygous translocation behaves like a dominant genetic marker that can be mapped like any other gene; however, what is actually mapped is the breakpoint of the translocation.

A translocation may also be nonreciprocal. A Robertsonian translocation is a type of nonreciprocal translocation in which the long arms of two acrocentric chromosomes are attached to a common centromere. In human beings, Robertsonian translocations that include chromosome 21 account for about 3 percent of all cases of Down syndrome, and the parents have a high risk of recurrence of Down syndrome in a subsequent child.

Malignant cells in many types of cancer contain specific types of chromosome abnormalities. Frequently, the break-points in the chromosomes coincide with the chromosomal location of one of a group of cellular oncogenes coding for cell growth factors. Abnormal oncogene expression is implicated in cancer. Viral oncogenes, found in certain cancer-causing viruses, are derived from cellular oncogenes. Cells also contain tumor-suppressor genes, the absence of which predisposes to cancer. The gene for retinoblastoma codes for a tumor-suppressor in the retina of

the eyes. The gene is dominant in predisposing to retinal malignancy but recessive at the cellular level. People who inherit one copy of the gene through the germ line develop retinal tumors when the gene becomes homozygous in cells in the retina. Homozygosity in the retina can result from any number of genetic events, including new mutation, deletion, chromosome loss and nondisjunction, and mitotic recombination.

Key Terms

acentric chromosome	fragile-X chromosome	retinblastoma
acrocentric chromosome	fragile-X syndrome	rhodopsin
adjacent-1 segregation	haploid	ring chromosome
adjacent-2 segregation	hexaploid	Robertsonian translocation
allopolyploid	imprinting	semisterility
alternate segregation	inversion	single-active-X principle
amniocentesis	inversion loop	submetacentric chromosome
aneuploid	karyotype	tandem duplication
autopolyploidy	Klinefelter syndrome	testis-determining factor (TDF)
Barr body	metacentric chromosome	tetraploid
cellular oncogene	monoploid	translocation
chimeric gene	monosomic	trinucleotide repeat
chromosome painting	mosaic	triplication
colchicine	octoploid	triploid
decaploid	oncogene	trisomic
deficiency	paracentric inversion	trisomy-X syndrome
deletion	pericentric inversion	trivalent
deutanopia	Philadelphia chromosome	tumor-suppressor gene
deuteranomaly	position effect	Turner syndrome
dicentric chromosome	position-effect variegation (PEV)	uncovering
dosage compensation	protanopia	unequal crossing-over
double-Y syndrome	protanomaly	univalent
Down syndrome	polyploidy	variegation

duplication	polysomy	viral oncogene
embryoid	pseudoautosomal	X inactivation
endoreduplication	reciprocal translocation	
euploid	red-green color blindness	

Review the Basics

- What is a metacentric chromosome? A submetacentric chromosome? An acrocentric chromosome?
- Why does an acentric chromosome fail to align at the metaphase plate and fail to move to one of the poles in anaphase? What type of abnormal chromosome forms a chromosomal "bridge" between the daughter cells at anaphase?
- How can a normal gamete have two sets of chromosomes? What possible gametes could be produced by a tetraploid plant with the genotype $AAaa BBbb$?
- Define each of the following terms: allopolyploid, aneuploid, deletion, duplication, paracentric inversion, pericentric inversion, reciprocal translocation, Robertsonian translocation.
- What does it mean to say that a deletion "uncovers" a recessive mutation?
- Which type of chromosome rearrangement can change a metacentric chromosome into a submetacentric chromosome? Which type of chromosome rearrangement can fuse two acrocentric chromosomes to make one metacentric chromosome?
- Inversions are often called "suppressors" of crossing-over. Is this term literally true? If not, what is meant by the term?
- What does *dosage compensation* mean with reference to X-linked genes?
- What are two broad classes of genes often associated with inherited cancers?

Guide to Problem Solving

Problem 1: The first artificial allotetraploid was created by the Russian agronomist G. D. Karpechenko in the 1930s by crossing the radish *Raphanus sativus* with the cabbage *Brassica oleracea*. Both species have a diploid chromosome number of 18. The initial F_1 hybrid was virtually sterile, but among the off-spring was a rare, fully fertile allotetraploid that he called *Raphanobrassica*. What is the chromosome number of the F_1 of the cross $R. sativus \times B. oleracea$? What is the chromosome number of *Raphanobrassica*? How many bivalents are formed in meiosis in *Raphanobrassica*?

Answer: Both *R. sativus* and *B. oleracea* produce gametes with 9 chromosomes, so the F_1 has 18 chromosomes. The allotetraploid results from a doubling of the F_1 chromosome complement, so *Raphanobrassica* has 36 chromosomes. Because each chromosome has a homolog, *Raphanobrassica* forms 18 bivalents.

Problem 2: Genes *a*, *b*, *c*, *d*, *e*, and *f* are closely linked in a chromosome, but their order is unknown. Three deletions in the region are found to uncover recessive alleles of the genes as follows:

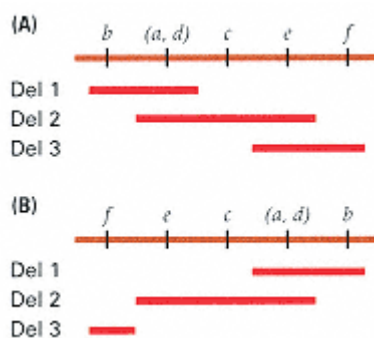
Deletion 1 uncovers *a*, *b*, and *d*.

Deletion 2 uncovers *a*, *d*, *c*, and *e*.

Deletion 3 uncovers *e* and *f*.

What is the order of the genes? In this problem, you will see that there is enough information to order most, but not all, of the genes. Suggest what experiments you might carry out to complete the ordering.

Answer: Problems of this sort are worked by noting that genes uncovered by a single deletion must be contiguous. The gene order is deduced from the overlaps between the deletions. The overlaps are *a* and *d* between the first and second deletions and *e* between the second and third. The gene order (as far as can be determined from these data) is diagrammed in part A of the accompanying illustration. The deletions are shown in red. Gene *b* is at the far left, then *a* and *d* (in unknown order), then *c*, *e*, and *f*. (Gene *c* must be to the left of *e*, because otherwise *c* would be uncovered by deletion 3.) Part B is a completely equivalent map with gene *b* at the right. The ordering can be completed with a three-point cross between *b*, *a*, and *d* or between *a*, *d*, and *c* or by examining additional deletions. Any deletion that uncovers either *a* or *d* (but not both) plus at least one other marker on either side would provide the information to complete the ordering.

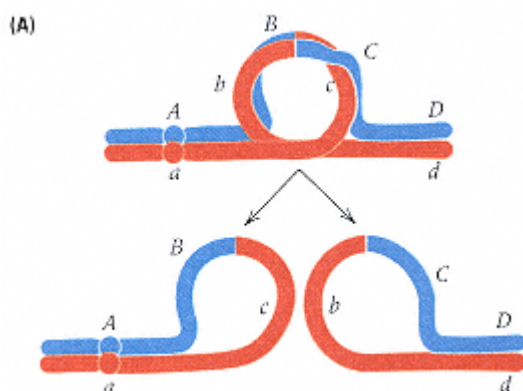


Problem 3: In each of the following cases, determine the consequences of a single crossover within the inverted region of a pair of homologous chromosomes with the gene order *A B C D* in one chromosome and *a c b d* in the other.

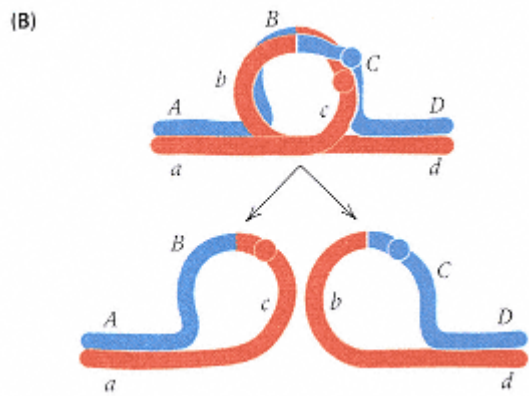
(a) The centromere is not included within the inversion.

(b) The centromere is included within the inversion.

Answer: These kinds of problems are most easily solved by drawing a diagram. The accompanying illustration shows how pairing within a heterozygous inversion results in a looped configuration. (Crossing-over occurs at the fourstrand stage of meiosis, but for simplicity only the chromatids participating in the crossover are diagrammed.) Part A illustrates the situation when the centromere is not included within the inversion. The crossover chromatids consist of a dicentric (two centromeres) and an acentric (no centromere), and the products are duplicated for the terminal region containing *A* and deficient for the terminal region containing *D*, or the other way around. The noncrossover chromatids (not shown) are the parental *A B C D* and *a c b d* monocentric chromosomes.



Part B illustrates the situation when the centromere is included in the inversion. The duplications and deficiencies are the same as in part A, but in this case both products are monocentric. As before, the noncrossover chromatids are the parental $A B C D$ and $a c b d$ configurations.



Analysis and Applications

7.1 How many Barr bodies would be present in each of the following human conditions?

- (a) Klinefelter syndrome
- (b) Turner syndrome
- (c) Down syndrome
- (d) XYY
- (e) XXX

Chapter 7 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

<http://www.jbpub.com/genetics>

Select the link to **Genetics: Principles and Analysis** and then choose the link to **GeNETics on the web**. You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. The original **Bar duplication** may itself have arisen from unequal crossing-over. You can access the details by searching for the keyword at this site. If assigned to do so, write a brief description of the proposed origin of the original duplication, and draw a diagram illustrating the events in the process.

2. Molecular studies of the red and green **opsin** genes indicate abnormalities in 15.7 percent of Caucasian men. This frequency is substantially smaller than the incidence of color blindness as shown by color-vision testing, which suggests that some chimeric genes created by unequal crossing-over may be compatible with normal color vision. More information about the molecular genetics and physiology of color vision can be found by searching the keyword site "protanopia" and "deutanopia." You can also find a description of the "Nagel anomaloscope" used in testing color vision. If assigned to do so, write a one-paragraph description of how this instrument works, and discuss some of its limitations.

3. There are many myths about **Down syndrome**. One is that people with Down syndrome are severely retarded. You can find more myths, and the truth about them, at the keyword site. If assigned to do so, prepare a double-column list giving a brief description of each myth and the correct story.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetic resources available on the World Wide Web. Select the **Mutable Site** for Chapter 7, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 7.

7.3 A plant species S coexists with two related species A and B. All species are interfertile. In meiosis, S forms 26 bivalents, and A and B form 14 and 12 bivalents, respectively. Hybrids between S and A form 14 bivalents and 12 univalents, and hybrids between S and B form 12 bivalents and 14 univalents. Suggest a probable evolutionary origin for species S.

7.5 A spontaneously aborted human fetus is found to have 45 chromosomes. What is the most probable karyotype? If the fetus survived, what genetic disorder would it have had?

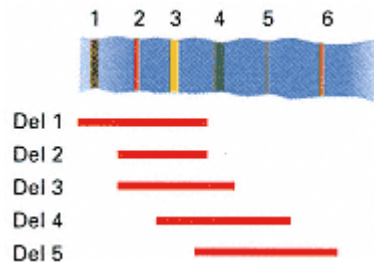
7.6 Color blindness in human beings is an X-linked trait. A man who is color-blind has a 45,X (Turner syndrome) daughter who is also color-blind. Did the nondisjunction that led to the 45,X child occur in the mother or the father? can you tell?

7.7 A phenotypically normal woman has a child with Down syndrome. The woman is found to have 45 chromosomes. What kind of chromosome abnormality can account for these observations? How many chromosomes does the affected child have? How does this differ from the usual chromosome number and karyotype of a child with Down syndrome?

7.8 A chromosome has the gene sequence A B C D E F G. What is the sequence following a C-through-E inversion? Following a C-through-E deletion? Two chromosomes with the sequences A B C D E F G and M N O P Q R S T U V undergo a reciprocal translocation after breaks in E–F and S–T. What are the possible products? Which products are genetically stable?

7.9 Recessive genes *a*, *b*, *c*, *d*, *e*, and *f* are closely linked in a chromosome, but their order is unknown. Three deletions in the region are examined. One deletion uncovers *a*, *d*, and *e*; another uncovers *c*, *d*, and *f*; and the third uncovers *b* and *c*. What is the order of the genes?

7.10 Six bands in a salivary gland chromosome of *Drosophila* are shown in the accompanying figure, along with the extent of five deletions (Del1–Del5).



Recessive alleles *a*, *b*, *c*, *d*, *e*, and *f* are known to be in the region, but their order is unknown. When the deletions are heterozygous with each allele, the following results are obtained:

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
Del 1	-	-	-	+	+	+
Del 2	-	+	-	+	+	+
Del 3	-	+	-	+	-	+
Del 4	+	+	-	-	-	+
Del 5	+	+	+	-	-	-

In this table, the - means that the deletion is missing the corresponding wildtype allele (the deletion uncovers the recessive allele) and + means that the corresponding wildtype allele is still present. Use these data to infer the position of each gene relative to the salivary gland chromosome bands.

7.11 A strain of corn that has been maintained by self-fertilization for many generations, when crossed with a normal strain, produces an F_1 in which many meiotic cells have dicentric chromatids and acentrics, and the amount of recombination in chromosome 6 is greatly reduced. However, in the original strain, no dicentrics or acentrics are found. What kind of abnormality in chromosome structure can explain these results?

7.12 Four strains of *Drosophila melanogaster* are isolated from different localities. The banding patterns of a particular region of salivary chromosome 2 have the following configurations (each letter denotes a band).

(a) a b f e d c g h i j

(b) a b c d e f g h i j

(c) a b f e h g i d c j

(d) a b f e h g c d i j

Assuming that part (c) is the ancestral sequence, deduce the evolutionary ancestry of the other chromosomes.

7.13 Two species of Australian grasshoppers coexist side by side. In meiosis, each has 8 pairs of chromosomes. When the species are crossed, the chromosomes in the hybrid form 6 pairs of chromosomes and one group of 4. What alteration in chromosomes structure could account for these results?

7.14 Why are translocation heterozygotes semisterile? Why are translocation homozygotes fully fertile? If a translocation homozygote is crossed with an individual that has normal chromosomes, what fraction of the F_1 is expected to be semisterile?

7.15 Curly wings (*Cy*) is a dominant mutation in the second chromosome of *Drosophila*. A *Cy*/+ male was irradiated with x rays and crossed with +/+ females, and the *Cy*/+ sons were mated individually with +/+ females. From one cross, the progeny were

curly males	146
wildtype males	0
curly females	0
wildtype females	163

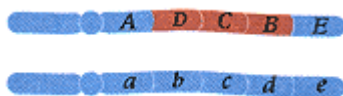
What abnormality in chromosome structure is the most likely explanation for these results? (Remember that crossing-over does not take place in male *Drosophila*.)

7.16 Yellow body (*y*) is a recessive mutation near the tip of the **X** chromosome of *Drosophila*. A wildtype male was irradiated with x rays and crossed with *yy* females, and one y^+ son was observed. This male was mated with *yy* females, and the offspring were

yellow females	256
yellow males	0
wildtype females	0
wildtype males	231

The yellow females were found to be chromosomally normal, and the y^+ males were found to breed in the same manner as their father. What type of chromosome abnormality could account for these results?

7.17 In the homologous chromosomes shown here, the red region represents an inverted segment of chromosome.



An individual of this genotype was crossed with an *a b c d e* homozygote. Most of the offspring were either A B C D E or *a b c d e*, but a few rare offspring were obtained that were A B c D E. What events occurring in meiosis in the inversion heterozygote can explain these rare progeny? Is the gene sequence in the rare progeny normal or inverted?

Challenge Problems

7.18 A wildtype strain of yeast, thought to be a normal haploid, was crossed with a different haploid strain carrying the mutation *his7*. This mutation is located in chromosome 2 and is an allele of a gene normally required for synthesis of the amino acid histidine. Among 15 tetrads analyzed from this cross, the following types of segregation were observed:

4 wildtype : 0 <i>his7</i>	4 tetrads
3 wildtype : 1 <i>his7</i>	4 tetrads
2 wildtype : 2 <i>his7</i>	1 tetrad

However, when the same wildtype strain was crossed with haploid strains with recessive markers on other chromosomes, segregation in the tetrads was always 2:2. What type of chromosome abnormality in the wildtype strain might account for the unusual segregation when the strain is mated with *his7*?

7.19 A strain of semisterile maize heterozygous for a reciprocal translocation between chromosomes 1 and 2 was crossed with chromosomally normal plants homozygous for the recessive mutations *brachytic* and *fine-stripe* on chromosome

1. When semisterile F_1 plants were crossed with plants of the *brachytic*, *fine-stripe* parental strain, the following phenotypes were found in a total of 682 F_2 progeny.

	Semisterile	Fertile
wildtype	333	19
<i>brachytic</i>	17	6
<i>fine-stripe</i>	1	8
<i>brachytic</i> , <i>fine-stripe</i>	25	273

What are the recombinant frequencies between *brachytic* and the translocation breakpoint and between *fine-stripe* and the translocation breakpoint?

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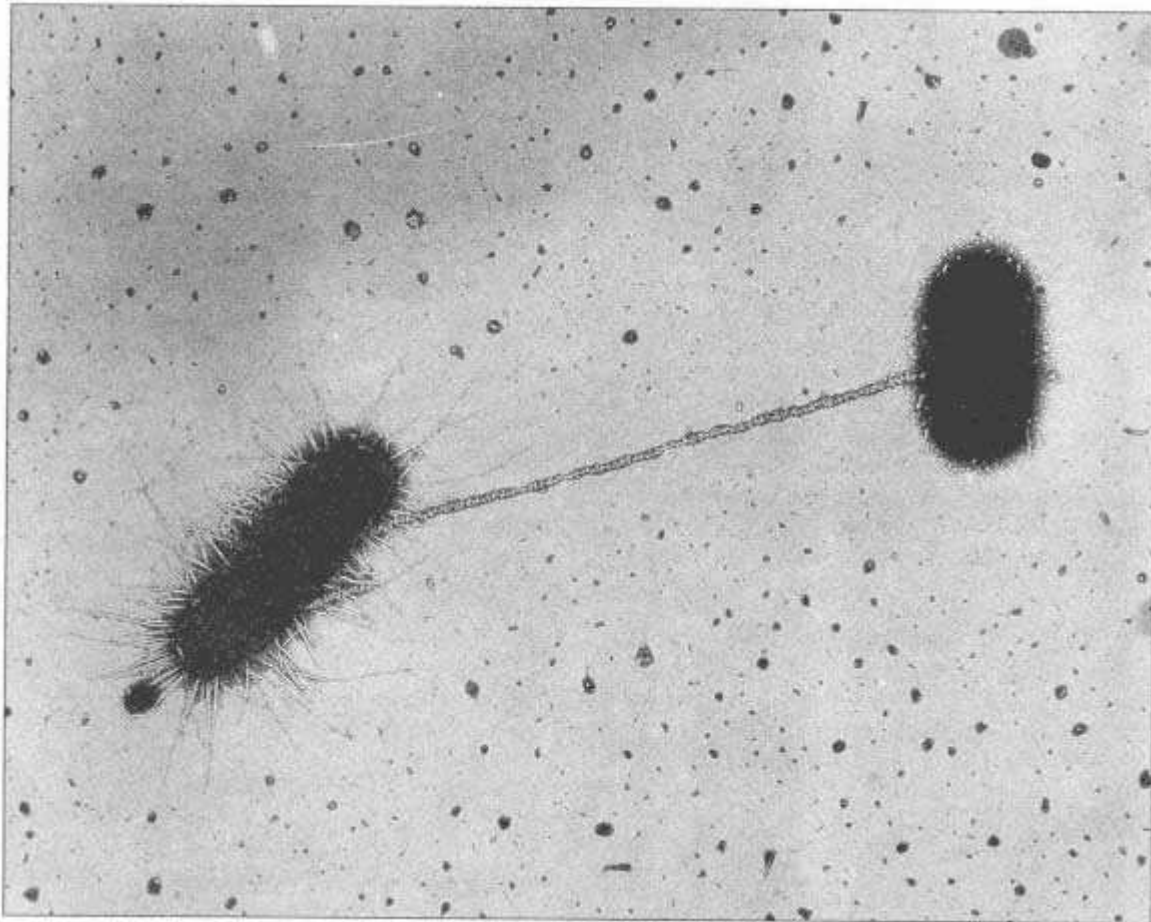
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Two cells of *Escherichia coli* caught in the act of mating (conjugation). The male cell is at the lower left, and the female cell, at the upper right. The connecting tube is an F pilus, encoded by genes in the F plasmid, through which DNA from the male cell enters the female cell. (The pilus shortens somewhat prior to DNA transfer.) In this case, for ease of visualization, the F pilus has been coated with particles of a male-specific bacteriophage. The male cell has numerous other appendages, not F pili, that are used in colonizing the intestine.

[Courtesy of C. C. Brinton, Jr., and J. Carnahan.]

Chapter 8— The Genetics of Bacteria and Viruses

CHAPTER OUTLINE

8-1 The Genetic Organization of Bacteria and Viruses

8-2 Bacterial Mutants

8-3 Bacterial Transformation

8-4 Conjugation

Plasmids

Hfr Cells

Time-of-Entry Mapping

F' Plasmids

8-5 Transduction

8-6 Bacteriophage Genetics

Plaque Formation and Phage Mutants

Genetic Recombination in Virulent Bacteriophages

Fine Structure of the rII Gene in Bacteriophage T4

8-7 Genetic Recombination in Temperate Bacteriophages

Lysogeny

Specialized Transducing Phage

8-8 Transposable Elements

Transposons in Genetic Analysis

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Challenge Problems

Further Reading

GeNETics on the web

PRINCIPLES

- Some bacteria are capable of DNA transfer and genetic recombination.

- In transformation, bacterial cells take up DNA from the surrounding medium and incorporate it into their genome by homologous recombination, replacing some host genes in the process.
- In *E. coli*, the F (fertility) plasmid can mobilize the chromosome for transfer to another cell in the process of conjugation.
- Some types of bacteriophages can incorporate bacterial genes and transfer them into new host cells in the process of transduction.
- DNA molecules from related bacteriophages that are present in the same host cell can undergo genetic recombination.
- Transposable elements and plasmids are widely used for genetic analysis and DNA manipulation in bacteria.

CONNECTIONS

CONNECTION: The Sex Life of Bacteria

Joshua Lederberg And Edward L. Tatum 1946
Gene Recombination In Escherichia Coli

CONNECTION: Is a Bacteriophage an "Organism"?

Alfred D. Hershey And Raquel Rotman 1948
Genetic Recombination Between Host-Range And Plaque-Type Mutants Of Bacteriophage In Single Bacterial Cells

CONNECTION: Artoo

Seymour Benzer 1955
Fine Structure Of A Genetic Region In Bacteriophage

In earlier chapters, we examined the genetic properties of representative eukaryotic organisms. The genomes of these organisms consist of multiple chromosomes. In the course of sexual reproduction, genotypic variation among the progeny is achieved both by random assortment of the chromosomes and by crossing-over, processes that take place in meiosis. Two important features of crossing-over in eukaryotes are that (1) it results in a reciprocal exchange of material between two homologous chromosomes and (2) both products of a single exchange can often be recovered in different progeny. The situation is quite different in prokaryotes, as we will see in this chapter.

Bacteria and viruses bring to traditional types of genetic experiments four important advantages over multicellular plants and animals. First, they are haploid, so dominance or recessiveness of alleles is not a complication in identifying genotype. Second, a new generation is produced in minutes rather than weeks or months, which vastly increases the rate of accumulation of data. Third, they are easy to grow in enormous numbers under controlled laboratory conditions, which facilitates molecular studies and the analysis of rare genetic events. Fourth, the individual members of these large populations are genetically identical; that is, each laboratory population is a **clone** of genetically identical cells.

8.1—

The Genetic Organization of Bacteria and Viruses

The organization of a typical prokaryotic cell is exemplified by the bacterial cell in Figure 8.1A. The genetic material is located in a region that lacks clear boundaries and is called the **nucleoid**. Contrast this relatively unstructured organization with the eukaryotic cell in Figure 8.1B. In eukaryotes, the genetic material is enclosed in the nucleus by a membrane envelope connected in special ways to the cytoplasm; eukaryotic cells also have other membrane systems that subdivide the cytoplasm into regions of specialized function.

A bacterial cell contains a major DNA molecule that almost never encounters another complete DNA molecule. Instead, genetic exchange is usually between a chromosomal fragment from one cell and an intact chromosome from another cell. Furthermore, a clear donor-recipient relationship exists: The donor cell is the source of a DNA fragment, which is transferred to the recipient cell by one of several mechanisms, and exchange of genetic material takes place in the recipient by means of reciprocal recombination between homologous DNA sequences.

Incorporation of a part of the donor DNA into the chromosome requires at least two exchange events, one at each end. Because the recipient molecule is circular, only an even number of exchanges results in a viable product. The usual outcome of these events is the recovery of *only one* of the crossover products. However, in some situations, the transferred DNA is also circular, and a single exchange results in total incorporation of the circular donor DNA into the chromosome of the recipient.

Three major types of genetic transfer are found in bacteria: *transformation*, in which a DNA molecule is taken up from the external environment and incorporated into the genome; *transduction*, in which DNA is transferred from one bacterial cell to another by a bacterial virus; and *conjugation*, in which donor DNA is transferred from one bacterial cell to another by direct contact. Recombination frequencies that result from these processes are used to produce genetic maps of bacteria. Although the maps are exceedingly useful, they differ in major respects from the types of maps obtained from crosses in eukaryotes because genetic maps in eukaryotes are based on frequencies of crossing-over in meiosis.

A virus is a small particle, considerably smaller than a cell, that is able to infect a susceptible cell and multiply within it to form a large number of progeny virus particles. Few, if any, organisms are not subject to viral infection. Many human diseases are caused by viruses, including influenza, measles, AIDS, and the common cold. Viruses that infect bacterial cells are called **bacteriophages** or simply **phages**. Most viruses consist of a single molecule of

genetic material enclosed in a protective coat composed of one or more kinds of protein molecules; however, their size, molecular constituents, and structural complexity vary greatly (Figure 8.2).

Isolated viruses possess no metabolic systems, so a virus can multiply only within a cell. It is "living" only in the sense that its genetic material directs its own multiplication; outside its host cell, a virus is an inert particle. Nevertheless, within an infected cell, many viruses and bacteriophages possess mechanisms by which their DNA molecules exchange genetic material resulting in genetic recombination. In the life cycle of a bacteriophage, a phage particle attaches to a host bacterium and injects its nucleic acid into the cell; the nucleic acid replicates many times; finally, newly synthesized nucleic acid molecules are packaged into protein shells (forming progeny phage), and then the particles are released from the cell (Figure 8.3). Except for new mutations, phage progeny from a bacterium infected

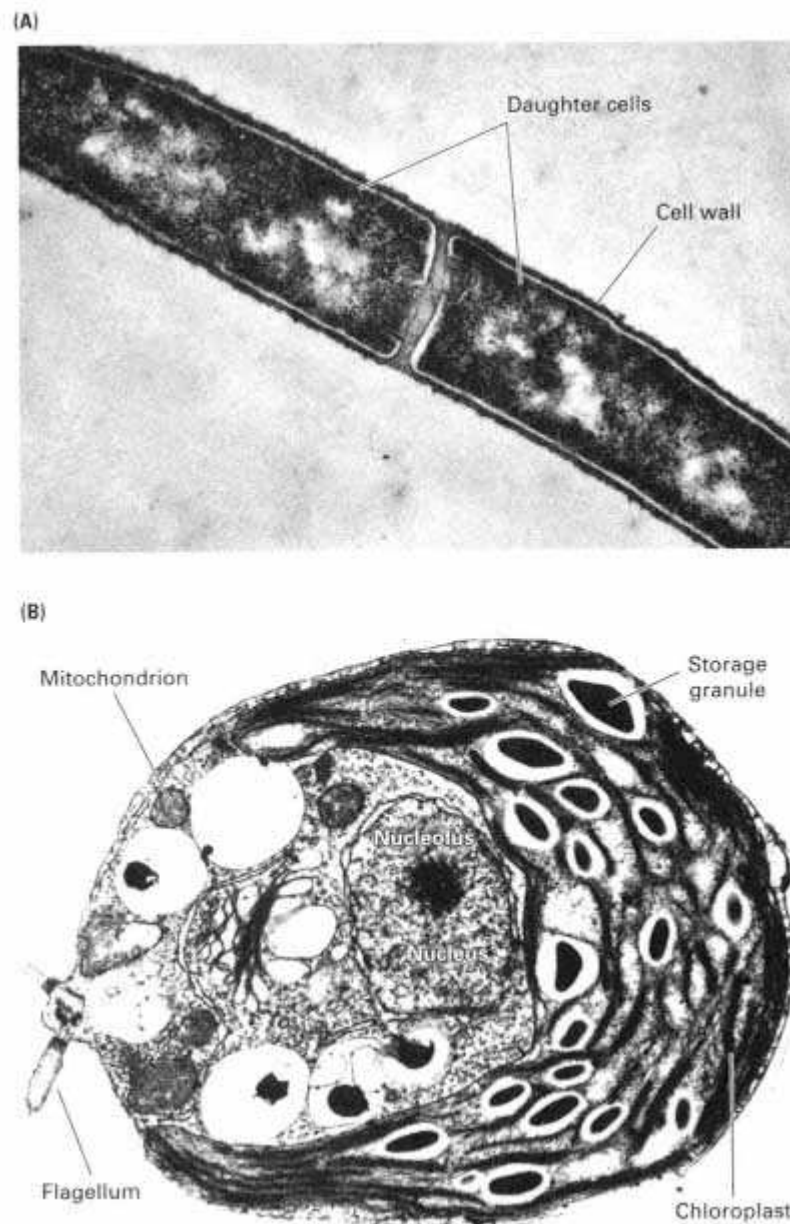


Figure 8.1

The organization of a prokaryotic cell and a eukaryotic cell. (A) An electron micrograph of a dividing bacterial cell, showing the dispersed genetic material (light areas). (B) An electron micrograph of a section through a cell of the eukaryotic alga *Tetraspora*, showing the membrane-bounded nucleus, the nucleolus (dark central body), and other membrane systems (chloroplasts and mitochondria) that subdivide the cytoplasm into regions of specialized function.

The dark sharply bounded regions in part B are starch storage granules, which store carbohydrate. Note the complexity of this structure compared with the simpler organization of the bacterial cell shown in part A.

[Part A, courtesy of A. Benichou-Ryter; part B, courtesy of Jeremy Pickett-Heaps.]

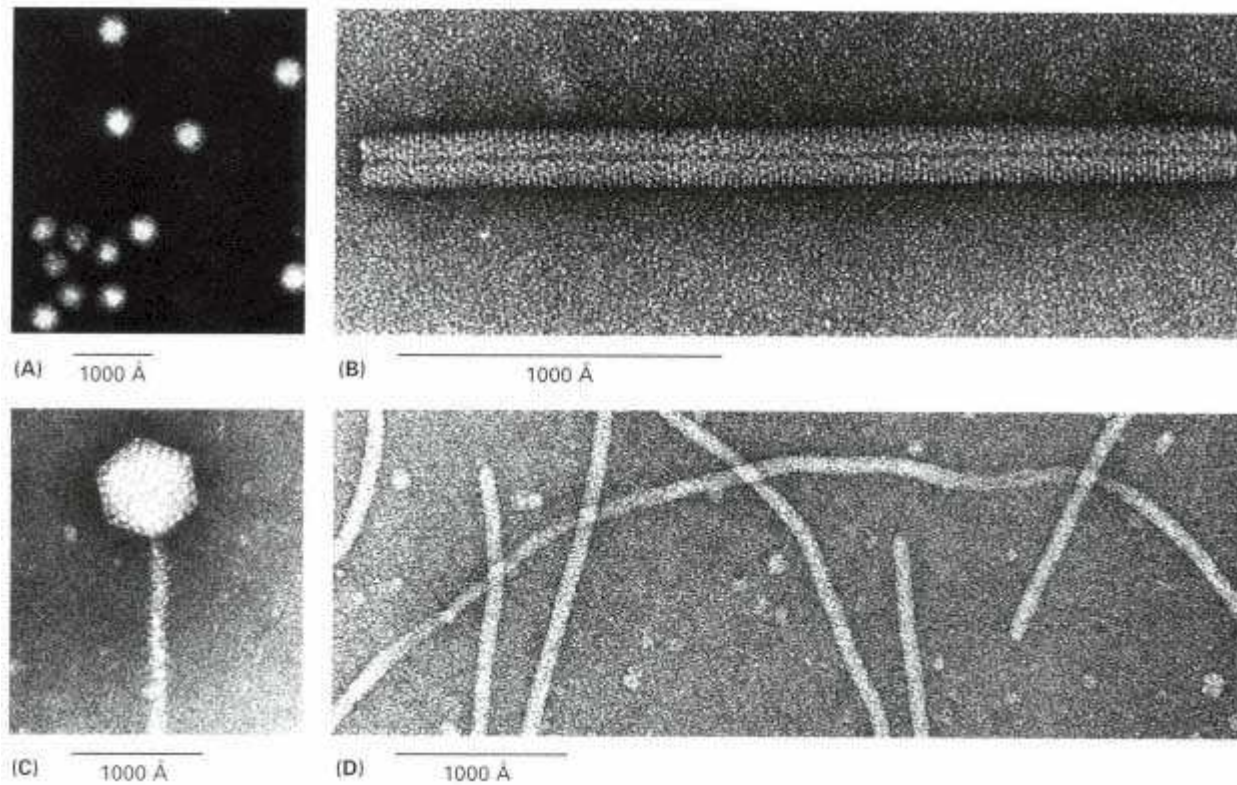


Figure 8.2
Electron micrographs of four different viruses: (A) poliovirus; (B) tobacco mosaic virus; (C) *E. coli* phage λ ; (D) *E. coli* phage M13. In each case, the length of the bar is 10^{-5} cm, or 1000 Å units.
[Courtesy of Robley Williams.]

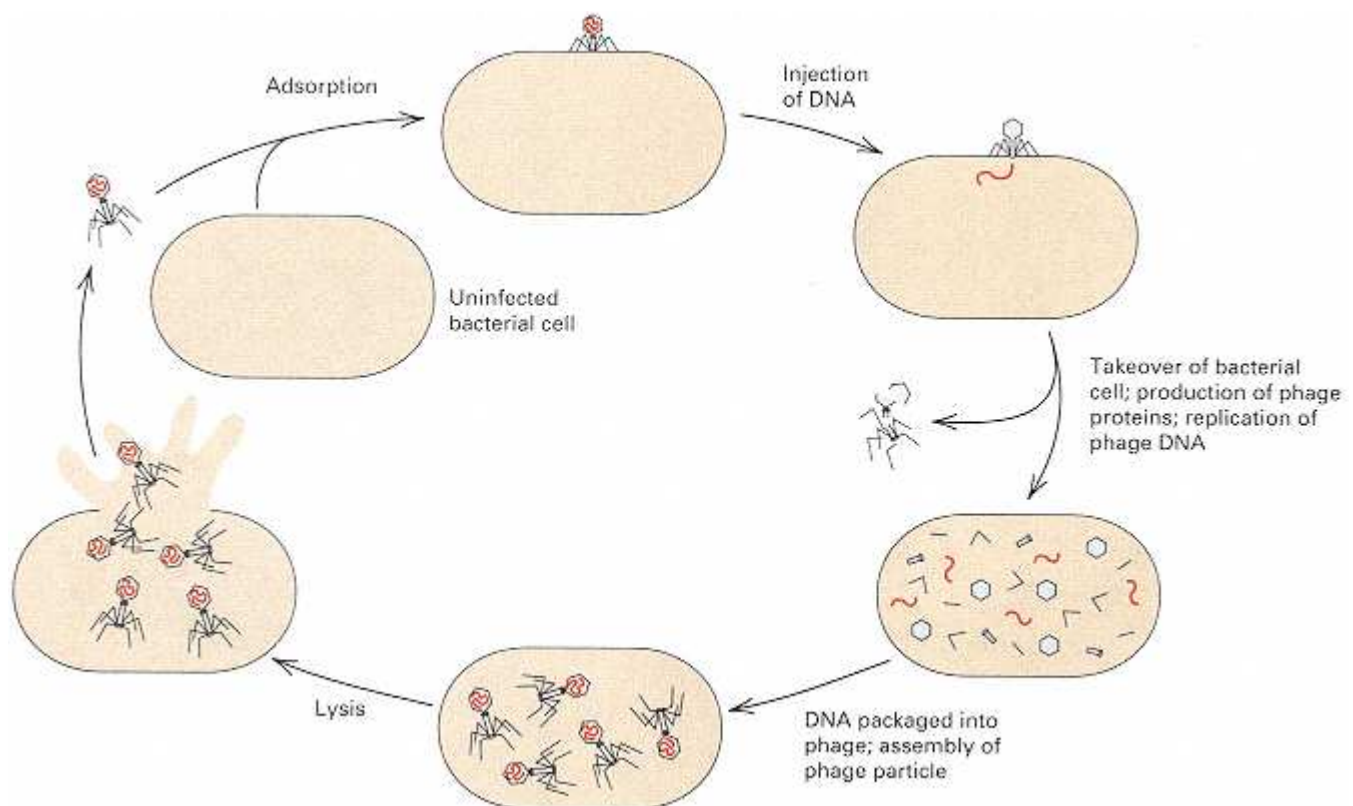


Figure 8.3
A schematic diagram of the life cycle of a typical bacteriophage.

by a single phage have the same genotype as the parental phage. However, if two phage particles with *different* genotypes infect a single bacterial cell, new genotypes can arise by genetic recombination. This process differs significantly from genetic recombination in eukaryotes in two ways: (1) the number of participating DNA molecules differs from one cell to the next, and (2) reciprocal recombinants are not always recovered in equal frequencies from a single infected cell. Some phages also possess systems that enable phage DNA to recombine with bacterial DNA. Phage-phage and phage-bacterium recombination are the topic of the second part of this chapter. The best-understood bacterial and phage systems are those of *E. coli*, and we will concentrate on these systems.

8.2— Bacterial Mutants

Bacteria can be grown in liquid medium or on the surface of a semisolid growth medium gelled with agar. Bacteria used in genetic analysis are usually grown on agar. A single bacterial cell placed on an agar medium will grow and divide many times, forming a visible cluster of cells called a *colony* (Figure 8.4). The number of bacterial cells present in a liquid culture can be determined by spreading a known volume of the culture on a solid medium and counting the number of colonies that form. Typical *E. coli* cultures contain up to 10^9 cells/ml. The appearance of colonies, or the ability or inability to form colonies on particular media can, in some cases, be used to identify the genotypes of bacterial cells.

As we have seen in earlier chapters, genetic analysis requires mutants; with bacteria, three types are particularly useful:

- **Antibiotic-resistant mutants** These mutants are able to grow in the presence of an antibiotic, such as streptomycin (Str) or tetracycline (Tet). For example, streptomycin-sensitive (Str-s) cells have the wildtype phenotype and fail to form colonies on medium that contains streptomycin, but streptomycin-resistant (Str-r) mutants can form colonies on such medium.
- **Nutritional mutants** Wildtype bacteria can synthesize most of the complex nutrients they need from simple



Figure 8.4

A petri dish with bacterial colonies that have formed on a solid medium. The heavy streaks of growth result from colonies so densely packed that there is no space between them.

molecules present in the growth medium. The wildtype cells are said to be **prototrophs**. The ability to grow in simple medium can be lost by mutations that disable the enzymes used in synthesizing the complex nutrients. Mutant cells are unable to synthesize an essential nutrient and cannot grow unless the required nutrient is supplied in the medium. Such a mutant bacterium is said to be an **auxotroph** for the particular nutrient. For example, a methionine auxotroph cannot grow on a **minimal medium** that contains only inorganic salts and a source of energy and carbon atoms (such as glucose), but the methionine auxotroph *can* grow if the minimal medium is supplemented with methionine.

- **Carbon-source mutants** Such mutant cells cannot utilize particular substances as sources of carbon atoms or of energy. For example, Lac⁻ mutants cannot utilize the sugar lactose for growth and are unable to form colonies on minimal medium that contains only lactose as the carbon source.

A medium on which all wildtype cells form colonies is called a **nonselective medium**. Mutants and wildtype cells may or may not be distinguishable by growth on a nonselective medium. If the medium allows growth of only one type of cell (either wildtype or mutant), then it is said to be **selective**. For example, a medium containing streptomycin is selective for the Str^r phenotype and selective against the Str^s phenotype; similarly, minimal medium containing lactose as the sole carbon source is selective for Lac⁺ cells and against Lac⁻ cells.

In bacterial genetics, phenotype and genotype are designated in the following way. A phenotype is designated by three letters, the first of which is capitalized, with a superscript + or - to denote presence or absence of the designated phenotype, and with *s* or *r* for sensitivity or resistance. A genotype is designated by lowercase italicized letters. Thus a cell unable to grow without a supplement of leucine (a leucine auxotroph) has a Leu⁻ phenotype, and this would usually result from a *leu*⁻ mutation. Often the - superscript is omitted, but using it prevents ambiguity.

8.3—

Bacterial Transformation

Bacterial transformation is a process in which recipient cells acquire genes from free DNA molecules in the surrounding medium. Transformation with purified DNA was the first experimental proof that DNA is the genetic material (Chapter 1). In these experiments, a rough-colony phenotype of *Streptococcus pneumoniae* was changed to a smooth-colony phenotype by exposure of the cells to DNA from a smooth-colony strain. In the laboratory, donor DNA is usually isolated from donor cells and then added to a suspension of recipient cells. In natural settings, such as soil, free DNA can become available by spontaneous breakage (lysis) of donor cells.

Transformation begins with uptake of a DNA fragment from the surrounding medium by a recipient cell and terminates with *one strand* of donor DNA replacing the homologous segment in the recipient DNA. Most bacterial species are probably capable of the recombination step, but many species have only a very limited ability to take up free DNA efficiently. Even in a species capable of transformation, DNA is able to penetrate only some of the cells in a growing population. However, many bacterial species can be made competent to take up DNA, provided that the cells are subjected to an appropriate chemical treatment (for example, treatment with CaCl₂).

Transformation is a convenient technique for gene mapping in some species. When DNA is isolated from a donor bacterium (Figure 8.5), it is invariably broken into small fragments. In most species, with suitable recipient cells and excess external DNA, transformation takes place at a frequency of about 1 transformed cell per 10³ cells. If two genes, *a* and *b*, used as genetic markers, are so widely separated in the donor chromosome that they are always contained in two different DNA fragments, then the probability of simultaneous trans-

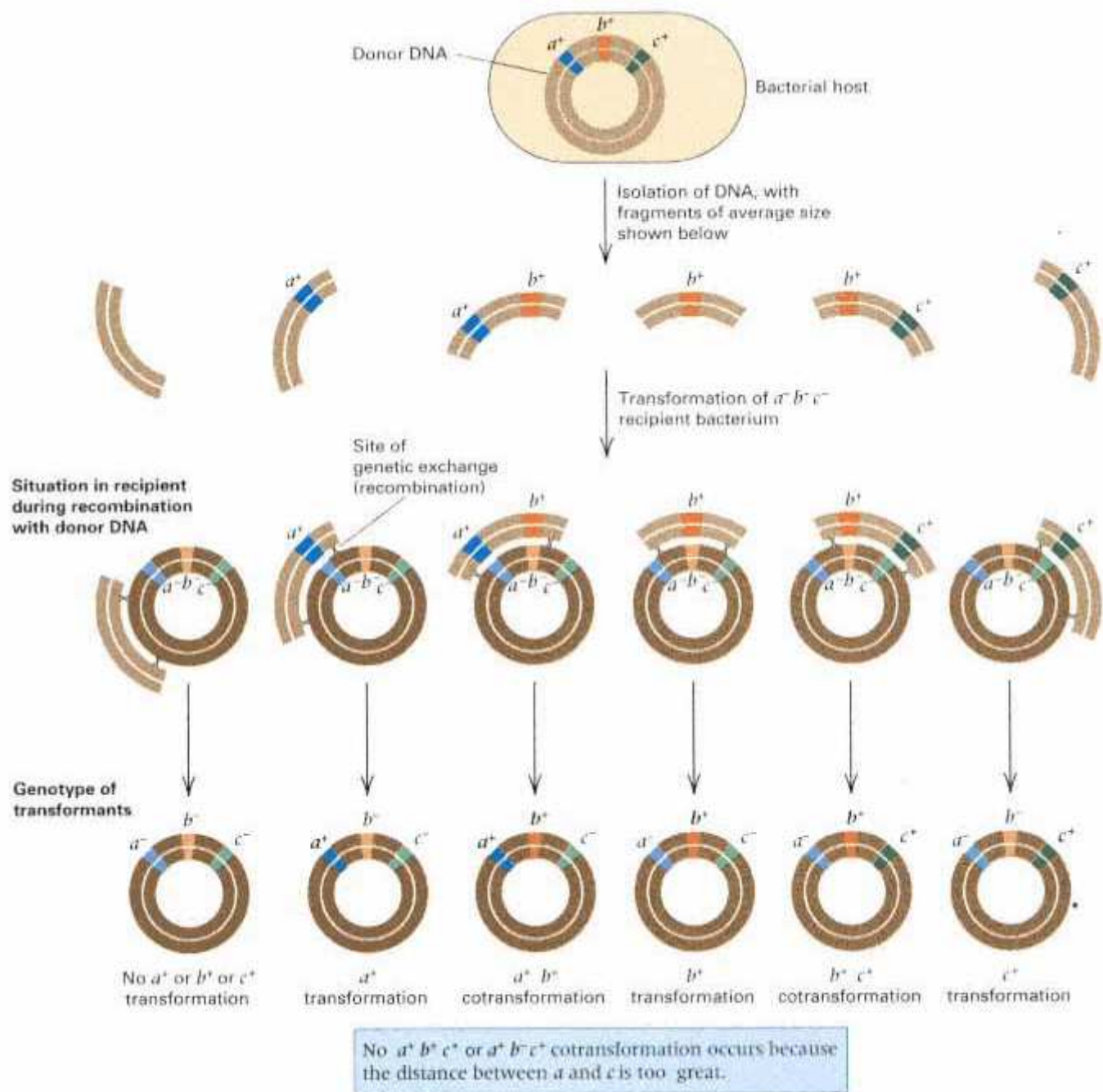


Figure 8.5

Cotransformation of linked markers. Markers a and b are near enough to each other that they are often present on the same donor fragment, as are markers b and c . Markers a and c are not near enough to undergo cotransformation. The gene order must therefore be $a b c$. The size of the transforming DNA, relative to that of the bacterial chromosome, is greatly exaggerated.

formation (**cotransformation**) of an $a^- b^-$ recipient into wildtype is the product of the probabilities of transformation of each genetic marker, or roughly $10^{-3} \times 10^{-3}$, which equals one $a^+ b^+$ transformant per 10^6 recipient cells. However, if the two genes are so near one another that they are often present in a single donor fragment, then the frequency of cotransformation is nearly the same as the

frequency of single-gene transformation, or one wildtype transformant per 10^3 recipients. The general principle is as follows:

Cotransformation of two genes at a frequency substantially greater than the product of the single-gene transformations implies that the two genes are close together in the bacterial chromosome.

Studies of the ability of various pairs of genes to be cotransformed also yield gene order. For example, if genes *a* and *b* can be cotransformed, and genes *b* and *c* can be cotransformed, but genes *a* and *c* cannot, the gene order must be *a b c* (Figure 8.5). Note that cotransformation frequencies are not equivalent to the recombination frequencies used in mapping eukaryotes, because they are determined by the size distribution of donor fragments and the likelihood of recombination between bacterial DNA molecules rather than by the occurrence of chiasmata in synapsed homologous chromosomes (Chapter 4).

8.4— Conjugation

Conjugation is a process in which DNA is transferred from a bacterial donor cell to a recipient cell by cell-to-cell contact. It has been observed in many bacterial species and is best understood in *E. coli*, in which it was discovered by Joshua Lederberg in 1951.

When bacteria conjugate, DNA is transferred to a recipient cell from a donor cell under the control of a set of genes that give the donor cell its transfer capability. These genes are often present in a nonchromosomal, circular DNA molecule called a **plasmid**. Plasmid-mediated recombination in *E. coli* usually results from the presence of a plasmid called the **F factor** or the **fertility factor**. In some instances, the F plasmid and similar types of plasmids can become incorporated into the bacterial chromosome so that the plasmid genes allowing DNA transfer are themselves part of the bacterial chromosome. In this case, the plasmid is said to have been **integrated** into the chromosome. Because it can exist either separate from the chromosome or incorporated into it, the F factor is an example of an **episome**, a term that refers to any genetic element that can exist free in the cell or as a segment of DNA integrated into the chromosome.

Conjugation begins with physical contact between a donor cell and a recipient cell. A tubular projection from the donor cell forms a passageway between the donor and recipient cells. Through this passageway, a copy of the donor DNA moves from the donor to the recipient. In the final stage, which requires recombination if the donor contains an integrated plasmid, a segment of the transferred donor DNA becomes part of the genetic complement of the recipient. If the donor contains a free plasmid, then only the plasmid DNA is transferred and takes up residence in the recipient.

Let us begin with a description of the genetic properties of plasmids and their transfer and then examine plasmid-mediated chromosomal transfer.

Plasmids

Plasmids are circular DNA molecules that are capable of replicating independently of the chromosome and range in size from a few kilobases to a few hundred kilobases (Figure 8.6). The F factor is approximately 100 kb in length and contains many genes for its maintenance in the cell and its transmission between cells. Plasmids have been observed in many bacterial species and are usually not essential for growth of the cells. For studying plasmids in the laboratory, a culture of cells derived from a single plasmid-containing cell is used; because plasmids replicate and are inherited, all cells of the culture contain the plasmid of interest. Plasmids not only contain a diversity of genes for their own maintenance, which are not present in the bacterial chromosome, but plasmids can also acquire chromosomal genes from their host by several mechanisms. The presence of certain plasmids in bacterial cells is made evident by phenotypic characteristics of the host cell conferred by genes in the plasmid. For example, a plasmid containing the *tet-r*

gene will make the recipient bacterial cell resistant to tetracycline.

Plasmids rely on the DNA-replication enzymes of the host cell for their reproduction, but *initiation* of replication is controlled by plasmid genes. The number of copies of a particular plasmid in a cell varies from one plasmid to the next, depending on its particular mode of regulation of initiation. High-copy-number plasmids are found in as many as 50 copies per host cell, whereas low-copy-number plasmids are present to the extent of 1 or 2 copies per cell. Plasmid DNA can be taken up by cells and become permanently established in the bacteria. The ability of plasmid DNA to transform cells genetically has made plasmids important in genetic engineering (Chapter 9).

From the point of view of bacterial genetics, the F plasmid is of greatest interest because of its role in mediating conjugation between cells of *E. coli*. Cells that contain F are donors and are designated F⁺ ("F plus"); those lacking F are recipients and are designated F⁻ ("F minus"). The F plasmid is a low-copy-number plasmid. A typical F⁺ cell contains one or two copies of F. These F plasmids replicate once per cell cycle and segregate to both daughter cells in cell division.

The F plasmid contains a set of genes for establishing conjugation between cells and for transferring DNA from donor to recipient. A copy of the F plasmid can be transferred in conjugation from an F⁺ cell to an F⁻ cell. Transfer is always accompanied by replication of the plasmid. Contact between an F⁺ and an F⁻ cell initiates rolling-circle replication of F (Section 5.4), which results in the transfer of a single-stranded linear branch of the rolling circle to the recipient cell. During transfer, DNA is synthesized in both donor and recipient (Figure 8.7). Synthesis in the donor replaces the transferred single strand, and synthesis in the recipient converts the transferred single strand into double-stranded DNA. When transfer is complete, the linear F strand becomes circular again in the recipient cell. Note that because one replica remains in the donor while the other is transferred to the recipient, after transfer *both cells contain F and can function as donors*. The transfer of F

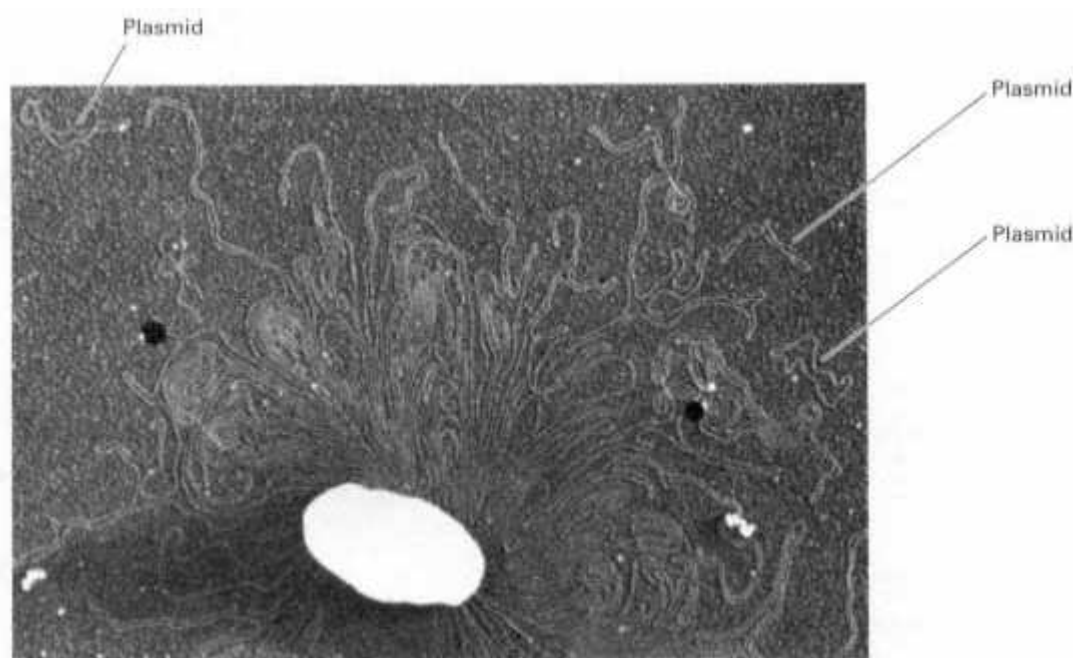


Figure 8.6
Electron micrograph of a ruptured *E. coli* cell showing released chromosomal DNA and several plasmid molecules.
[Courtesy of David Dressler and Huntington Potter.]

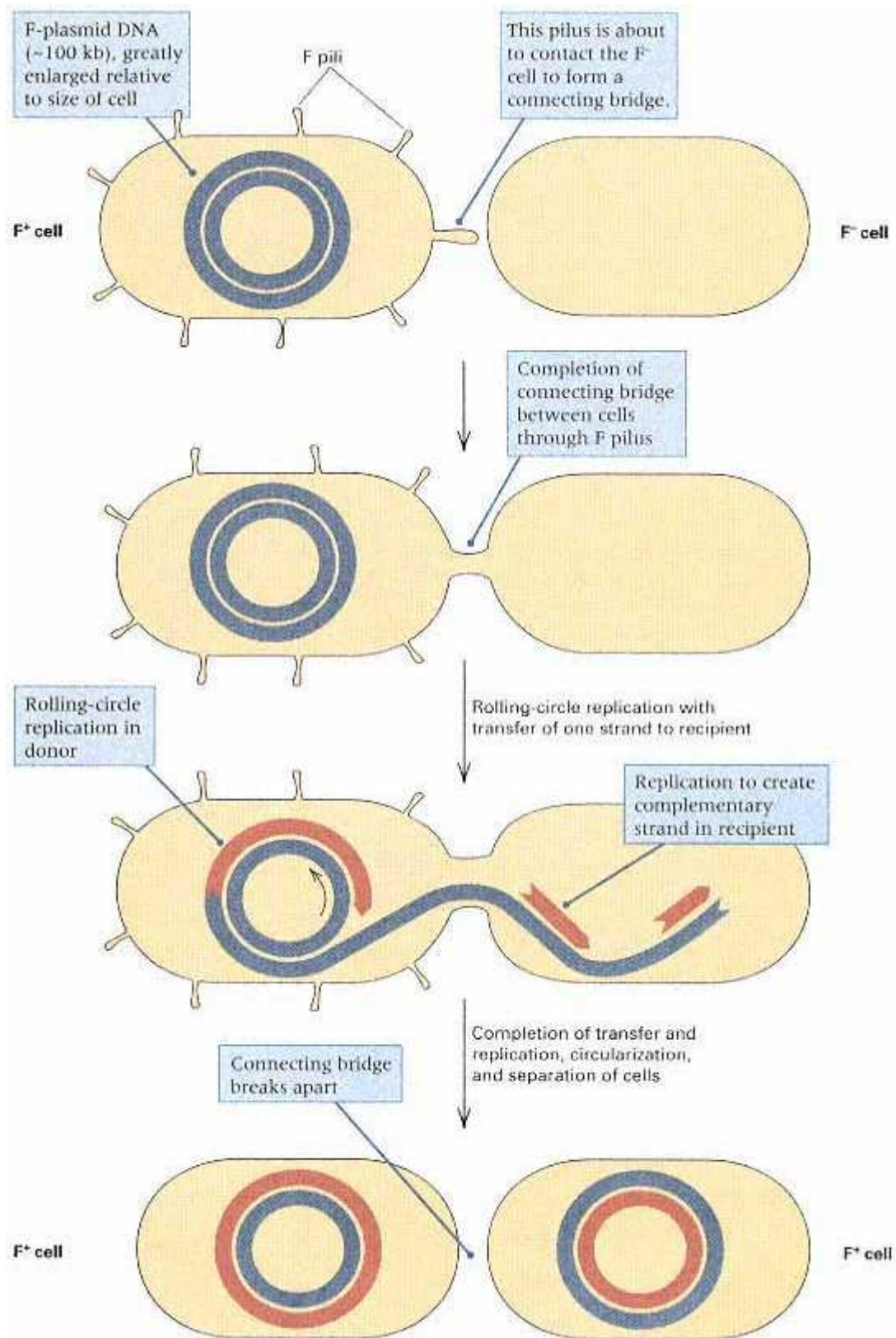


Figure 8.7

Transfer of F from an F^+ to an F^- cell. Pairing of the cells triggers rolling-circle replication. Pink represents DNA synthesized during pairing. For clarity, the bacterial chromosome is not shown, and the plasmid is drawn overly large; the plasmid is in fact much smaller than a bacterial chromosome.

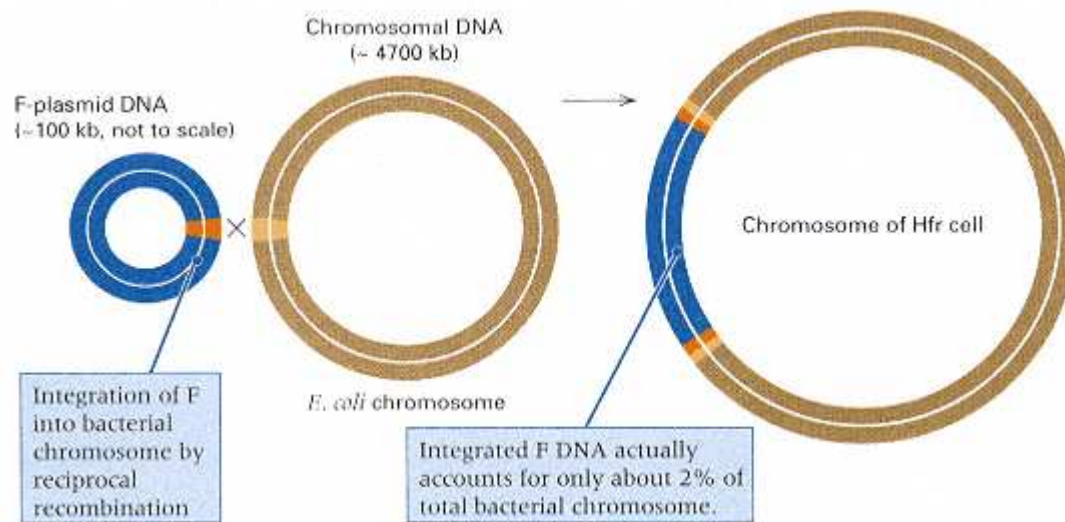


Figure 8.8

Integration of F (blue circle) by recombination between a nucleotide sequence in F and a homologous sequence in the bacterial chromosome. The F plasmid DNA is shown greatly enlarged relative to the size of the bacterial chromosome. In reality, the length of the DNA molecule in F is about 2 percent of the length of the bacterial chromosome.

requires only a few minutes. In laboratory cultures, if a small number of donor cells is mixed with an excess of recipient cells, F spreads throughout the population in a few hours, and all cells ultimately become F^+ . Transfer is not so efficient under natural conditions, and only about 10 percent of naturally occurring *E. coli* cells contain the F factor.

Hfr Cells

The F plasmid occasionally becomes integrated into the *E. coli* chromosome by an exchange between a sequence in F and a sequence in the chromosome (Figure 8.8). The bacterial chromosome remains circular, though enlarged about 2 percent by the F DNA. Integration of F is an infrequent event, but single cells containing integrated F can be isolated and cultured. The cells in such a strain are called **Hfr cells**. Hfr stands for **high frequency of recombination**, which refers to the relatively high frequency with which donor genes are transferred to the recipient. Integrated F mediates the transfer of DNA from the bacterial chromosome in an Hfr cell; thus a replica of part of the bacterial chromosome, as well as part of the plasmid, is transferred to the F^- cell.

The $Hfr \times F^-$ conjugation process is illustrated in Figure 8.9. The stages of transfer are much like those by which F is transferred to F^- cells: coming together of donor and recipient cells, rolling-circle replication in the donor cell, and conversion of the transferred single-stranded DNA into double-stranded DNA by lagging-strand synthesis in the recipient. However, in the case of Hfr matings, the transferred DNA does not become circular and is not capable of further replication in the recipient because the transferred F factor is not complete. The replication and associated transfer of the chromosomal DNA are controlled by the integrated F and are initiated in the Hfr chromosome at the same point in F at which replication and transfer begin within an unintegrated F plasmid. A part of F is the first DNA transferred, chromosomal genes are transferred next, and the remaining part of F is the last DNA to enter the recipient. Because the conjugating cells usually break apart long before the entire bacterial chromosome is transferred, the final segment of F is almost never transferred into the recipient.

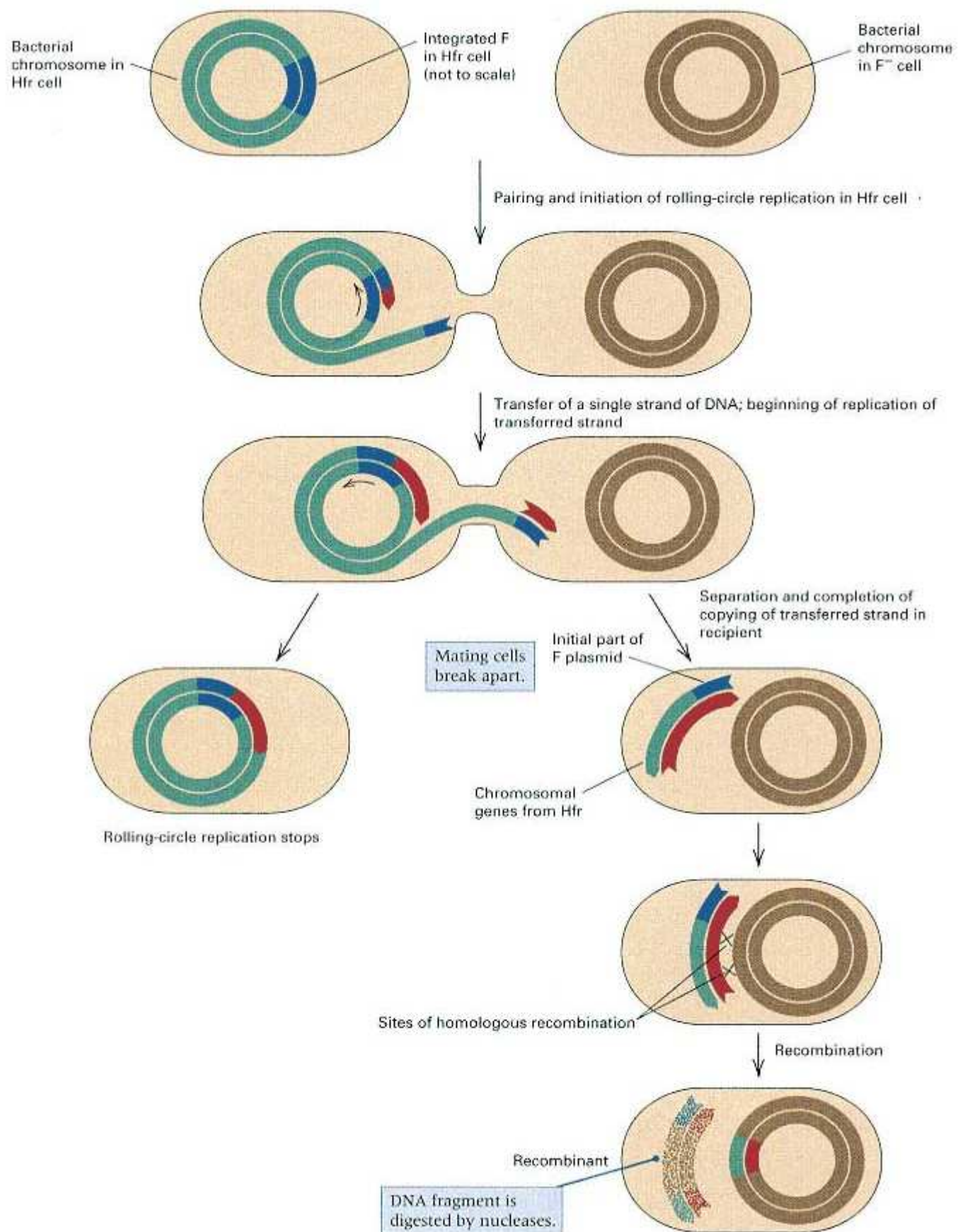


Figure 8.9

Stages in the transfer and production of recombinants in an Hfr x F⁻ mating. Pairing initiates rolling-circle replication within the F sequence in the Hfr cell, resulting in the transfer of a single strand of DNA. The single strand is converted into double-stranded DNA in the recipient. The mating cells usually break apart before the entire chromosome is transferred. Recombination takes place between the Hfr fragment and the F⁻ chromosome and leads to recombinants containing genes from the Hfr chromosome. Note that only a part of F is transferred. This part of F is not incorporated into the recipient chromosome. The recipient remains F⁻.

Several differences between F transfer and Hfr transfer are notable.

- It takes 100 minutes under the usual conditions for an entire bacterial chromosome to be transferred, in contrast with about 2 minutes for the transfer of F. The difference in time is a result of the relative sizes of F and the chromosome (100 kb versus 4700 kb).
- During transfer of Hfr DNA into a recipient cell, the mating pair usually breaks apart before the entire chromosome is transferred. Under usual conditions, several hundred genes are transferred before the cells separate.
- In a mating between Hfr and F⁻ cells, the F⁻ recipient remains F⁻ because cell separation usually takes place before the final segment of F is transferred.
- In Hfr transfer, some regions in the transferred DNA fragment become incorporated into the recipient chromosome. The incorporated regions replace homologous regions in the recipient chromosome. The result is that some F⁻ cells become recombinants containing one or more genes from the Hfr donor cell. For example, in a mating between Hfr *leu*⁺ and F⁻ *leu*⁻, some F⁻ *leu*⁺ cells arise. However, *the genotype of the donor Hfr cell remains unchanged*.

Genetic analysis requires that recombinant recipients be identified. Because the recombinants derive from recipient cells, a method is needed to eliminate the donor cells. The usual procedure is to employ an F⁻ recipient containing an allele that can be selected. Genes that confer antibiotic resistance are especially useful for this purpose. For instance, after a mating between Hfr *leu*⁺ *str*^{-s} and F⁻ *leu*⁻ *str*^{-r} cells, the Hfr *Str*^{-s} cells can be selectively killed by plating the mating mixture on medium containing streptomycin. A selective medium that lacks leucine can then be used to distinguish between the nonrecombinant and the recombinant recipients. The F⁻ *leu*⁻ parent cannot grow in medium that lacks leucine, but recombinant F⁻ *leu*⁺ cells can grow because they possess a *leu*⁺ gene. Only recombinant recipients—that is, cells having the genotype *leu*⁺ *str*^{-r}—form colonies on a selective medium containing streptomycin and lacking leucine. The selected allele should be located at such a place in the chromosome that most mating cells will have broken apart before the selected gene is transferred, and the selected allele must not be present in the Hfr cell. The selective agent can then be used to select the F⁻ cells and eliminate the Hfr donors.

When a mating is done in this way, the transferred marker that is selected by the growth conditions (*leu*⁺ in this case) is called a **selected marker**, and the marker used to prevent growth of the donor (*str*^{-s} in this case) is called the **counterselected marker**. Selection and counterselection are necessary in bacterial matings because recombinants constitute only a small proportion of the entire population of cells (in spite of the name "high frequency of recombination").

Time-of-Entry Mapping

Genes can be mapped by Hfr × F⁻ matings. However, the genetic map is quite different from all maps that we have seen so far in that it is not a linkage map but a transfer-order map. It is obtained by deliberate interruption of DNA transfer in the course of mating—for example, by violent agitation of the suspension of mating cells in a kitchen blender. The time at which a particular gene is transferred can be determined by breaking the mating cells apart at various times and noting the earliest time at which breakage no longer prevents recombinants from appearing. This procedure is called the **interrupted-mating technique**. When this is done with Hfr × F⁻ matings, the number of recombinants of any particular allele increases with the time during which the cells are in contact. This phenomenon is illustrated in Table 8.1. The reason for the increase is that different Hfr × F⁻ pairs initiate conjugation and chromosome transfer at slightly different times.

A greater understanding of the transfer process can be obtained by observing the

Table 8.1 Data showing the production of Leu^+ Str^r recombinants in a cross between $\text{Hfr } \text{leu}^+ \text{str}^s$ and $\text{F}^- \text{leu}^- \text{str}^r$ cells when mating is interrupted at various times

Minutes after mating	Number of Leu^+ Str^r recombinants per 100 Hfr cells
0	0
3	0
6	6
9	15
12	24
15	33
18	42
21	43
24	43
27	43

Note: Minutes after mating means minutes after the Hfr and F^- cell suspensions are mixed. Extrapolation of the recombination data to a value of zero recombinants indicates that the earliest time of entry of the leu^+ marker is 4 minutes.

results of a mating with several genetic markers. For example, consider the mating

$$\text{Hfr } a^+ b^+ c^+ d^+ e^+ \text{str}^s \times \text{F}^- a^- b^- c^- d^- e^- \text{str}^r$$

in which a^- cells require nutrient A, b^- cells require nutrient B, and so forth. At various times after mixing of the cells, samples are agitated violently and then plated on a series of media that contain streptomycin and different combinations of the five substances A through E (in each medium, one of the five is left out). Colonies that form on the medium lacking A are $a^+ \text{str}^r$, those growing without B are $b^+ \text{str}^r$, and so forth. All of these data can be plotted on a single graph to give a set of curves, as shown in Figure 8.10A. Four features of this set of curves are notable.

1. The number of recombinants in each curve increases with length of time of mating.
2. For each marker, there is a time (the **time of entry**) before which no recombinants are detected.
3. Each curve has a linear region that can be extrapolated back to the time axis, defining the time of entry of each gene a^+ , b^+ , . . . , e^+ .
4. The number of recombinants of each type reaches a maximum, the value of which decreases with successive times of entry.

The explanation for the time-of-entry phenomenon is the following: All donor cells do not start transferring DNA at the same time, so the number of recombinants increases with time. Transfer begins at a particular point in the Hfr chromosome (the replication origin of F). Genes are transferred in linear order to the recipient, and the time of entry of a gene is the time at which that gene first enters a recipient in the population. Separation of a mating pair prevents further transfer and limits the number of recombinants seen at a particular time.

The times of entry of the genes used in the mating just described can be placed on a map, as shown in Figure 8.10B. The numbers on this map and the others are genetic distances between the markers, *measured as minutes*

between their times of entry. Mating with another F^- with genotype $b^- e^- f^- g^- h^- str-r$ could be used to locate the three genes f , g , and h . Data for the second recipient would yield a map such as that shown in Figure 8.10C. Because genes b and e are common to both maps, the two maps can be combined to form a more comprehensive map, as shown in Figure 8.10D.

Studies with different Hfr strains (Figure 8.10E) also are informative. It is usually found that different Hfr strains are distinguishable by their origins and directions of transfer, which indicates that F can integrate at numerous sites in the chromosome and in two different orientations. Combining the maps obtained with different Hfr strains yields a composite map that is *circular*, as illustrated in Figure 8.10F. The circularity of the map is a result of the circularity of the *E. coli* chromosome in F^- cells and the multiple points of integration

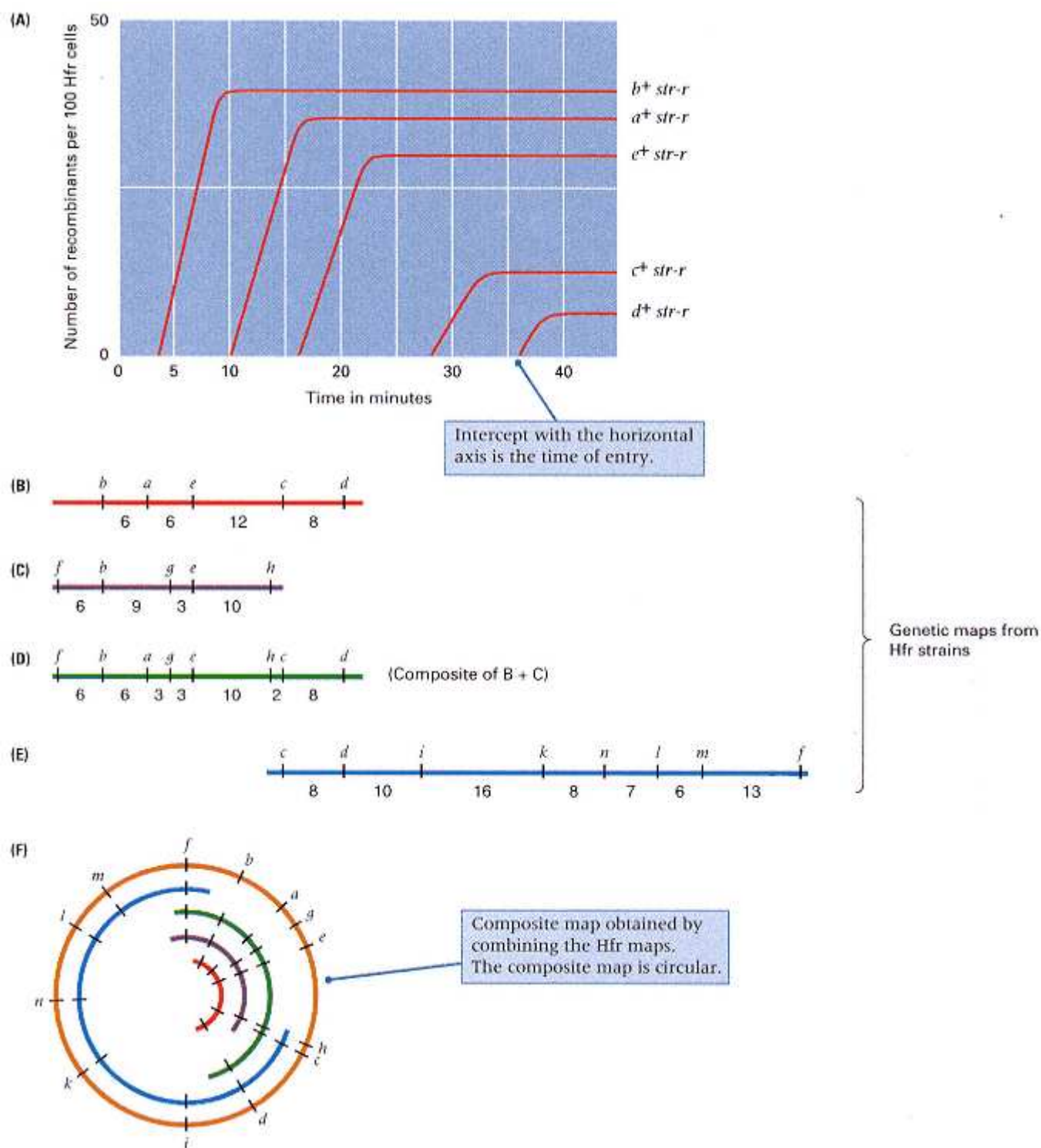


Figure 8.10
Time-of-entry mapping. (A) Time-of-entry curves for one Hfr strain. (B) The linear map derived from the data in part A. (C) A linear map obtained with the same Hfr but with a different F^- strain containing the alleles $b^- e^- f^- g^- h^-$. (D) A composite map formed from the maps in parts B and C. (E) A linear map from another Hfr strain. (F) The circular map (gold) obtained by combining the two maps (green and blue) of parts D and E.

Connection The Sex Life of Bacteria

Joshua Lederberg and Edward L. Tatum 1946
Yale University, New Haven, Connecticut <i>Gene Recombination in Escherichia coli</i>
Since their discovery in the nineteenth century, bacteria were considered "things apart"—unlike other organisms in fundamental ways. Lederberg and Tatum's discovery of what at first appeared to be a conventional sexual cycle was a sensation, completely unexpected. It meant that bacteria could be considered "genetic organisms" along with yeast, <i>Neurospora</i>, <i>Drosophila</i>, and other genetic favorites. For this and related discoveries, Lederberg and Tatum were awarded the 1958 Nobel Prize along with George W. Beadle. In this excerpt, you will note that the authors discuss bacterial recombination as requiring a cell fusion that would bring both parental genomes together. This interpretation shows that it is possible to make exactly the right observation, and realize its significance, but not quite grasp what is really going on. The conclusion that bacterial recombination involved unidirectional transfer was reached much later, after the discovery of Hfr strains and the development of the interrupted-mating technique.
Analysis of mixed cultures of nutritional mutants has revealed the presence of new types which strongly suggest the occurrence of a sexual process in the bacterium <i>Escherichia coli</i>. The mutants consist of strains which differ from their parent wildtype strain K-12, in lacking the ability to synthesize growth factors. As a result of these deficiencies they will only grow in media supplemented with their specific nutritional requirements. In these mutants single nutritional requirements are established as single mutational steps under the influence of x-rays or ultraviolet light. By successive treatments, strains
These types can most reasonably be interpreted as instances of the assortment of genes in new combinations
with several requirements have been obtained. In the recombination studies here reported, two triple mutants have been used, one requiring threonine, leucine and thiamin, the other requiring biotin, phenylalanine and cystine. The strains were grown in mixed culture in complete medium. The cells were washed with sterile water and inoculated heavily into synthetic agar medium, to which various supplements had been added to allow the growth of colonies of various nutritional types. This procedure readily allows the detection of very small numbers of cell types different from the parental forms. The only new types found in "pure" cultures of the individual mutants were occasional forms which had reverted for a single factor, giving strains which required only two of the original three substances. In mixed cultures, however, a variety of types has been found. These include wild-type strains with no growth-factor deficiencies and single mutant types requiring only thiamin or phenylalanine . . . These types can most reasonably be interpreted as instances of the assortment of genes in new combinations. In order that various genes may have the opportunity to recombine, a cell fusion would be required . . . The fusion presumably occurs only rarely, since in the cultures investigated only one cell in a million can be classified as a recombinant type . . . These experiments imply the occurrence of a sexual process in the bacterium <i>Escherichia coli</i>.
Source: <i>Nature</i> 158:558

of the F plasmid; if F could integrate at only one site and in one orientation, the map would be linear.

A great many such mapping experiments have been carried out, and the data have been combined to provide an accurate map of approximately 2000 genes throughout the *E. coli* chromosome. Figure 8.11 is a map of the chromosome of *E. coli* containing a sample of the mapped genes. Both the DNA molecule and the genetic map are circular. The entire chromosome requires 100 minutes to be transferred (it usually breaks first), so the total map length is 100 minutes. In the outer circle, the arrows indicate the direction of transcription and the coding region

included in each transcript. The purple arrowheads show the origin and direction of transfer of a number of Hfr strains. Transfer from HfrC, for example, goes counterclockwise starting with *purE* *acrA* *lac*.

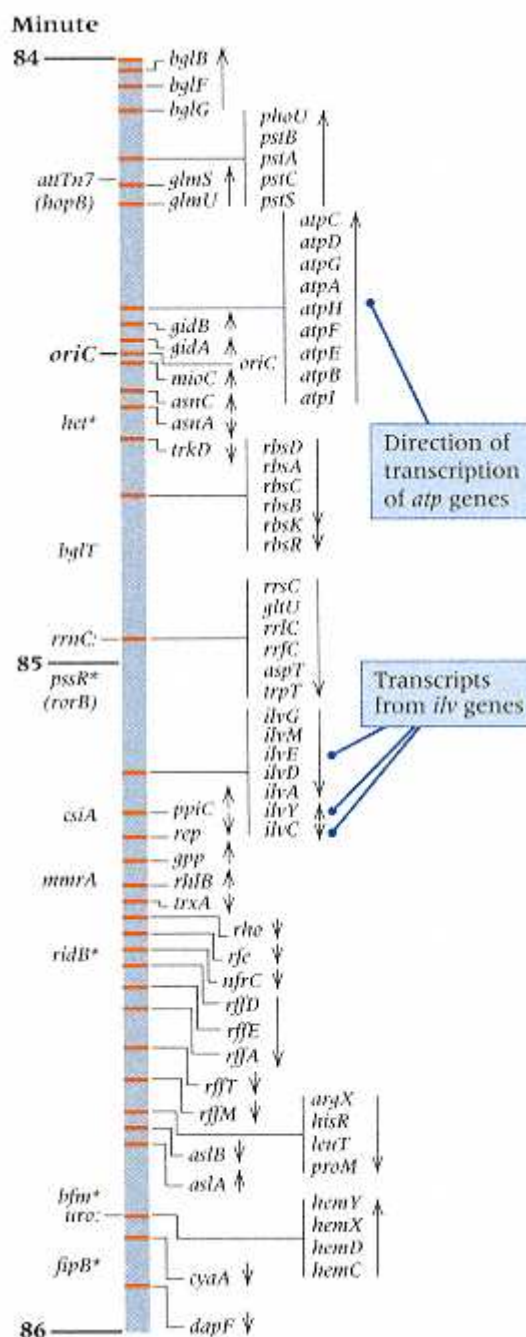


Figure 8.12

A more dense genetic map showing genes in the region between 84 and 86 minutes of the *E. coli* chromosome. Arrows indicate the direction of transcription and length of the messenger RNA for each gene or gene cluster. Symbols to the left of the light blue bar indicate untranscribed DNA sites and genes for which the direction of transcription is not known. For example, *attTn7* is an insertion site for the transposon Tn7.

[From M. K. B. Berlyn, K. B. Low, and K. E. Rudd. 1996. In *Escherichia coli and Salmonella: Cellular and Molecular Biology*, 2nd ed. (F. C. Neidhardt, R. Curtiss III, J. L. Ingraham, E. C. C. Lin, K. B. Low, B. Magasanik, W. Reznikoff, M. Riley, M. Schaechter, and H. E. Umbarger, eds.) Washington, DC: American Society for Microbiology.]

Greater detail of part of the *E. coli* map for the 2 minutes between minutes 84 and 86 is shown in Figure 8.12. Genes and gene clusters transcribed as a unit are indicated by the arrows pointing in the direction of transcription, shown to the right of the blue line. To the left of the line are either regions that are not transcribed or genes in

which the direction of transcription is unknown. The symbol *oriC* represents the origin of DNA replication of the *E. coli* chromosomes.

***F'* Plasmids**

Occasionally, F is excised from Hfr DNA by an exchange between the same sequences used in the integration event. However, in some cases, the excision process is not a precise reversal of integration. Instead, breakage and reunion take place between nonhomologous sequences at the boundary of F and nearby chromosomal DNA (Figure 8.13). Aberrant excision creates a plasmid containing a fragment of chromosomal DNA, which is called an **F' plasmid** ("F prime"). By the use of Hfr strains having different origins of transfer, F' plasmids with chromosomal segments from many regions of the chromosome have been isolated. These elements are extremely useful because they render any recipient cell diploid for the region of the chromosome carried by the plasmid. These diploid regions make possible dominance tests and gene-dosage tests (studies of the effects on gene expression of increasing the number of copies of a gene). Because only a part of the genome is diploid, cells containing an F' plasmid are **partial diploids**, also called **merodiploids**. Examples of genetic analysis using F' plasmids will be offered in Chapter 11 in a discussion of the *E. coli lac* genes.

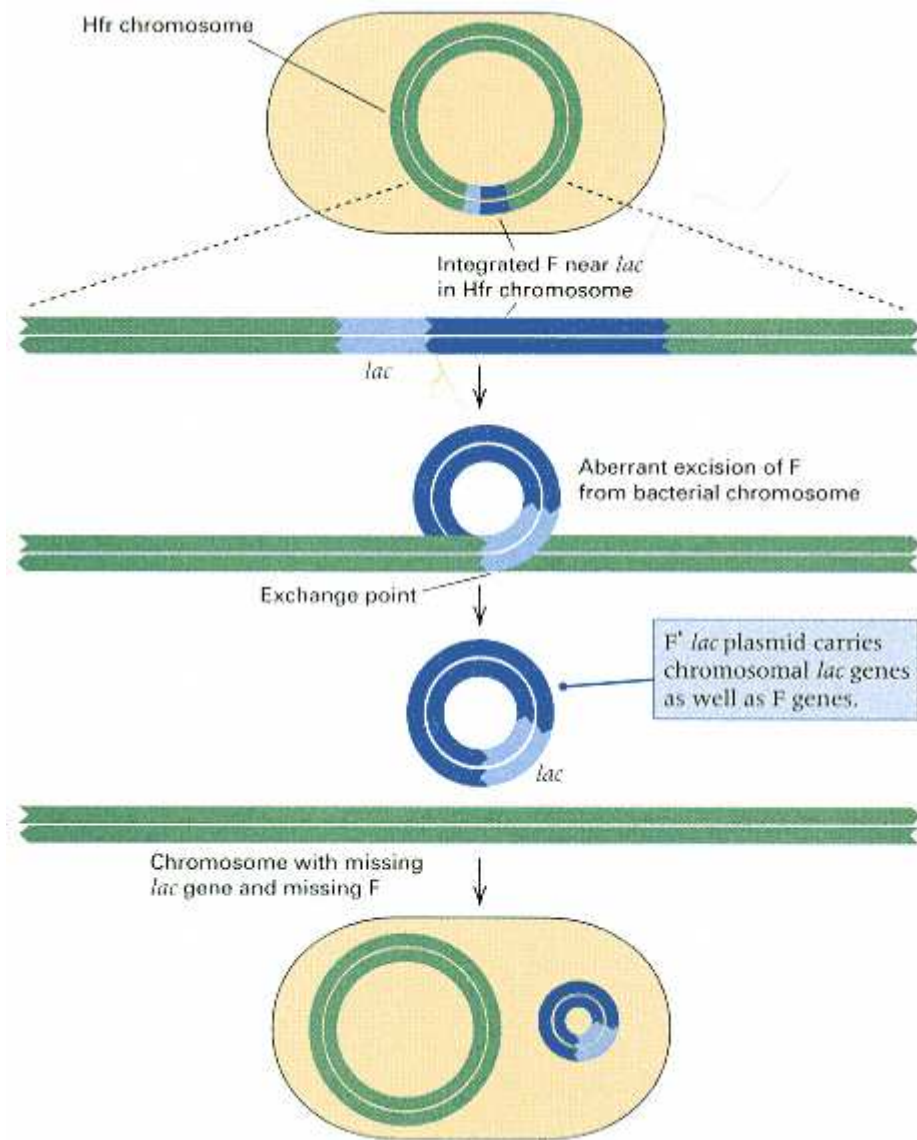


Figure 8.13
Formation of an F' *lac* plasmid by aberrant excision of F from an Hfr chromosome.
Breakage and reunion are between nonhomologous regions.

8.5— Transduction

In the process of **transduction**, a bacterial DNA fragment is transferred from one bacterial cell to another by a phage particle containing the bacterial DNA. Such a particle is called a **transducing phage**. Two types of transducing phages are known—generalized and specialized. A **generalized transducing phage** produces some particles that contain only DNA obtained from the host bacterium, rather than phage DNA; the bacterial DNA fragment can be derived from *any* part of the bacterial chromosome. A **specialized transducing phage** produces particles that contain both phage and bacterial genes linked in a single phage DNA molecule, and the bacterial genes are obtained from a *particular* region of the bacterial chromosome. In this section, we consider *E. coli* phage P1, a wellstudied generalized transducing phage. Specialized transducing particles will be discussed in Section 8.7.

During infection by P1, the phage makes a nuclease that cuts the bacterial DNA into fragments. Single fragments of bacterial DNA comparable in size to P1 DNA are occasionally packaged into phage particles in place of P1 DNA. The positions of the nuclease cuts in the host

chromosome are random, so a transducing particle may contain a fragment derived from any region of the host DNA. A large population of P1 phages will contain a few particles carrying any bacterial gene. On the average, any particular gene is present in roughly one transducing particle per 10^6 viable phages. When a transducing particle adsorbs to a bacterium, the bacterial DNA contained in the phage head is injected into the cell and becomes available for recombination with the homologous region of the host chromosome. A typical P1 transducing particle contains from 100 kb to 115 kb of bacterial DNA.

Let us now examine the events that follow infection of a bacterium by a generalized transducing particle obtained, for example, by growth of P1 on wildtype *E. coli* containing a *leu*⁺ gene (Figure 8.14). If such a particle adsorbs to a bacterial cell of *leu*⁻ genotype and injects the DNA that it contains into the cell, then the cell survives because the phage head contained only bacterial genes and no phage genes. A recombination event exchanging the *leu*⁺ allele carried by the phage for the *leu*⁻ allele carried by the host converts the genotype of the host cell from *leu*⁻ into *leu*⁺. In such an experiment, typically about 1 *leu*⁻ cell in 10^6 becomes *leu*⁺. Such frequencies are easily detected on selective growth medium. For example, if the infected cell is placed on solid medium that lacks leucine, it is able to multiply and a *leu*⁺ colony forms. A colony does not form unless recombination inserted the *leu*⁺ allele.

The fragment of bacterial DNA contained in a transducing particle is large enough to include about 50 genes, so transduction provides a valuable tool for genetic linkage studies of short regions of the bacterial genome. Consider a population of P1 prepared from a *leu*⁺ *gal*⁺ *bio*⁺ bacterium. This sample contains particles able to transfer any of these alleles to another cell; that is, a *leu*⁺ particle can transduce a *leu*⁻ cell to *leu*⁺, or a *gal*⁺ particle can transduce a *gal*⁻ cell to *gal*⁺. Furthermore, if a *leu*⁻ *gal*⁻ culture is infected with phage, both *leu*⁺ *gal*⁻ and *leu*⁻ *gal*⁺ bacteria are produced. However, *leu*⁺ *gal*⁺ colonies do not arise because the *leu* and *gal* genes are too far apart to be included in the same DNA fragment (Figure 8.15A).

The situation is quite different with a recipient cell with genotype *gal*⁻ *bio*⁻, because the *gal* and *bio* genes are so closely linked that both genes are sometimes present in a single DNA fragment carried in a transducing particle—namely, a *gal*⁺ -*bio*⁺ particle (Figure 8.15B). However, not all *gal*⁺ transducing particles also include *bio*⁺, nor do all *bio*⁺ particles include *gal*⁺. The probability of both markers being in a single particle—and hence the probability of simultaneous transduction of both markers (**cotransduction**)—depends on how close to each other the genes are. The closer they are, the greater the frequency of cotransduction. Cotransduction of the *gal*⁺ -*bio*⁺ pair can be detected by plating infected cells on the appropriate growth medium. If *bio*⁺ transductants are selected by spreading the infected cells on a glucose-containing medium that lacks biotin, then both *gal*⁺ *bio*⁺ and *gal*⁻ *bio*⁺ colonies will grow. If these colonies are tested for the *gal* marker, then 12 percent are found to be *gal*⁺ *bio*⁺ and the rest *gal*⁻ *bio*⁺; similarly, if *gal*⁺ transductants are selected, then about 12 percent are found to be *gal*⁺ *bio*⁺. In other words, the **frequency of cotransduction** of *gal* and *bio* is 12 percent, which means that 12 percent of all transducing particles that contain one gene also include the other.

Studies of cotransduction can be used to map closely linked genetic markers by means of three-factor crosses analogous to those described in Chapter 4. That is, P1 is grown on wildtype bacteria and used to transduce cells that carry a mutation of each of three closely linked genes. Cotransductants containing various pairs of wildtype alleles are examined. The gene located in the middle can be identified because its wildtype allele is almost always cotransduced with the wildtype alleles of the genes that flank it. For example, in Figure 8.15B, a genetic marker located between *gal*⁺ and *bio*⁺ will almost always be present in *gal*⁺ *bio*⁺ transductants.

How is the frequency of cotransduction between genes related to their map distance in minutes in the standard *E. coli* map? The theoretical relation depends on the size of a molecule in a transducing phage relative to the size of the entire chromosome. For bacteriophage P1, the distance between

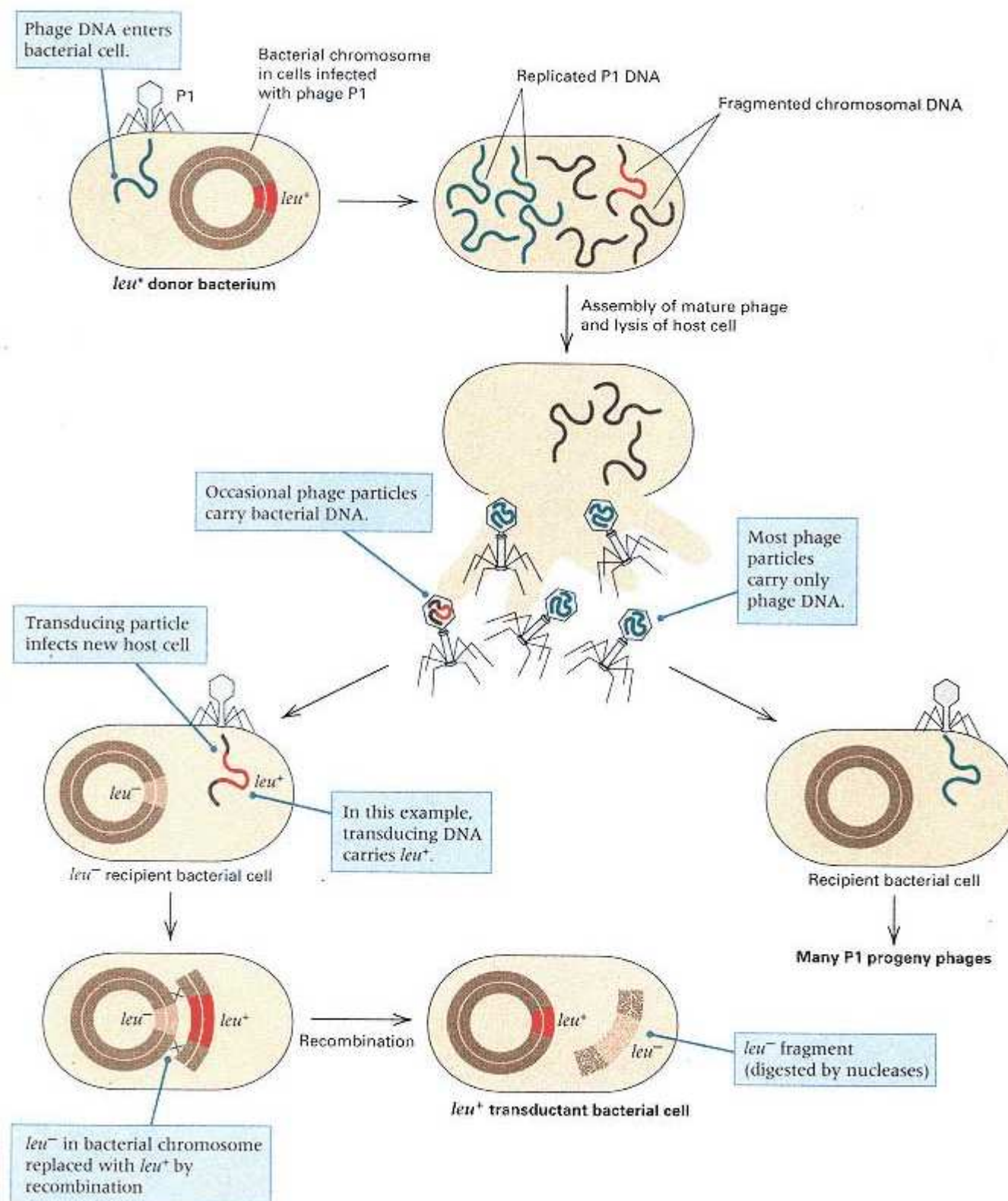


Figure 8.14

Transduction. Phage P1 infects a *leu*⁺ donor, yielding predominantly normal P1 phages with an occasional one carrying bacterial DNA instead of phage DNA. If the phage population infects a bacterial culture, then the normal phages produce progeny phages, whereas the transducing particle yields a transductant. Note that the recombination step requires two crossovers. For clarity, double-stranded phage DNA is drawn as a single line.

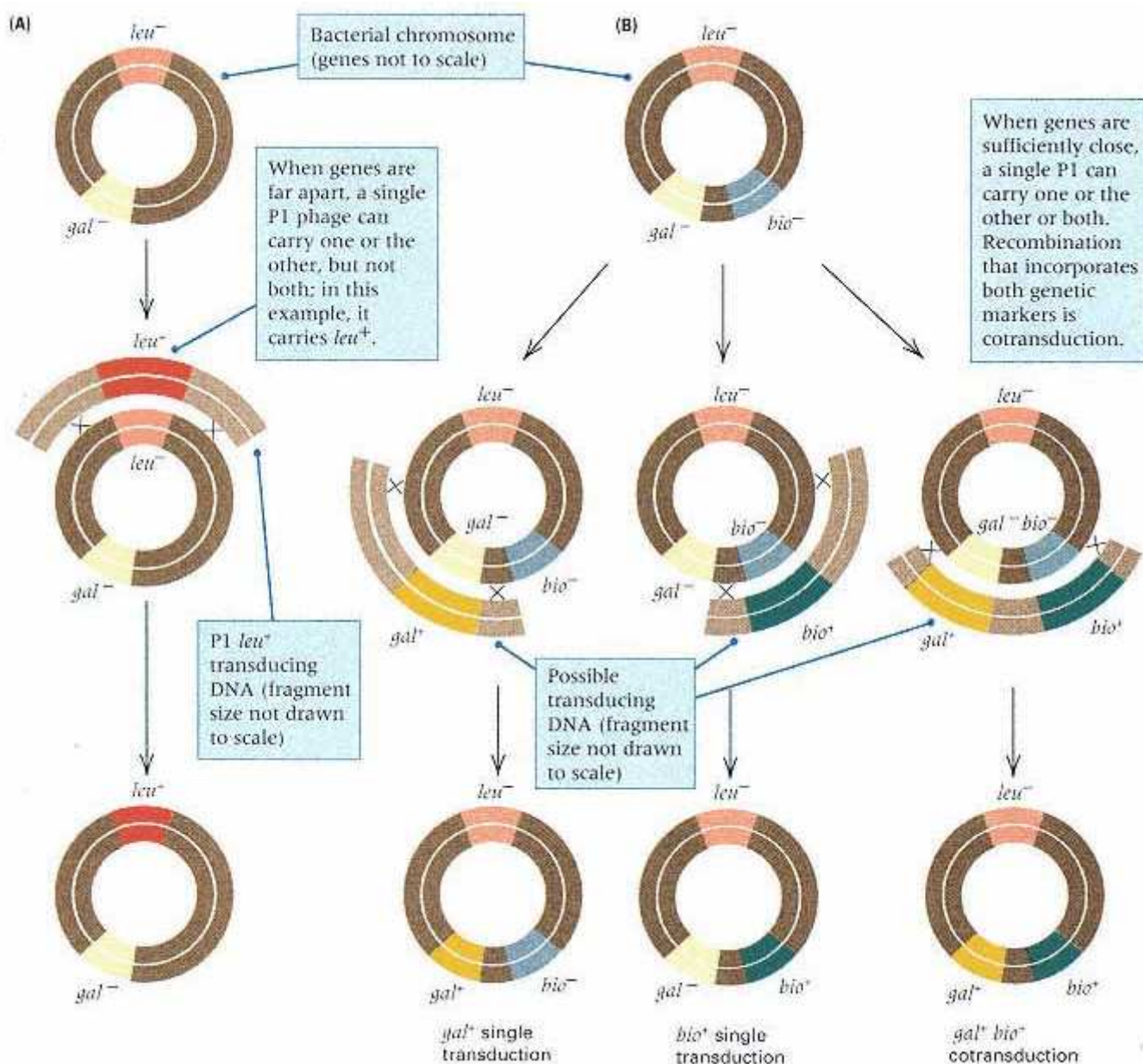


Figure 8.15

Demonstration of linkage of the *gal* and *bio* genes by cotransduction. (A) A P1 transducing particle carrying the *leu*⁺ allele can convert a *leu*⁻ *gal*⁻ cell into a *leu*⁺ *gal*⁻ genotype (but cannot produce a *leu*⁺ *gal*⁺ genotype). (B) The transductants that could be formed by three possible types of transducing particles—one carrying *gal*⁺, one carrying *bio*⁺, and one carrying the linked alleles *gal*⁺ *bio*⁺. The third type results in cotransduction. The distance between *gal* and *bio*, relative to that between *leu* and *gal*, is greatly exaggerated, and the size of the DNA fragment in a transducing particle, relative to the size of the bacterial chromosome, is not drawn to scale.

markers in minutes on a conjugational map is related to the frequency of cotransduction approximately as follows:

$$\text{Map distance in minutes} = 2 - 2 \times (\text{Cotransduction frequency})^{1/3}$$

The implication of this equation is that the frequency of cotransduction falls off very rapidly as the distance between the markers increases. In the formula, the shortest distance for which the cotransduction frequency equals 0 is 2 minutes; hence any markers separated by more than 2 minutes will not be cotransduced by phage P1. For *gal* and *bio*, which have a cotransduction frequency of about 12 percent, the distance equals 1 minute (see Figure 8.11).

Connection Is a Bacteriophage an "Organism"?

Alfred D. Hershey and Raquel Rotman 1948

Washington University,
St. Louis, Missouri

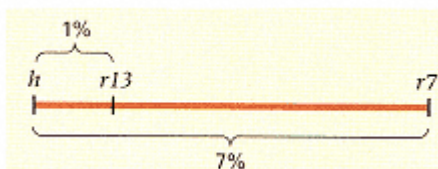
Genetic Recombination Between Host-range and Plaque-type Mutants of Bacteriophage in Single Bacterial Cells

Are bacteriophages "organisms"? Well . . . yes and no. If by the term organism you mean a cell or group of cells that function together as a whole to maintain life and its activities, then bacteriophages are not organisms. By organism you might also mean "anything resembling a living thing in its complexity of structure and function." In this sense bacteriophages are organisms. Even though they require a bacterial host to survive and reproduce, their reproductive cycle conforms in all respects to that of cellular organisms, including DNA as their genetic material, semiconservative replication of DNA, heritable mutations, and, as shown first in this report, a process of recombination. Hershey and Rotman were quite unsure whether phage recombination resulted from physical exchange of DNA molecules or whether, when two molecules recombined, both reciprocal products of the exchange were produced. Later research showed that the exchanges are reciprocal and that they are physical exchanges.

We have shown that any two of several mutants of the bacterial virus T2 interact with each other, in bacterial cells infected with both, to give rise to wildtype and double mutant genetic recombinants . . . In principle, the experimental technique we have to describe is very similar to genetic crossing. One starts with a pair of mutants, each corresponding to a mutant haploid germ cell differing from wildtype by a different unit change. Bacterial cells are infected with both members of the pair, and during viral growth the pair interact to produce viral progeny corresponding to germ cells of a new generation, but now including some individuals differing from wildtype

The analogy to other genetic recombination is obvious, and it is natural to look for a common mechanism.

by both unit changes, and other individuals differing from wildtype not at all. The analogy to other genetic recombination is obvious, and it is natural to look for a common mechanism . . . The crosses between *h* and *r* mutants yield the linkage system shown below.



In the diagram, *h* refers to bacterial host range and *r13* and *r7* to two different, rapidly lysing mutants. The data show that the *h* locus is very closely linked to *r13* (less than 1 percent of wildtype), and that the linkage relation between *r13* and *r7* is 6 percent . . . The results further show that the two recombinants [wildtype and double mutant] appear in equal numbers in any one cross, and that pairs of reverse crosses yield equal numbers of recombinants. It is these relations that increase the resemblance to simple types of Mendelian segregation . . . A hypothesis is proposed according to which one visualizes genetic interaction not between two viral particles, but between two sets of independently multiplying chromosome-like structures. Genetic exchange occurs either by reassortment of these structures, or by something like crossing-over between homologous pairs, depending on the structural relation between the genetic factors concerned. The interpretation made brings the linkage relations into superficial agreement with the requirements of linear structure, but there is little evidence that the

genetic exchanges are reciprocal, and accordingly little evidence that they are material exchanges.

Source: *Genetics* 34: 44–71

8.6—

Bacteriophage Genetics

The life cycles of phages fit into two distinct categories—the lytic and the lysogenic cycles. In the **lytic cycle**, phage nucleic acid enters a cell and replicates repeatedly, the bacterium is killed, and hundreds of phage progeny result (see Figure 8.3). All phage species can undergo a lytic cycle; a phage capable *only* of lytic growth is called **virulent**. In the alternative, **lysogenic cycle**, no progeny particles are produced, the bacterium survives, and a phage DNA molecule is transmitted to each bacterial daughter cell. In most cases, transmission of this kind is accomplished by integration of the phage chromosome into the bacterial chromosome. A phage capable of such a life cycle is called **temperate**.

Several phage particles can infect a single bacterium, and the DNA of each

particle can replicate. In a multiply infected cell in which a lytic cycle is underway, there are genetic exchanges between phage DNA molecules. If the infecting particles carry different mutations as genetic markers, then recombinant phages result. Recombination in phage may be **general recombination**, which means that it can take place anywhere along two homologous DNA molecules. General recombination will be considered in this section. In temperate phages there is a second type of recombination called **site-specific recombination** because it takes place only between a particular site in the phage and a particular site in the bacterial chromosome. The end result of site-specific recombination is not the production of recombinant phage chromosomes but the joining of phage and bacterial DNA molecules. This process is described in Section 8.7.

Plaque Formation and Phage Mutants

Phages are easily detected because in a lytic cycle, an infected cell breaks open (**lysis**) and releases phage particles to the growth medium. The formation of plaques can be observed as outlined in Figure 8.16A. A large number of bacteria (about 10^8) are

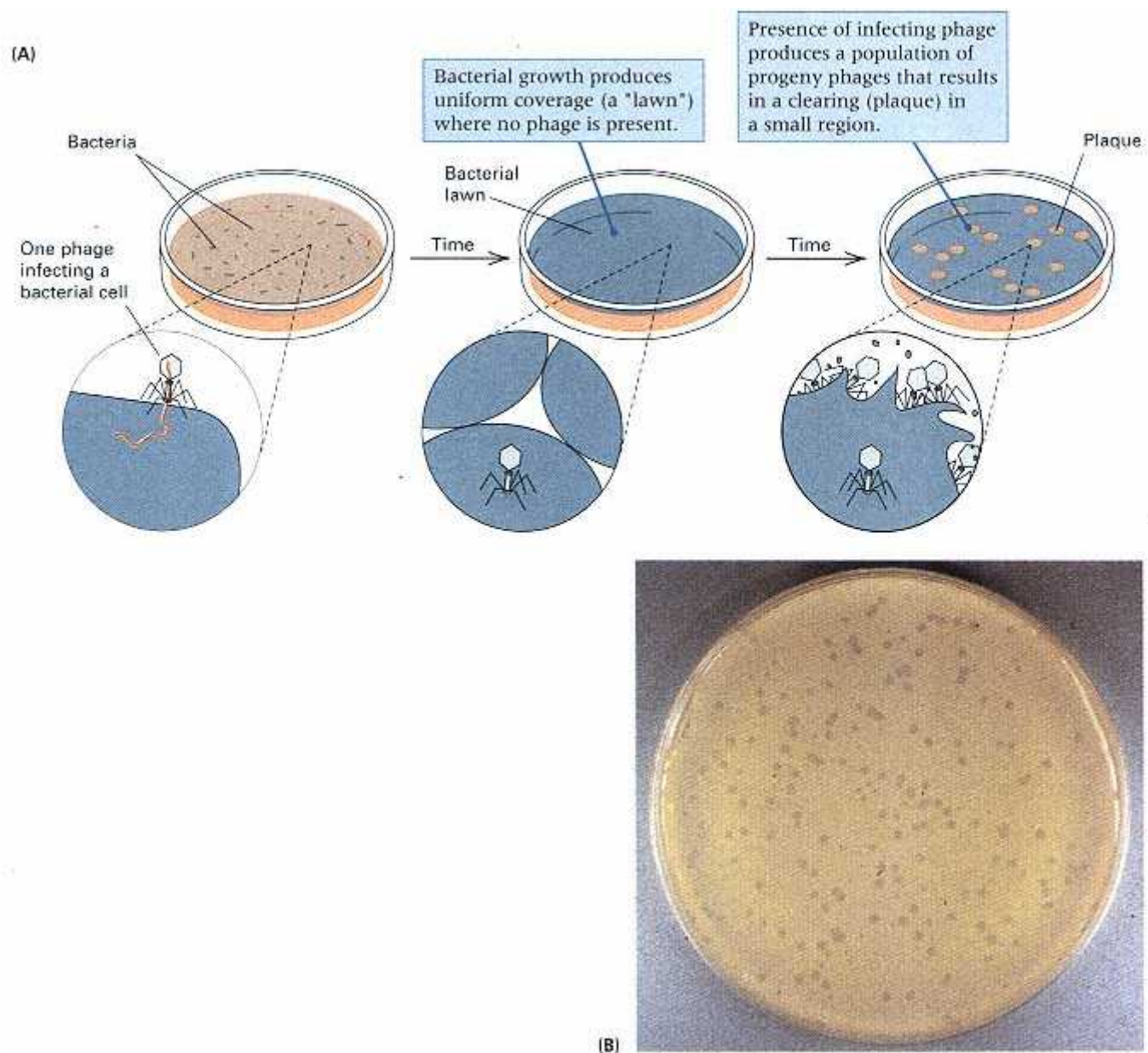


Figure 8.16

Plaque formation. (A) In the absence of a phage, bacterial cells grow and form a translucent lawn. Bacterial cells deposited in the vicinity of a phage are infected and lyse. Progeny phages diffusing outward from the original site infect other cells and cause their lysis. Because of phage infection and lysis, no bacteria can grow in a small region around the site of each phage particle originally present in the medium. The area devoid of bacteria remains transparent and is called a plaque. (B) Large plaques in a lawn of *E. coli* formed by infection with a mutant of bacteriophage λ . Each plaque results from an initial infection by a single bacteriophage.

placed on a solid medium. After a period of growth, a continuous turbid layer of bacteria results. If a phage is present at the time the bacteria are placed on the medium, it adsorbs to a cell, and shortly afterward, the infected cell lyses and releases many phages. Each of these progeny adsorbs to a nearby bacterium, and after another lytic cycle, these bacteria in turn release phages that can infect still other bacteria in the vicinity. (Progeny phage remain close to their site of origin because their size prevents diffusion in agar medium.) These cycles of infection continue, and after several hours, the phages destroy all of the bacteria in a localized area, giving rise to a clear, transparent region—a **plaque**—in the otherwise turbid layer of confluent bacterial growth. Phages can multiply only in growing bacterial cells, so exhaustion of nutrients in the growth medium limits phage multiplication and the size of the plaque. Because a plaque is a result of an initial infection by one phage particle, the number of individual phages originally present on the medium can be counted.

The genotypes of phage mutants can be determined by studying the plaques. In some cases, the appearance of the plaque is sufficient. For example, phage mutations that decrease the number of phage progeny from infected cells often yield smaller plaques. Large plaques can be produced by mutants that cause premature lysis of infected cells, so that each round of infection proceeds more quickly (Figure 8.16B). Another type of phage mutation can be identified by the ability or inability of the phage to form plaques on a particular bacterial strain.

Genetic Recombination in Virulent Bacteriophages

If two phage particles with different genotypes infect a single bacterium, then some phage progeny are genetically recombinant. Figure 8.17 shows plaques from progeny of a mixed infection with *E. coli* phage T4 mutants. The *r*⁻ (*rapid lysis*) allele results in large plaques, and the *h*⁻ (*host range*) allele results in clear plaques. The cross is written as

$$\begin{array}{l} r^{-} h^{+} \text{ (large turbid plaque)} \times \\ r^{+} h^{-} \text{ (small clear plaque)} \end{array}$$

Four plaque types can be seen in Figure 8.17. Two—the large turbid plaque and the small clear plaque—correspond to the phenotypes of the parental phages. The other two phenotypes—the large clear plaque

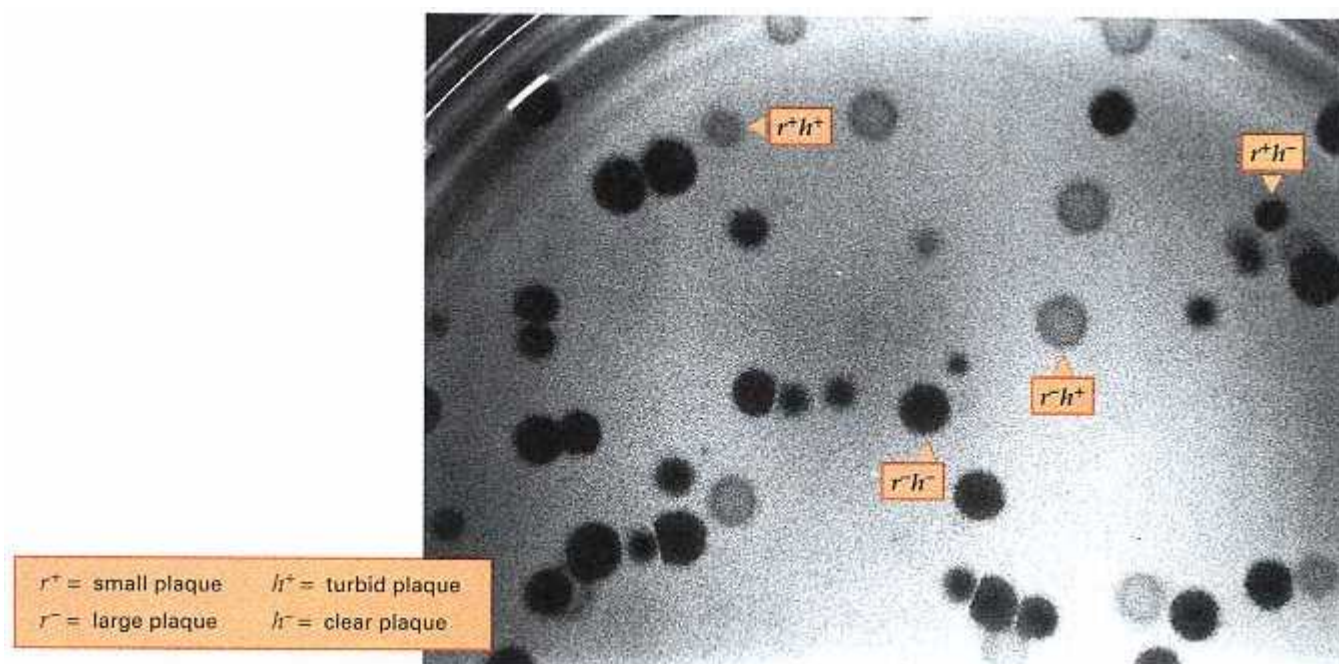


Figure 8.17

A phage cross is performed by infecting host cells with both parental types of phage simultaneously. This example shows the progeny of a cross between T4 phages of genotypes *r*⁻ *h*⁺ and *r*⁺ *h*⁻ when both parental phages infect cells of *E. coli*. The arrowheads point to plaques formed from progeny phages of the indicated genotypes.

[Courtesy of Leslie Smith and John W. Drake.]

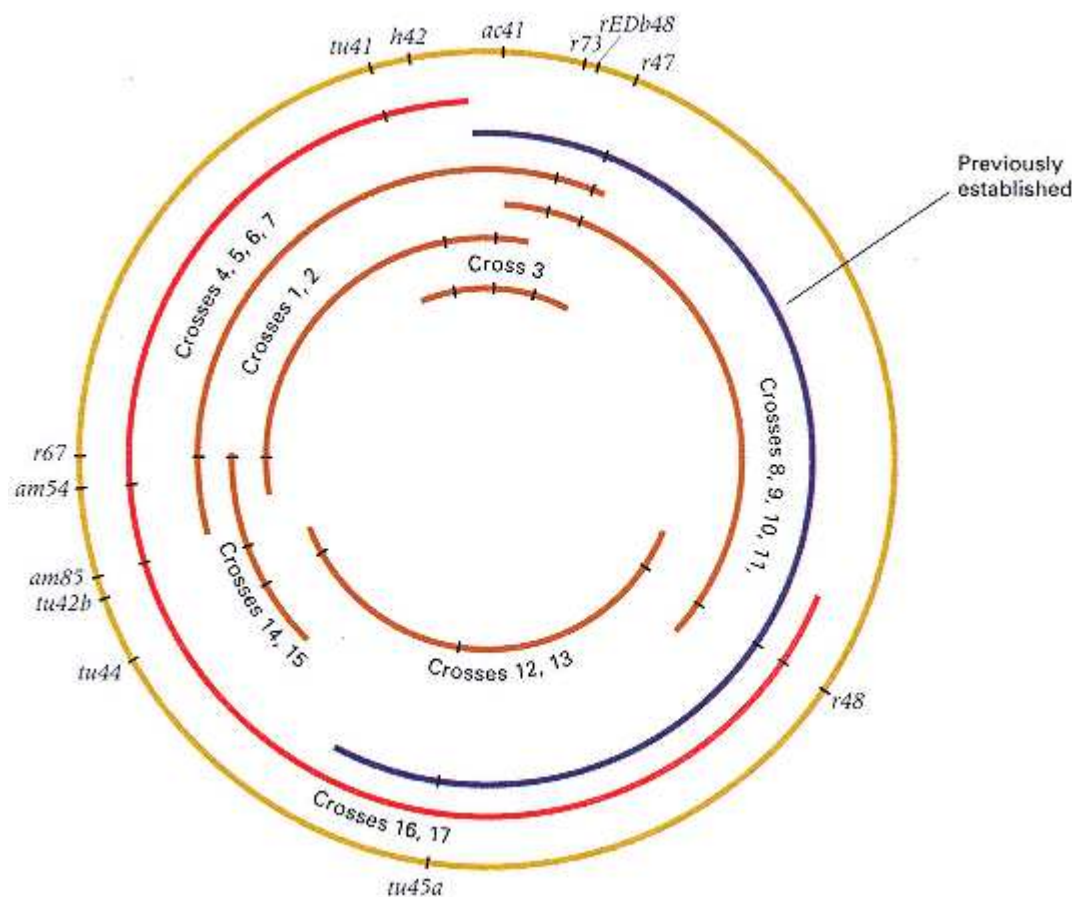


Figure 8.18

Circular genetic map of phage T4. Each arc connects three or four markers that were mapped in the crosses indicated. The circular map results when all data are considered together.

[After G. Streisinger, R. S. Edgar, and G. H. Denhardt. 1964. *Proc. Natl. Acad. Sci. USA* 51: 775.]

and the small turbid plaque—are recombinants that correspond to the genotypes $r^- h^-$ and $r^+ h^+$, respectively. When many bacteria are infected, approximately equal numbers of reciprocal recombinant types are usually found among the progeny phage. In an experiment like that in Figure 8.17, in which each of the four genotypes yields a different phenotype of plaque morphology, the number of each of the genotypes can be counted by examining each of the plaques that is formed. The recombination frequency, expressed as a percentage, is defined as

$$\text{Recombination frequency} = \frac{\text{Number of recombinant phage}}{\text{Total number of phage}} \times 100$$

Recombination frequencies can be used to estimate map distances, just as they are in eukaryotes. Early mapping experiments indicated that mutations in T4 mapped in three separate clusters. However, all three clusters showed linkage to one another. In elegant experiments with three-point crosses, George Streisinger and colleagues demonstrated in 1964 that the genetic map for T4 phage is actually circular.

In each cross, they mapped three or four genetic markers with respect to one another and proceeded systematically through the entire T4 genome, eventually demonstrating all of the linkages shown in Figure 8.18. Many additional genes were identified and mapped later by other researchers (Figure 8.19), and the results were fully consistent with the circular map. In Figure 8.19, the regions indicated in the innermost circle are the three clusters of T4 markers that had initially been identified and mapped. The outermost circle in Figure 8.19 presents a much larger set of markers

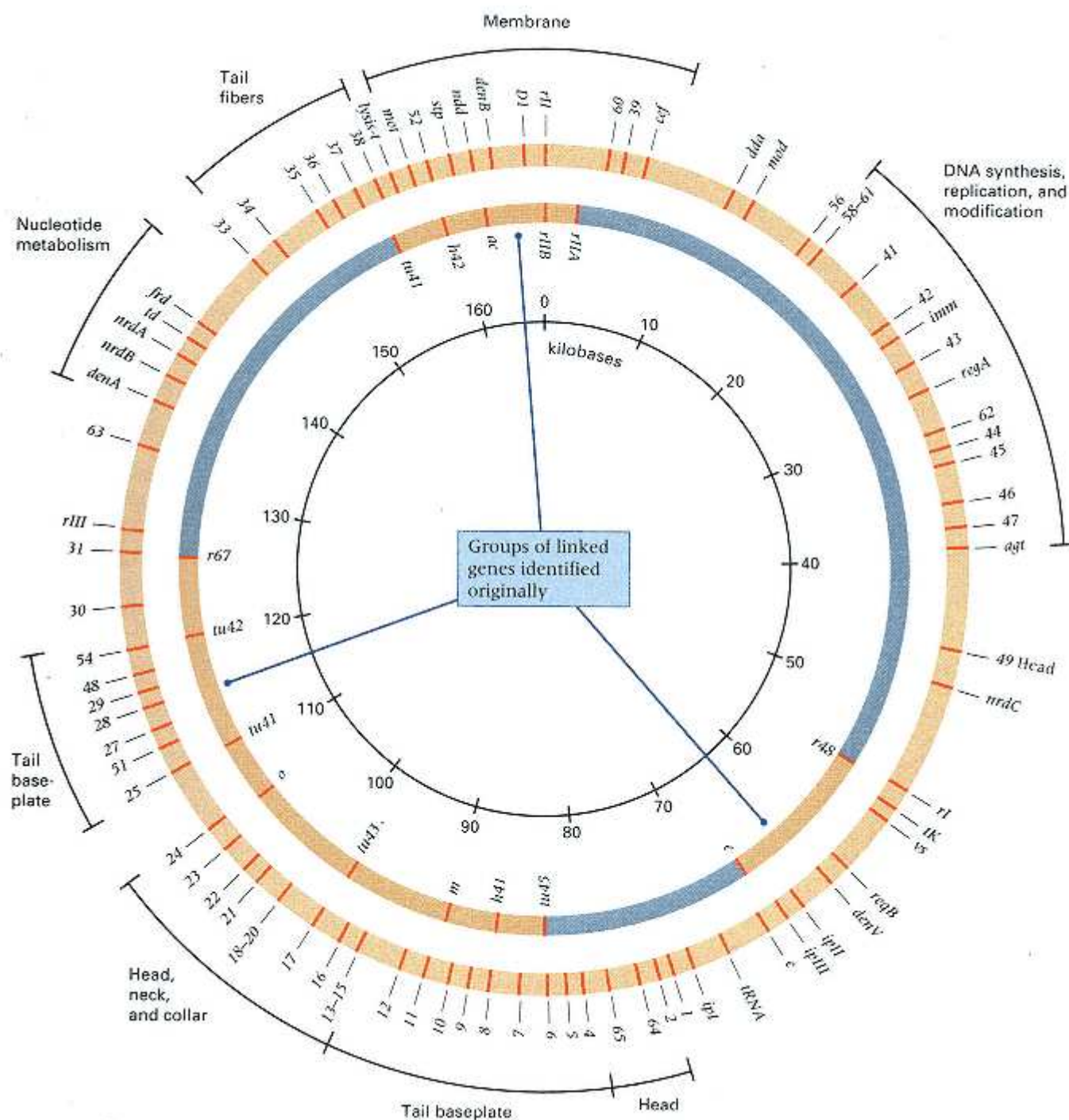


Figure 8.19

Circular genetic map of bacteriophage T4 with additional markers. The middle circle contains the markers mapped and used in the earliest T4 experiments. The units along the innermost circle are kilobases; 0 kb is at a fixed point between the *rIIA* and *rIIB* cistrons. The outer circle contains many genes defined by so-called conditional mutations, which cause a mutant phenotype under one condition (for example, at a high temperature) but not under other conditions (for example, at a low temperature).

Many of the genes show clustering according to their functions.

[After W. B. Wood and H. R. Revel. 1976. *Bacteriol. Rev.* 40:847.]

Connection Artoo

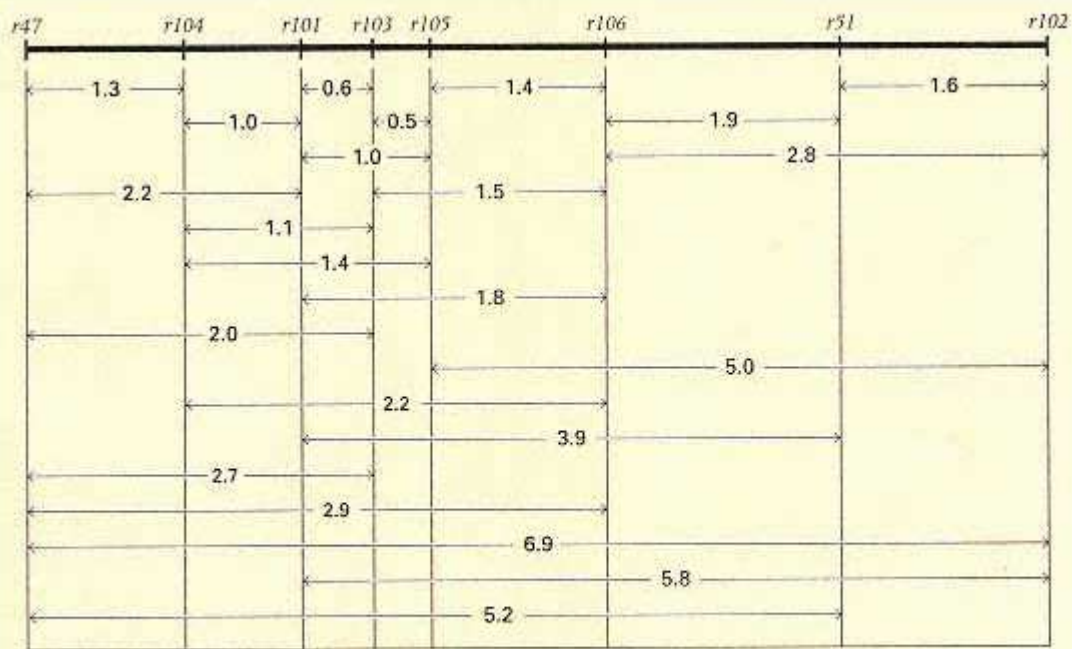
Seymour Benzer 1955

Purdue University,
West Lafayette, Indiana
Fine Structure of a Genetic Region in Bacteriophage

Just as every fan of "Star Wars" can identify the lovable robot Artoo-Detoo (a.k.a. R2-D2), every geneticist can identify the gene rII. The rII gene in bacteriophage T4 was the first experimental example of genetic fine structure. Benzer used the special property that rII mutants cannot grow on E. coli strain K but can grow on strain B to examine recombination between different nucleotides within the rII gene. He demonstrated that the rII gene was divisible by recombination. (It is now known that, in principle, recombination can take place between any adjacent nucleotides.) But if the gene can be subdivided by recombination, then what is a gene, anyway? If two different mutations can undergo recombination whether or not they are in the same gene, then how can one decide, experimentally, whether two different mutations are, or are not, alleles? Benzer realized that the key experimental operation in the definition of allelism was not recombination but rather the complementation test. This is a rare paper with two great ideas in it: recombination within a gene, and the use of the complementation test to determine experimentally whether two different mutations are, or are not, alleles of the same gene.

The phenomenon of genetic recombination provides a powerful tool for separating mutations and discerning their positions along a chromosome. When it comes to very close neighboring mutations, a difficulty arises, since the closer two mutations lie to one another, the smaller is the probability that recombination between them will occur. Therefore, failure to observe recombinant types ordinarily does not justify the conclusion that the two mutations are inseparable . . . A high degree of resolution can best be achieved if there is available a selective feature for the detection of small proportions of recombinants. Such a feature is offered by the case of the *rII* mutants of T4 bacteriophage described in this paper. The wildtype phage produces plaques on either of two bacterial hosts, *Escherichia coli* strain B or strain K, while a mutant of the *rII* group produces plaques only on B. Therefore, if a cross is made between two different *rII* mutants any wildtype recombinants which arise, even in proportions as low as 10^{-8} , can be detected by plating on strain K . . . In this way, a series of eight *rII* mutants of T4 have been crossed with each other. The results of these crosses are given in the figure below.

The distances are only roughly additive; there is some systematic deviation in the sense that a long distance tends to be smaller than the sum of its component shorter ones, [which is accounted for by multiple recombination events] . . . Thus, while all *rII* mutants in this set fall into a small portion of the phage linkage map, it is possible to seriate [order] them unambiguously, and their positions *within* the region are well scattered . . . *Test for allelism.* The functional relatedness of two closely linked mutations causing similar defects may be tested by constructing diploid heterozygotes containing the two mutations . . . The *transform*, containing one of the mutations in each chromosome, may or may not produce the wild phenotype. If it does [complementation], it is concluded that the two mutations in question are located in separate functional units. [They are alleles of different genes.] . . . In order to characterize a unit of genetic "function," it is necessary to define what function is meant . . . On the basis of phenotype tests of *trans* configuration heterozygotes [complementation tests], the *rII* region can be subdivided into two functionally separable segments. Each segment may have the "function" of specifying the sequence of amino acids in a polypeptide chain.



← $r = 1\%$ →

Source: *Proceedings of the National Academy of Sciences of the USA* 41: 344-354

establishing the overall circularity of the genetic map beyond doubt. The T4 genetic map in Figure 8.19 also indicates that genes in T4 show extensive clustering according to the function for which they are required. For example, there is a large cluster of genes for DNA replication in the upper right quadrant, and there is a cluster of genes for phage head components near the bottom.

In view of the circular nature of the T4 genetic map, it came as quite a surprise to find that the DNA molecule in a T4 phage particle is a single *linear* molecule. This discovery was completely unexpected and at first seemed inconsistent with the genetic data. However, the discrepancy was resolved by the finding that the very ends of the phage T4 DNA are duplicated, or have **terminal redundancy**. Because of the redundancy, each molecule is about 2 percent longer than would be expected. As the DNA is replicating inside the cell, recombination between each of the duplicated ends of one T4 genome with homologous sequences in other T4 genomes results in products much longer than can be contained in a T4 head (Figure 8.20). Such concatenated molecules are formed because recombination in the T4 genome is very frequent, averaging about 20 recombination events per chromosome. When the DNA is packaged, it is enzymatically cleaved into "headful" packages consisting of about 102 percent of the minimum length of the T4 genome; hence the duplication at the ends. Because of the headful packaging mechanism, each T4 DNA molecule is terminally redundant. Moreover, in a whole population of DNA molecules present in a phage sample, except for the short terminal redundancy, each of the molecules is also a **circular permutation** of the others. (A molecule is a circular permutation of another if the other sequence can be created by joining its ends and cleaving at an internal position.) Because each T4 molecule is a circular permutation, different molecules begin at different points in the DNA sequence, but they

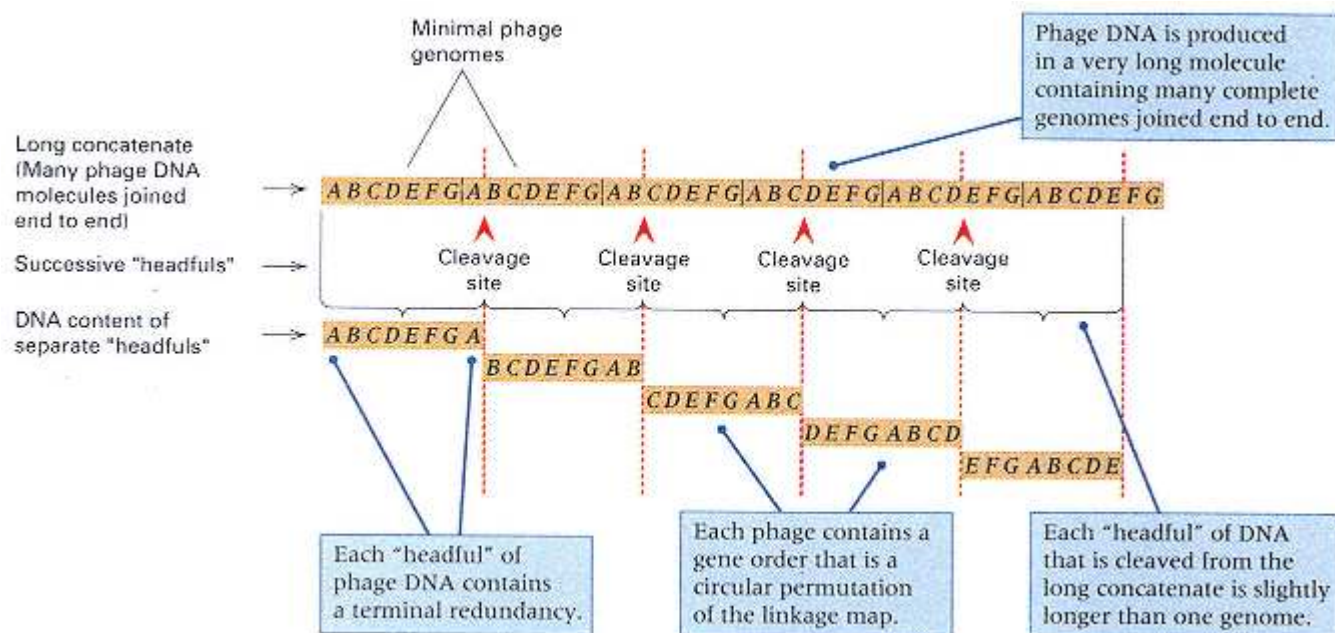


Figure 8.20

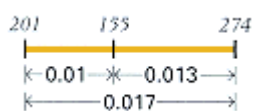
Terminal redundancy and circular permutation of the phage T4 linear DNA molecule. A long concatenate, or series of molecules linked together, is formed inside the cell by genetic recombination. Each successive headful of DNA is slightly longer than unit size and contains a terminal redundancy. Except for the terminal redundancy, the molecules are also circular permutations of each other.

always incorporate a little more than one complete genome. The properties of terminal redundancy and circular permutation account for the circular genetic map.

Fine Structure of the *rII* Gene in Bacteriophage T4

In Section 4.7, we described experiments with *Drosophila* that showed that recombination within a gene could take place. A genotype heterozygous for two different mutant alleles of the same gene could, at low frequency, generate recombinant chromosomes carrying either both mutations or neither mutation. These experiments gave the first indication of intragenic recombination and the first evidence that genes have fine structure. Other studies of genes were performed in the years that followed this experiment, but none could equal the finestructure mapping of the *rII* gene in bacteriophage T4 carried out by Seymour Benzer. Using novel genetic mapping techniques that reduced the number of required crosses from more than half a million to several thousand, Benzer succeeded in mapping 2400 independent mutations in the *rII* locus of phage T4.

Wildtype T4 bacteriophage is able to multiply in *E. coli* strains B and K12(λ) and gives small ragged plaques. Mutations in the *rII* gene of T4 yield large round plaques on strain B but are completely unable to propagate in strain K12(λ). If *E. coli* cells of strain B are all infected with two different *rII* mutants, then recombination between the mutants can be detected, even if the frequency is very low, by taking advantage of the inability of *rII* mutants to grow on K12(λ). Plating the progeny phages on K12(λ) selects for growth of the *rII*⁺ recombinant progeny, because only these recombinants can grow. Furthermore, because very large numbers of progeny phage can be examined (numbers of 10¹⁰ bacteriophages/ml are not unusual), even very low frequencies of recombination can be detected. Typical results for three different *rII* mutations might yield a map like the following:



where the numbers above the line designate *rII* alleles and the numbers below are map distances given as two times the percent of *rII*⁺, namely

$$\text{Map Distance} = \frac{2 \times \left[\frac{\text{Number of plaques per phage on strain K12 } (\lambda)}{\text{Number of plaques per phage on strain B}} \right] \times 100}{1} \times 100$$

Some mutations failed to recombine with several mutations, each of which recombined with the others. These were interpreted to be deletion mutations, because they prevented recombination with two or more "point" mutations known to be at different sites in the gene. Each deletion eliminated a part of the bacteriophage genome, including a region of the *rII* gene. The use of deletions greatly simplified the ordering and mapping of thousands of mutations.

Figure 8.21 depicts the array of deletion mutations used for mapping. Deletion mapping is based on the presence or absence of recombinants; each cross yields a yes-or-no answer and so avoids many of the ambiguities of genetic maps based on frequencies of recombination. In any cross between an unknown "point" mutation (for example, a simple nucleotide substitution) and one of the deletions, the presence of wildtype progeny means that the point mutation is outside of the region missing in the deletion. (The reciprocal product of recombination, which carries the deletion plus the point mutation, is not detected in these experiments.) On the other hand, if the point mutation is present in the region missing in the deletion, then wildtype recombinant progeny cannot be produced. Because each cross clearly reveals whether a particular mutation is within the region missing in the deletion, deletion mapping also substantially reduces the amount of work needed to map a large number of mutations.

The series of crosses that would be made in order to map a particular *rII* mutation is presented in Figure 8.22, in which the large intervals A1 through A6 plus B are the same as those in Figure 8.21. To illustrate the method, suppose that a particular mutation being examined is located in the region denoted A4. This mutation would

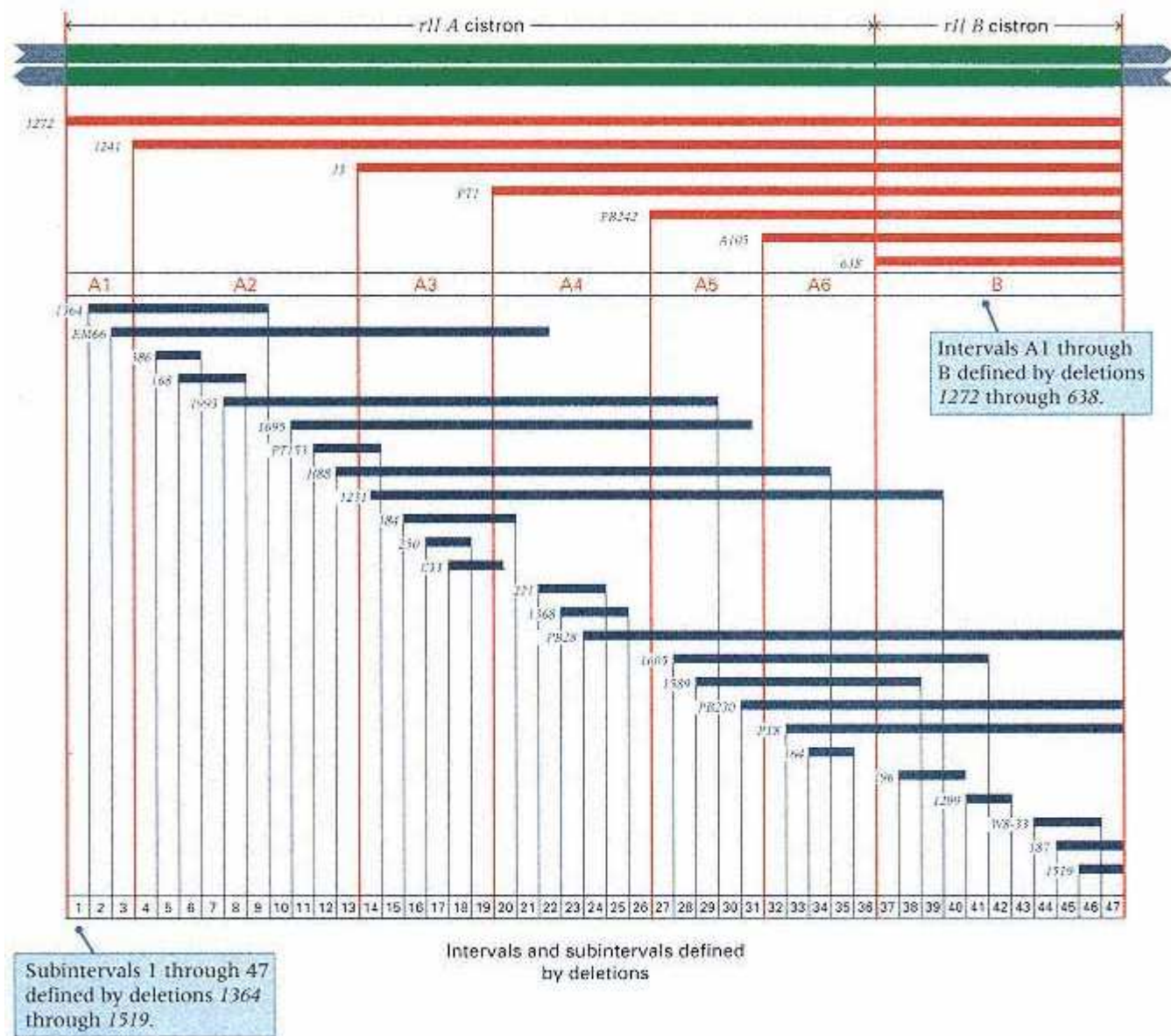


Figure 8.21

The array of deletion mutations used to divide the *rII* locus of bacteriophage T4 into 7 regions and 47 smaller subregions. The extent of each deletion is indicated by a horizontal bar. Any deletion endpoint used to establish a boundary between regions or subregions is indicated with a vertical line.

[After S. Benzer, 1961. *Proc. Natl. Acad. Sci. USA* 47:403.]

fail to yield wildtype recombinants in crosses with the large deletions *r1272*, *r1241*, *rJ3*, and *rPT1*, but it would yield wildtype recombinant progeny in crosses with the large deletions *rPB242*, *rA105*, and *r638*. Conversely, any mutation that yielded the same pattern of outcomes in crosses with the large deletions would be assigned to region A4. Still finer resolution of the genetic map within region A4 is made possible by the set of deletions shown at the bottom of Figure 8.22, the endpoints of which define seven subregions (a through g) within A4. For example, a mutation in region A4 that yields wildtype recombinants with the deletion *r1368* but not with *r221* would be assigned to the c subregion. At the finest level of resolution, mutations

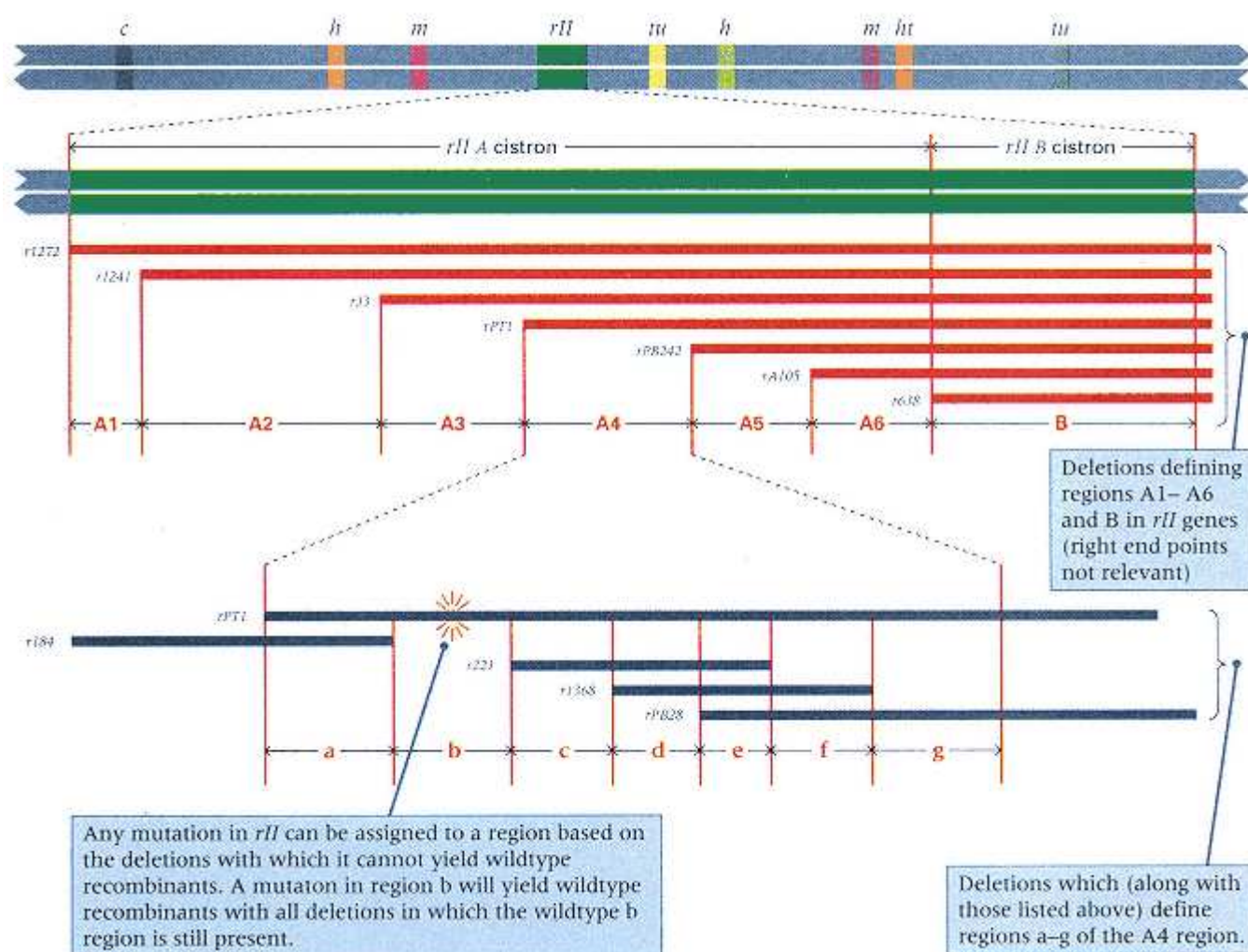


Figure 8.22

The location of *rII* in relation to other genetic markers in a linear version of the phage T4 genetic map. Major subdivisions of the *rII* region are defined by the left ends of seven deletion mutations. All seven deletions extend through the right end of the *rII* region. Genetic material deleted is indicated in red. Further subdivision of the A4 region is made possible by the additional deletions shown at the bottom of the figure. For simplicity, only the deleted regions are shown.

within a subregion are ordered by crosses among themselves. In phage T4, mutant sites that are very close can be separated by recombination, because on the average, 1 percent recombination corresponds to a distance of about 100 base pairs. Hence, any two mutations that fail to recombine can be assigned to the same site within the gene. The genetic map generated for a large number of independent *rII* mutations is given in Figure 8.23.

The *rII* mutation and mapping studies were important because they gave experimental support to these conclusions:

- Genetic exchange can take place within a gene and probably between any pair of adjacent nucleotides.

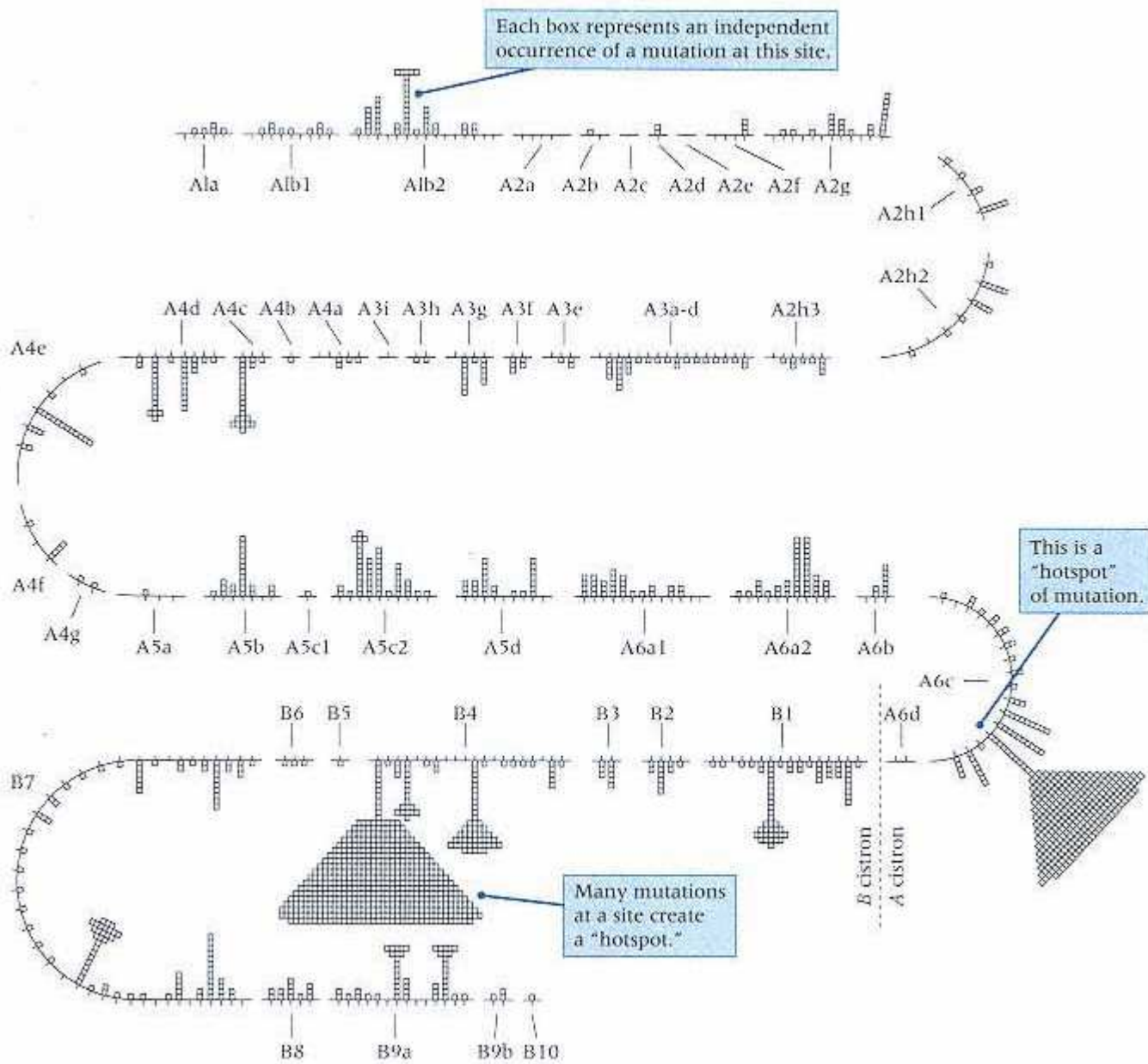


Figure 8.23

Genetic map of the *rII* locus of phage T4. Each small square indicates a separate, independent occurrence of a mutation at the indicated site. The arrangement of sites within each A or B segment is arbitrary.
[After S. Benzer. 1961. *Proc. Natl. Acad. Sci. USA* 47:403.]

- Mutations are not produced at equal frequencies at all sites within a gene. For example, the 2400 *rII* mutations were located at only 304 sites. One of these sites accounted for 474 mutations (Figure 8.23); a site that shows such a high frequency of mutation is called a **hotspot** of mutation. At sites other than hotspots, mutations were recovered only once or a few times.

The *rII* analysis was also important because it helped to distinguish experimentally between three distinct meanings of the word *gene*. Most commonly, the word *gene* refers to a unit of function. Physically, this corresponds to a protein-coding segment of DNA. Benzer assigned the term **cistron** to this unit of function, and the term is still occasionally used. The unit of function is normally defined experimentally by a